



Engineering Mechanics

Lecture 01: Introduction

By
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Indian Institute of Technology Jodhpur

Course Title:	Engineering Mechanics	Course Code:	MEL1010
Department:	Mechanical / CIE	L-T-P [C]:	2-1-0 [3]

Monday: 01:00 PM – 02:00 PM
Tuesday: 01:00 PM – 02:00 PM
Thursday: 01:00 PM – 02:00 PM (Tutorial)

Monday: 04:00 PM – 05:00 PM
Tuesday: 08:00 AM – 09:00 AM
Thursday: 08:00 AM – 09:00 AM (Tutorial)

Attendance: 75% mandatory attendance

Evaluation Policy

- Tutorials: 15 Marks
- Assignment: 15 Marks
 - Quiz 1: 05 Marks
 - Quiz 2: 05 Marks
 - Quiz 3: 05 Marks
 - Project: 7.5 Marks
- Essay/Article writing: 2.5 Marks
- Mid Semester Exam : 20 Marks
 - End Term Exams: 40 Marks

- **Statics [10 Lectures]:** Basic dimensions in Mechanics, Law of dimensional homogeneity, Vector and Scalar Quantities, Elements of vector algebra. Moment of a force about a point/axis, Couple, Free Body Diagram, Equations of Equilibrium, Static indeterminacy, Trusses and Beams, Coulomb Friction, Surface contact friction, Transmission of power through belt. Pappus-Guldinus Theorems, second moments and product of Area, Principal axes.
- **Dynamics of Particles [6 Lectures]:** Kinematics of particles, Velocity and acceleration in terms of path variables, simple relative motion, motion of particle relative to a pair of translating axes Newton's laws of rectangular coordinates/rectilinear translation, cylindrical coordinate/Central force motion. Conservation of Mechanical Energy, Work-energy equations, Center of mass based Kinetic energy, Impulse and Momentum relation of particles, Moment of momentum equations-single particle/system of particles
- **Dynamics of Rigid bodies [12 Lectures]:** Translation/Rotation of rigid bodies, Chasles' theorem, time derivative of vector for different references. Parallel axis theorems, Rotational Pure rotation of a body of revolution about its axis of revolution/combined with translation. Three dimensional rotation, moment of inertia tensor, relation between angular momentum and torque in three dimensions, Motion in non-inertial frames, centrifugal force, Coriolis force/acceleration, rate of change of vector in inertial and rotating frames.

• Reference Books

1. Shames, I., (2003), *Engineering Mechanics*, Prentice Hall
2. Gross, D., Hauger, W. and Schröder J., (2012), *Engineering Mechanics*, Springer
3. **Meriam, J. L., Kraige, L. G., (2002), *Engineering Mechanics*, John Wiley & Sons**
4. Harbola, M K., (2013), *Engineering Mechanics*, CENGAGE Learning

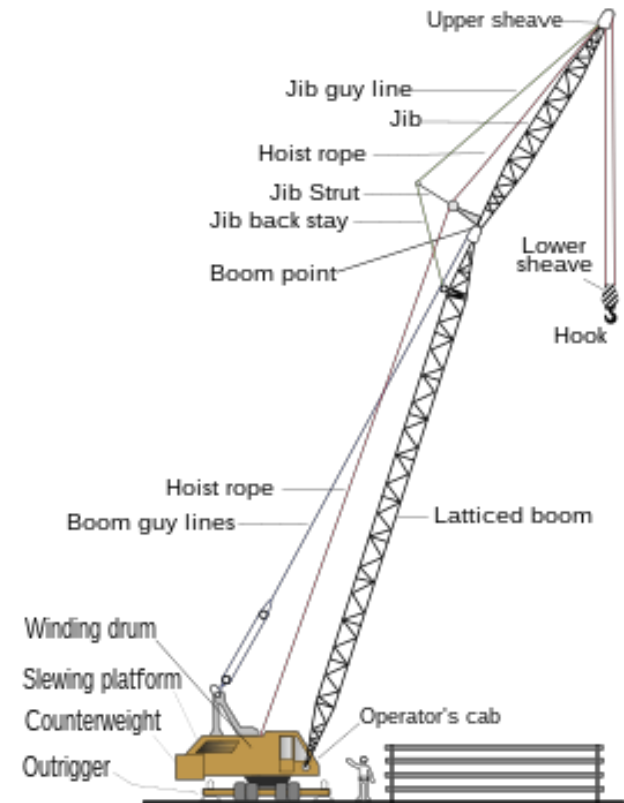
• Self-Learning Material

1. Harbola, M K., *Engineering Mechanics*, NPTEL Course Material, Department of Physics, Indian Institute of Technology Kanpur, <https://nptel.ac.in/courses/115104094/>
2. Lewin, W. H. G., *Physics I: Classical Mechanics* - Fall 1999, MIT, http://videolectures.net/mit801f99_physics_classical_mechanics/

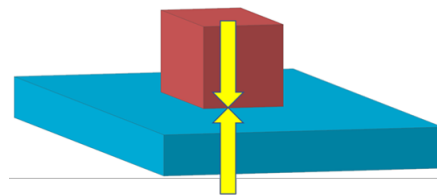
- **Mechanics** is the physical science which deals with the effects of forces on objects.
- State of rest or motion of bodies/ objects, that are subjected to the action of forces
- Sub-divided into three branches
 - a) Rigid-body
 - b) Deformable-body
 - c) Fluid-mechanics

- **Space** is the geometric region occupied by bodies whose positions are described by linear and angular measurements relative to a coordinate system. For 3-D problems, 3 independent coordinates are needed. For 2-D problems, only 2 coordinates are required.
- **Time** is the measure of the succession of events and is a basic quantity in dynamics. Time is not directly involved in the analysis of statics problems.
- **Mass** is a measure of the inertia of a body, which is its resistance to a change of velocity. Mass can also be thought of as the quantity of matter in a body. The mass of a body affects the gravitational attraction force between it and other bodies. Mass appears in many applications in statics.
- **Force** is the action of one body on another. A force tends to move a body in the direction of its action. The action of a force is characterized by its **magnitude**, by the **direction** of its action, and by its **point of application**.

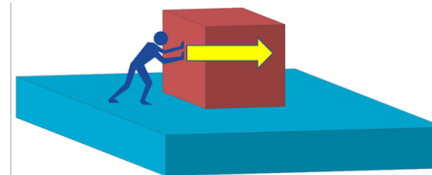
- A **particle** is a body of negligible dimensions. In the mathematical sense, a particle is a body whose dimensions are considered to be near zero so that we may analyze it as a mass concentrated at a point. We often choose a particle as a differential element of a body. We may treat a body as a particle when its dimensions are irrelevant to the description of its position or the action of forces applied to it.
- **Rigid body.** A body is considered rigid when the change in distance between any two of its points is negligible for the purpose at hand. For instance, the calculation of the tension in the cable which supports the boom of a mobile crane under load is essentially unaffected by the small internal deformations in the structural members of the boom.



- A. **Static**: Deals with equilibrium of bodies, either in rest or motion (with a constant velocity)
- B. **Dynamic**: Deals with the accelerated motion of bodies

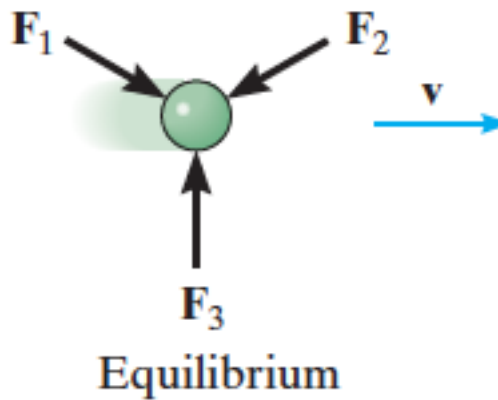


Statics



Dynamics

- A particle originally at rest, or moving in a straight line with constant velocity, tends to remain at **rest or continues to move with *uniform velocity*** if there is **no unbalanced force** acting on it.



Statics

- The **acceleration** of a particle is **proportional to the vector sum of forces** acting on it and is **in the direction of this vector sum**.

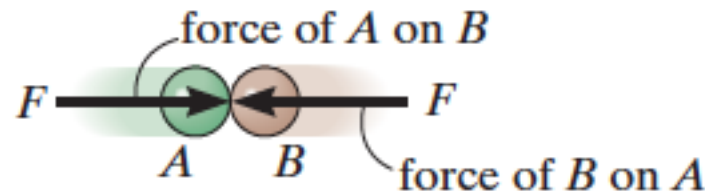
$$F = ma$$



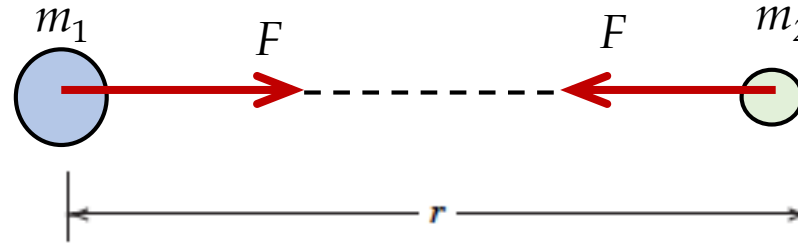
Accelerated motion

Dynamics

- The **forces of action and reaction** between interacting bodies are **equal in magnitude, opposite in direction, and *collinear***.



Constraint Forces



$$F = G \frac{m_1 m_2}{r^2}$$

Where,

F : the mutual force of attraction between two particles

G : a universal constant known as the constant of gravitation, $6.673(10^{-11}) \text{ m}^3/(\text{kg.s}^2)$, based on experimental evidence

$m_1 m_2$: the masses of the two particles

r : the distance between the centres of the particles

$$W = mg, \quad \text{where } g = \left(G \frac{M}{r^2} \right)$$

Where

G : a universal constant known as the constant of gravitation,
 $6.673(10^{-11}) \text{ m}^3/(\text{kg}.\text{s}^2)$.

M : the masses of the earth

m : the masses of the body on the surface of earth

r : distance between the earth's centre and the body

The standard value for gravitational acceleration 'g' is

$$\text{SI units} \quad g = 9.806 \, 65 \text{ m/s}^2$$

$$\text{U.S. units} \quad g = 32.1740 \text{ ft/sec}^2$$

This value is at sea level and at a 45° latitude.

Units

- In mechanics we use **four fundamental quantities** called dimensions.
- These are **length (space), mass, force, and time**.

QUANTITY	DIMENSIONAL SYMBOL
Mass	M
Length	L
Time	T
Force	F

- The **units** used to measure these quantities cannot all be chosen independently
- It must be consistent with **Newton's second law**

SI Units

- The **International System of Units**, abbreviated SI (from the French, *Système International d'Unités*),
- Accepted throughout the world, and is a **modern version of the metric system**.
- By international agreement, SI units **will in time replace** other systems.

Base units

QUANTITY	DIMENSIONAL SYMBOL
Mass	M
Length	L
Time	T
Force	F

SI UNITS		
UNIT		SYMBOL
Base units	kilogram	kg
	meter	m
	second	s
	newton	N

Derived Unit

- **Newton** (N) is derived unit of force
- $\text{Force (N)} = \text{mass (kg)} \times \text{acceleration (m/s}^2\text{)}$
 - $N = \text{kg} \cdot \text{m/s}^2$ (MLT⁻²)
 - Thus, **1 newton is the force required** to give a mass of 1 kg an acceleration of 1 m/s^2 .

QUANTITY	DIMENSIONAL SYMBOL
Mass	M
Length	L
Time	T
Force	F

SI UNITS		
	UNIT	SYMBOL
Base units	kilogram	kg
	meter	m
	second	s
	newton	N

U.S. Customary Units

- The U.S. customary, or British system of units, also called **the Foot-Pound-Second (FPS) system**,
- It has common system in both business and industry in most of the **English-speaking countries**.
- System will in time be **replaced by SI units**,

QUANTITY	DIMENSIONAL SYMBOL
Mass	M
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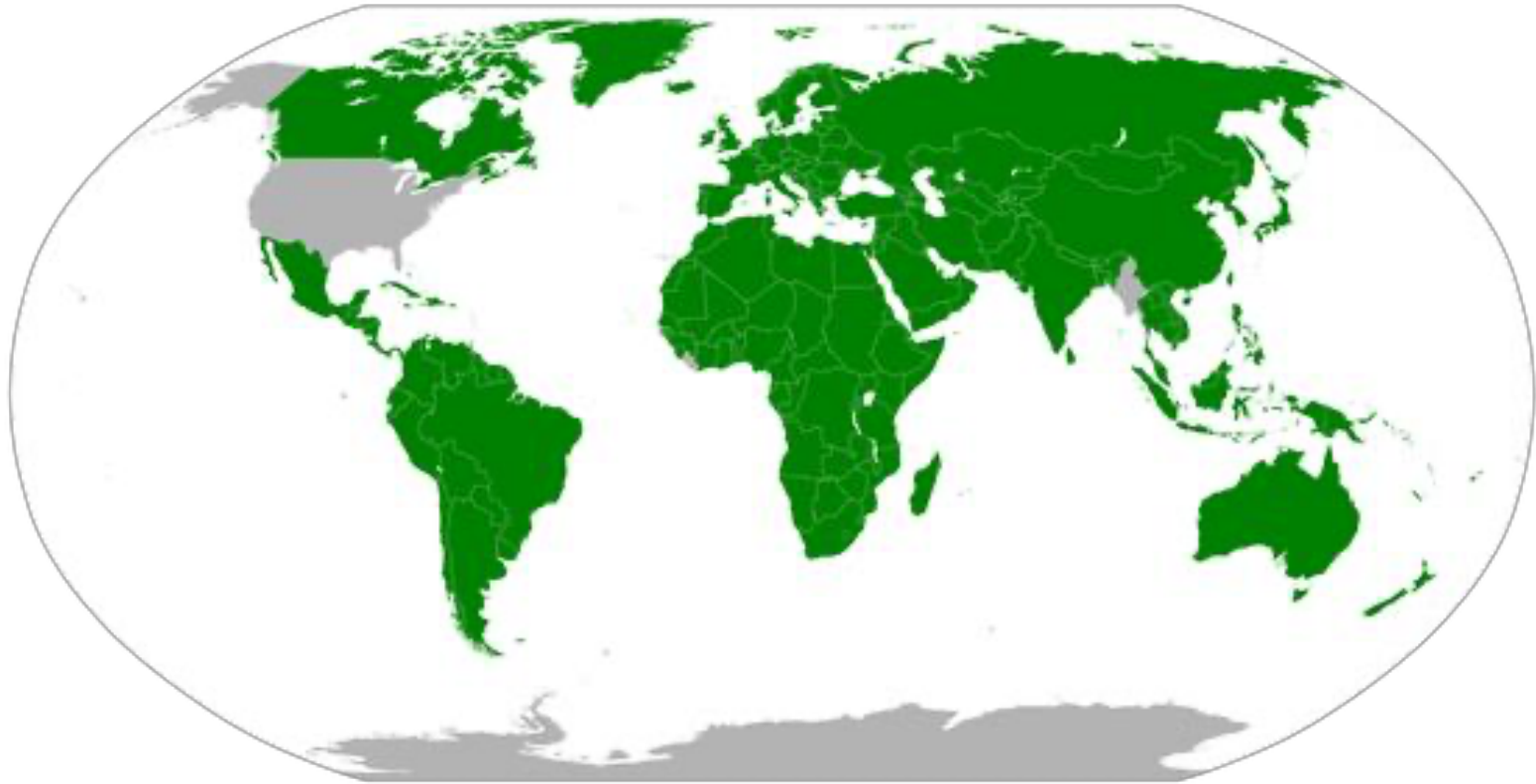
U.S. CUSTOMARY UNITS	
UNIT	SYMBOL
slug	—
Base units	foot
	second
	pound
	ft
	sec
	lb

U.S. Customary Units

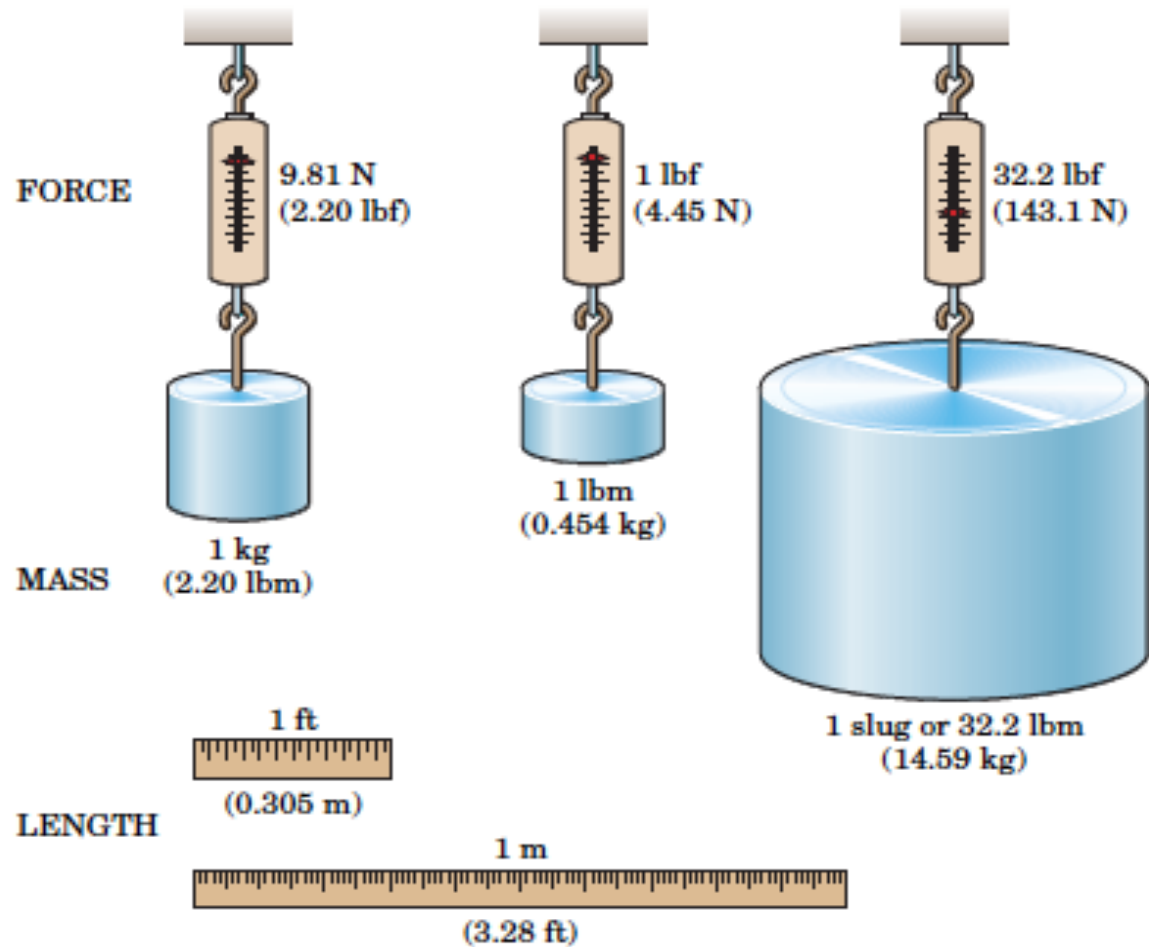
- *pound* is unit of force (basic unit)
- *slug* is unit of mass (derived unit)
- $\text{Force (lb)} = \text{mass (slug)} \times \text{acceleration (ft/sec}^2)$
 - $\text{slug} = \text{lb} \cdot \text{sec}^2 / \text{ft}$ (FL^{-1}S^2)
- Therefore, 1 *slug* is the mass which is given an acceleration of 1 ft/s^2 when acted on by a force of 1 *lb*.

QUANTITY	DIMENSIONAL SYMBOL
Mass	M
Length	L
Time	T
Force	F

U.S. CUSTOMARY UNITS	
UNIT	SYMBOL
slug	—
Base units {	foot ft second sec pound lb



Unit Conversion



Primary Standards

- **Primary standards** for the measurements of **mass**, **length**, and **time** have been established by international agreement using some reference



Mass

- The kilogram is defined as the mass of a **specific platinum iridium cylinder**
- It is kept at the **International Bureau of Weights and Measures** near Paris, France.
- An accurate copy is kept in the United States at the **National Institute of Standards and Technology** (NIST).



The standard kilogram

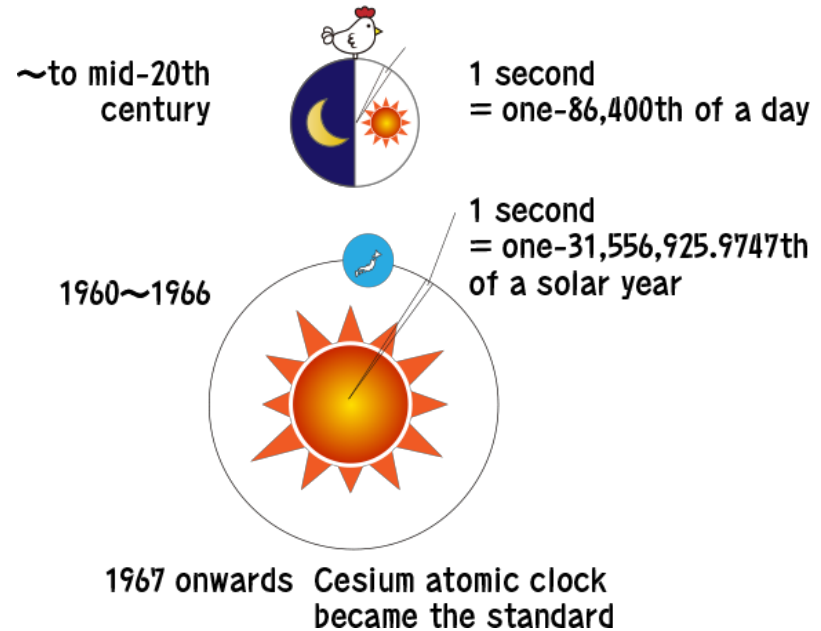
Length

- Meter: Originally, **one ten-millionth** of the **distance from the pole to the equator** along the meridian through Paris
- Later defined as length of a **specific platinum-iridium bar** kept at the **International Bureau of Weights and Measures**.
- The **difficulty of accessing** the bar and **reproducing** accurate measurements
- length for the meter is now defined as **distance traveled by light in a vacuum in $(1/299\,792\,458)$ second**

Time

- The second was originally defined as the fraction $1/(86\ 400)$ of the mean solar day.
- Irregularities in the earth's rotation led to difficulties, and a more accurate and reproducible standard has been adopted.
- Atomic clock: The duration of 9192 631 770 periods of the radiation of a specific state of the Cesium-133 atom.

The changing definition of a second



Problem Solving in Statics

Part A: Formulate the problem

Part B: Develop the solution

Part A: Formulate the problem

- a) State the **given data**.
- b) State the **desired result**.
- c) State your **assumptions** and **approximations**.

Part B: Develop the solution

- (a) **Draw diagrams** that you need to understand the relationships.
- (b) **State the governing principles** to be applied for solution.
- (c) **Make calculations.**
- (d) Check **consistency of calculations** with the accuracy of data.
- (e) **Use consistent units** throughout calculations.
- (f) Ensure that answers are reasonable in terms of **magnitudes, directions, common sense**, etc.
- (g) Draw **conclusions.**

Making Appropriate Assumptions

- While constructing an **idealized mathematical model** for a given engineering problem, certain **approximations** will always be involved.
- It is often necessary to **neglect small distances, angles, or forces** compared with large distances, angles, or forces for simplification.
- **Force** is distributed over a **small area** may be considered it to be a **concentrated force**

Using Graphics

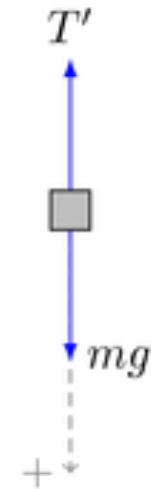
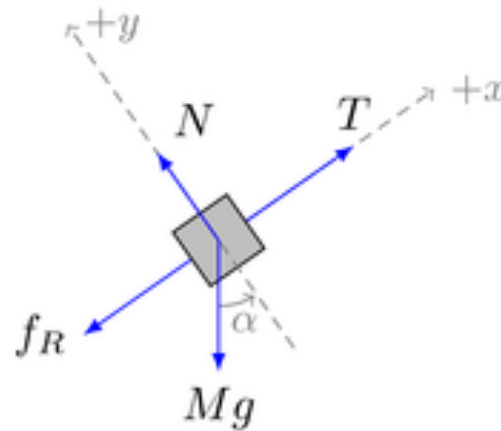
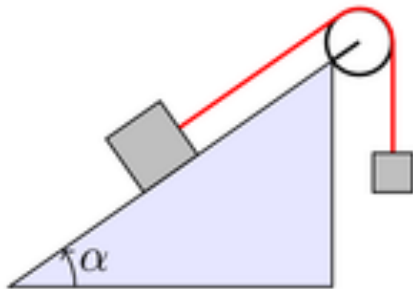
Graphics is an important analytical tool for three reasons:

1. Graphical representation helps in physical interpretation, especially when we must visualize three-dimensional problems.
2. Graphical solution to problems is more easy and practical than mathematical solution and an aid in our thought processes.
3. Charts or graphs are valuable aids for representing results in a form which is easy to understand.

The Free-Body Diagram

- Used in applying the principles of mechanics to **analyse forces acting on a body**.
- it is essential to **isolate the body** from all other bodies to get **complete and accurate account** of all **forces** acting on this
- The diagram of such an isolated body with all external forces acting on it is called a **freebody diagram**.

Free Body Diagram



Numerical Values versus Symbols

- We may use **numerical values to represent quantities**, or use **algebraic symbols and leave the answer as a formula**.
- Numerical Values:
 - Useful when we need to **know the magnitude** of each term.
- Symbols
 - helps to focus on the **connection** between the **physical situation** and its related **mathematical description**.
 - Symbolic solution be used for obtaining **answers to the same type of problem**, but having different units or numerical values.
 - A symbolic solution **enables** us to make a **dimensional check at every step**, which is more difficult when numerical values used

Dimensional Homogeneity

- In any equation representing a physical situation, the **dimensions of every term** on both sides of the equation **must be the same**.
- This property is called dimensional homogeneity

Solution Methods

- Obtain **mathematical solutions by hand**.
- Obtain **graphical solutions** for certain problems.
- Solve problems **by computer** when
 - a **large number of equations** must be solved, a parameter variation must be studied,
 - an **intractable equation** must be solved.

THANK YOU

Engineering Mechanics Lecture- II

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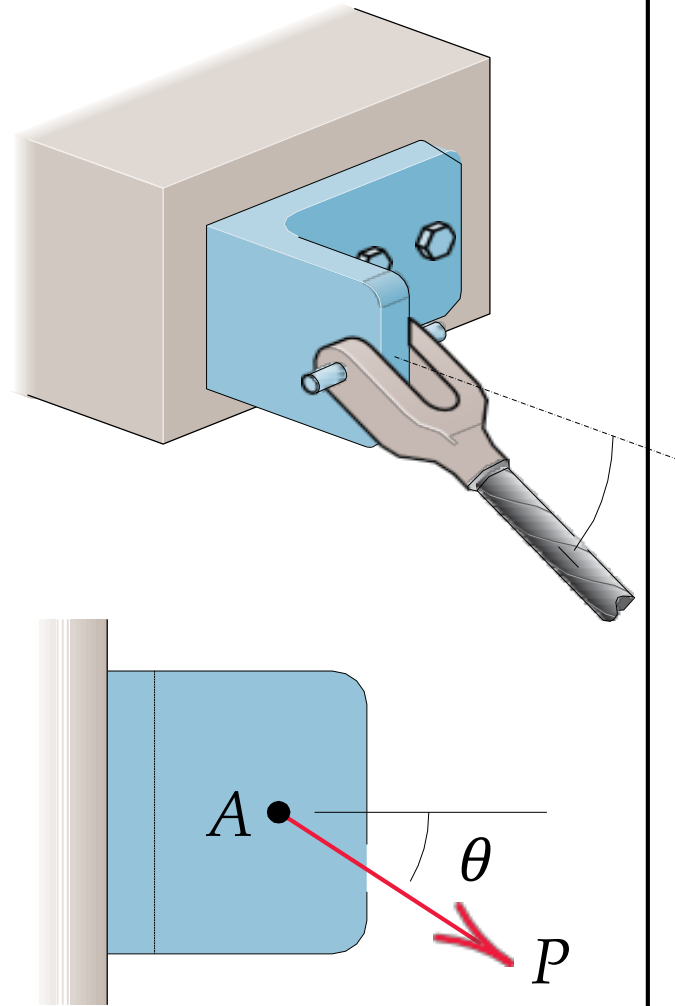
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Force Systems



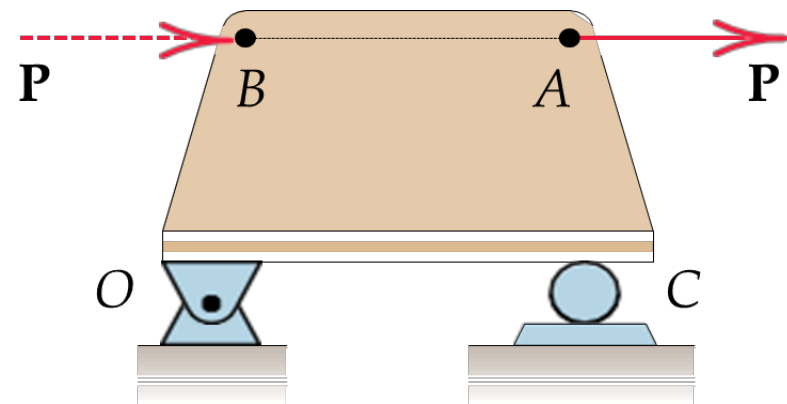
Force

- Force: an action of one body on another
- force is a vector quantity, because its effect depends on the direction as well as on the magnitude of the action
- the complete specification of the action of a force must include its *magnitude*, *direction*, and *point of application*.
- External and Internal Effects



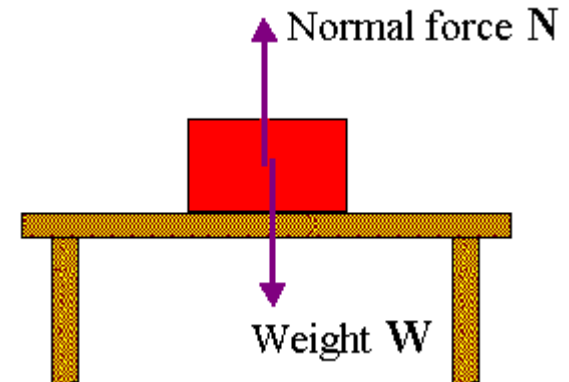
Principle of Transmissibility

- A force may be applied at any point on its given line of action without altering the resultant effects of the force external to the rigid body on which it acts.
- Force may be considered as Sliding vector
- the force $\mathbf{P}$ acting on the rigid plate in may be applied at A or at B or at any other point on its line of action, and the net external effects of $\mathbf{P}$ on the bracket will not change.
- The external effects are the force exerted on the plate by the bearing support at O and the force exerted on the plate by the roller support at C .



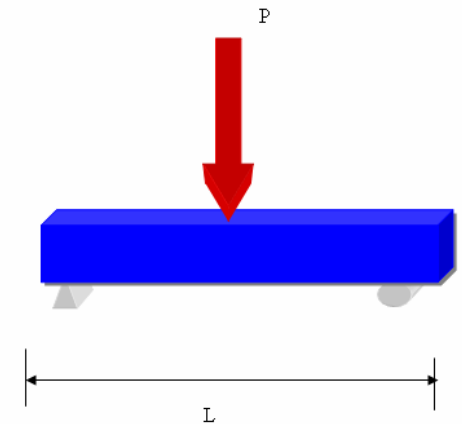
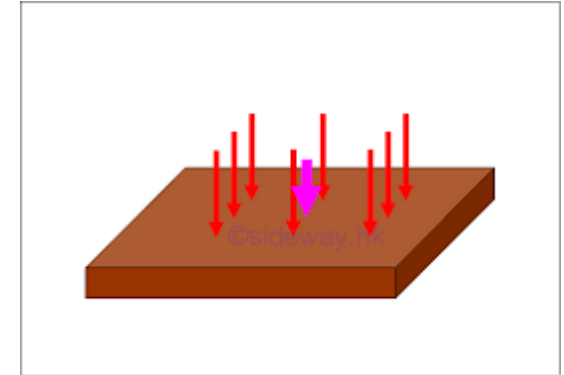
Force Classification

- **Contact Force** is produced by direct physical contact, e.g., force exerted on a body by a supporting surface
- **Body Force:** is generated by virtue of the position of a body within a force field, e.g., weight



Force Classification

- **Distributed force:** Force distributed over a region.
 - Every contact force is actually applied over a finite area and is therefore really a distributed force.
- **Concentrated force:** Force concentrated at a point
 - when the dimensions of the area are very small compared with the other dimensions of the body, we may consider the force to be concentrated at a point with negligible loss of accuracy.

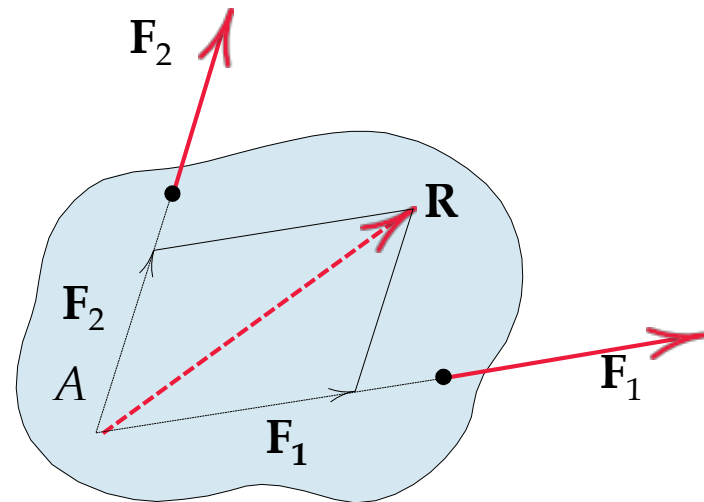
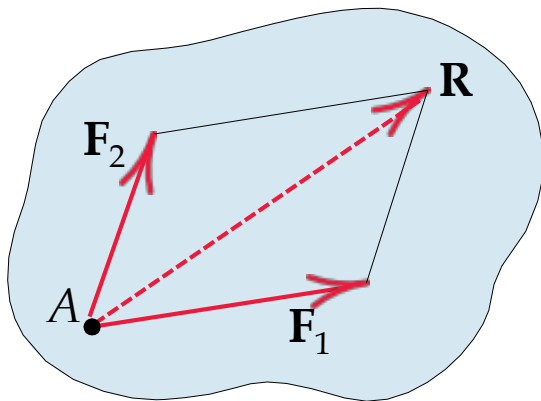


Action and Reaction

- According to Newton's third law, the action of a force is always accompanied by an equal and opposite reaction.
- Distinguish between the action and the reaction in a pair of forces.
- we first isolate the body in question and then identify the force exerted on that body (not the force exerted by the body).
- It is very easy to mistakenly use the wrong force of the pair unless we distinguish carefully between action and reaction.

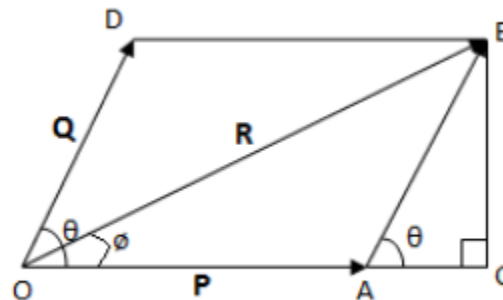
Concurrent Forces

- Two or more forces are said to be concurrent at a point if their lines of action intersect at that point.

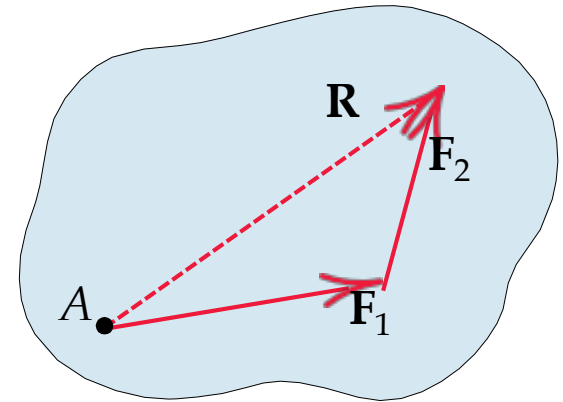
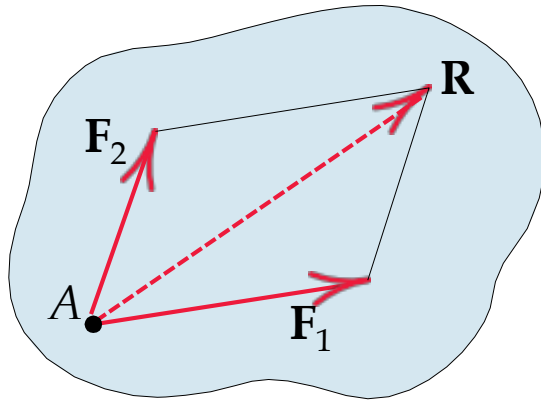


- **PARALLELOGRAM LAW OF FORCES**

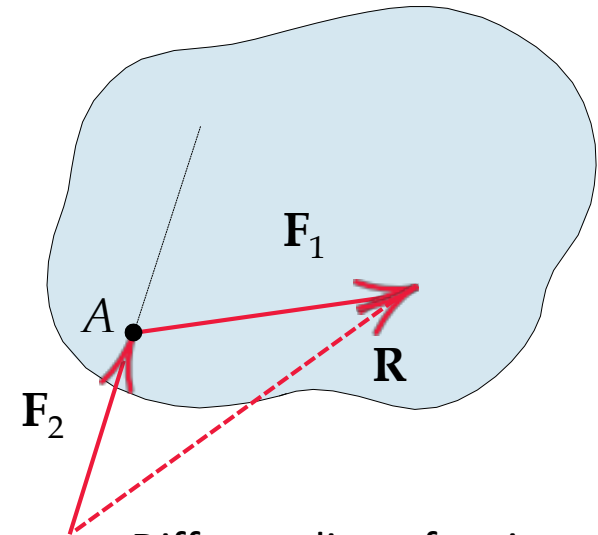
- It states that if two concurrent forces, acting simultaneously on a body be represented in magnitude and direction by the two sides of a parallelogram then their resultant may be represented in magnitude and direction by the diagonal of the parallelogram, drawn from the same point.



Concurrent Forces



Same line of action

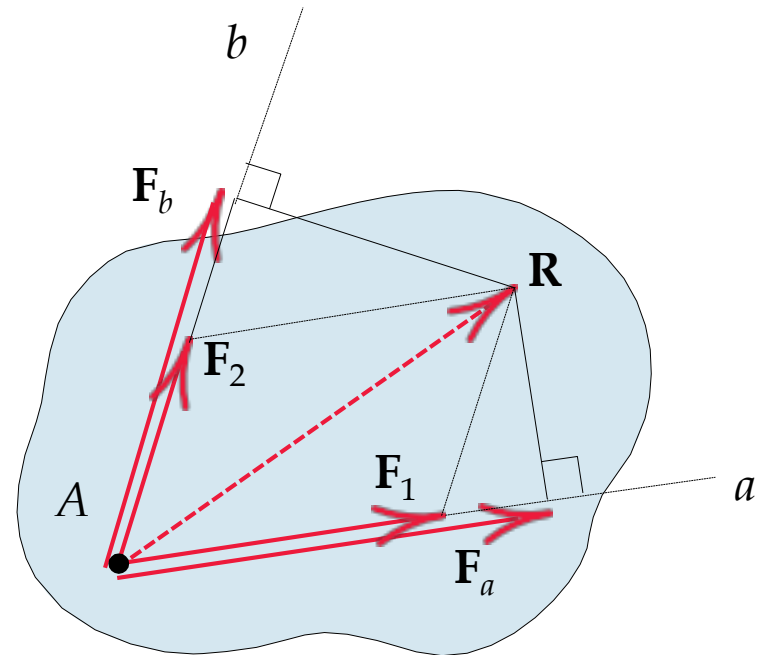


Different line of action

Vector Components

- we often need to replace a force by its vector components in directions which are convenient for a given application.

$$\mathbf{R} = \mathbf{F}_1 + \mathbf{F}_2$$



Components vs Projections

RECTANGULAR COMPONENTS

- The most common two-dimensional resolution of a force vector is into rectangular components.

$$\mathbf{F} = \mathbf{F}_x + \mathbf{F}_y$$

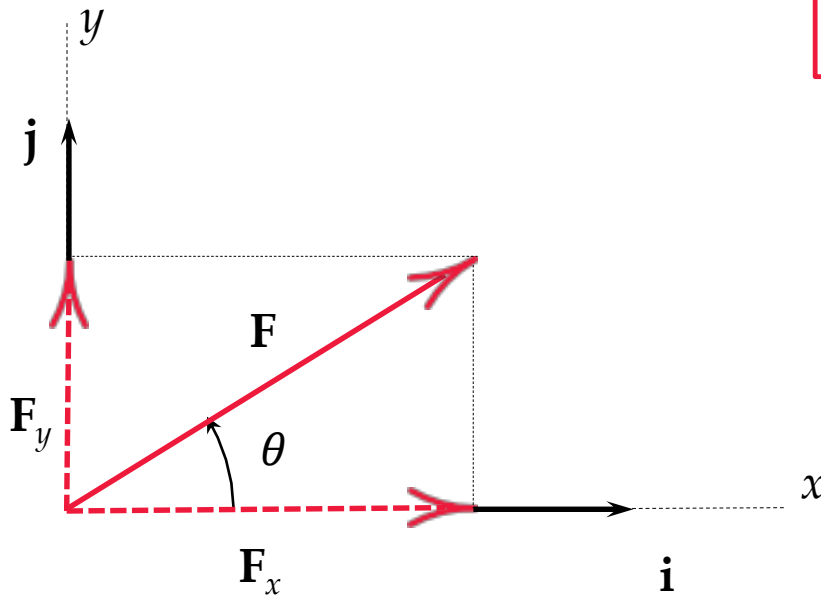
$$\mathbf{F} = F_x \mathbf{i} + F_y \mathbf{j}$$

$$F = \sqrt{(F_x)^2 + (F_y)^2}$$

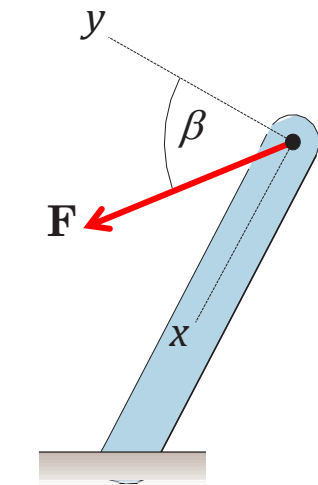
$$F_x = F \cos \theta$$

$$F_y = F \sin \theta$$

$$\theta = \tan^{-1} (F_y / F_x)$$

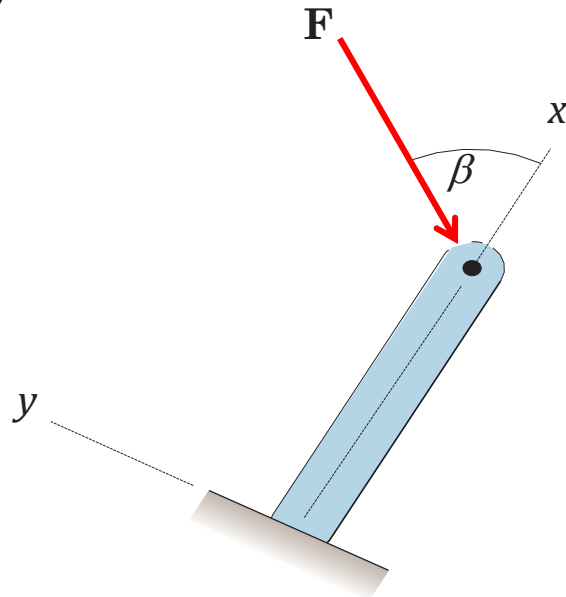


Determining the Components of a Force



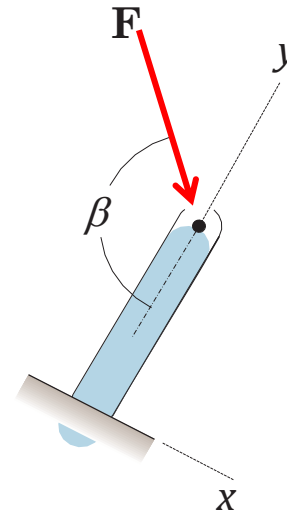
$$F_x = F \sin \beta$$

$$F_y = F \cos \beta$$



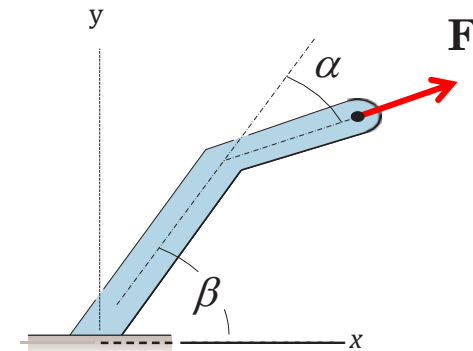
$$F_x = -F \cos \beta$$

$$F_y = -F \sin \beta$$



$$F_x = F \sin (\pi - \beta)$$

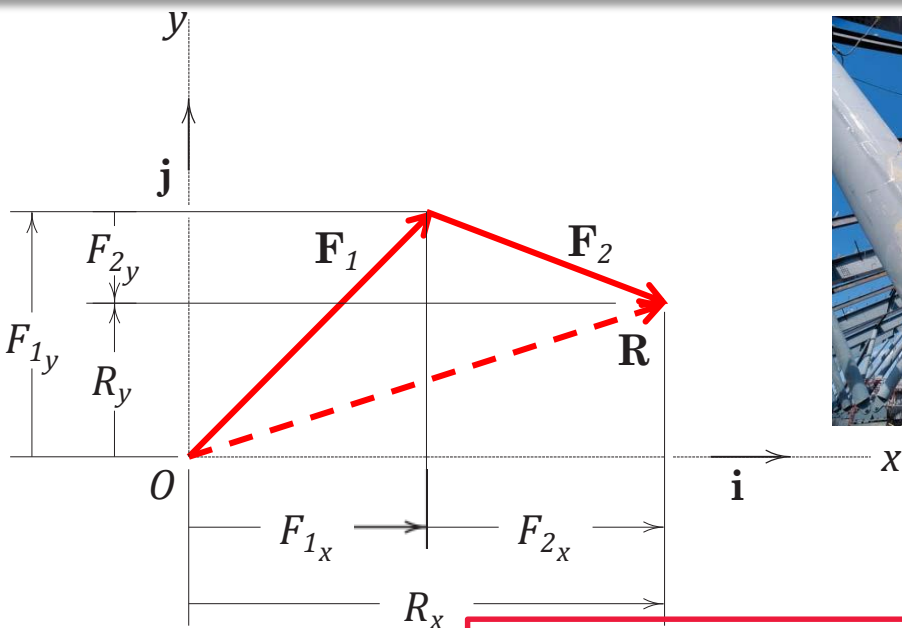
$$F_y = -F \cos (\pi - \beta)$$



$$F_x = F \cos (\beta - \alpha)$$

$$F_y = F \sin (\beta - \alpha)$$

Determining the Components of Force



$$\mathbf{F}_1 = F_{1x}\mathbf{i} + F_{1y}\mathbf{j}$$

$$\mathbf{F}_2 = F_{2x}\mathbf{i} + F_{2y}\mathbf{j}$$

$$\mathbf{R} = R_x\mathbf{i} + R_y\mathbf{j}$$

$$\mathbf{R} = \mathbf{F}_1 + \mathbf{F}_2$$

$$R_x\mathbf{i} + R_y\mathbf{j} = (F_{1x}\mathbf{i} + F_{1y}\mathbf{j}) + (F_{2x}\mathbf{i} + F_{2y}\mathbf{j})$$

$$R_x\mathbf{i} + R_y\mathbf{j} = (F_{1x} + F_{2x})\mathbf{i} + (F_{1y} + F_{2y})\mathbf{j}$$

$$R_x = (F_{1x} + F_{2x}) = \sum F_{ix}$$

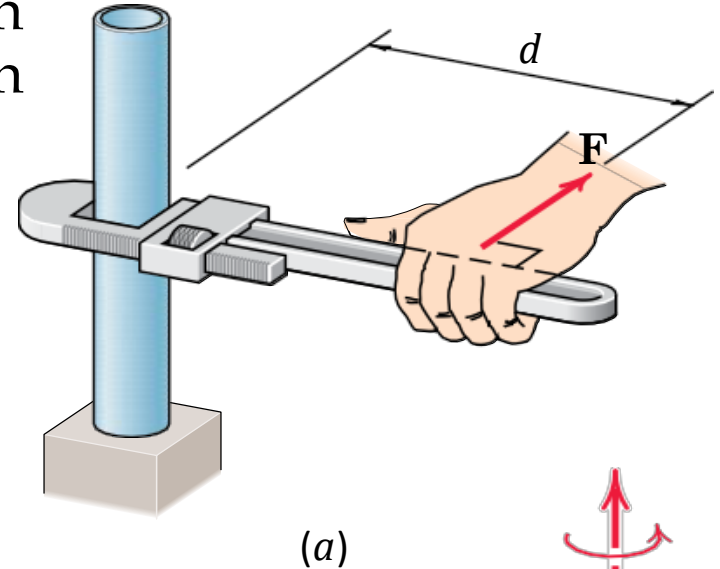
$$R_y = (F_{1y} + F_{2y}) = \sum F_{iy}$$

Moment

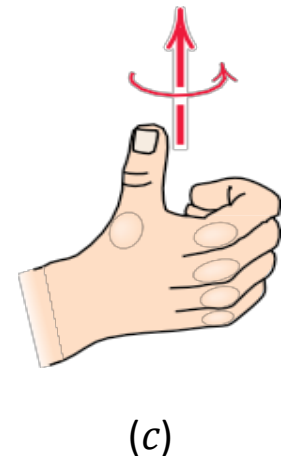
- Force can tend to rotate a body about an axis. This rotational tendency is known as the moment M of the force.
- Moment is also referred to as torque.

Magnitude of moment

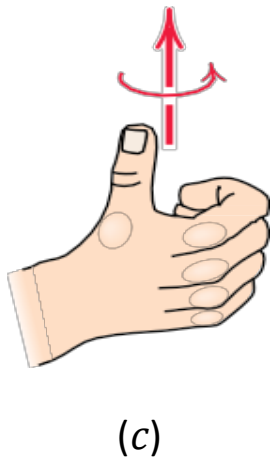
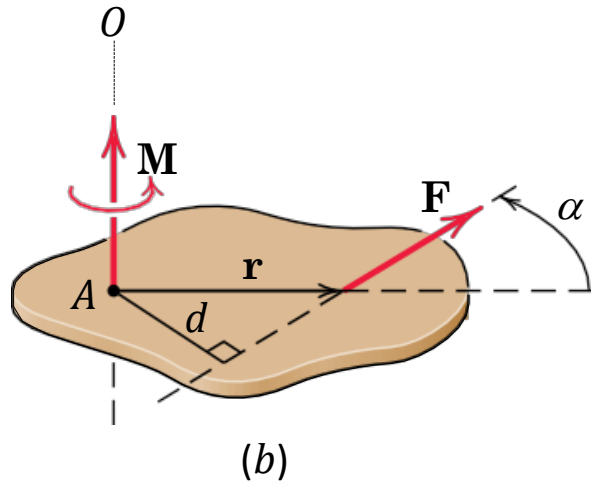
$$M = Fd$$



Direction



General 2D case



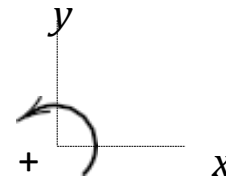
$$M = Fd$$

$$d = r \sin \alpha$$

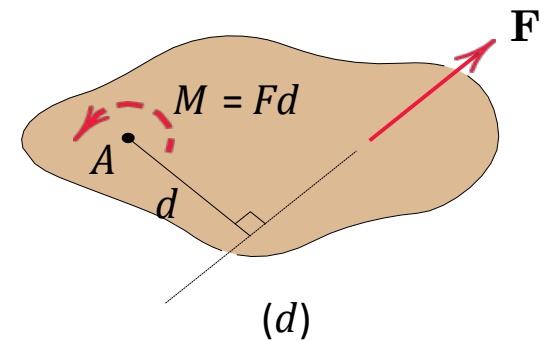
$$M = Fr \sin \alpha$$

$$M = |\mathbf{r} \times \mathbf{F}|$$

$$\mathbf{M} = \mathbf{r} \times \mathbf{F}$$



Anti clockwise +ve



Varignon's Theorem

- The moment of a force about any point is equal to the sum of the moments of the components of the force about the same point.

Varignon's Theorem

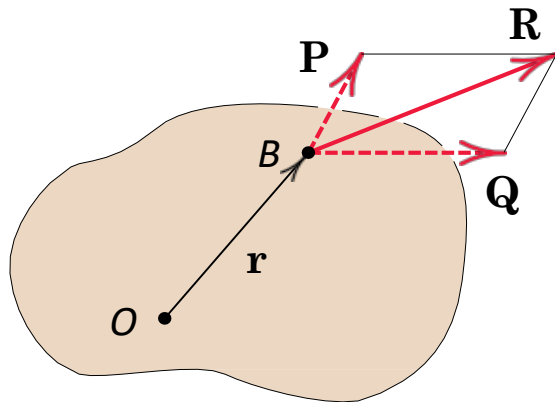
$$\mathbf{M}_O = \mathbf{r} \times \mathbf{R}$$

$$\mathbf{M}_O = \mathbf{r} \times (\mathbf{P} + \mathbf{Q})$$

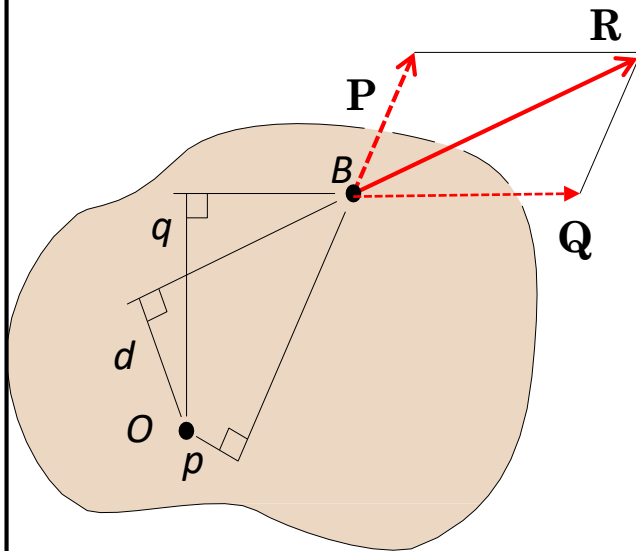
Using the distributive law for cross products

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{P} + \mathbf{r} \times \mathbf{Q}$$

$$M_O = Pp - Qq = -Rd$$



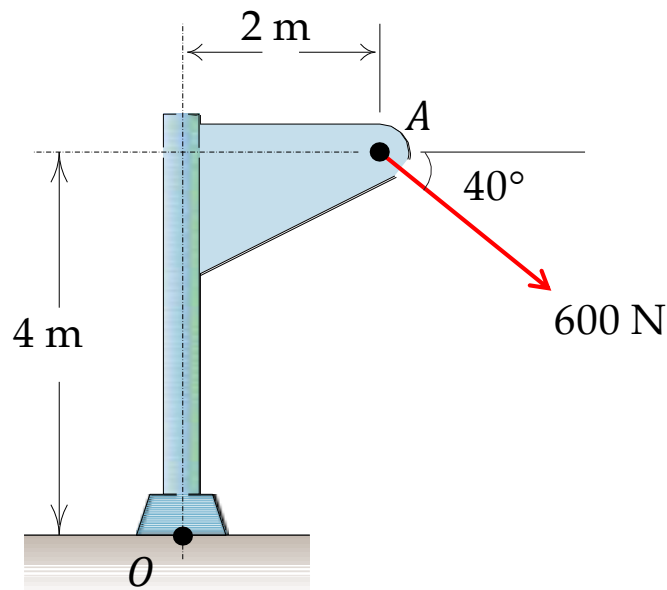
(a)



(b)

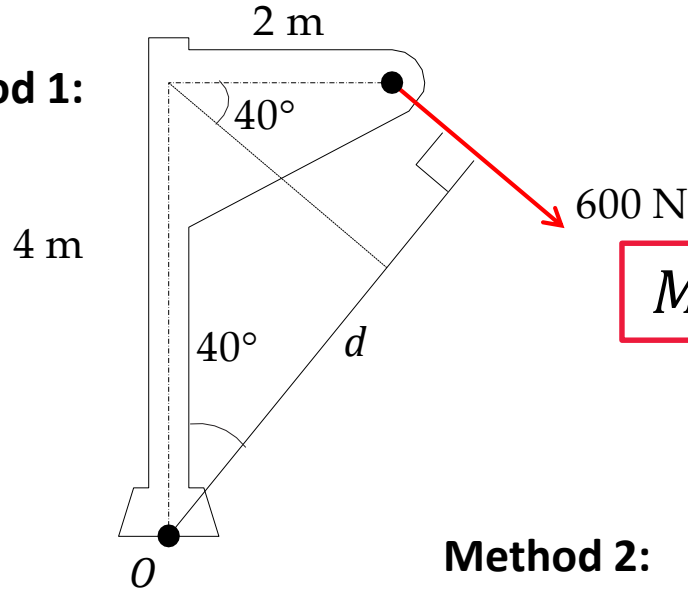
Example 1

- Calculate the magnitude of the moment about the base point O of the 600-N force in five different ways.



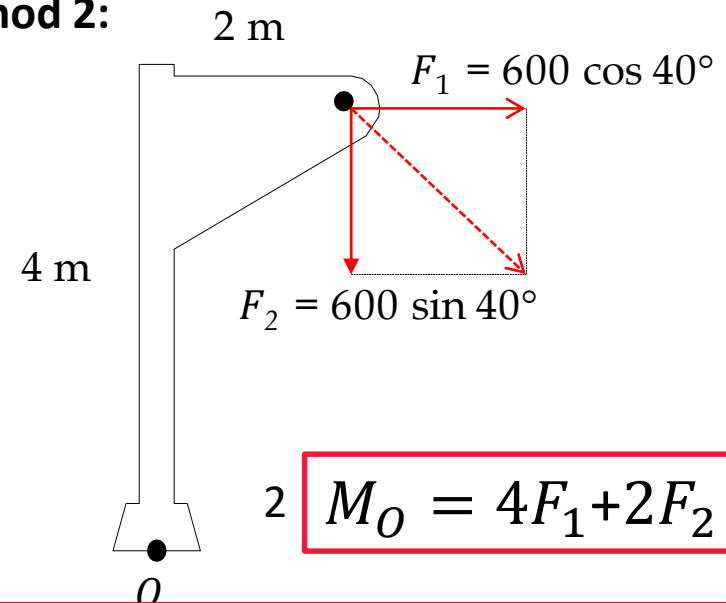
Example 1

Method 1:

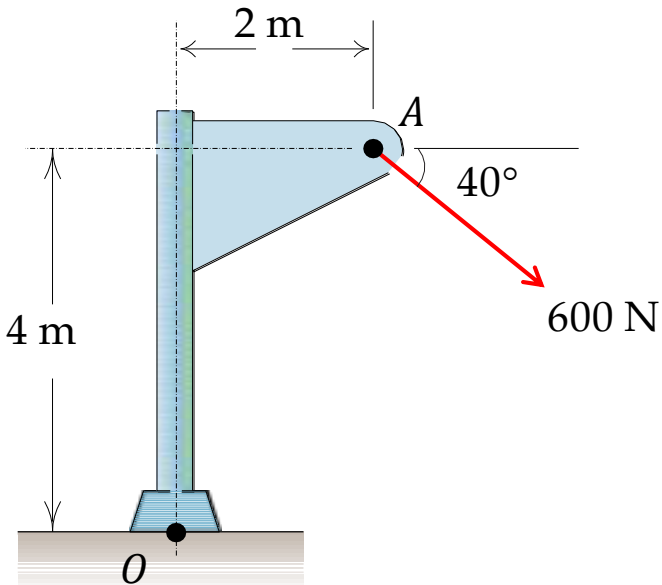


$$M_O = Fd$$

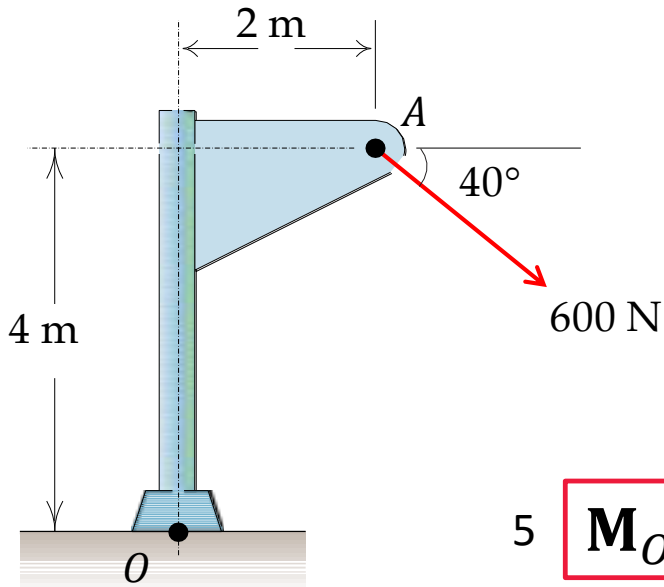
Method 2:



$$2 M_O = 4F_1 + 2F_2$$



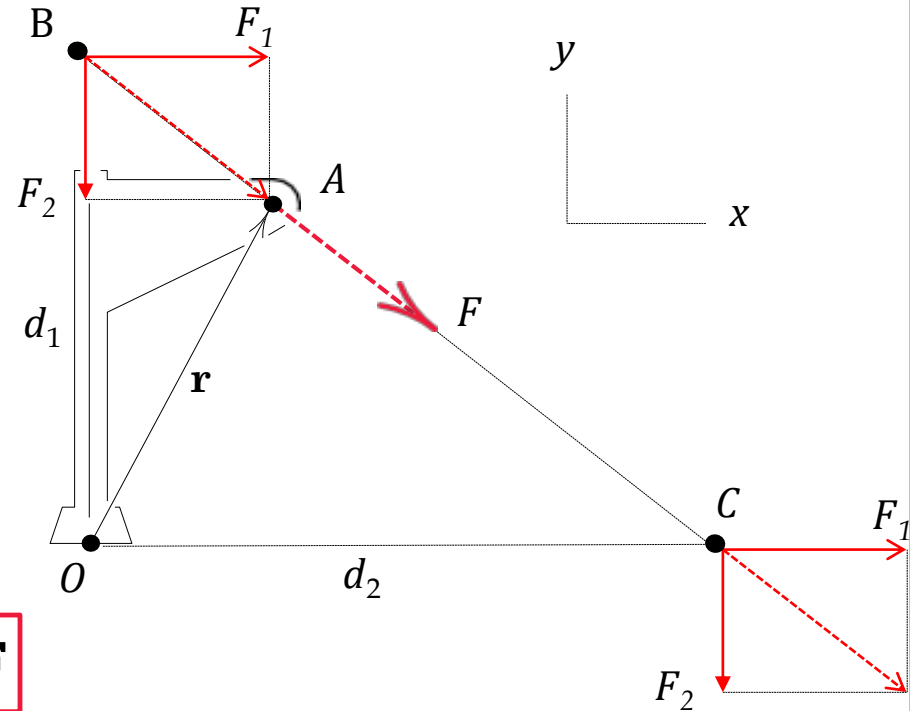
Example 1



3 $M_O = F_1 d_1$

5 $\mathbf{M}_O = \mathbf{r} \times \mathbf{F}$

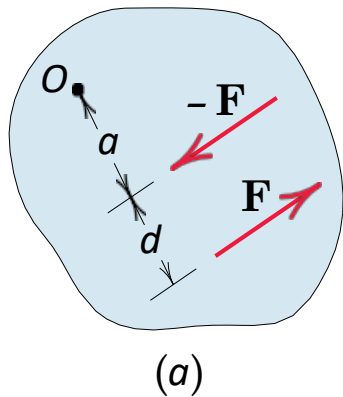
Methods 3, 4 and 5:



4 $M_O = F_2 d_2$

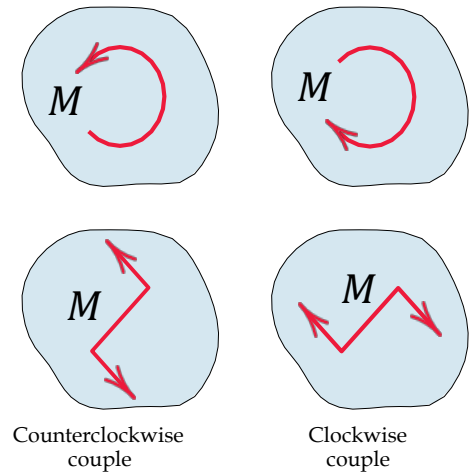
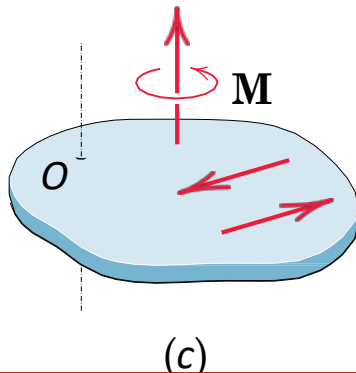
Couple

- The moment produced by two equal, opposite, and noncollinear forces is called a couple



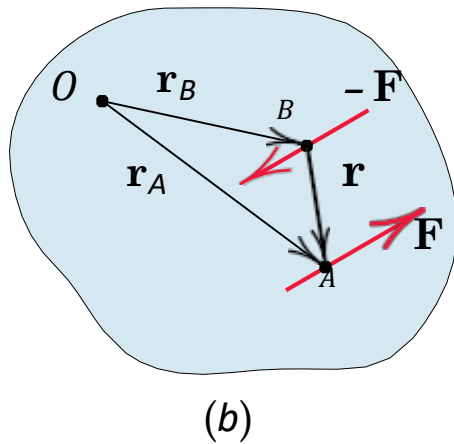
$$M_O = F(a + d) - Fa$$

$$M_O = Fd$$



(d)

Couple: A general case

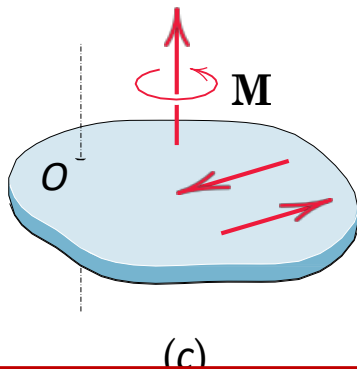


$$\mathbf{M}_O = \mathbf{r}_A \times \mathbf{F} + \mathbf{r}_B \times -\mathbf{F}$$

$$\mathbf{M}_O = (\mathbf{r}_A - \mathbf{r}_B) \times \mathbf{F}$$

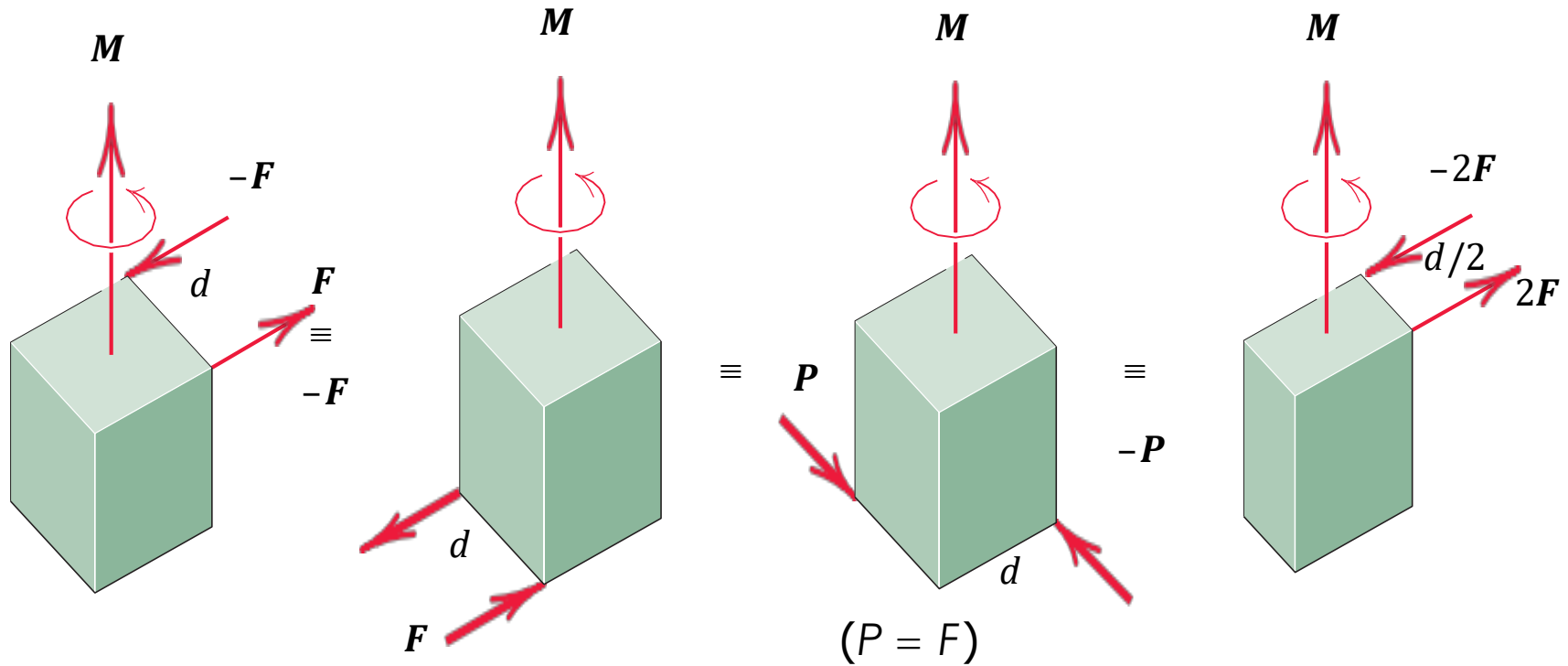
$$(\mathbf{r}_A - \mathbf{r}_B) = \mathbf{r}$$

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F}$$

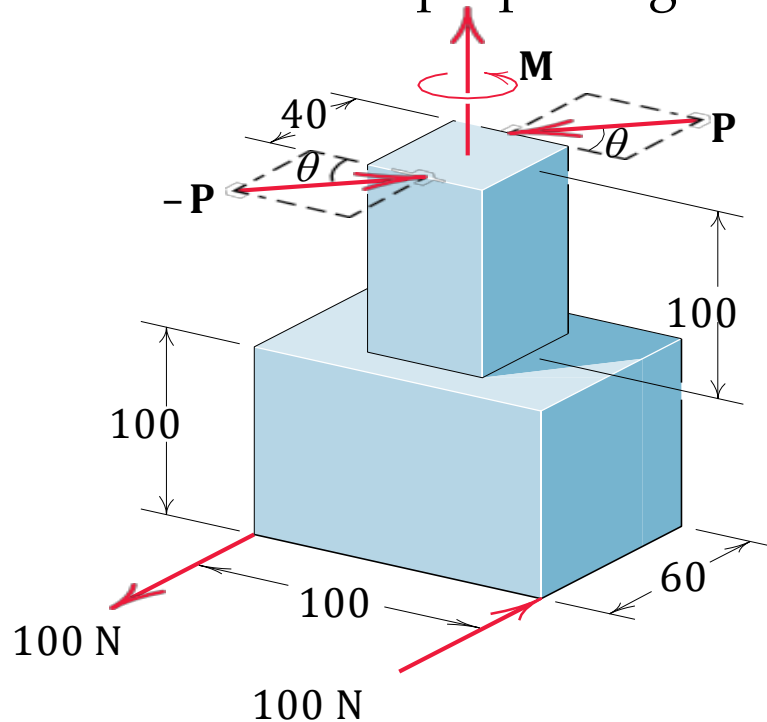


Independent from O and hence free vector

Equivalent Couples

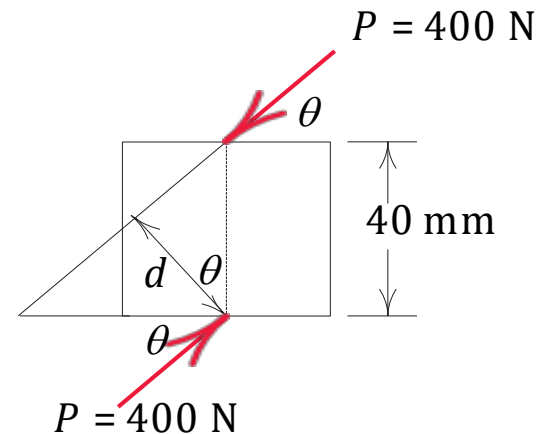


Example 2: The rigid structural member is subjected to a couple consisting of the two 100-N forces. Replace this couple by an equivalent couple consisting of the two forces P and $-P$, each of which has a magnitude of 400 N. Determine the proper angle.



Dimensions in millimeters

$$M_O = 100 \times 0.1$$

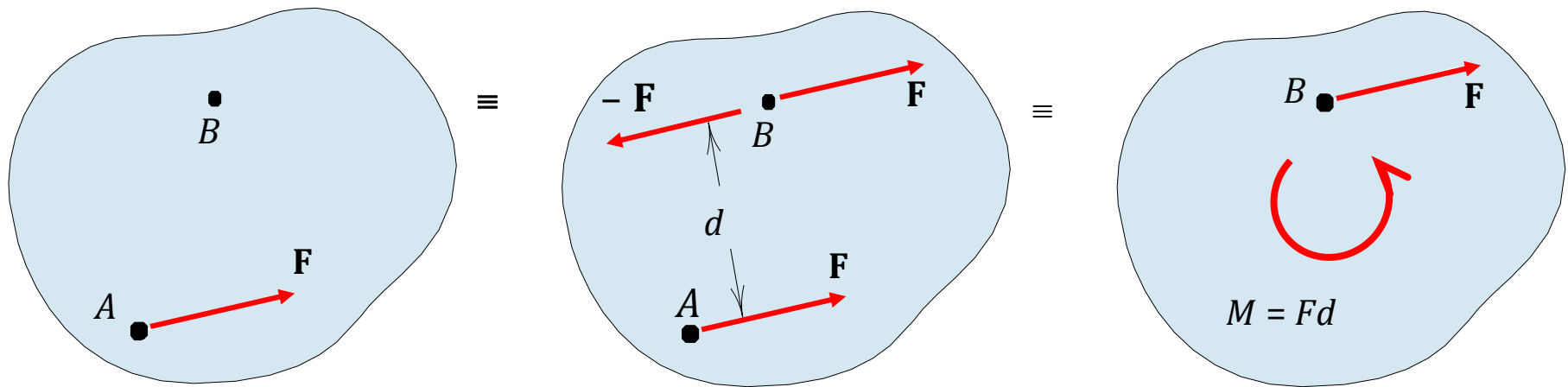


$$M_O = 400 \times 0.04 \cos \theta$$

Get θ by equating both equations

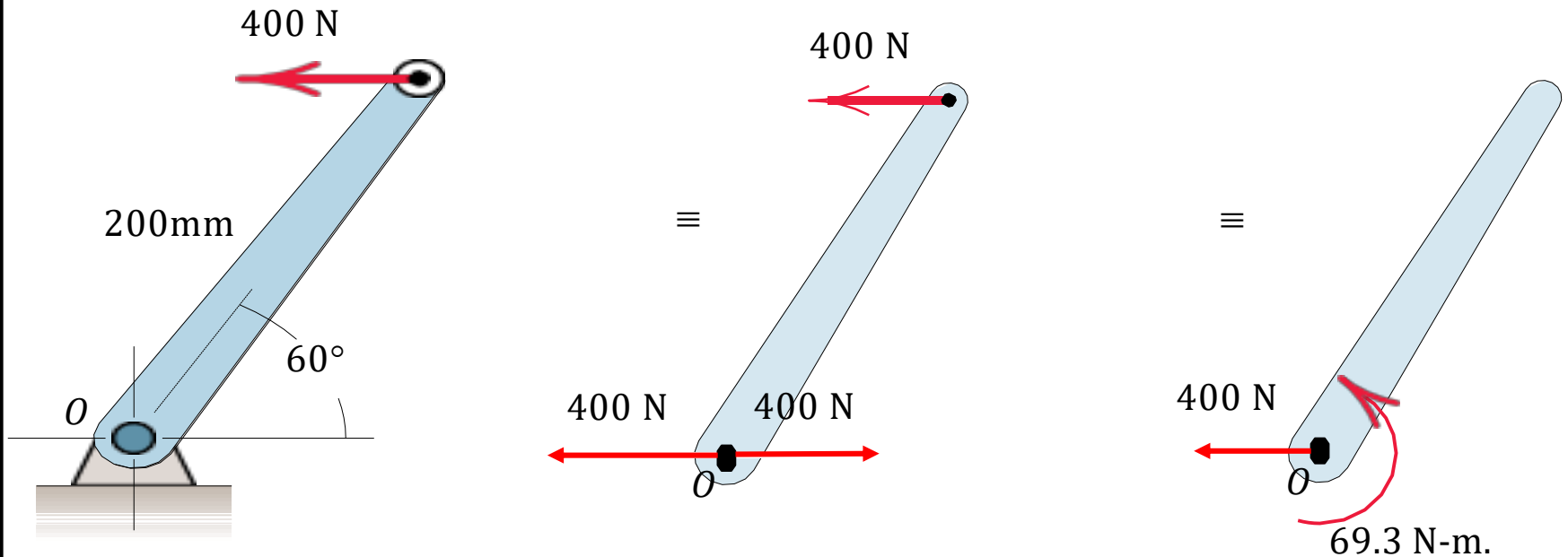
Force-Couple Systems

- A force can be replaced of by a force and a couple and referred to as a force-couple system.



$$M = Fd$$

Example 3: Replace the horizontal 400-N force acting on the lever by an equivalent system consisting of a force at O and a couple.



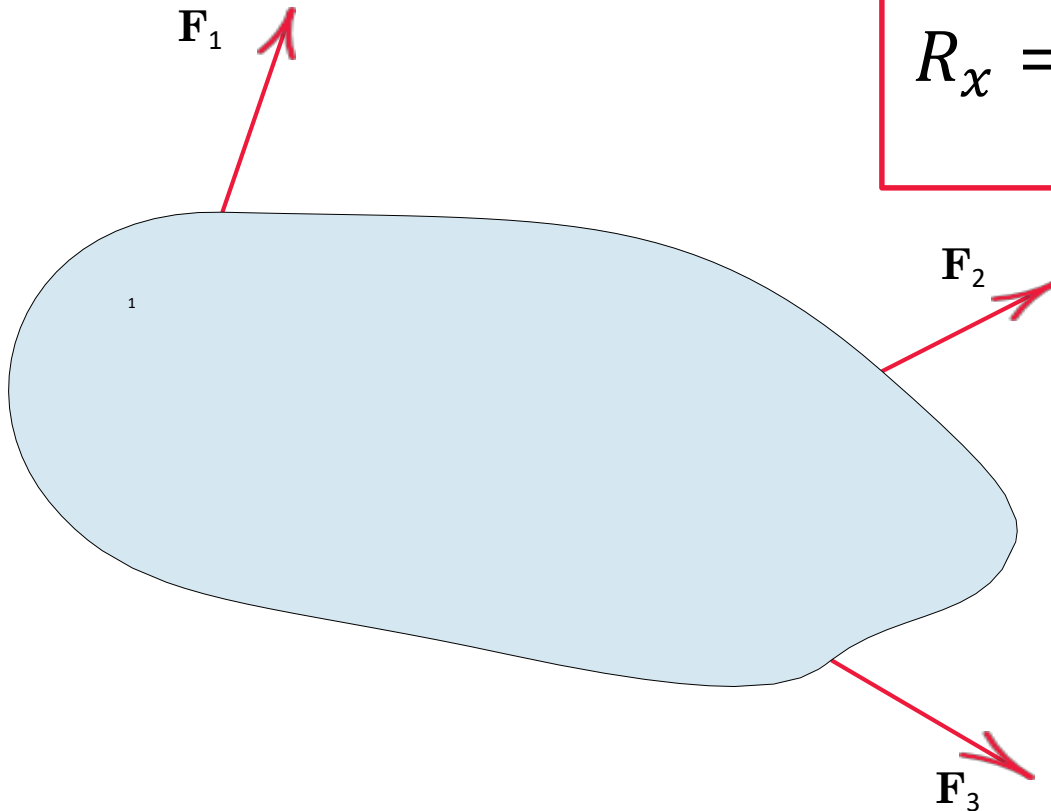
$$M_O = 400 \times 0.2 \sin 60$$

Resultant: Method 1

$$\mathbf{R} = \mathbf{F}_1 + \mathbf{F}_2 + \cdots + \mathbf{F}_n$$

$$R_x = \sum F_{ix}$$

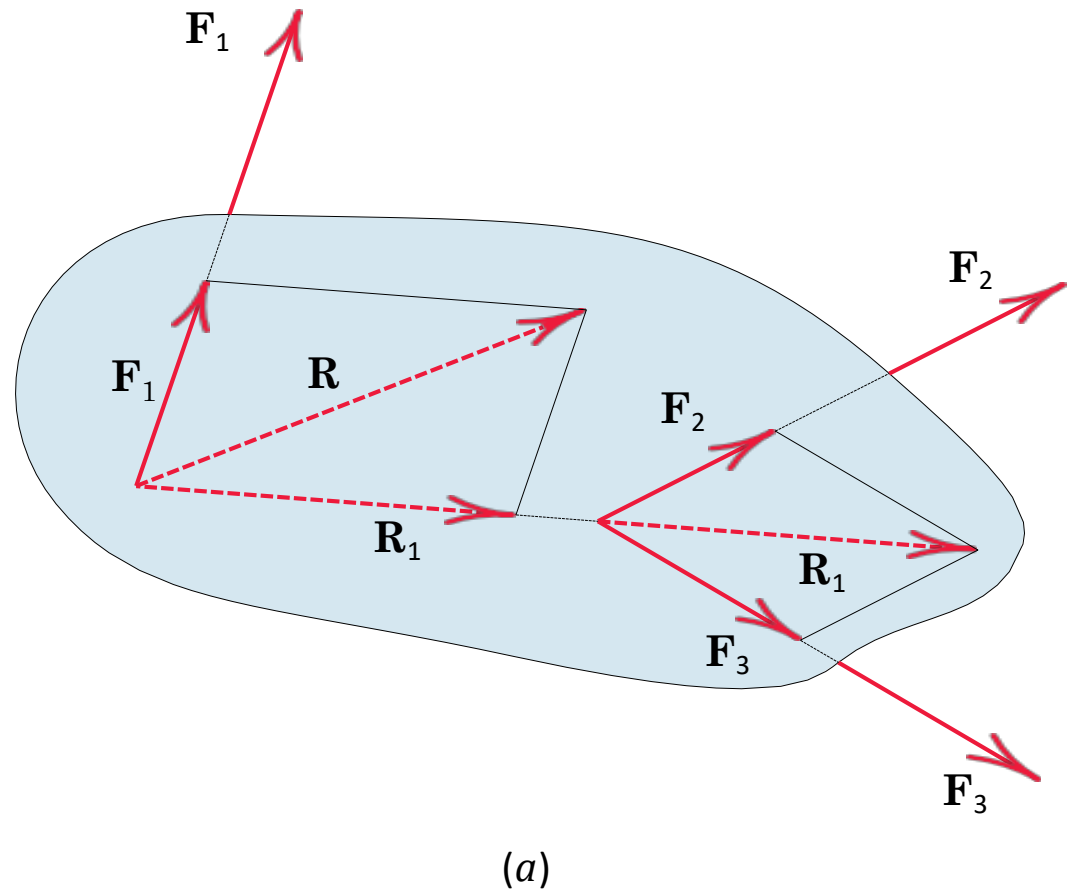
$$R_y = \sum F_{iy}$$



$$R = \sqrt{(R_x)^2 + (R_y)^2}$$

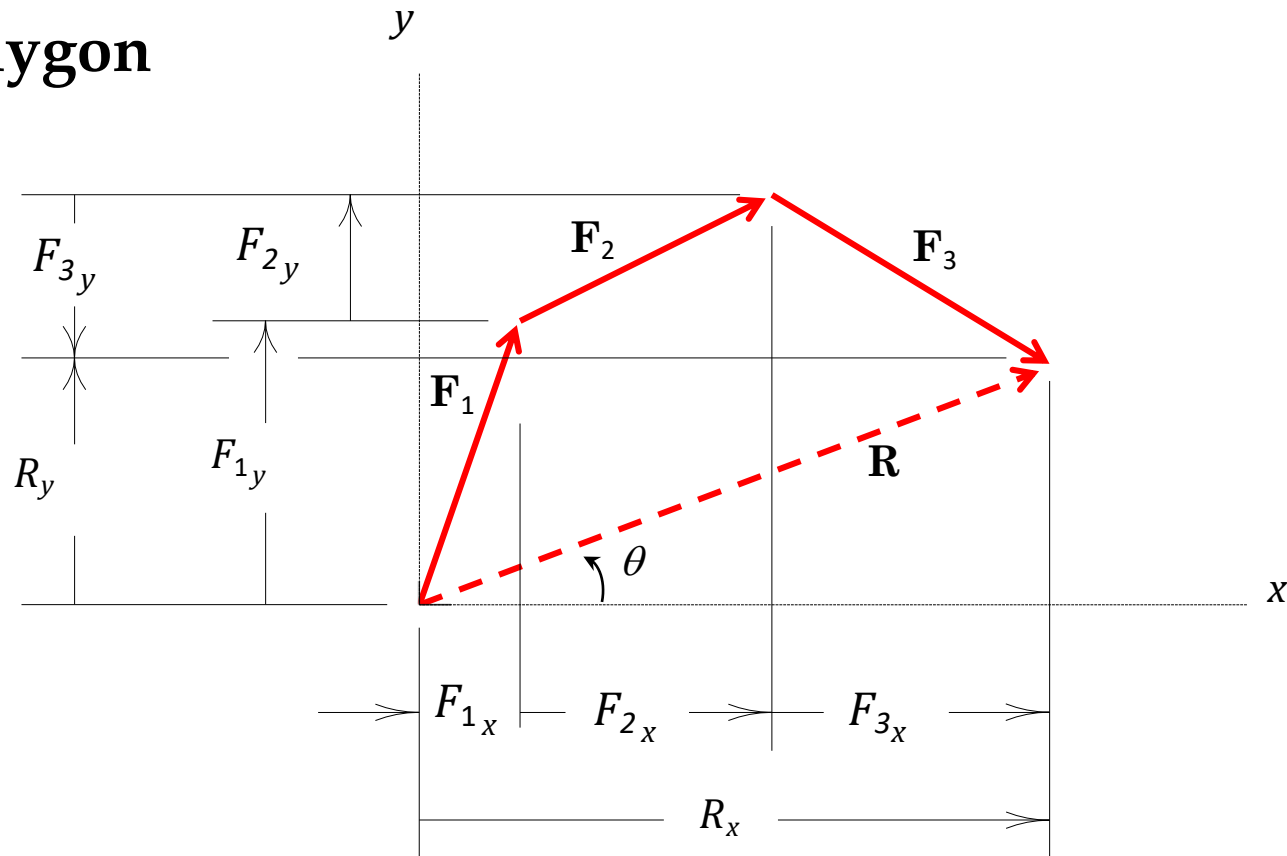
$$\theta = \tan^{-1} (R_y/R_x)$$

Resultant: Method 2 Parallelogram



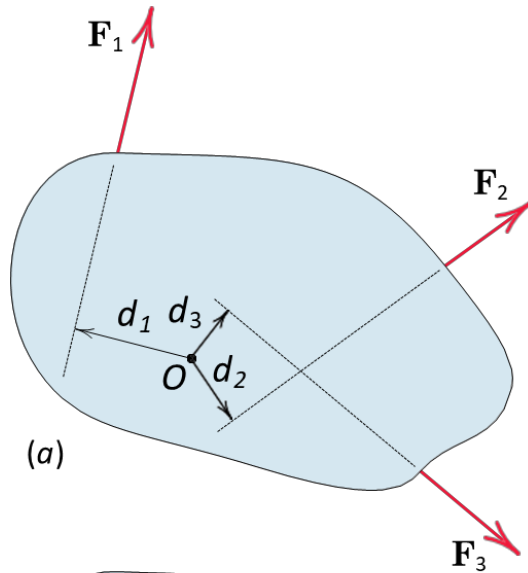
Resultant: Method 2

Force Polygon

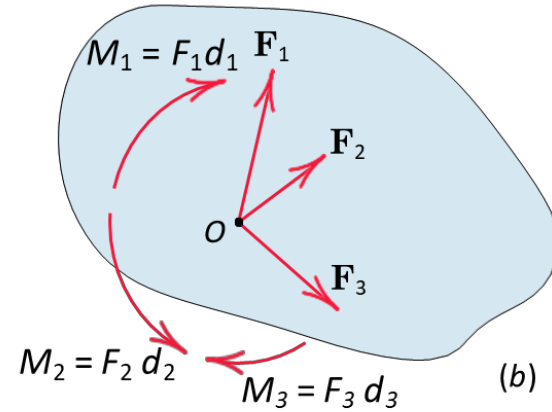


(b)

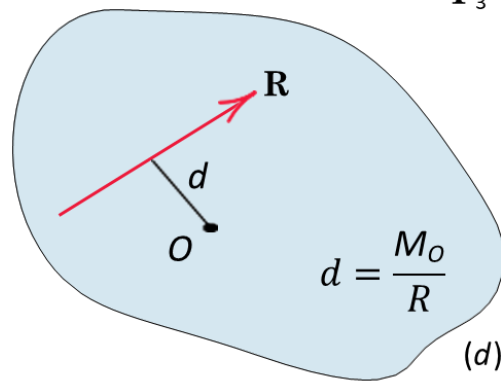
Resultant: Method 3



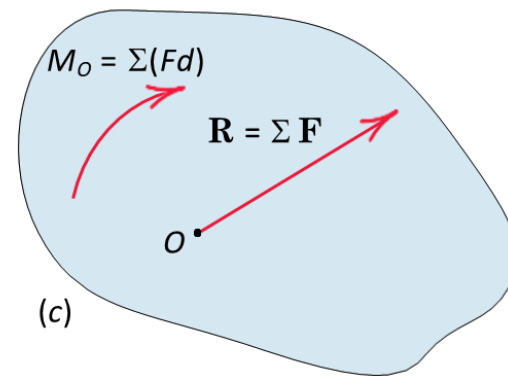
(a)



(b)



(d)



(c)

Principle of Moments

- The moment of the resultant force about any point O equals the sum of the moments of the original forces of the system about the same point.

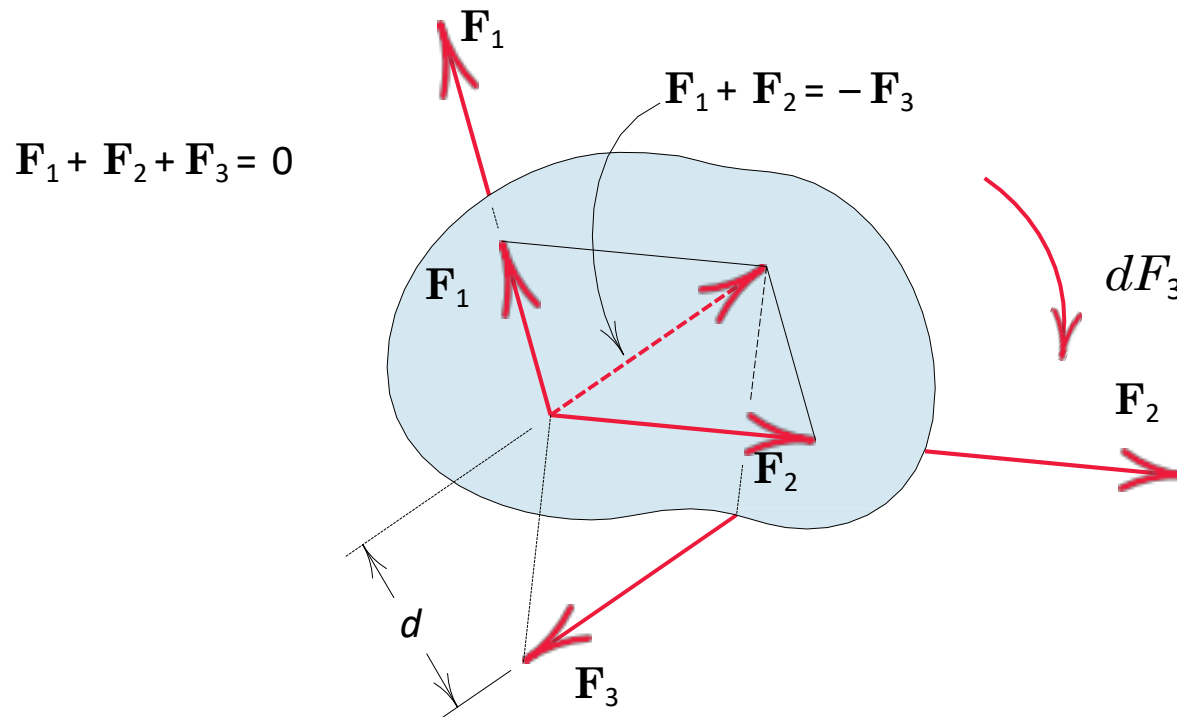
$$Rd = M_O$$

$$M_O = \sum F_i d_i$$

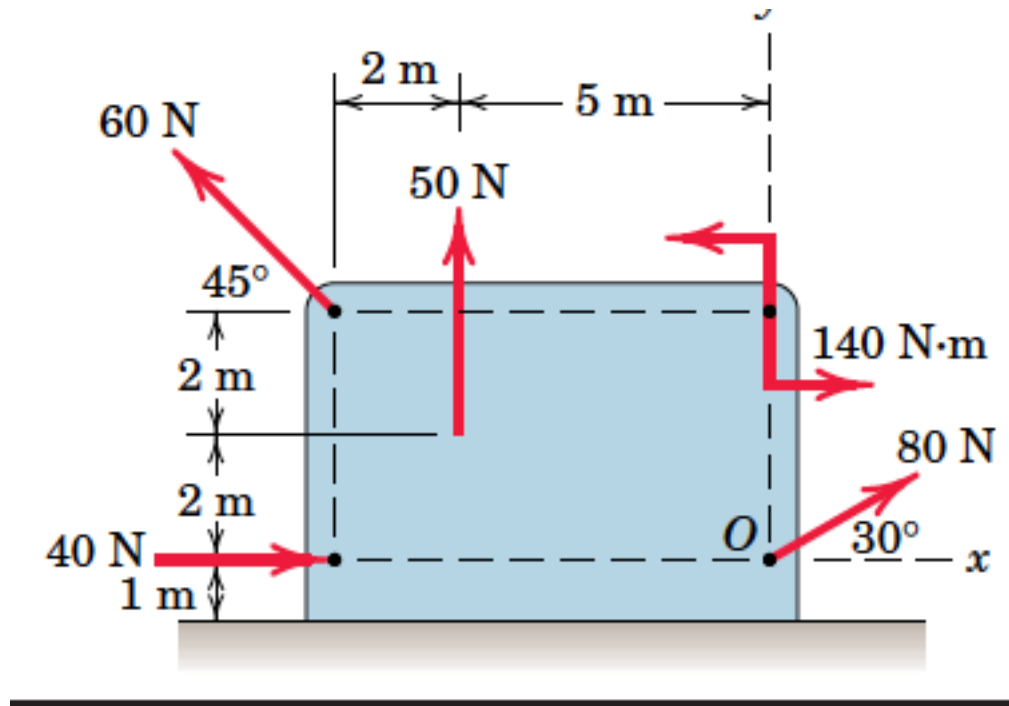
$$\mathbf{R} = \sum \mathbf{F}_i$$

Resultant: An observation

- If the resultant force R for a given force system is zero, the resultant of the system need not be zero because the resultant may be a couple.

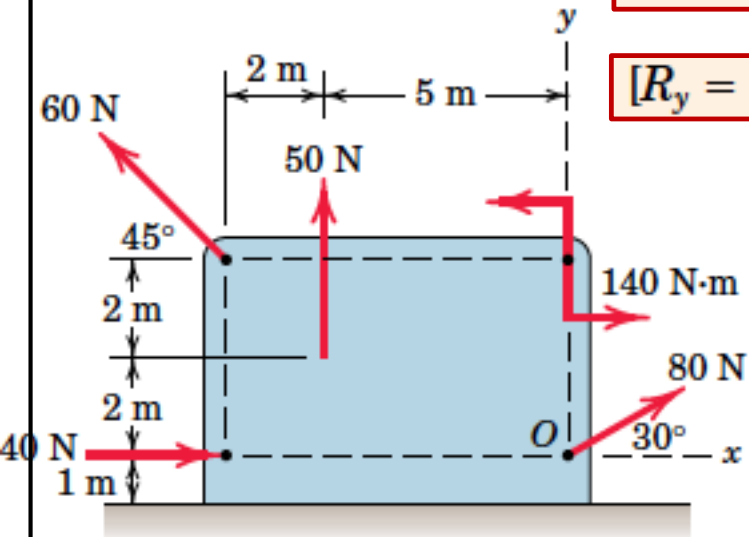


Example 4: Determine the resultant of the four forces and one couple which act on the plate shown.



Example 4

Part 1: Resultant



$$[R_x = \Sigma F_x]$$

$$R_x = 40 + 80 \cos 30^\circ - 60 \cos 45^\circ = 66.9 \text{ N}$$

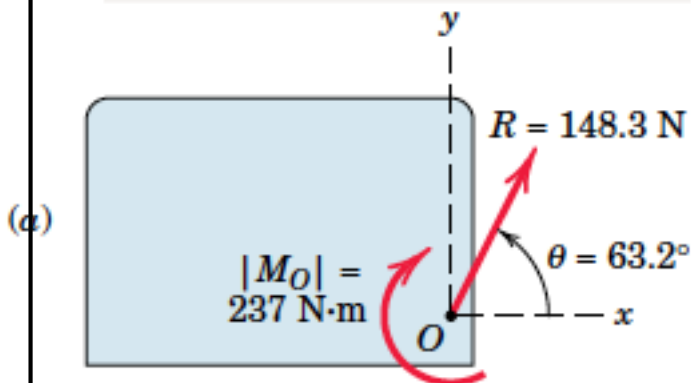
$$[R_y = \Sigma F_y]$$

$$R_y = 50 + 80 \sin 30^\circ + 60 \cos 45^\circ = 132.4 \text{ N}$$

$$R = \sqrt{(66.9)^2 + (132.4)^2} = 148.3 \text{ N}$$

$$\theta = \tan^{-1} \frac{132.4}{66.9} = 63.2^\circ$$

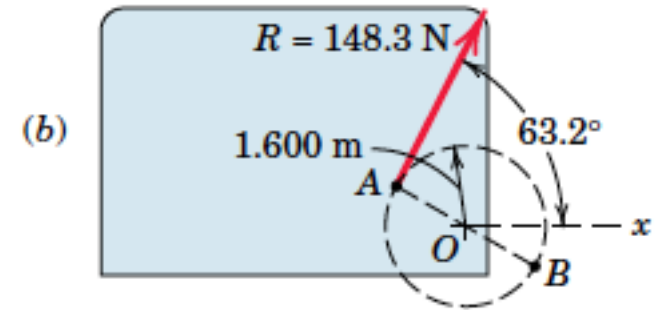
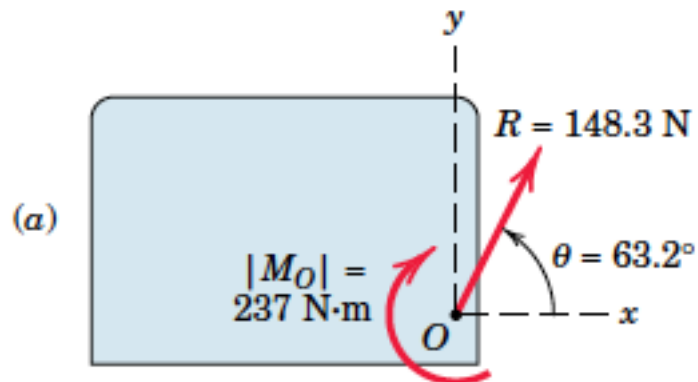
Part 2: Moment



$$[M_O = \Sigma (Fd)]$$

$$\begin{aligned} M_O &= 140 - 50(5) + 60 \cos 45^\circ(4) - 60 \sin 45^\circ(7) \\ &= -237 \text{ N}\cdot\text{m} \end{aligned}$$

Example 4

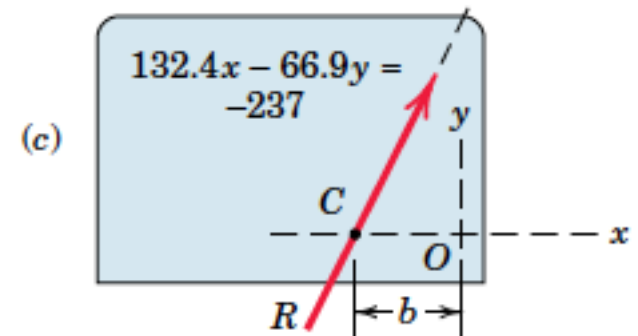


Part 3: Replacing force and moment by a force

$$[Rd = |M_O|]$$

$$148.3d = 237$$

$$d = 1.600 \text{ m}$$



Part 4: Intercept of force on x axis

$$R_y b = |M_O|$$

$$b = \frac{237}{132.4} = 1.792 \text{ m}$$

THANK YOU

Engineering Mechanics

Lecture – III

3 Dimensional Force Systems

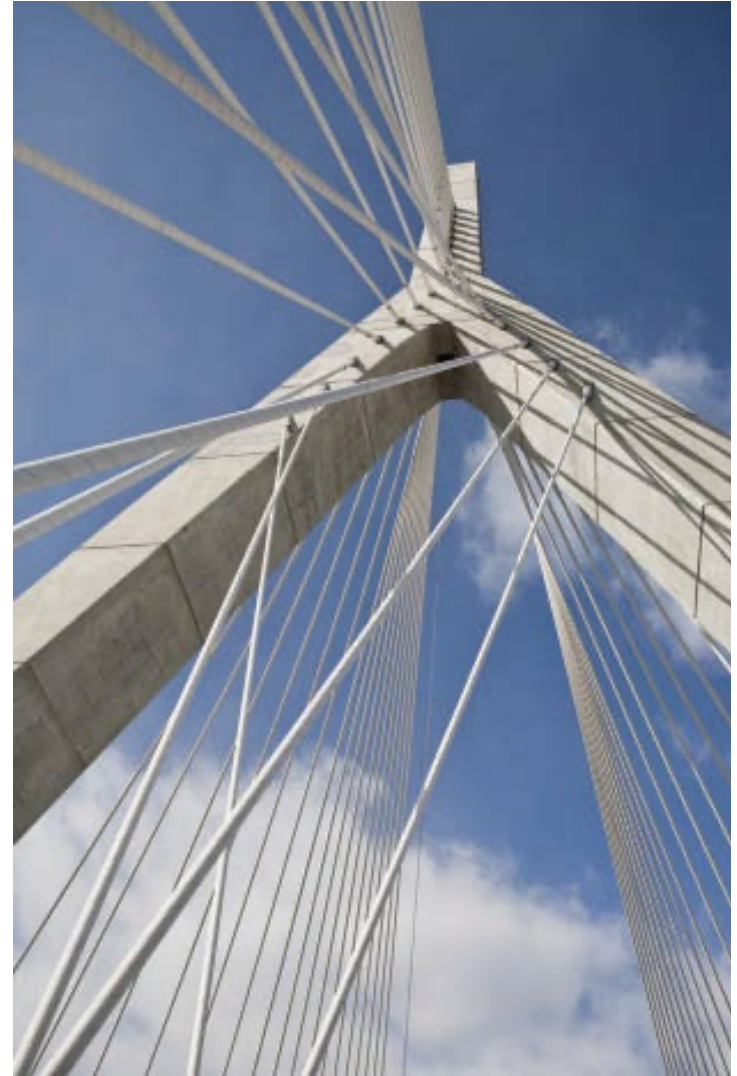
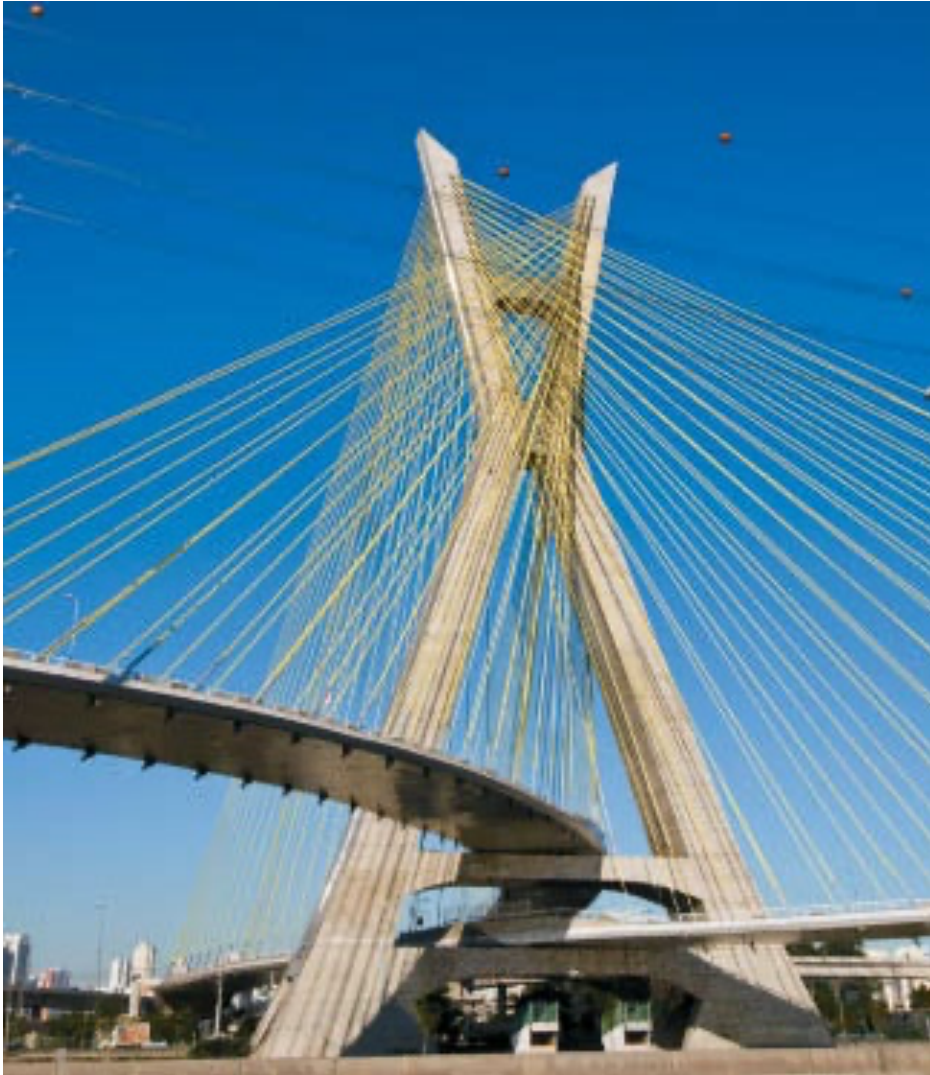
By

Dr. Bhupendra Singh

Assistant Professor

Civil and Infrastructure Engineering

Indian Institute of Technology Jodhpur



RECTANGULAR COMPONENTS

Direction cosine representation

$$F_x = F \cos \theta_x$$

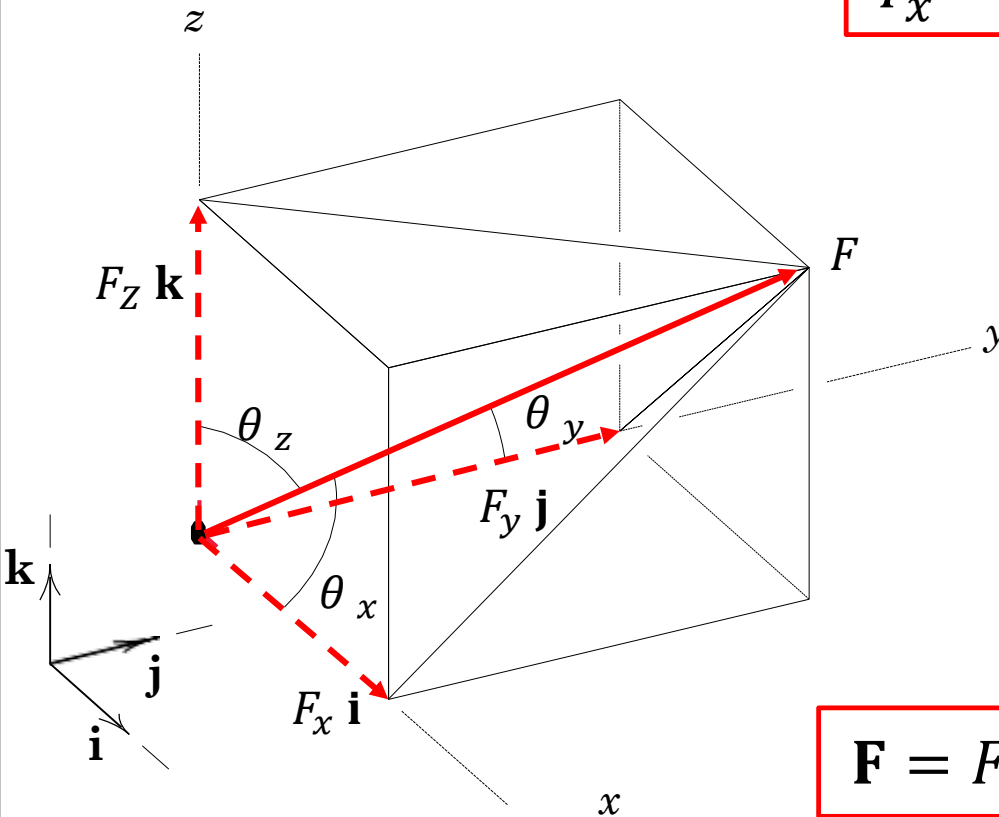
$$F_y = F \cos \theta_y$$

$$F_z = F \cos \theta_z$$

$$F = \sqrt{F_x^2 + F_y^2 + F_z^2}$$

$$\mathbf{F} = F_x \mathbf{i} + F_y \mathbf{j} + F_z \mathbf{k}$$

$$\mathbf{F} = F(\cos \theta_x \mathbf{i} + \cos \theta_y \mathbf{j} + \cos \theta_z \mathbf{k})$$



RECTANGULAR COMPONENTS

$$\mathbf{F} = F(\cos \theta_x \mathbf{i} + \cos \theta_y \mathbf{j} + \cos \theta_z \mathbf{k})$$

$$\mathbf{F} = F(l\mathbf{i} + m\mathbf{j} + n\mathbf{k})$$

$$l = \cos \theta_x$$

$$m = \cos \theta_y$$

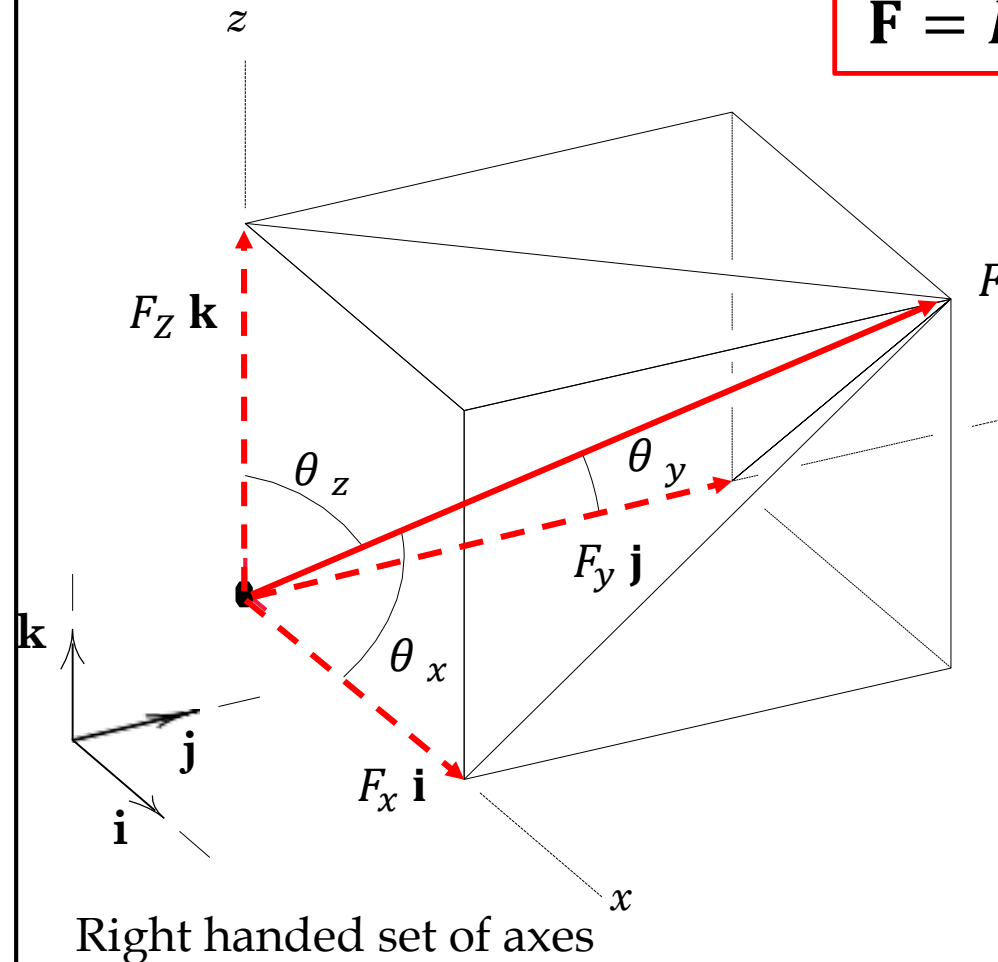
$$n = \cos \theta_z$$

$$l^2 + m^2 + n^2 = 1$$

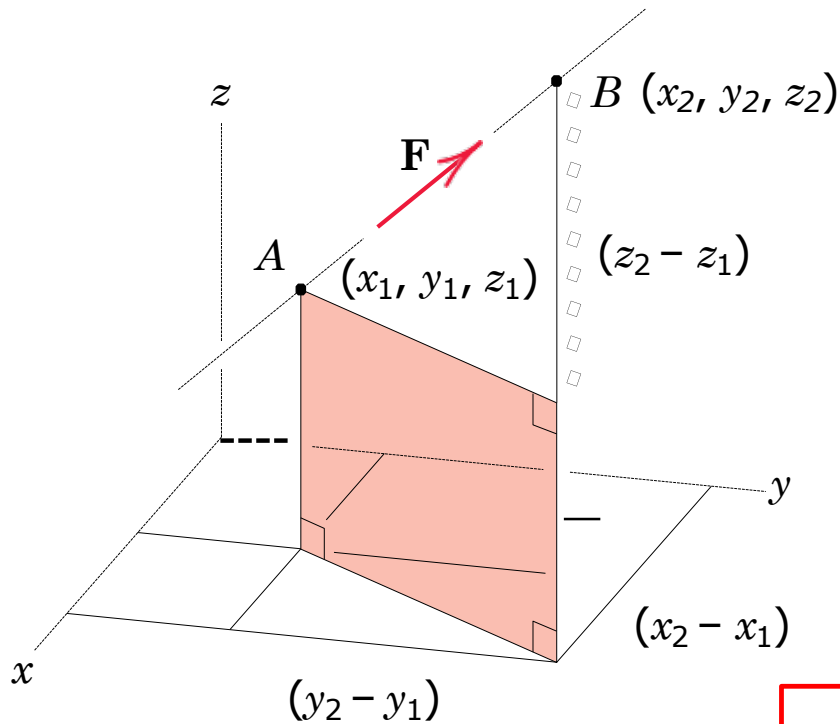
Force in terms of unit vector

$$\mathbf{F} = F\mathbf{n}_F$$

$$\mathbf{n}_F = (l\mathbf{i} + m\mathbf{j} + n\mathbf{k})$$



Direction Specification by two points

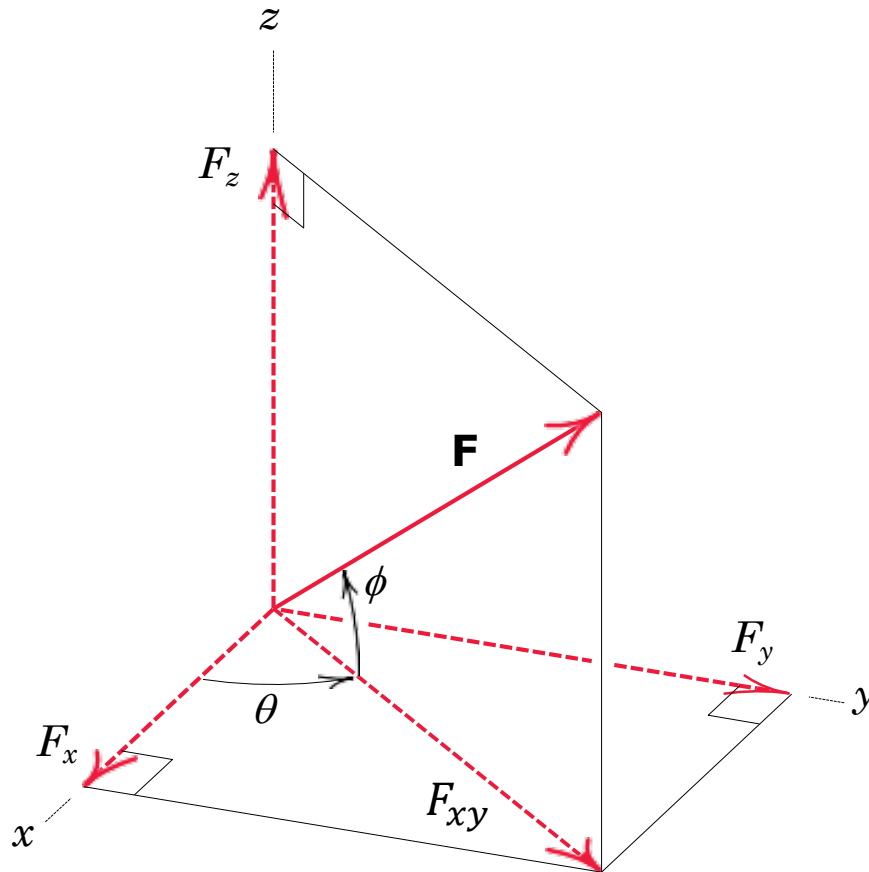


$$\mathbf{F} = F \mathbf{n}_F$$

$$\mathbf{n}_F = \frac{\overrightarrow{AB}}{|\overrightarrow{AB}|}$$

$$\mathbf{n}_F = \frac{(x_2 - x_1)\mathbf{i} + (y_2 - y_1)\mathbf{j} + (z_2 - z_1)\mathbf{k}}{\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}}$$

Direction Specification by two angles



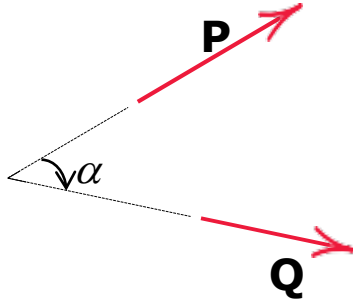
$$\mathbf{F} = F_x \mathbf{i} + F_y \mathbf{j} + F_z \mathbf{k}$$

$$F_z = F \sin \varphi$$

$$F_{xy} = F \cos \varphi$$

$$F_x = F_{xy} \cos \theta = F \cos \varphi \cos \theta$$

$$F_y = F_{xy} \sin \theta = F \cos \varphi \sin \theta$$

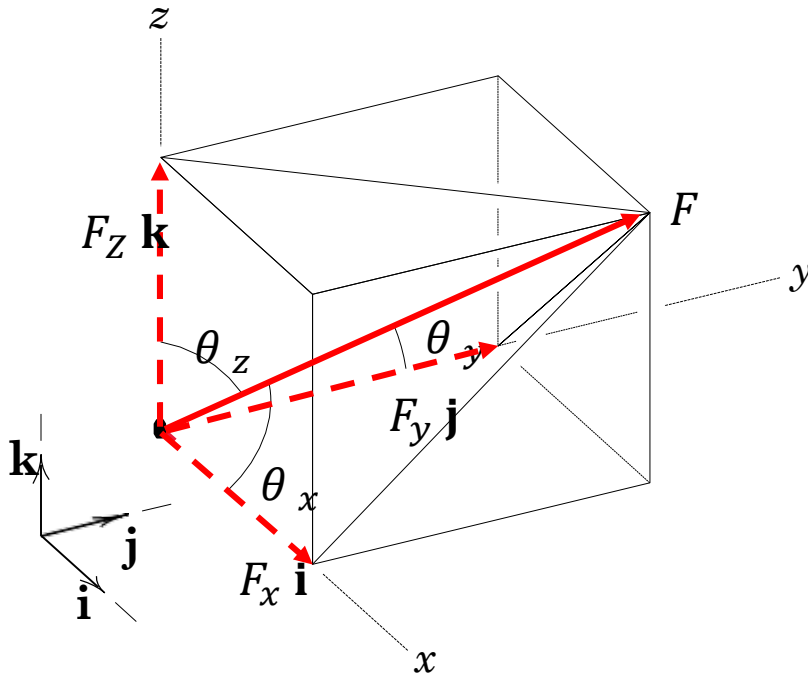


Dot Product

$$\mathbf{P} \cdot \mathbf{Q} = (P \cos \alpha) Q$$

$$\mathbf{P} \cdot \mathbf{Q} = PQ \cos \alpha$$

Projection of F along x



$$\mathbf{F} \cdot \mathbf{i} = F \cos \theta_x$$

Projection of Force (F)

Projection of F along the axis defined by unit vector $\mathbf{n}$

$$\mathbf{F} = F(l\mathbf{i} + m\mathbf{j} + n\mathbf{k})$$

$$\mathbf{n} = \alpha\mathbf{i} + \beta\mathbf{j} + \gamma\mathbf{k}$$

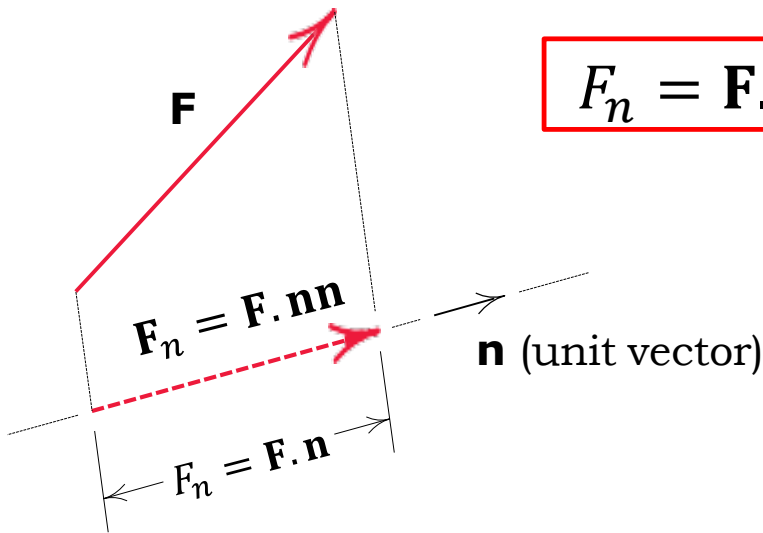
$$F_n = \mathbf{F} \cdot \mathbf{n} = F(l\mathbf{i} + m\mathbf{j} + n\mathbf{k}) \cdot (\alpha\mathbf{i} + \beta\mathbf{j} + \gamma\mathbf{k})$$

$$F_n = F(l\alpha + m\beta + n\gamma)$$

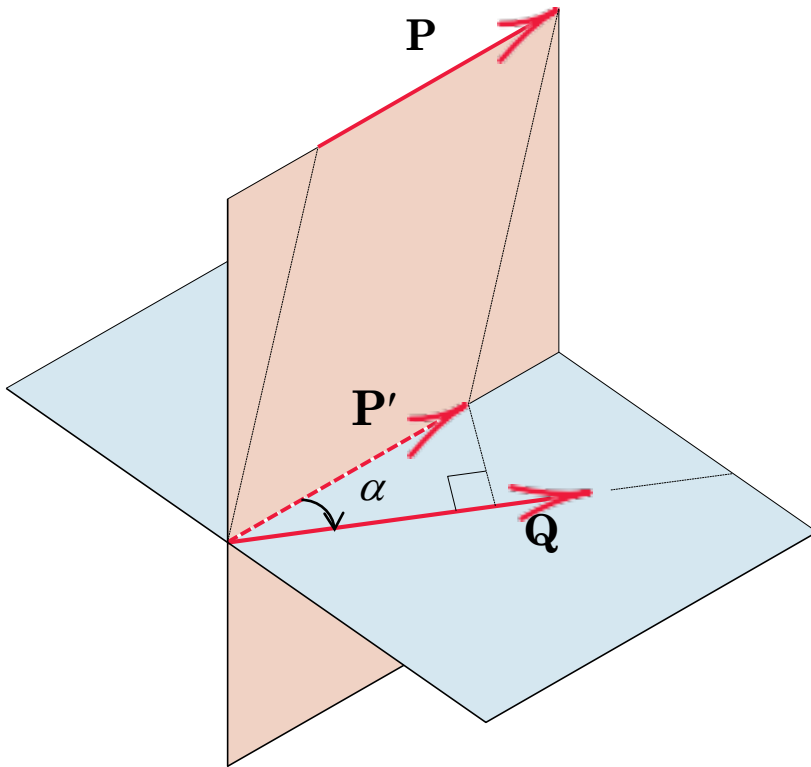
Since

$$\mathbf{i} \cdot \mathbf{i} = \mathbf{j} \cdot \mathbf{j} = \mathbf{k} \cdot \mathbf{k} = 1$$

$$\mathbf{i} \cdot \mathbf{j} = \mathbf{j} \cdot \mathbf{i} = \mathbf{i} \cdot \mathbf{k} = \mathbf{k} \cdot \mathbf{i} = \mathbf{k} \cdot \mathbf{j} = \mathbf{j} \cdot \mathbf{k} = 0$$



Angle between Two Vectors



$$\mathbf{P} \cdot \mathbf{Q} = PQ \cos \alpha$$

$$\theta = \cos^{-1} \frac{\mathbf{P} \cdot \mathbf{Q}}{PQ}$$

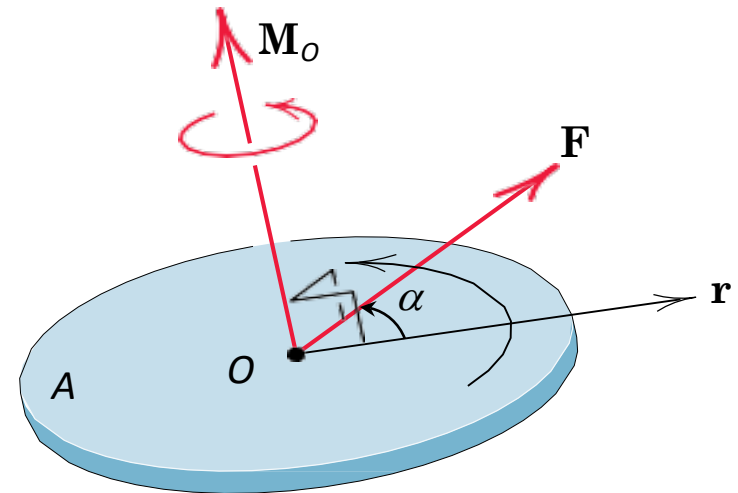
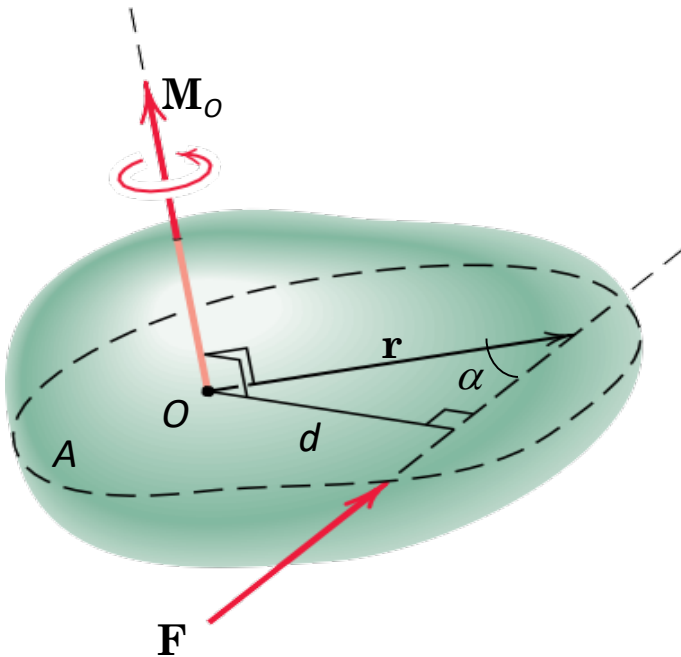
This also true for $\mathbf{P} \cdot \mathbf{n}$

$$\theta = \cos^{-1} \frac{\mathbf{F} \cdot \mathbf{n}}{F}$$

Moment in 3 Dimensions

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F}$$

$\mathbf{r}$: vector from O to any point on line of action of $\mathbf{F}$

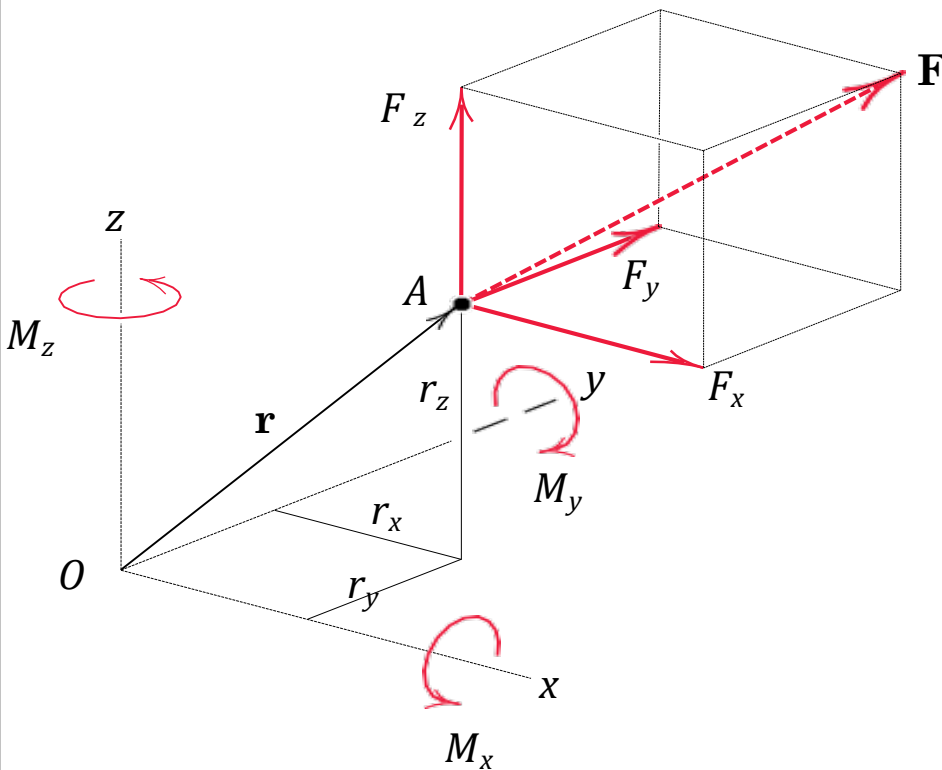


Evaluating the Cross Product

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F}$$

$$\mathbf{M}_O = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ r_x & r_y & r_z \\ F_x & F_y & F_z \end{vmatrix}$$

$$\mathbf{M}_O = (r_y F_z - r_z F_y) \mathbf{i} + (r_z F_x - r_x F_z) \mathbf{j} + (r_x F_y - r_y F_x) \mathbf{k}$$

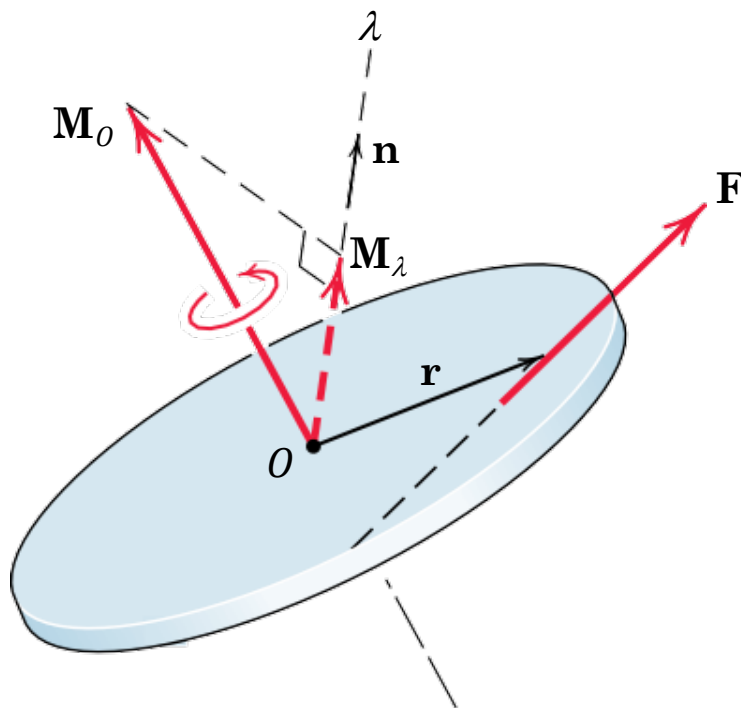


$$M_x = (r_y F_z - r_z F_y)$$

$$M_y = (r_z F_x - r_x F_z)$$

$$M_z = (r_x F_y - r_y F_x)$$

Moment about an Arbitrary Axis



$$\mathbf{M}_\lambda = [(\mathbf{r} \times \mathbf{F}) \cdot \mathbf{n}] \mathbf{n}$$

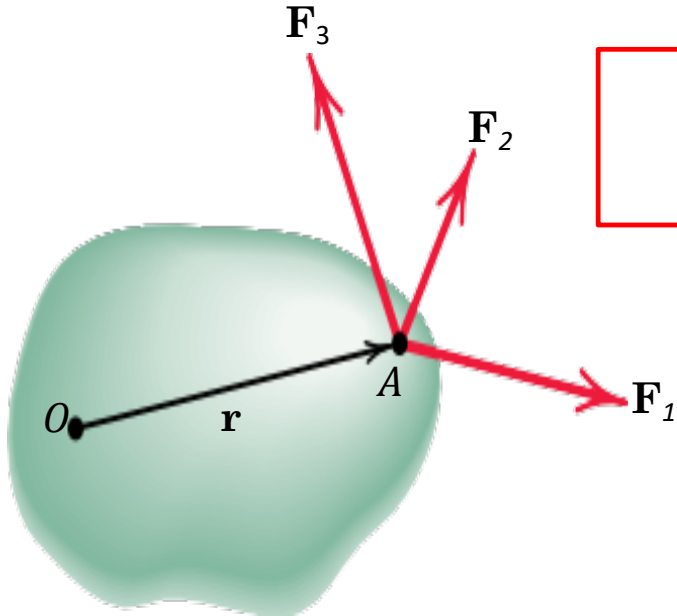
Triple Scalar product

$$|\mathbf{M}_\lambda| = M_\lambda = \begin{vmatrix} r_x & r_y & r_z \\ F_x & F_y & F_z \\ \alpha & \beta & \gamma \end{vmatrix}$$

Varignon's Theorem in Three Dimensions

$$\mathbf{r} \times \mathbf{F}_1 + \mathbf{r} \times \mathbf{F}_2 + \mathbf{r} \times \mathbf{F}_3 + \dots \\ = \mathbf{r} \times (\mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \dots) = \mathbf{r} \times \Sigma \mathbf{F}$$

$$\mathbf{M}_O = \Sigma(\mathbf{r} \times \mathbf{F}) = \mathbf{r} \times \mathbf{R}$$



Couples in Three Dimensions

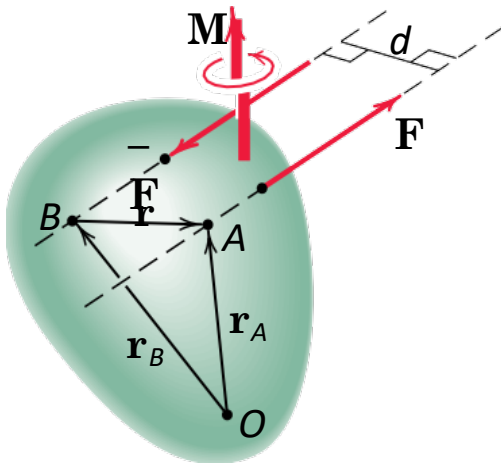
Taking moment of $\mathbf{F}$ and $-\mathbf{F}$ about point O

$$\mathbf{M}_O = \mathbf{r}_A \times \mathbf{F} + \mathbf{r}_B \times (-\mathbf{F})$$

$$\mathbf{M}_O = (\mathbf{r}_A - \mathbf{r}_B) \times \mathbf{F}$$

$$\mathbf{r} = (\mathbf{r}_A - \mathbf{r}_B)$$

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F}$$

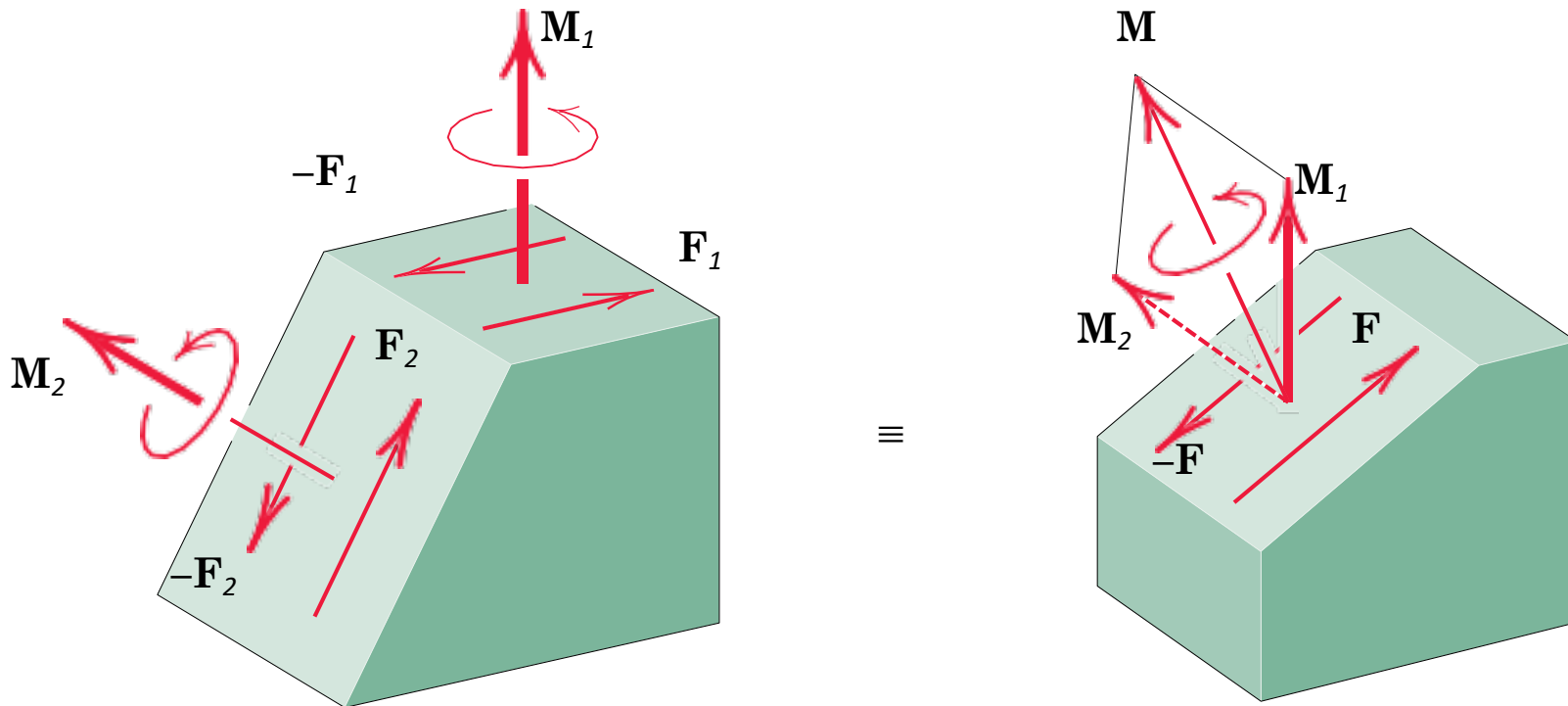


Independent from O and hence free vector

$\mathbf{r}$: vector from any point on line of action of $-\mathbf{F}$ to any point on line of action of $\mathbf{F}$

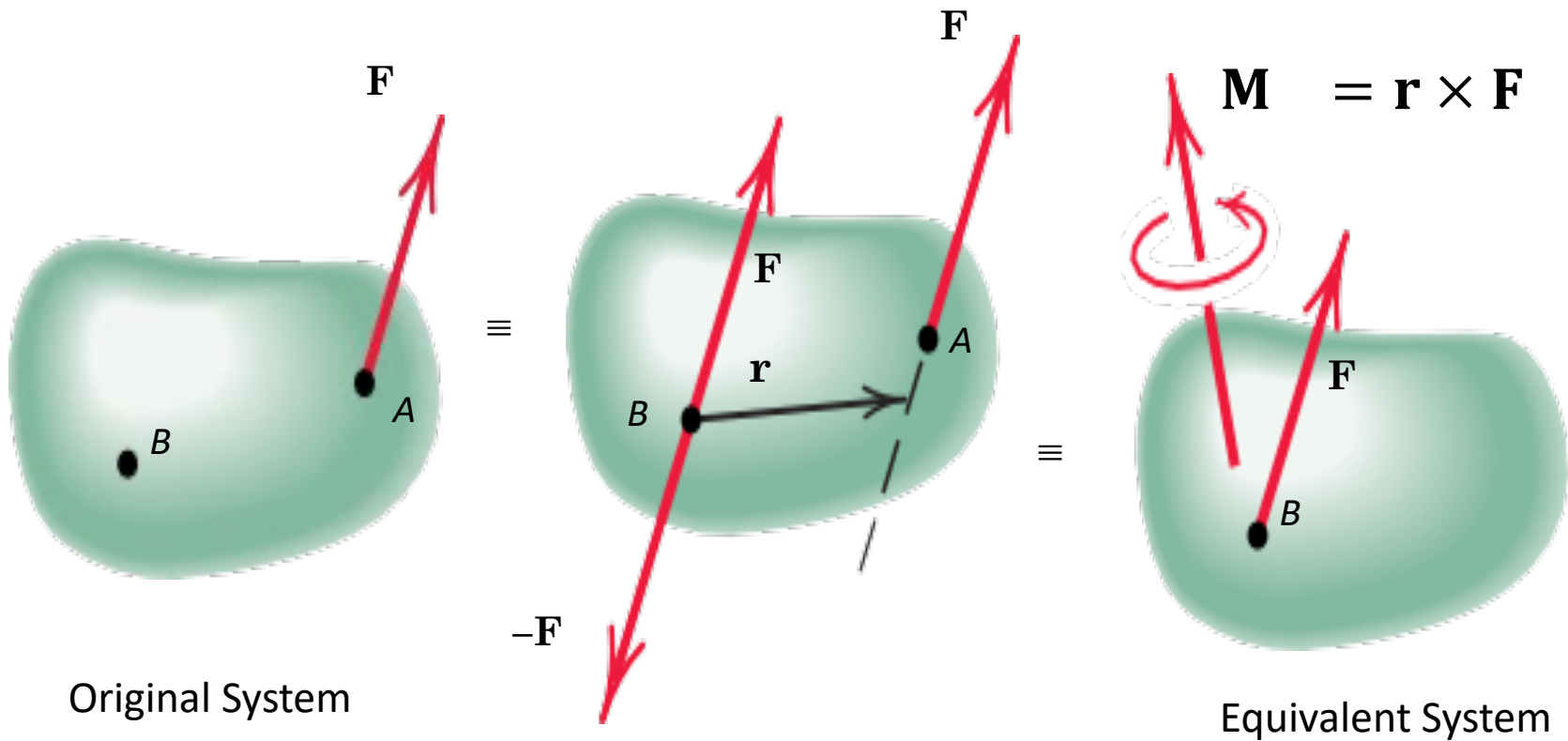
Note: The moment of a couple is a *free vector*, whereas the moment of a force about a point is a *sliding vector* whose direction is along the axis through the point.

Summation of Couples

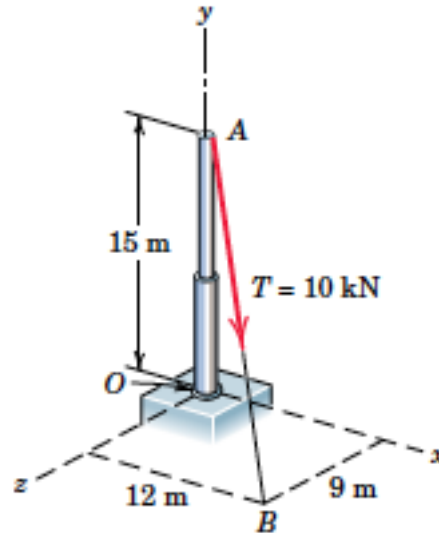


All rule of vector addition apply

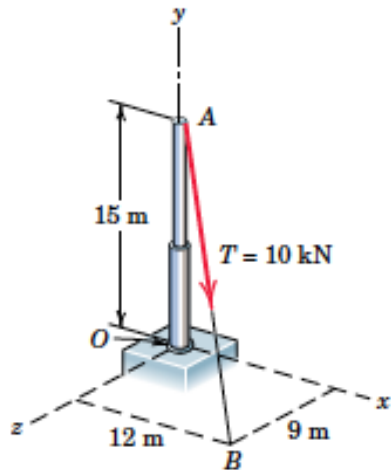
Force-couple system in D



- Example 1: A tension T of magnitude 10 kN is applied to the cable attached to the top A of the rigid mast and secured to the ground at B . Determine the moment M_z of T about the z -axis passing through the base O .



Solution:



$$\mathbf{T} = T\mathbf{n}_{AB} = 10 \left[\frac{12\mathbf{i} - 15\mathbf{j} + 9\mathbf{k}}{\sqrt{(12)^2 + (-15)^2 + (9)^2}} \right]$$

$$= 10(0.566\mathbf{i} - 0.707\mathbf{j} + 0.424\mathbf{k}) \text{ kN}$$

$$[\mathbf{M}_O = \mathbf{r} \times \mathbf{F}]$$

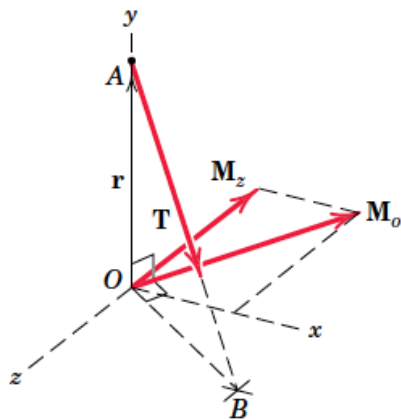
$$\mathbf{M}_O = 15\mathbf{j} \times 10(0.566\mathbf{i} - 0.707\mathbf{j} + 0.424\mathbf{k})$$

$$= 150(-0.566\mathbf{k} + 0.424\mathbf{i}) \text{ kN}\cdot\text{m}$$

$$M_z = \mathbf{M}_O \cdot \mathbf{k}$$

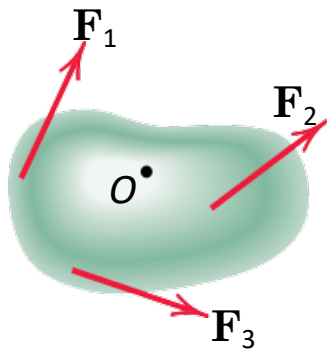
$$M_z = 150(-0.566\mathbf{k} + 0.424\mathbf{i}) \cdot \mathbf{k}$$

$$M_z = -84.9 \text{ kN}\cdot\text{m}.$$

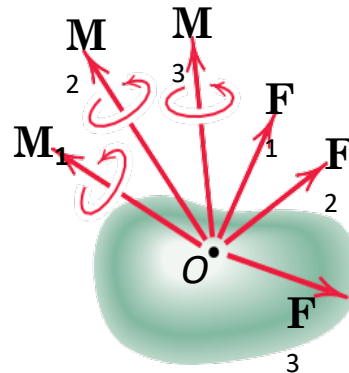


Resultant

Original System

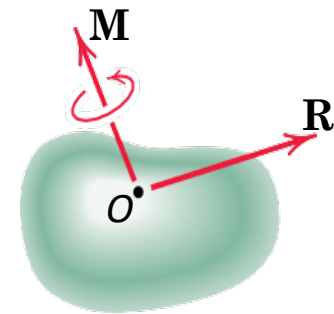


≡



≡

Equivalent System



$$\mathbf{R} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \dots = \Sigma \mathbf{F}$$

$$\mathbf{M} = \mathbf{M}_1 + \mathbf{M}_2 + \mathbf{M}_3 + \dots = \Sigma(\mathbf{r} \times \mathbf{F})$$

$$R_x = \Sigma F_x$$

$$R_y = \Sigma F_y$$

$$M_x = \Sigma(\mathbf{r} \times \mathbf{F})_x$$

$$M_y = \Sigma(\mathbf{r} \times \mathbf{F})_y$$

$$R_z = \Sigma F_z$$

$$M_z = \Sigma(\mathbf{r} \times \mathbf{F})_z$$

$$R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2 + (\Sigma F_z)^2}$$

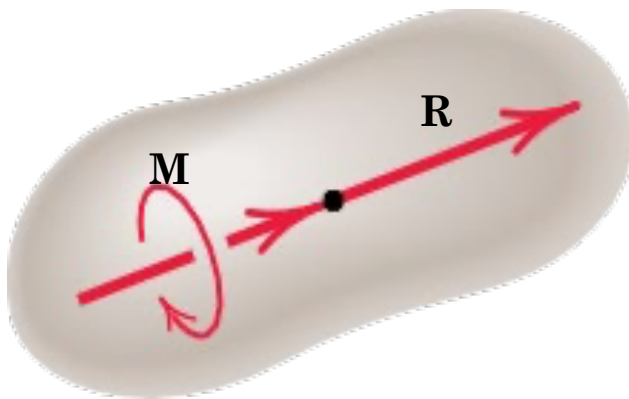
$$M = \sqrt{M_x^2 + M_y^2 + M_z^2}$$

Special Cases

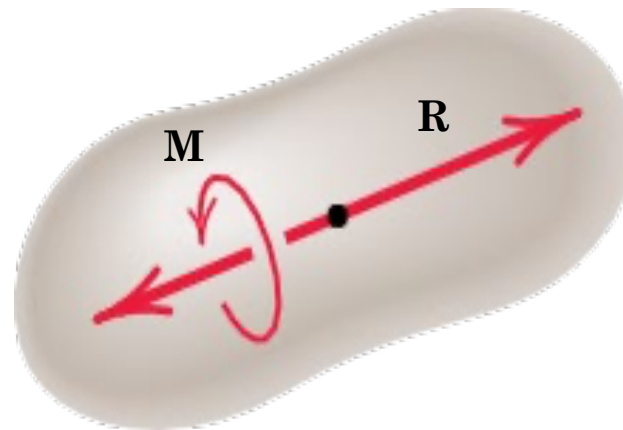
- **Concurrent Forces**
 - No moments about the point of concurrency.
- **Parallel Forces**
 - Magnitude of the parallel resultant force $\mathbf{R}$ is simply the magnitude of the algebraic sum of the given forces.
- **Coplanar Forces**
 - Similar to 2D cases

Wrench Resultant

- When the resultant couple vector M is parallel to the resultant force R , the resultant is called a wrench

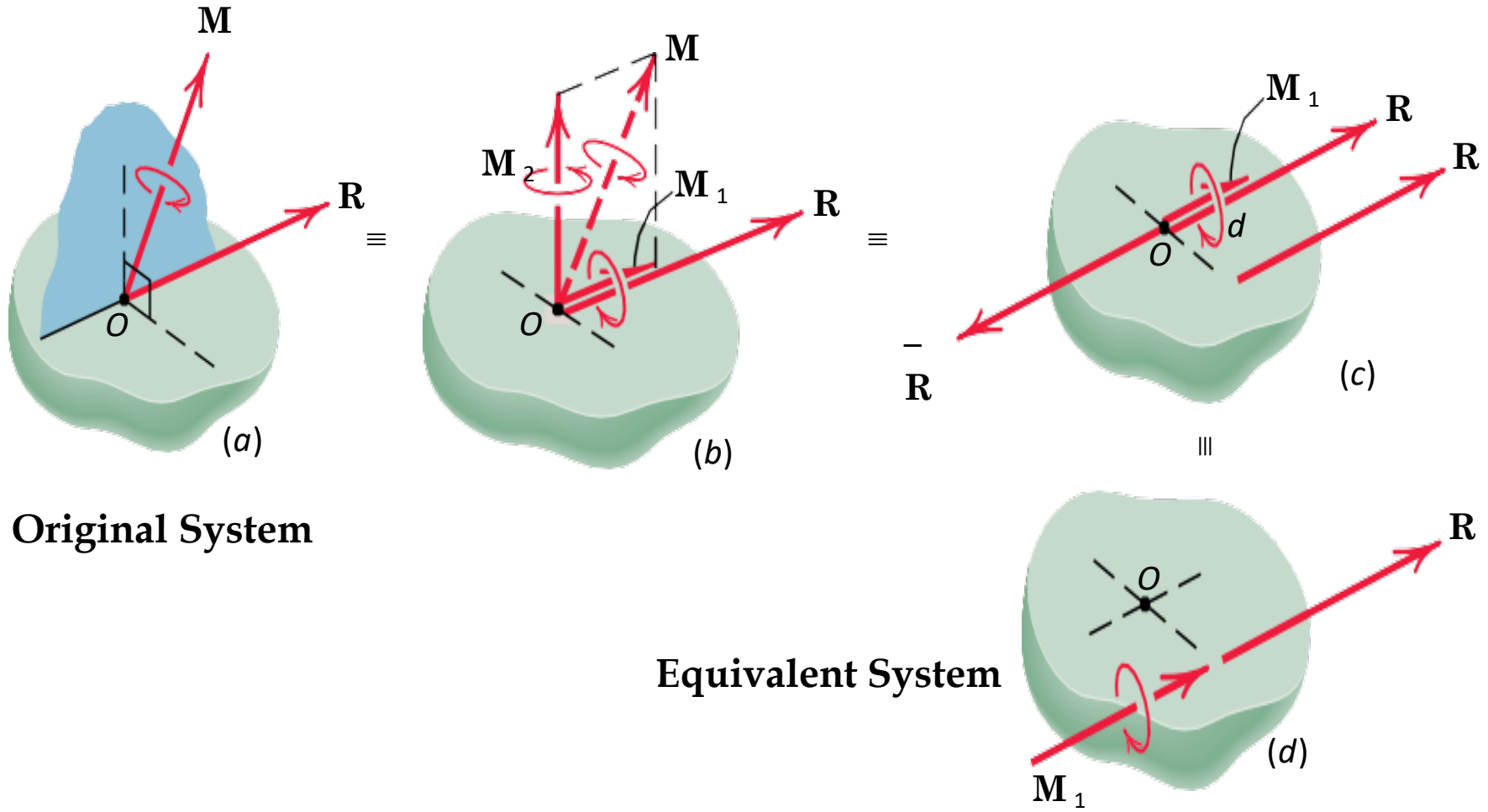


Positive wrench

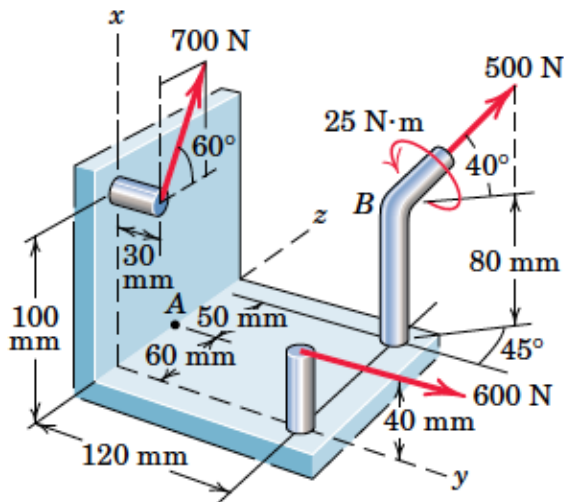


Negative wrench

Wrench Resultant



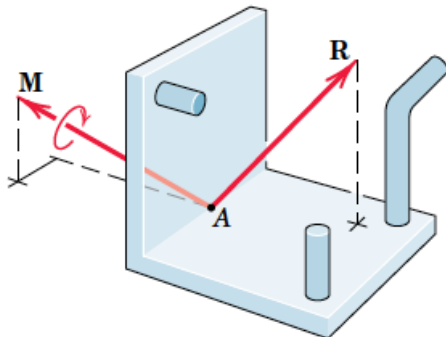
- Example 2: Replace the two forces and the negative wrench by a single force R applied at A and the corresponding couple M .



$$\begin{aligned}
 [R_x = \Sigma F_x] \quad R_x &= 500 \sin 40^\circ + 700 \sin 60^\circ = 928 \text{ N} \\
 [R_y = \Sigma F_y] \quad R_y &= 600 + 500 \cos 40^\circ \cos 45^\circ = 871 \text{ N} \\
 [R_z = \Sigma F_z] \quad R_z &= 700 \cos 60^\circ + 500 \cos 40^\circ \sin 45^\circ = 621 \text{ N}
 \end{aligned}$$

$$\mathbf{R} = 928\mathbf{i} + 871\mathbf{j} + 621\mathbf{k} \text{ N}$$

$$R = \sqrt{(928)^2 + (871)^2 + (621)^2} = 1416 \text{ N}$$



Example 2

$$[\mathbf{M} = \mathbf{r} \times \mathbf{F}]$$

$$\mathbf{M}_{500} = (0.08\mathbf{i} + 0.12\mathbf{j} + 0.05\mathbf{k}) \times 500(\mathbf{i} \sin 40^\circ + \mathbf{j} \cos 40^\circ \cos 45^\circ + \mathbf{k} \cos 40^\circ \sin 45^\circ)$$

$$\mathbf{M}_{500} = 18.95\mathbf{i} - 5.59\mathbf{j} - 16.90\mathbf{k} \text{ N}\cdot\text{m}$$

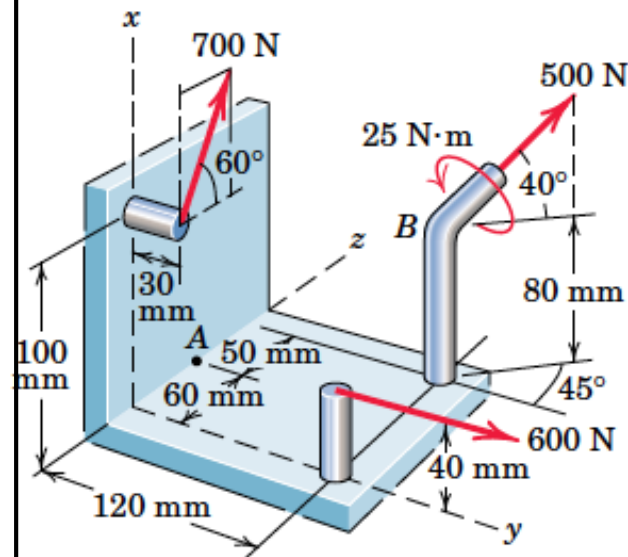
$$\begin{aligned} \mathbf{M}_{600} &= (600)(0.060)\mathbf{i} + (600)(0.040)\mathbf{k} \\ &= 36.0\mathbf{i} + 24.0\mathbf{k} \text{ N}\cdot\text{m} \end{aligned}$$

$$\begin{aligned} \mathbf{M}_{700} &= (700 \cos 60^\circ)(0.030)\mathbf{i} - [(700 \sin 60^\circ)(0.060) \\ &\quad + (700 \cos 60^\circ)(0.100)]\mathbf{j} - (700 \sin 60^\circ)(0.030)\mathbf{k} \\ &= 10.5\mathbf{i} - 71.4\mathbf{j} - 18.19\mathbf{k} \text{ N}\cdot\text{m} \end{aligned}$$

$$\begin{aligned} \mathbf{M}' &= 25.0(-\mathbf{i} \sin 40^\circ - \mathbf{j} \cos 40^\circ \cos 45^\circ - \mathbf{k} \cos 40^\circ \sin 45^\circ) \\ &= -16.07\mathbf{i} - 13.54\mathbf{j} - 13.54\mathbf{k} \text{ N}\cdot\text{m} \end{aligned}$$

$$\mathbf{M} = 49.4\mathbf{i} - 90.5\mathbf{j} - 24.6\mathbf{k} \text{ N}\cdot\text{m}$$

$$M = \sqrt{(49.4)^2 + (90.5)^2 + (24.6)^2} = 106.0 \text{ N}\cdot\text{m}$$



THANK YOU

Engineering Mechanics

Lecture-IV

Equilibrium in 2D

By

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Assistant Professor

Civil and Infrastructure Engineering

Indian Institute of Technology Jodhpur

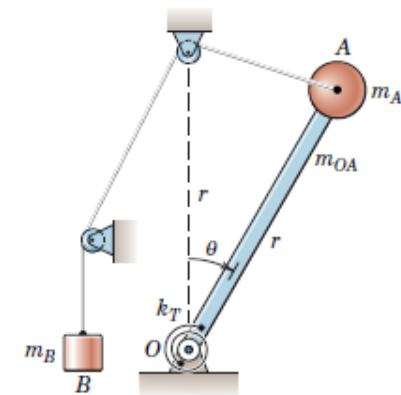
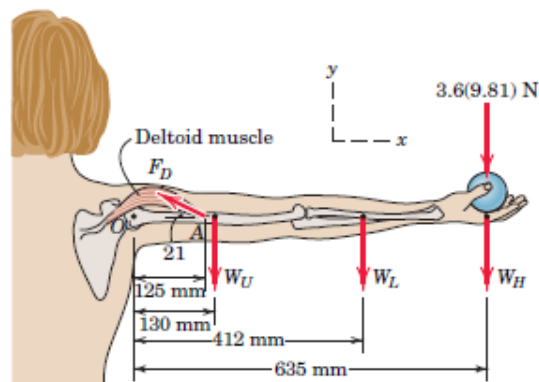
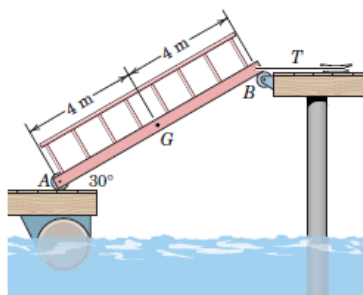
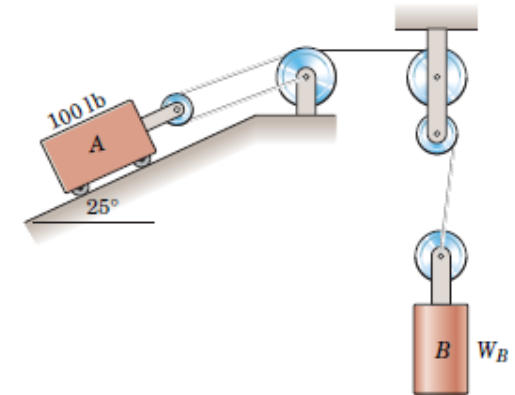
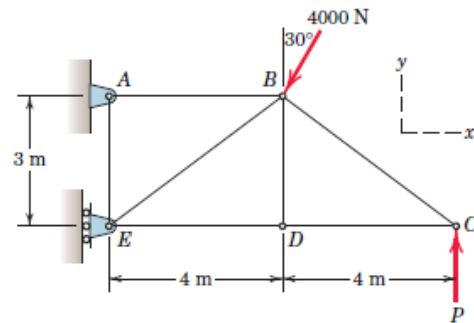
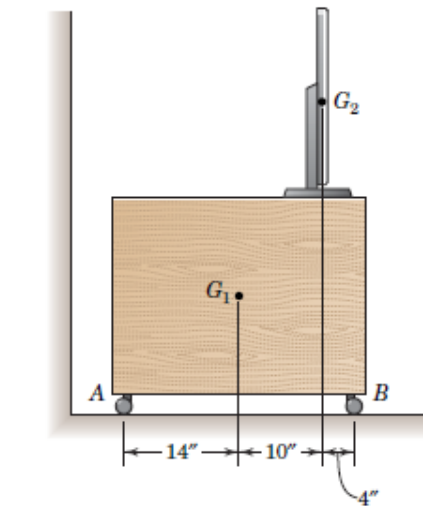
Equilibrium

- When a body is in equilibrium, the resultant of *all* forces acting on it is zero.
- Thus, the resultant force **R** and the resultant couple **M** are both zero, and we have the equilibrium equations

$$\mathbf{R} = \Sigma \mathbf{F} = 0$$

$$\mathbf{M} = \Sigma \mathbf{M} = 0$$

EQUILIBRIUM IN TWO DIMENSIONS



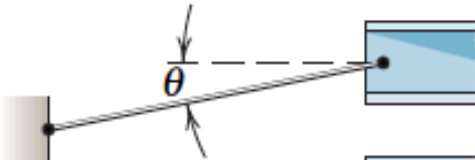
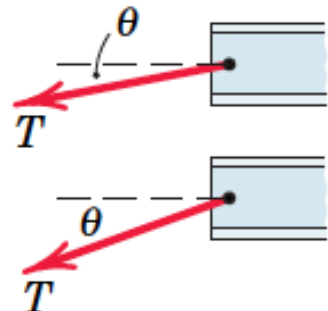
Free-body diagram

- The free-body diagram is the most important single step in the solution of problems in mechanics.

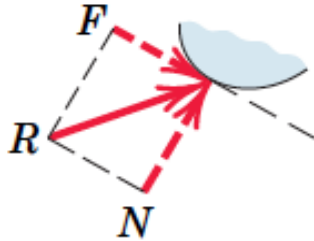
CONSTRUCTION OF FREE-BODY DIAGRAMS

- *Step 1.* Decide which **system** to isolate.
- *Step 2.* Next **isolate the chosen system** by drawing a diagram which represents its *complete external boundary*.
- *Step 3.* Identify and represent **all forces which act on the isolated system** as applied by the removed contacting and attracting bodies
- *Step 4.* Show the **choice of coordinate axes** directly on the diagram. Pertinent **dimensions** may also be represented for convenience.

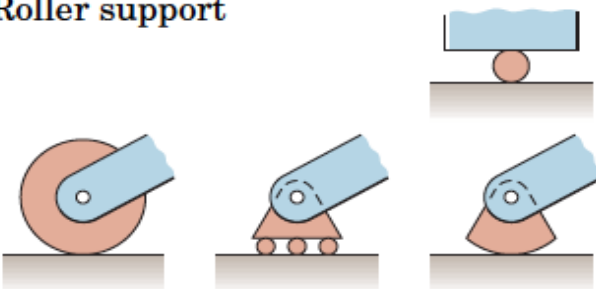
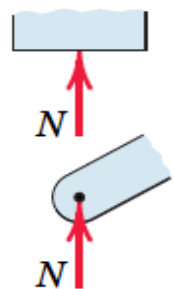
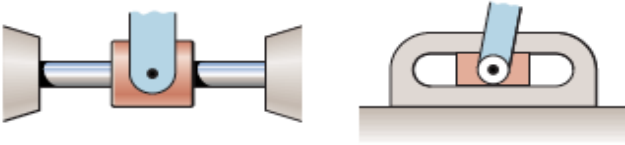
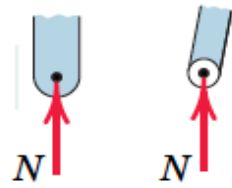
Action of force in 2D

MODELING THE ACTION OF FORCES IN TWO-DIMENSIONAL ANALYSIS	
Type of Contact and Force Origin	Action on Body to Be Isolated
1. Flexible cable, belt, chain, or rope Weight of cable negligible  Weight of cable not negligible 	 Force exerted by a flexible cable is always a tension away from the body in the direction of the cable.

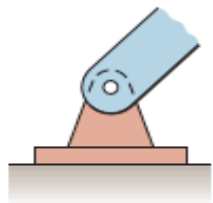
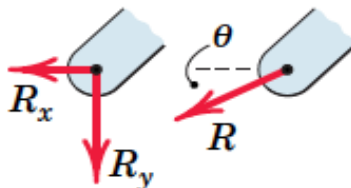
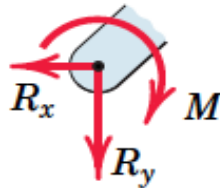
Action of force in 2D

MODELING THE ACTION OF FORCES IN TWO-DIMENSIONAL ANALYSIS	
Type of Contact and Force Origin	Action on Body to Be Isolated
2. Smooth surfaces 	 Contact force is compressive and is normal to the surface.
3. Rough surfaces 	 Rough surfaces are capable of supporting a tangential component F (frictional force) as well as a normal component N of the resultant contact force R.

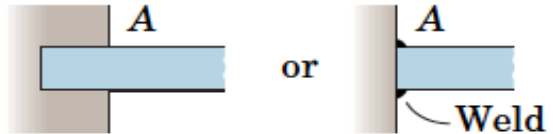
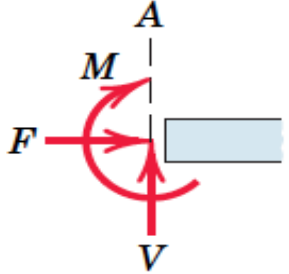
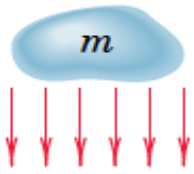
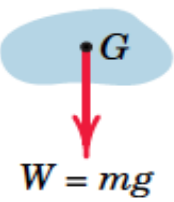
Action of force in 2D

MODELING THE ACTION OF FORCES IN TWO-DIMENSIONAL ANALYSIS	
Type of Contact and Force Origin	Action on Body to Be Isolated
4. Roller support 	 Roller, rocker, or ball support transmits a compressive force normal to the supporting surface.
5. Freely sliding guide 	 Collar or slider free to move along smooth guides; can support force normal to guide only.

Action of force in 2D

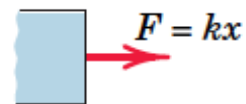
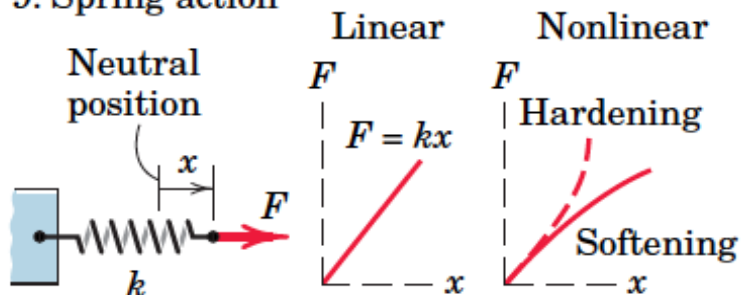
MODELING THE ACTION OF FORCES IN TWO-DIMENSIONAL ANALYSIS (<i>cont.</i>)	
Type of Contact and Force Origin	Action on Body to Be Isolated
6. Pin connection 	Pin free to turn  Pin not free to turn  A freely hinged pin connection is capable of supporting a force in any direction in the plane normal to the pin axis. We may either show two components R_x and R_y or a magnitude R and direction θ. A pin not free to turn also supports a couple M.

Action of force in 2D

MODELING THE ACTION OF FORCES IN TWO-DIMENSIONAL ANALYSIS	
Type of Contact and Force Origin	Action on Body to Be Isolated
7. Built-in or fixed support 	 A built-in or fixed support is capable of supporting an axial force F, a transverse force V (shear force), and a couple M (bending moment) to prevent rotation.
8. Gravitational attraction 	 The resultant of gravitational attraction on all elements of a body of mass m is the weight $W = mg$ and acts toward the center of the earth through the center of gravity G.

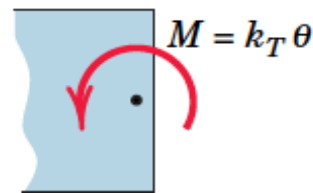
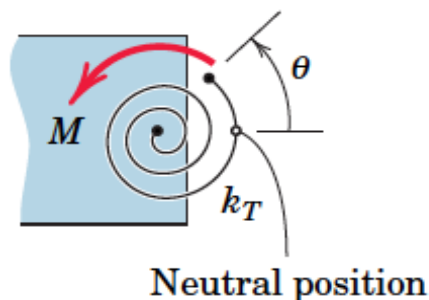
Action of force in 2D

9. Spring action



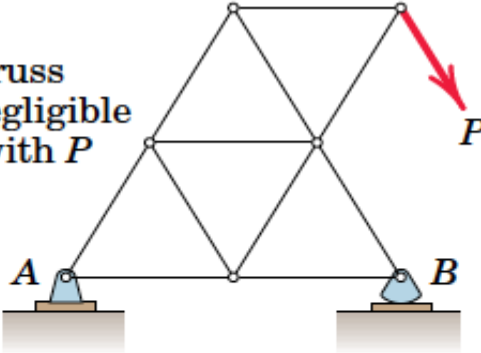
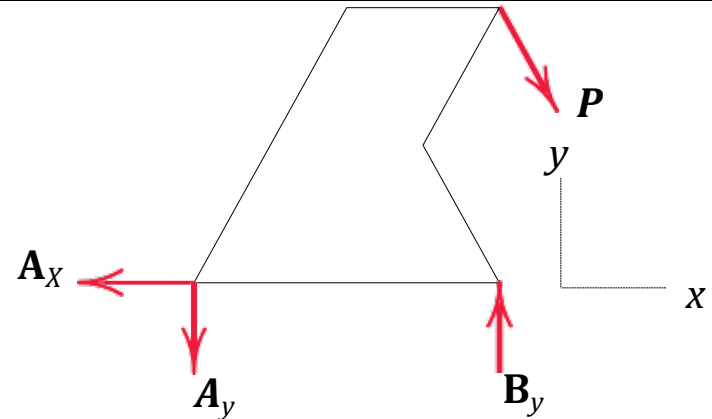
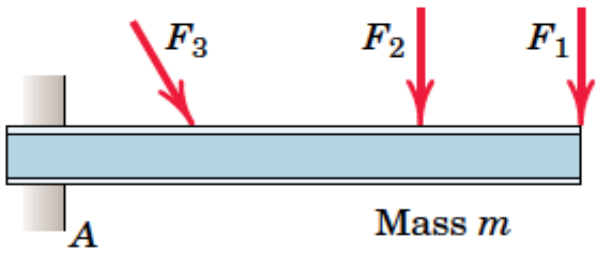
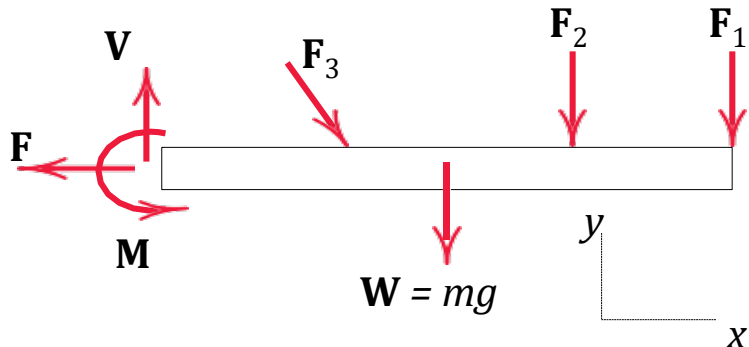
Spring force is tensile if the spring is stretched and compressive if compressed. For a linearly elastic spring the stiffness k is the force required to deform the spring a unit distance.

10. Torsional spring action



For a linear torsional spring, the applied moment M is proportional to the angular deflection θ from the neutral position. The stiffness k_T is the moment required to deform the spring one radian.

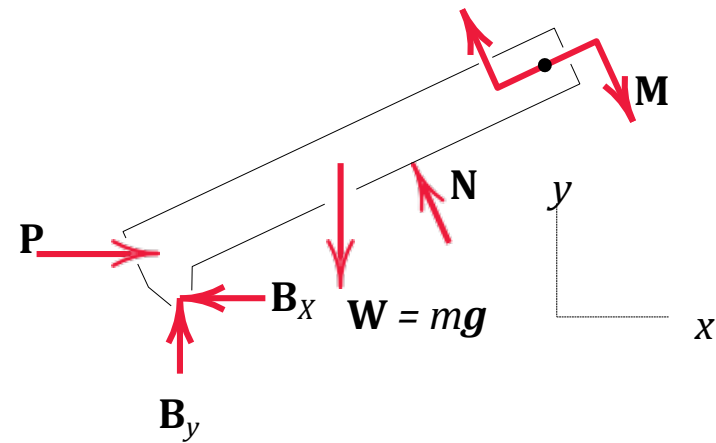
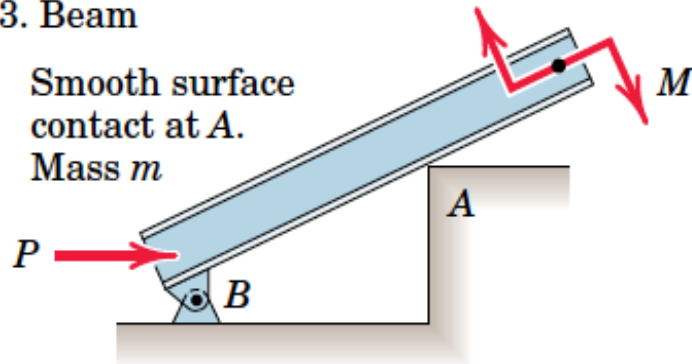
Freebody Diagram in 2D

SAMPLE FREE-BODY DIAGRAMS	
Mechanical System	Free-Body Diagram of Isolated Body
1. Plane truss Weight of truss assumed negligible compared with P 	
2. Cantilever beam 	

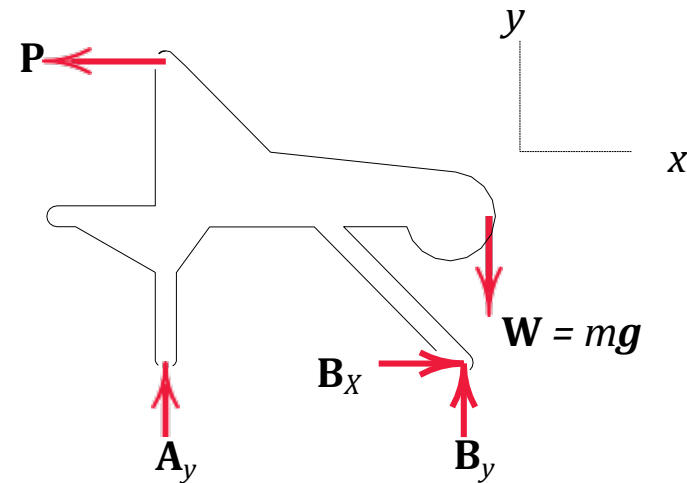
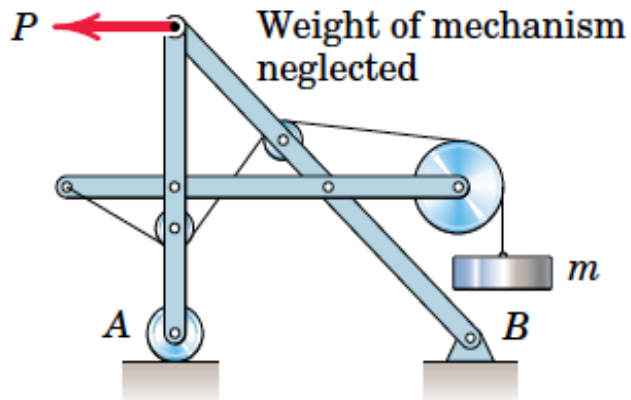
Freebody Diagram in 2D

3. Beam

Smooth surface
contact at A.
Mass m



4. Rigid system of interconnected bodies analyzed as a single unit



EQUILIBRIUM CONDITIONS (2D)

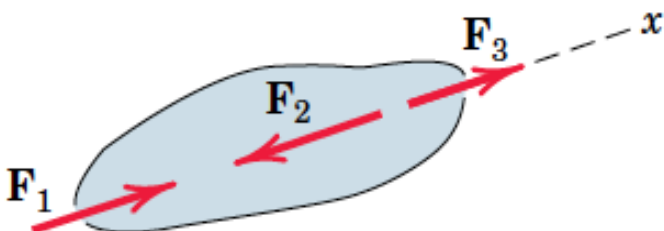
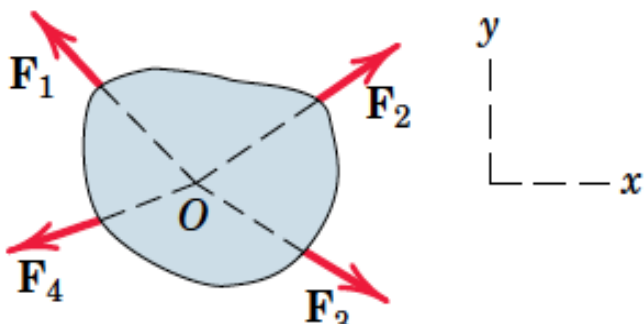
- **Equilibrium:** the condition in which the resultant of all forces and moments acting on a body is zero
- which in two dimensions may be written in scalar form as

$$\Sigma F_x = 0$$

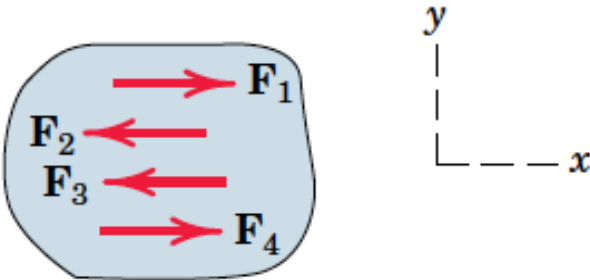
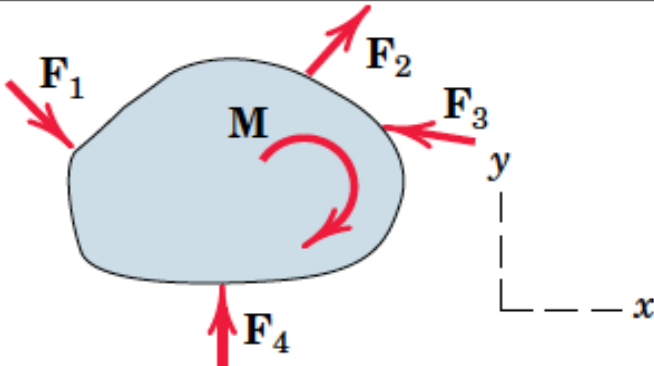
$$\Sigma F_y = 0$$

$$\Sigma M_o = 0$$

Categories of equilibrium

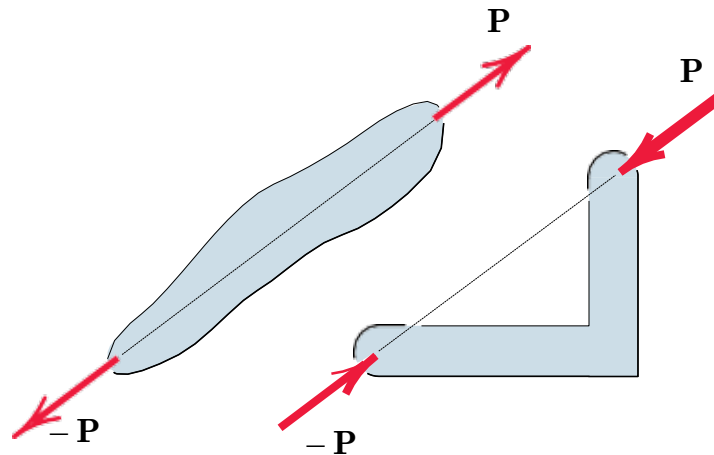
CATEGORIES OF EQUILIBRIUM IN TWO DIMENSIONS		
Force System	Free-Body Diagram	Independent Equations
1. Collinear		$\Sigma F_x = 0$
2. Concurrent at a point		$\Sigma F_x = 0$ $\Sigma F_y = 0$

Categories of equilibrium

CATEGORIES OF EQUILIBRIUM IN TWO DIMENSIONS		
Force System	Free-Body Diagram	Independent Equations
3. Parallel		$\Sigma F_x = 0 \quad \Sigma M_z = 0$
4. General		$\Sigma F_x = 0 \quad \Sigma M_z = 0$ $\Sigma F_y = 0$

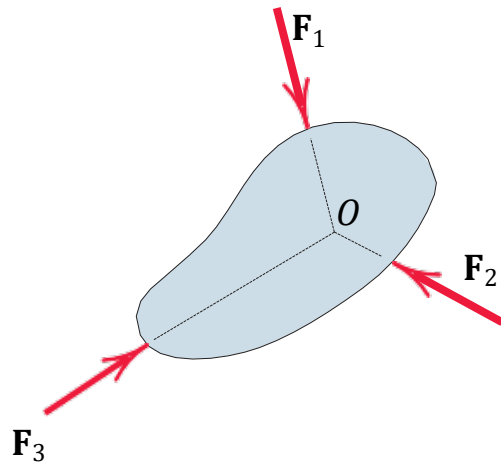
Two- Force Members

- A *two-force member* to be in equilibrium, the forces must be *equal, opposite, and collinear*

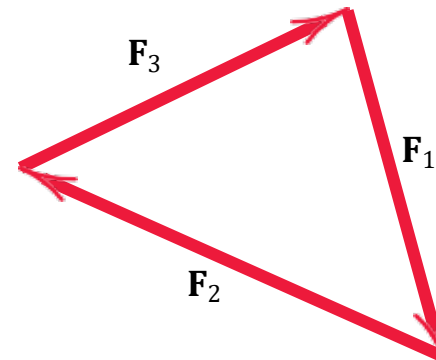


Three-Force Members

- We see that equilibrium requires the lines of action of the three forces to be *concurrent*.
- If they were *not concurrent*, then one of the forces would exert a resultant *moment* about the point of intersection of the other two



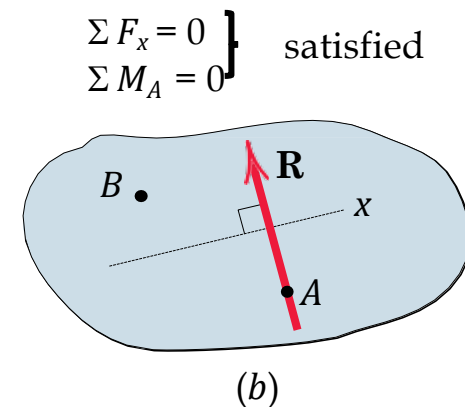
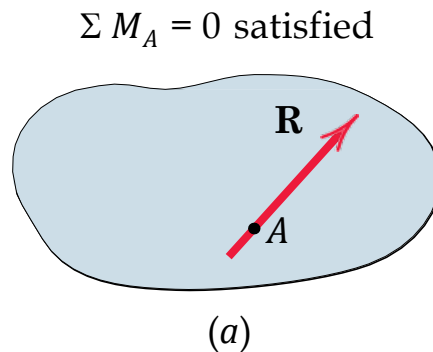
$$\Sigma \mathbf{F} = 0$$



2nd Alternative Equilibrium Equations

1. $\Sigma M_A = 0$; Resultant may still exist, cannot be a couple, but must be a force R passing through A .
2. $\Sigma F_x = 0$; Resultant may still exist, not only must pass through A , but also must be perpendicular to the x -direction as shown.
3. $\Sigma M_B = 0$; Resultant can not exist if B is any point such that the line AB is not perpendicular to the x -direction

These form second set of alternative Equilibrium Equations.

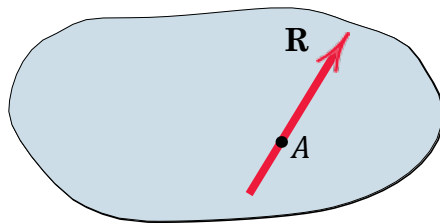


3rd Alternative Equilibrium Equations

1. $\Sigma M_A = 0$; Resultant may still exist
2. $\Sigma M_B = 0$; Resultant may still exist
3. $\Sigma M_C = 0$; Resultant can not exist if A, B, and C are any three points not on the same straight line.

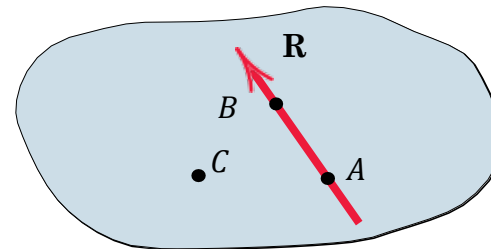
These form third set of alternative Equilibrium Equations.

$\Sigma M_A = 0$ satisfied



(c)

$\Sigma M_A = 0$
 $\Sigma M_B = 0$ } satisfied



(d)

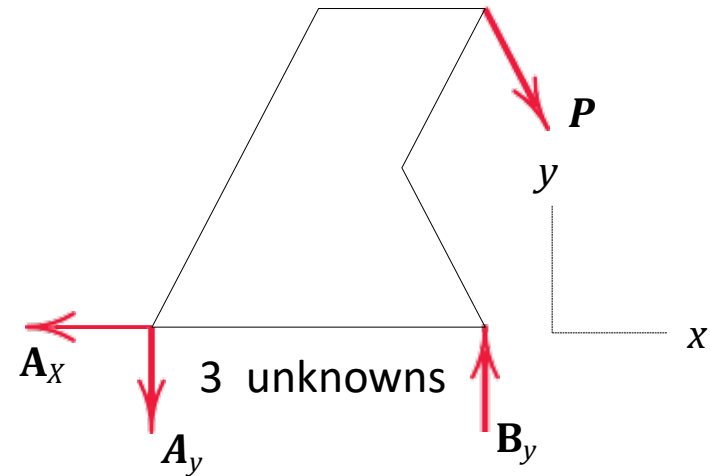
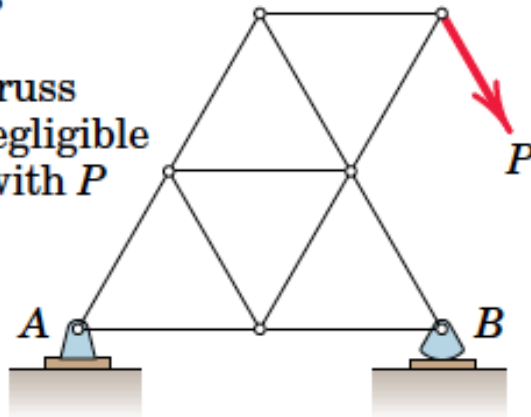
Constraints and Statically Determinacy

- The **equilibrium equations** are both necessary and sufficient conditions to establish the equilibrium of a body.
- Whether the equations are **adequate to determine all the unknowns** depends on the **characteristics of the constraints** provided by its supports.
- ***Constraints*** mean the restriction of movement.

Number of unknowns

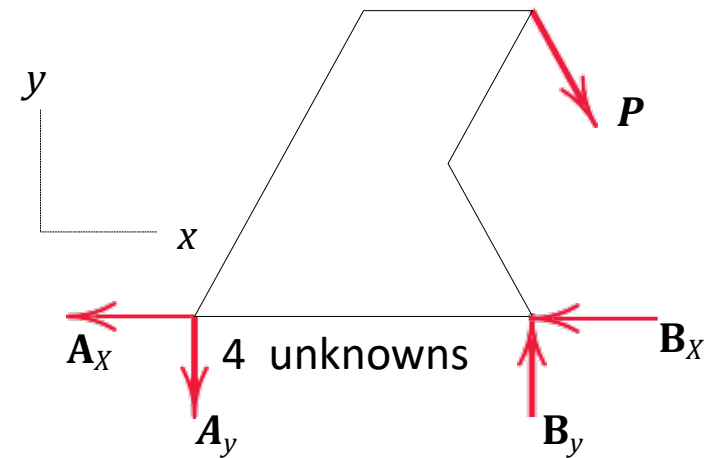
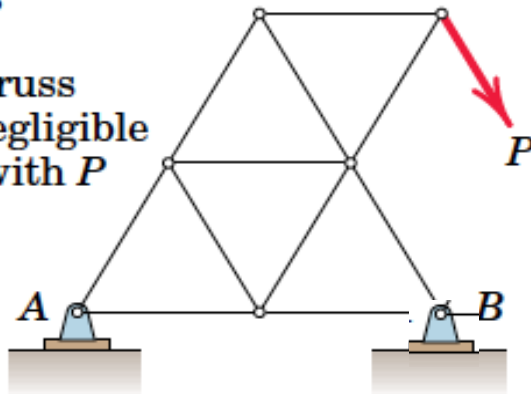
1. Plane truss

Weight of truss
assumed negligible
compared with P



1. Plane truss

Weight of truss
assumed negligible
compared with P



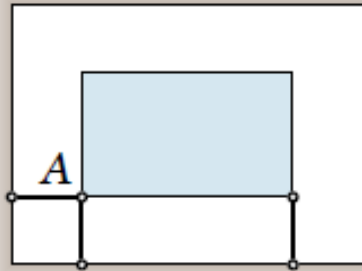
Statically indeterminate

- A rigid body, or rigid combination of elements treated as a single body, which possesses **more external supports or constraints than are necessary to maintain an equilibrium** position is called *statically indeterminate*.
- Supports which can be removed without destroying the equilibrium condition are said to be *redundant*.
- *Degree of statical indeterminacy*: the total number of unknown external forces, minus the number of available independent equations of equilibrium.

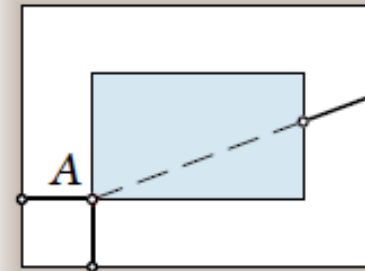
Statically Determinate

- Bodies which are supported by the minimum number of constraints necessary to ensure an equilibrium configuration are called *statically determinate*,
- for such bodies the equilibrium equations are sufficient to determine the unknown

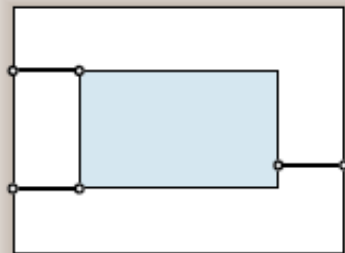
Adequacy of Constraints



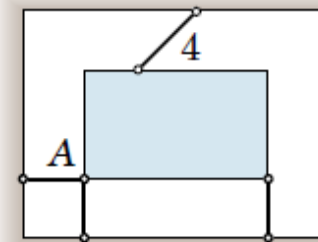
(a) Complete fixity
Adequate constraints



(b) Incomplete fixity
Partial constraints



(c) Incomplete fixity
Partial constraints

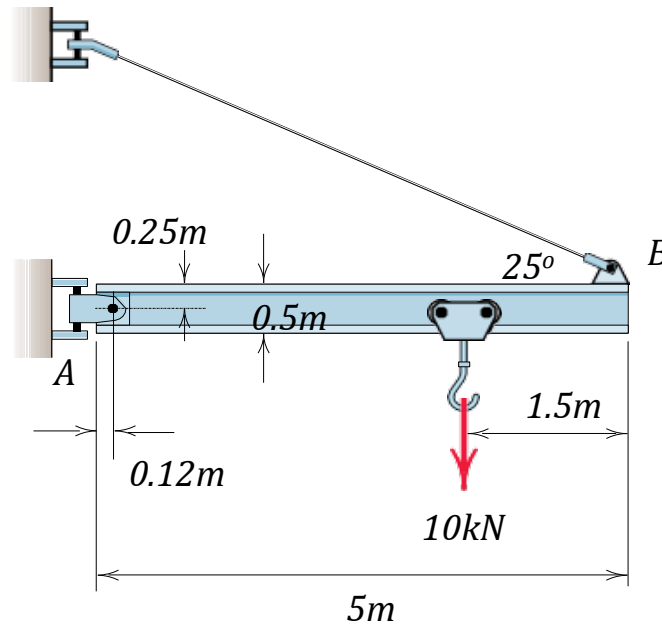


(d) Excessive fixity
Redundant constraint

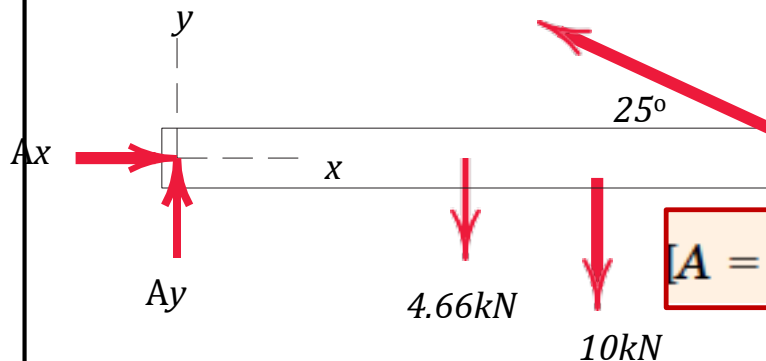
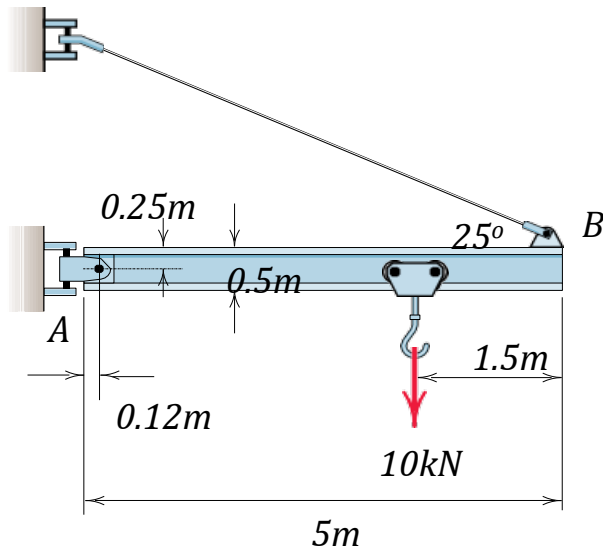
APPROACH TO SOLVING PROBLEMS

1. **Identify** clearly the quantities which are **known** and **unknown**
2. Make an unambiguous **choice of the body to be isolated** and draw its complete free-body diagram
3. **Choose** a convenient set of **reference axes**, badly! always using righthanded axes when vector cross products are employed.
4. **Identify and state** the applicable **force and moment principles** or equations which govern the equilibrium conditions of the problem.
5. **Match** the number of **independent equations** with the number of **unknowns** in each problem.
6. **Carry out the solution** and check the results

- **Example 1:** Determine the magnitude T of the tension in the supporting cable and the magnitude of the force on the pin at A for the jib crane shown. The beam AB is a standard 0.5-m I-beam with a mass of 95 kg per meter of length.



Solution:



$$[\Sigma M_A = 0]$$

$$(T \cos 25^\circ)0.25 + (T \sin 25^\circ)(5 - 0.12) - 10(5 - 1.5 - 0.12) - 4.66(2.5 - 0.12) = 0$$

$$T = 19.61 \text{ kN}$$

$$[\Sigma F_x = 0]$$

$$A_x - 19.61 \cos 25^\circ = 0$$

$$A_x = 17.77 \text{ kN}$$

$$[\Sigma F_y = 0]$$

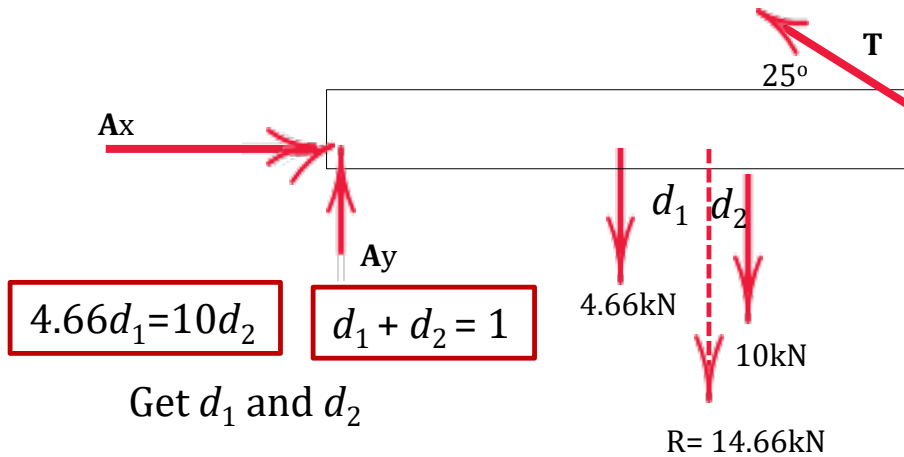
$$A_y + 19.61 \sin 25^\circ - 4.66 - 10 = 0$$

$$A_y = 6.37 \text{ kN}$$

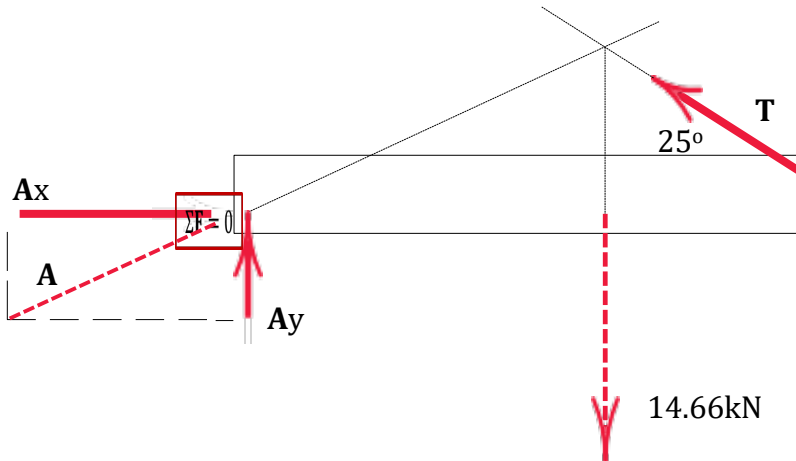
$$[A = \sqrt{A_x^2 + A_y^2}]$$

$$A = \sqrt{(17.77)^2 + (6.37)^2} = 18.88 \text{ kN}$$

Step 1: Find resultant of 4.66 kN and 10kN

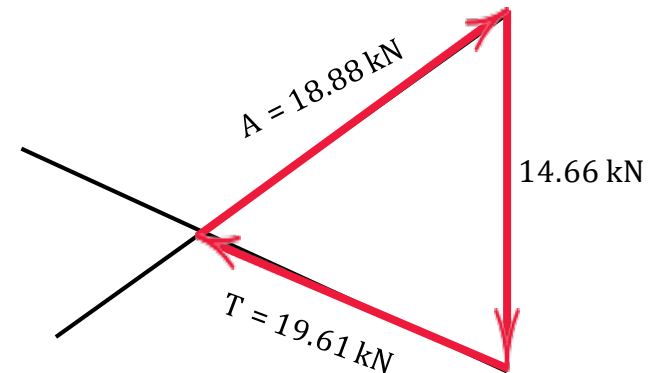


Step 2: Find direction of A



Graphical Approach

Step 3: Find magnitude of T and F



$$\Sigma F = 0$$

THANK YOU

Engineering Mechanics

Lecture 5: Equilibrium in 3-D

By
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EQUILIBRIUM IN THREE DIMENSIONS

- EQUILIBRIUM CONDITIONS

$$\Sigma \mathbf{F} = 0$$

$$\Sigma F_x = \Sigma \mathbf{F} \cdot \mathbf{i} = 0$$

$$\Sigma F_y = \Sigma \mathbf{F} \cdot \mathbf{j} = 0$$

$$\Sigma F_z = \Sigma \mathbf{F} \cdot \mathbf{k} = 0$$

$$\Sigma \mathbf{M} = 0$$

$$\Sigma M_x = \Sigma \mathbf{M} \cdot \mathbf{i} = 0$$

$$\Sigma M_y = \Sigma \mathbf{M} \cdot \mathbf{j} = 0$$

$$\Sigma M_z = \Sigma \mathbf{M} \cdot \mathbf{k} = 0$$

Moment due to forces can be obtained as $\Sigma(\mathbf{r} \times \mathbf{F})$

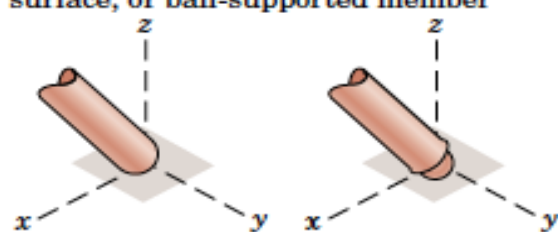
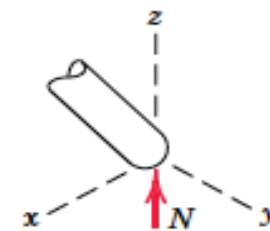
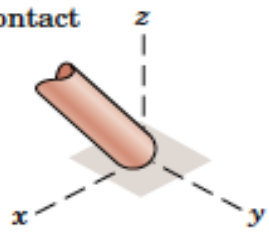
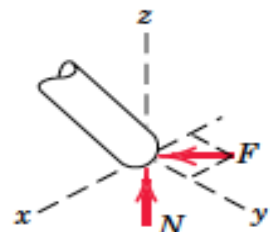
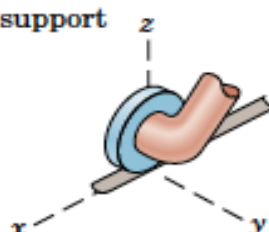
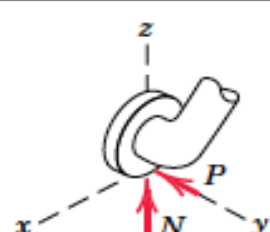
The six scalar relationships are independent conditions

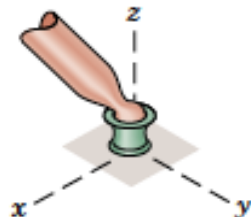
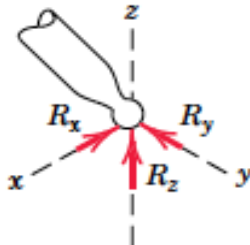
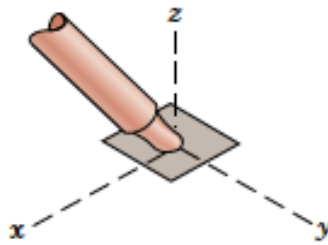
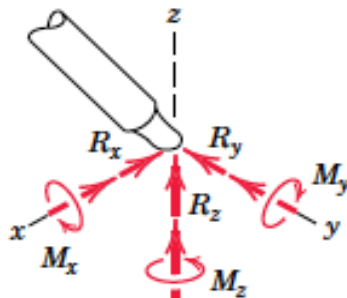
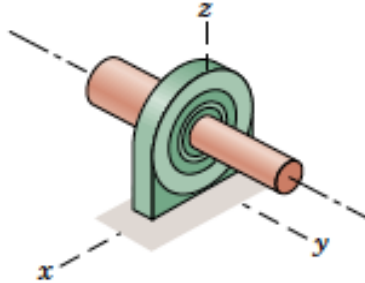
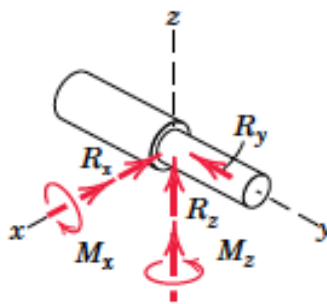
Caution: Use Right hand frame

Freebody Diagram

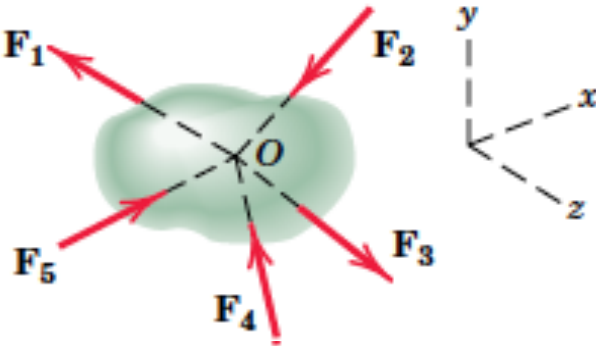
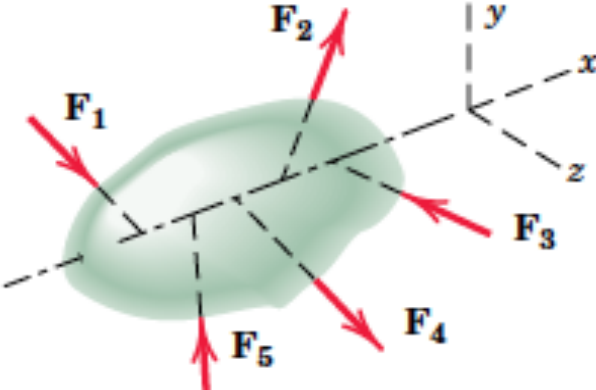
- The free-body diagram is the **most important single step** in the solution of problems in 3D.
- The essential purpose of the free-body diagram is **to develop a reliable picture of the physical action of all forces** (and couples if any) acting on a body.

Action of Force in 3D

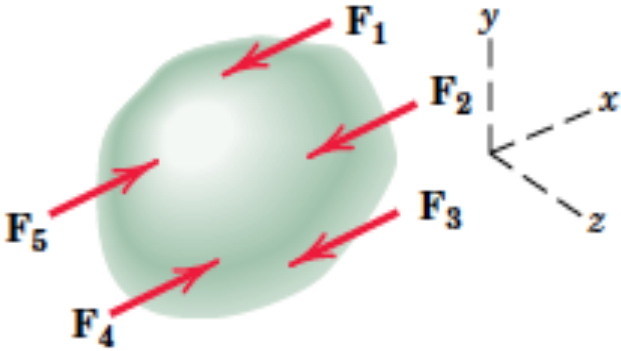
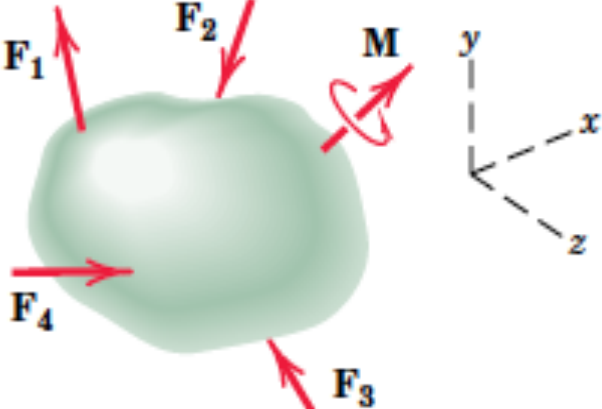
MODELING THE ACTION OF FORCES IN THREE-DIMENSIONAL ANALYSIS	
Type of Contact and Force Origin	Action on Body to Be Isolated
1. Member in contact with smooth surface, or ball-supported member 	 Force must be normal to the surface and directed toward the member.
2. Member in contact with rough surface 	 The possibility exists for a force F tangent to the surface (friction force) to act on the member, as well as a normal force N.
3. Roller or wheel support with lateral constraint 	 A lateral force P exerted by the guide on the wheel can exist, in addition to the normal force N.

4. Ball-and-socket joint 	 A ball-and-socket joint free to pivot about the center of the ball can support a force $\mathbf{R}$ with all three components.
5. Fixed connection (embedded or welded) 	 In addition to three components of force, a fixed connection can support a couple $\mathbf{M}$ represented by its three components.
6. Thrust-bearing support 	 Thrust bearing is capable of supporting axial force R_y as well as radial forces R_x and R_z. Couples M_x and M_z must, in some cases, be assumed zero in order to provide statical determinacy.

Categories of Equilibrium

CATEGORIES OF EQUILIBRIUM IN THREE DIMENSIONS		
Force System	Free-Body Diagram	Independent Equations
1. Concurrent at a point		$\Sigma F_x = 0$ $\Sigma F_y = 0$ $\Sigma F_z = 0$
2. Concurrent with a line		$\Sigma F_x = 0$ $\Sigma F_y = 0$ $\Sigma F_z = 0$ $\Sigma M_y = 0$ $\Sigma M_z = 0$

Categories of Equilibrium

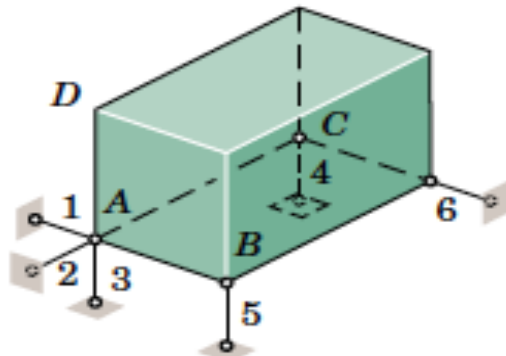
3. Parallel		$\Sigma F_x = 0$ $\Sigma M_y = 0$ $\Sigma M_z = 0$
4. General		$\Sigma F_x = 0$ $\Sigma F_y = 0$ $\Sigma F_z = 0$ $\Sigma M_x = 0$ $\Sigma M_y = 0$ $\Sigma M_z = 0$

Constraints and Statical Determinacy

- The **six scalar relations** necessary and sufficient conditions to establish equilibrium,
- **Do not necessarily provide** all of the information required to calculate the unknown forces acting **in a 3D equilibrium** situation

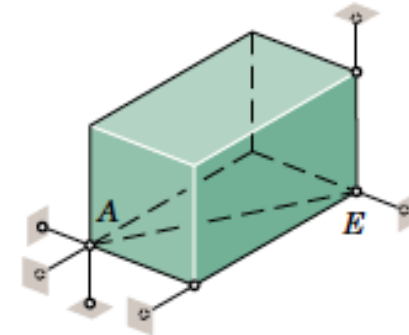
Adequacy of Constraints

Adequate Constraints



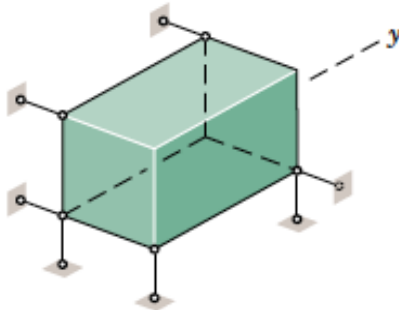
(a) Complete fixity
Adequate constraints

Partial Constraints: Can rotate about AE



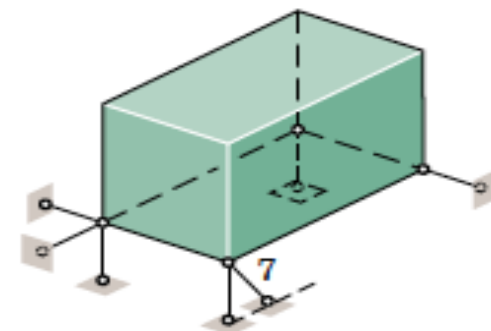
(b) Incomplete fixity
Partial constraints

Partial Constraints: Can translate along y



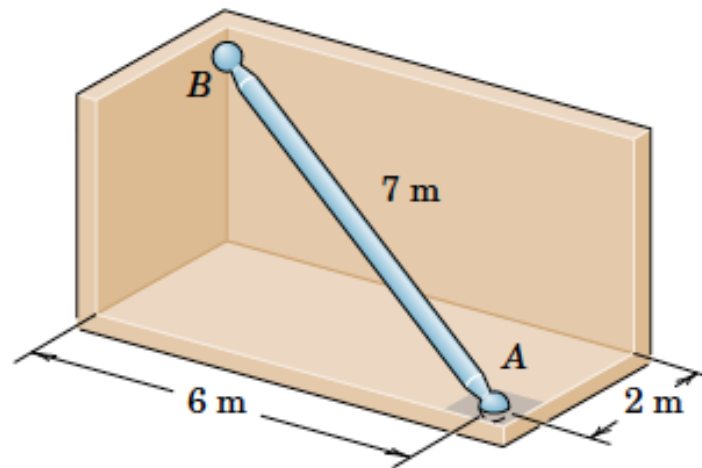
(c) Incomplete fixity
Partial constraints

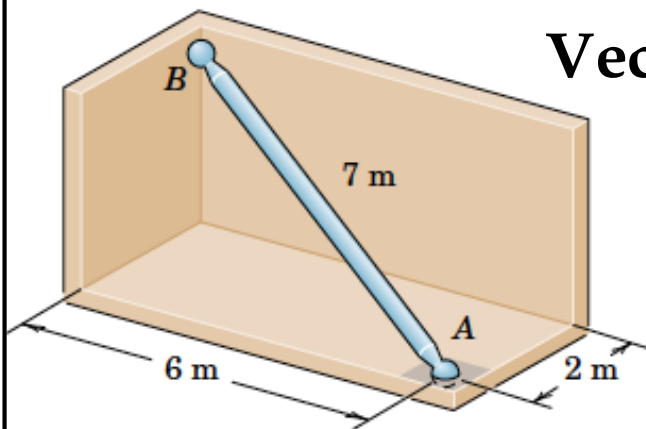
Redundant Constraints



(d) Excessive fixity
Redundant constraints

- Example: The uniform 7-m steel shaft has a mass of 200 kg and is supported by a ball and socket joint at A in the horizontal floor. The ball end B rests against the smooth vertical walls as shown. Compute the forces exerted by the walls and the floor on the ends of the shaft.





Vector Approach

$$W = mg = 200(9.81) = 1962 \text{ N}$$

$$7 = \sqrt{2^2 + 6^2 + h^2}$$

$$h = 3 \text{ m}$$

$$\mathbf{r}_{AB} = -2\mathbf{i} - 6\mathbf{j} + 3\mathbf{k} \text{ m}$$

$$\mathbf{r}_{AG} = -1\mathbf{i} - 3\mathbf{j} + 1.5\mathbf{k} \text{ m}$$

$$[\Sigma \mathbf{M}_A = 0]$$

$$\mathbf{r}_{AB} \times (\mathbf{B}_x + \mathbf{B}_y) + \mathbf{r}_{AG} \times \mathbf{W} = 0$$

$$(-2\mathbf{i} - 6\mathbf{j} + 3\mathbf{k}) \times (B_x\mathbf{i} + B_y\mathbf{j}) + (-\mathbf{i} - 3\mathbf{j} + 1.5\mathbf{k}) \times (-1962\mathbf{k}) = 0$$

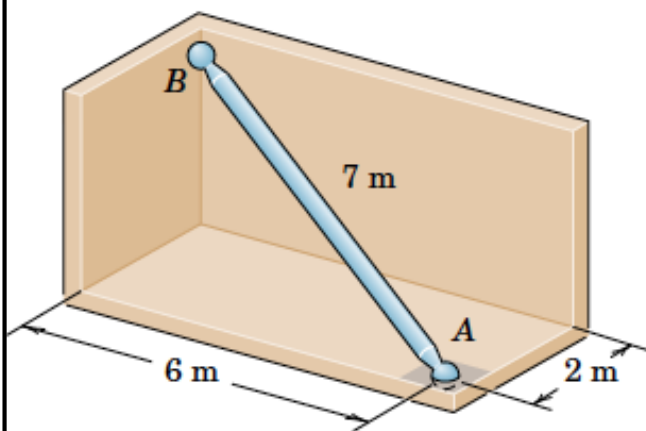
$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 & -6 & 3 \\ B_x & B_y & 0 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & -3 & 1.5 \\ 0 & 0 & -1962 \end{vmatrix} = 0$$

$$(-3B_y + 5890)\mathbf{i} + (3B_x - 1962)\mathbf{j} + (-2B_y + 6B_x)\mathbf{k} = 0$$

$$B_x = 654 \text{ N}$$

$$B_y = 1962 \text{ N}$$

Vector Approach



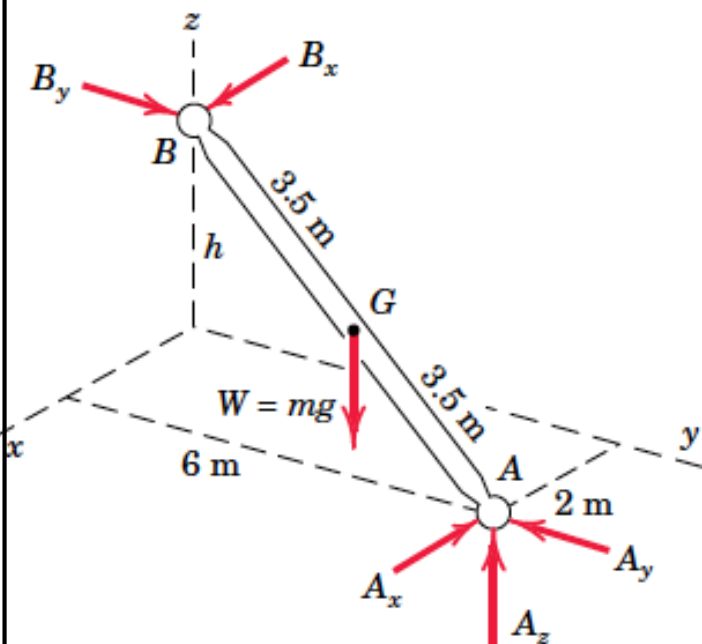
$$[\Sigma \mathbf{F} = 0]$$

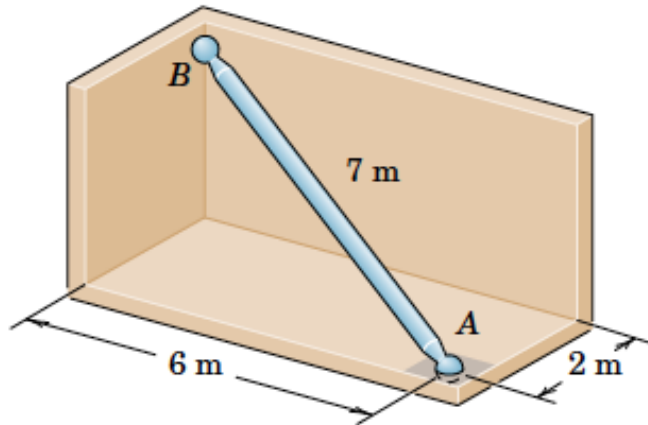
$$(654 - A_x)\mathbf{i} + (1962 - A_y)\mathbf{j} + (-1962 + A_z)\mathbf{k} = 0$$

$$A_x = 654 \text{ N} \quad A_y = 1962 \text{ N} \quad A_z = 1962 \text{ N}$$

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

$$= \sqrt{(654)^2 + (1962)^2 + (1962)^2} = 2850 \text{ N}$$





Scalar Approach

$$[\Sigma M_{A_x} = 0]$$

$$1962(3) - 3B_y = 0$$

$$B_y = 1962 \text{ N}$$

$$[\Sigma M_{A_y} = 0]$$

$$-1962(1) + 3B_x = 0$$

$$B_x = 654 \text{ N}$$

$$[\Sigma F_x = 0]$$

$$-A_x + 654 = 0$$

$$A_x = 654 \text{ N}$$

$$[\Sigma F_y = 0]$$

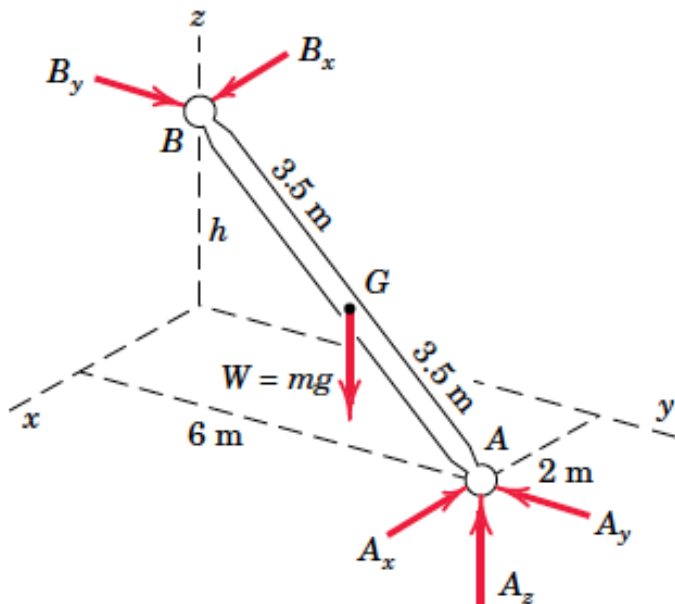
$$-A_y + 1962 = 0$$

$$A_y = 1962 \text{ N}$$

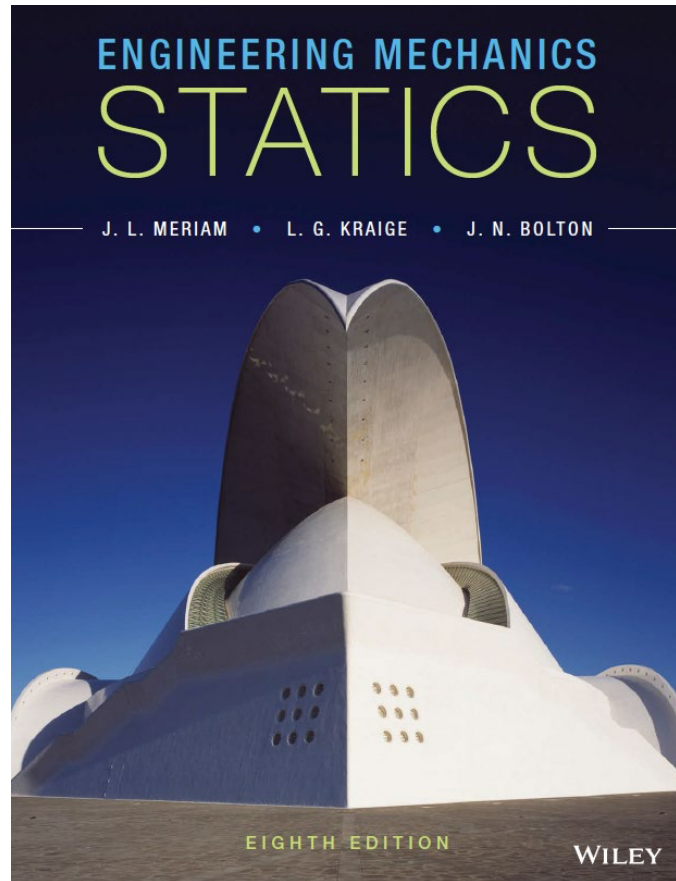
$$[\Sigma F_z = 0]$$

$$A_z - 1962 = 0$$

$$A_z = 1962 \text{ N}$$



References :



THANK YOU

Engineering Mechanics

Lecture 6

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Civil and Infrastructure Engineering

Indian Institute of Technology Jodhpur

Structure

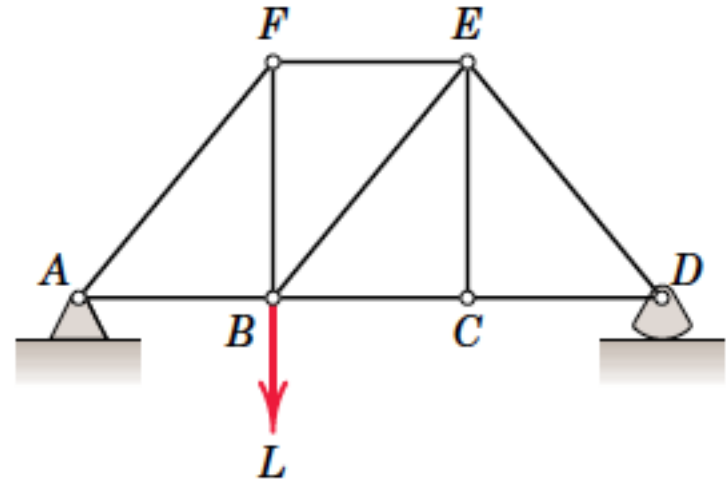
- Any **connected system of members** built to **support or transfer forces** and to safely **withstand the loads** applied to it.
- Type of Structures
 - Truss
 - Frame
 - Machine
- In this chapter, we consider only **statically determinate** structures

Trusses

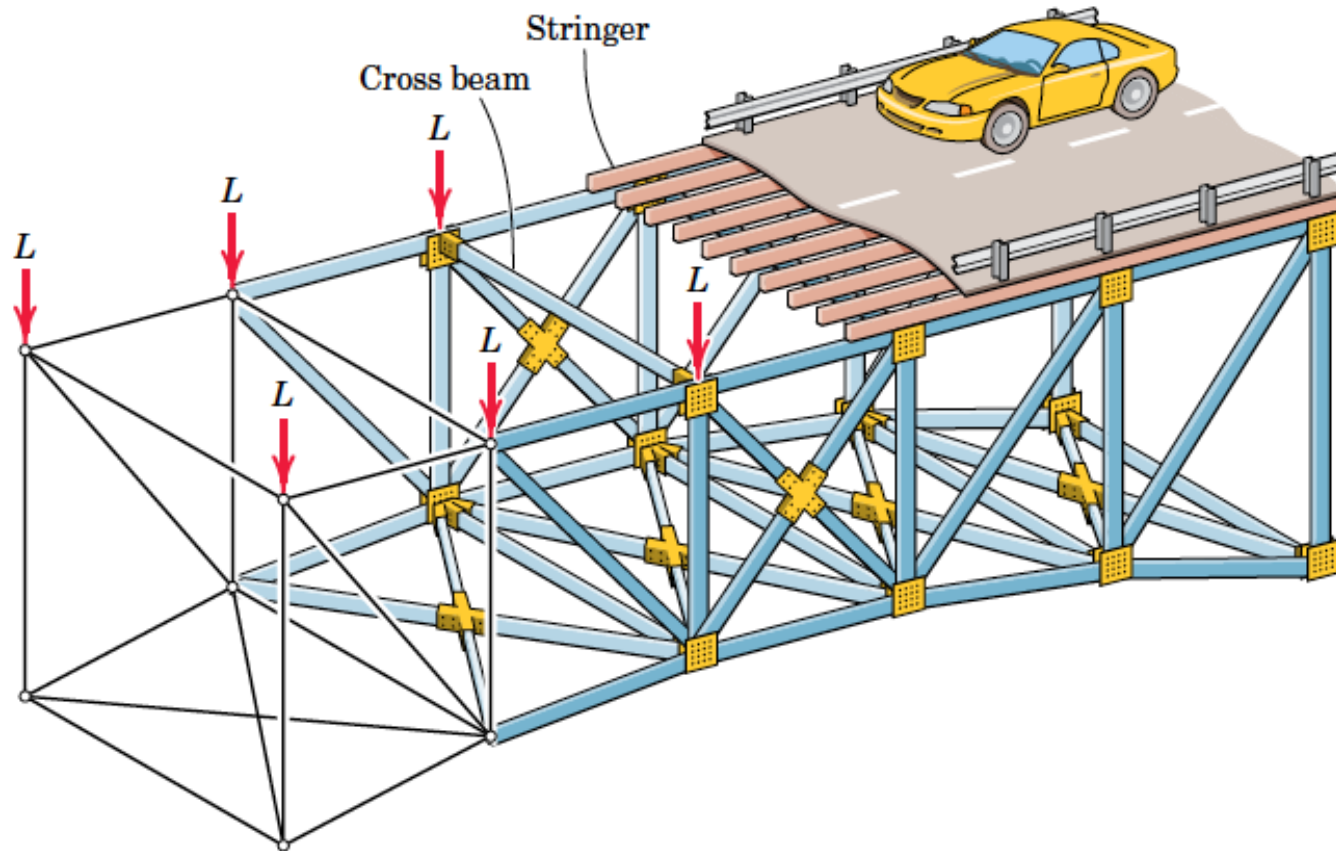


Plane Truss

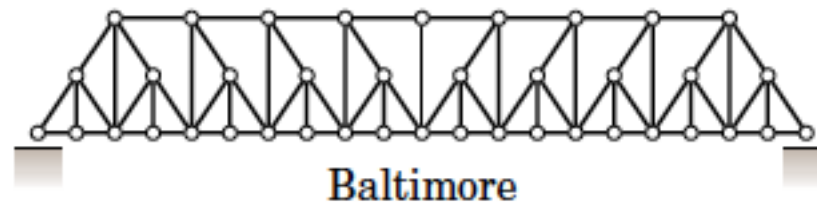
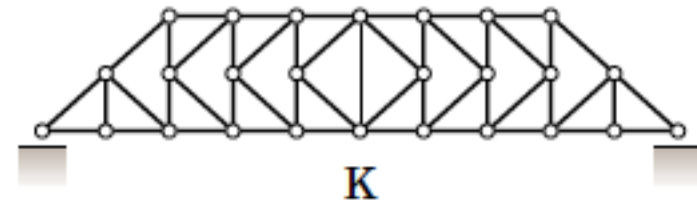
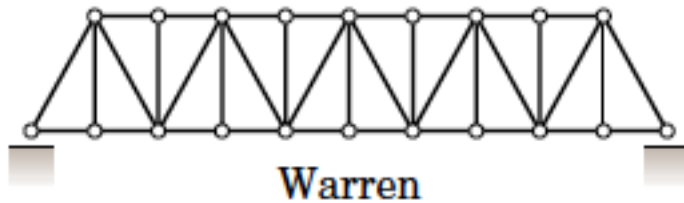
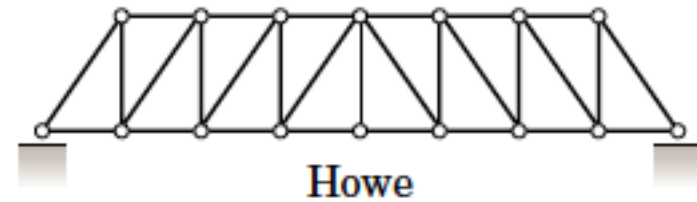
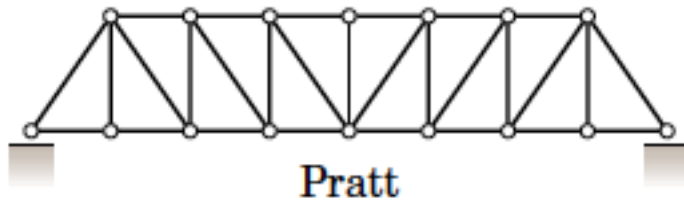
- A framework composed of **members joined at their ends** to form a rigid structure is called a truss.
- Examples: **Bridges, roof supports, derricks**, and other such structures



Section of a typical bridge structure

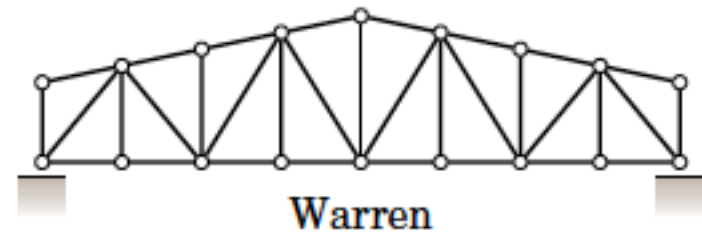
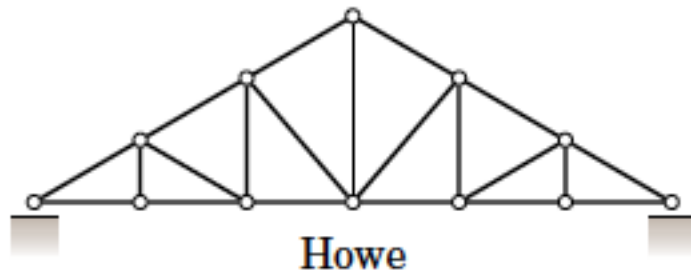
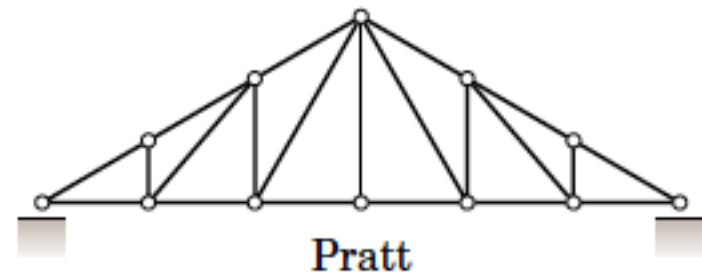
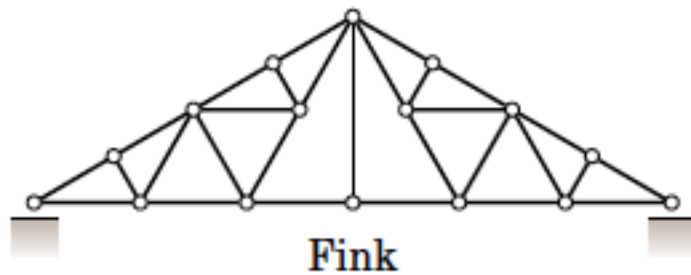


Commonly Used Bridge Trusses



Commonly Used Bridge Trusses

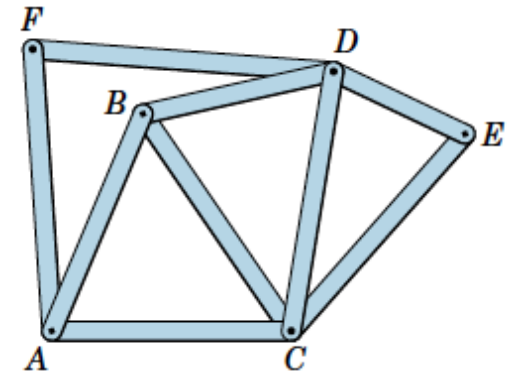
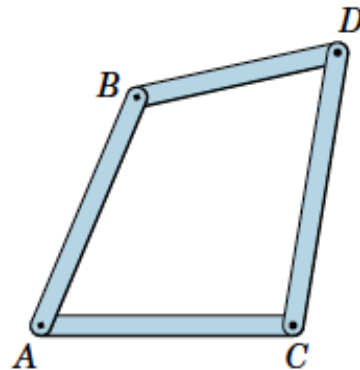
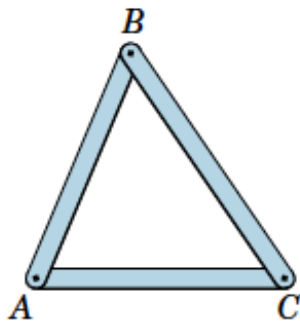
Commonly Used Roof Trusses



Commonly Used Roof Trusses

Simple Trusses

- The basic element of a plane truss is **the triangle**. It constitute a **rigid frame**.
- **Four or more bars** pin-jointed to form a polygon of as many sides constitute a **nonrigid frame**
- We can make nonrigid frame stable, **by adding a diagonal bar** joining A and D or B and C and thereby forming two triangles.
- Structures built from a **basic triangle** in the manner described are known as **simple trusses**.

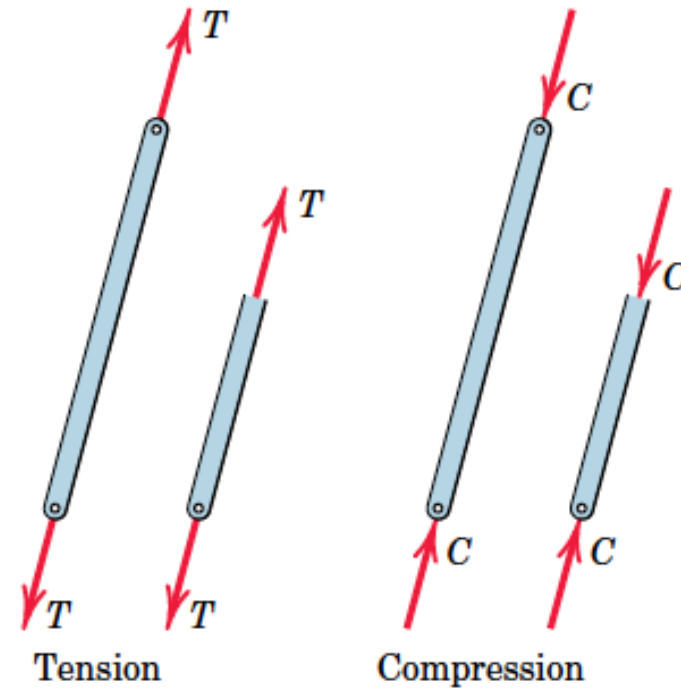


Design of a Truss

- First **determine the forces** in the various members and then
- Select **appropriate sizes and structural shapes** to withstand the forces.
- This chapter focuses on **determine the forces in the various members**

Assumption

- All members to be two-force members
- The two forces are equal, opposite, and collinear for equilibrium
- The member may be in **tension or compression**



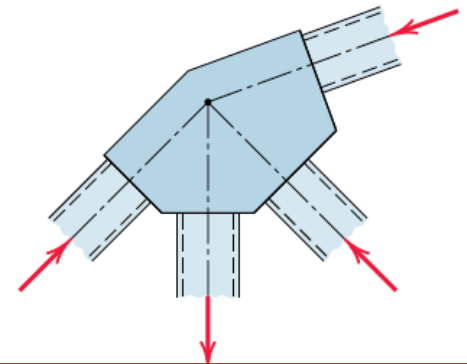
Two-Force Members

Assumption

- **Weight** of the member is **small** compared with the force it supports.
- If not we can **replace the weight W** of the member by **two forces, each $W/2$** if the member is uniform, each end of the member
- This gives the **correct result for the average tension or compression** along the member but will **not account for the effect of bending** of the member.

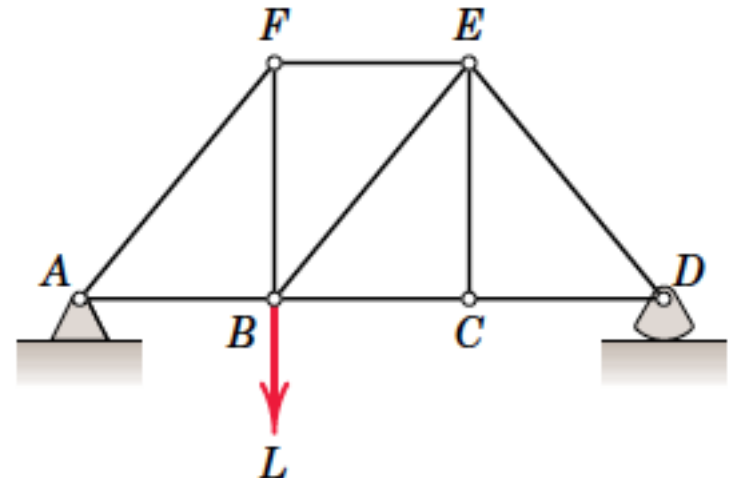
Truss Connections and Supports

- All **external forces** are applied at the pin connections
- When **welded or riveted connections** are used to join structural members,
- we may usually assume that the connection is a **pin joint** if the centrelines of the members are **concurrent** at the joint



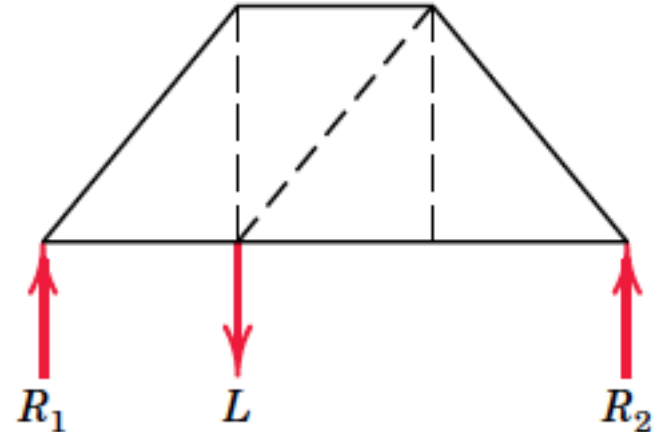
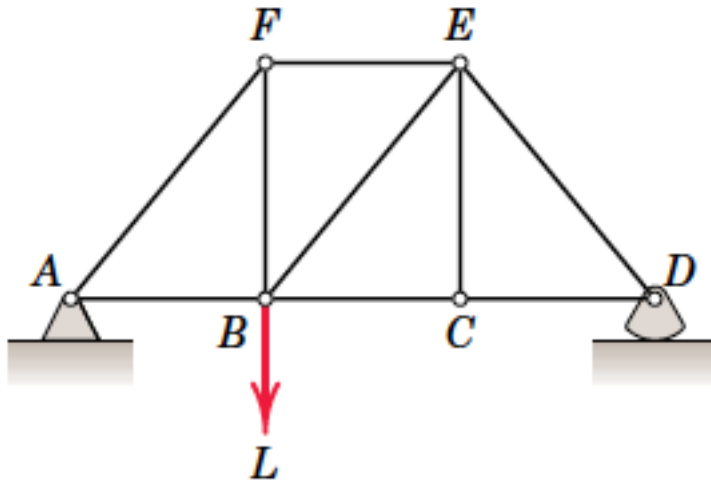
Truss Connections and Supports

- A **roller**, or some kind of **slip joint** is used at one of the supports to provide **for expansion and contraction due to temperature changes and for deformation from applied loads**.
- Trusses and frames in which no such provision is made are **statically indeterminate**



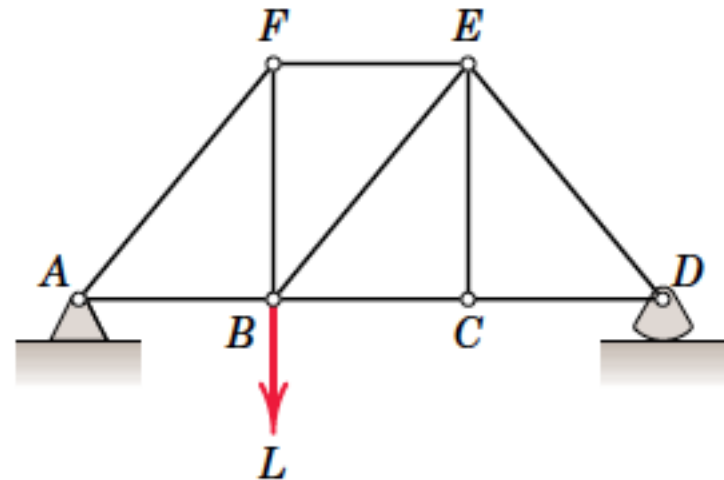
Force Analysis of Simple Truss

- Step 1 : Find external reactions by applying the equilibrium equations to the truss as a whole.



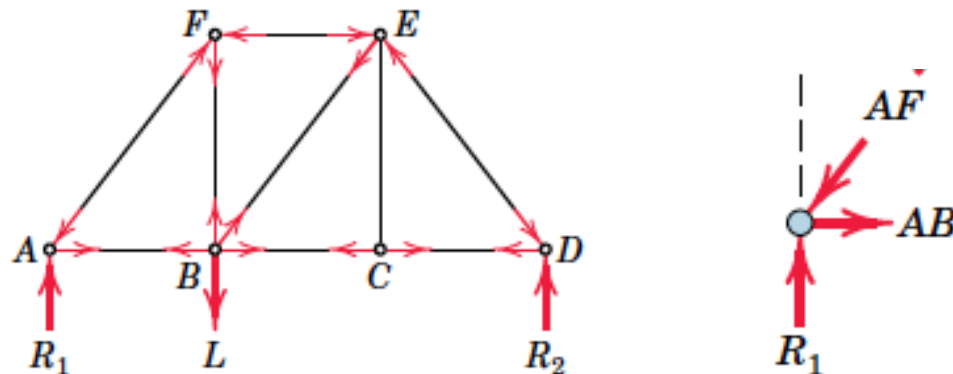
Force Analysis of Simple Truss

- Step 2 : Finding the forces in the members of a truss
 - Method of joint or
 - Method of section



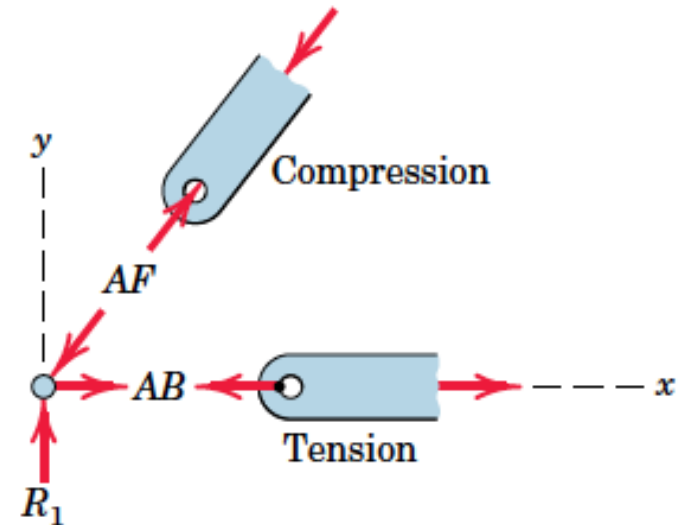
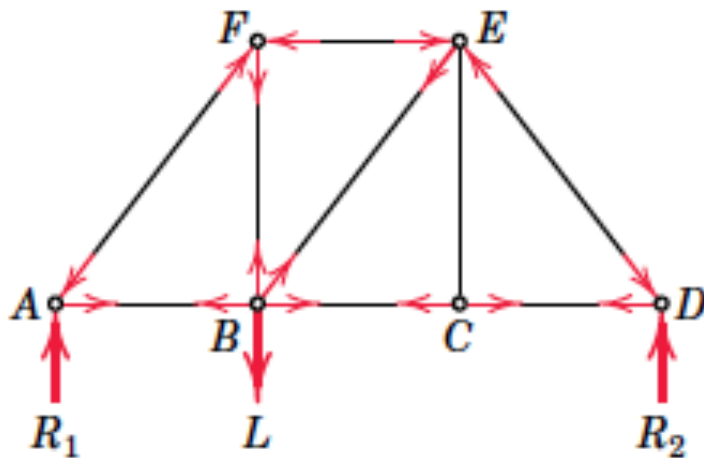
Method of Joint

- It consists of satisfying the conditions of equilibrium for the forces acting on the **connecting pin of each joint**.
- The method therefore deals with the **equilibrium of concurrent forces**, and
- only **two independent** equilibrium equations are involved.

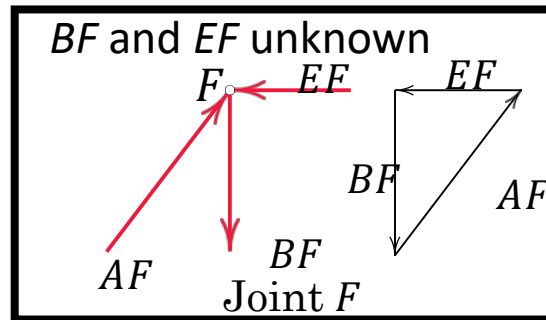
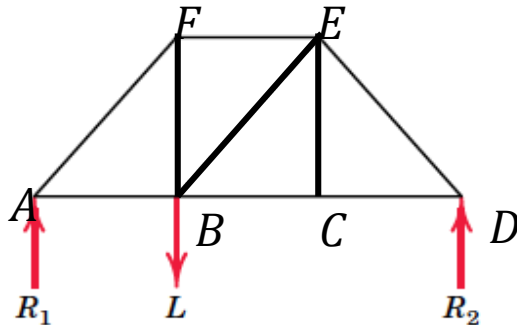


Method of Joints

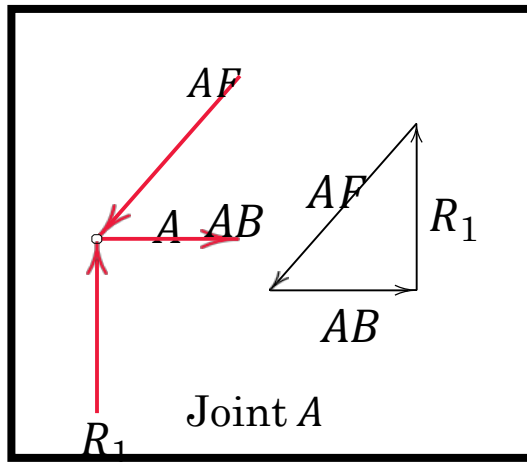
- We begin the analysis with any joint where at least **one known load** exists and where **not more than two unknown forces** are present.
- The solution may be started with the **pin at the left end**.



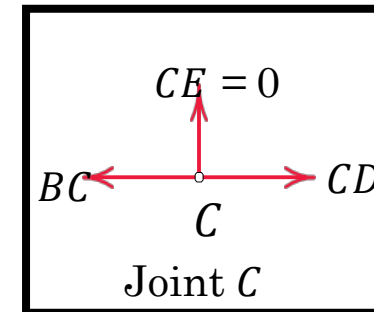
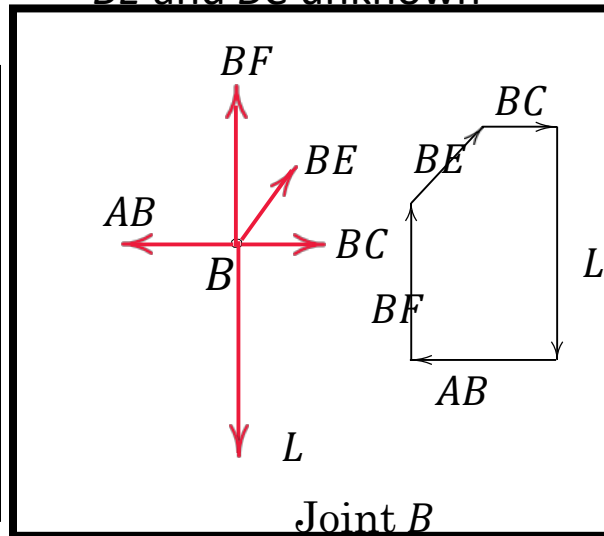
Approach: Solving for Joints A, F, B, C



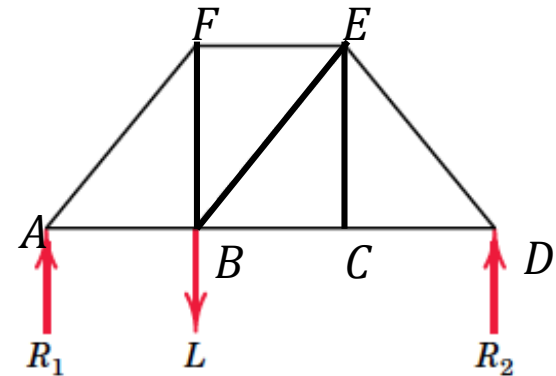
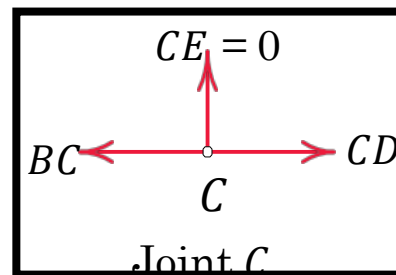
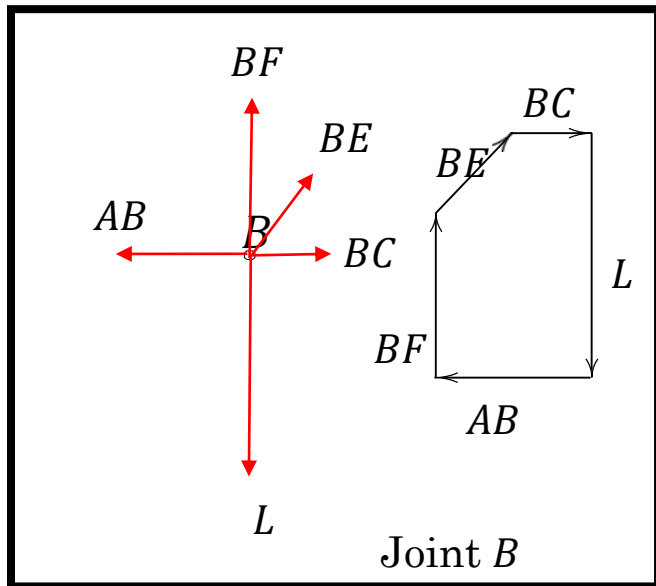
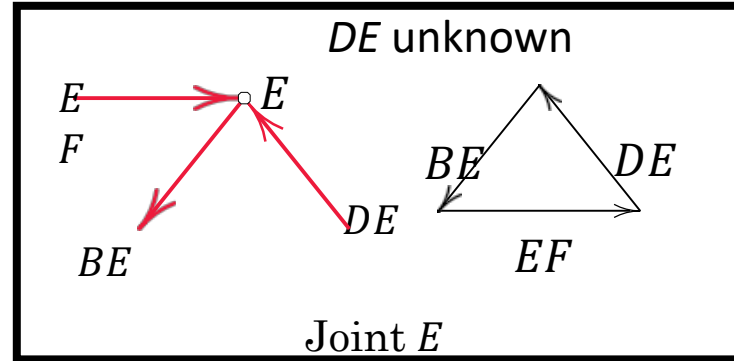
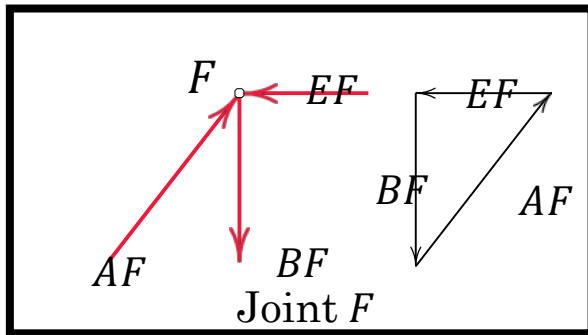
AF and AB unknown



BE and BC unknown

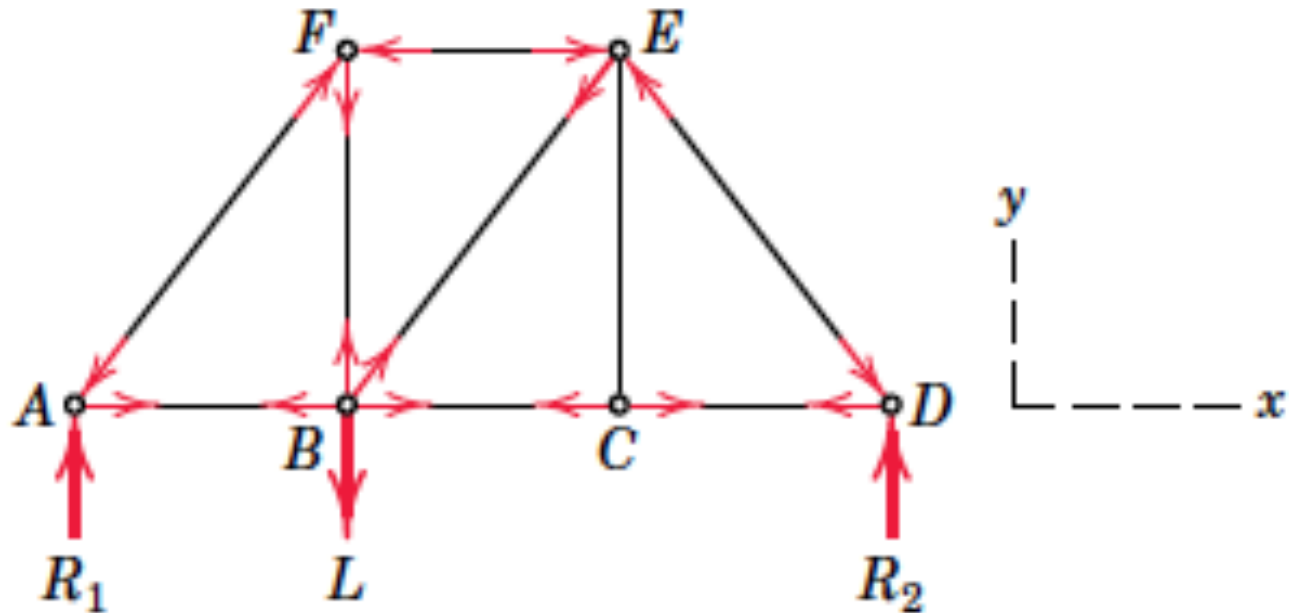


Approach : Solving for Joint E



C

Approach: Forces at Joints



Internal and External Redundancy

Truss is Statically indeterminate when

1. **More external supports** than are necessary to ensure equilibrium,
 - the extra supports constitute external redundancy.
2. **More internal members** than are necessary to prevent collapse then the
 - extra members constitute internal redundancy and the truss is again

Condition for determinacy

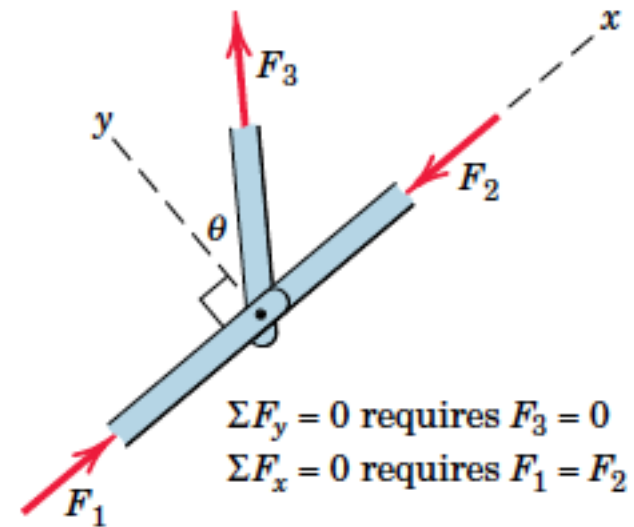
- Equilibrium of each joint by 2 equations hence for j joints total $2j$ equations
- truss composed of m two-force members and having the maximum of three unknown support reactions, there are in all $m + 3$ unknowns
- For any plane truss, the equation $m + 3 = 2j$ will be satisfied if the truss is statically determinate internally.

Internal Redundancy

- For triangle $m=3$, $j=3$ and $m + 3 = 2j = 6$
- If $m + 3 > 2j$, there are more members than independent equations, and the truss is statically indeterminate internally
- If $m + 3 < 2j$, there is a deficiency of internal members, and the truss is unstable and will collapse under load.

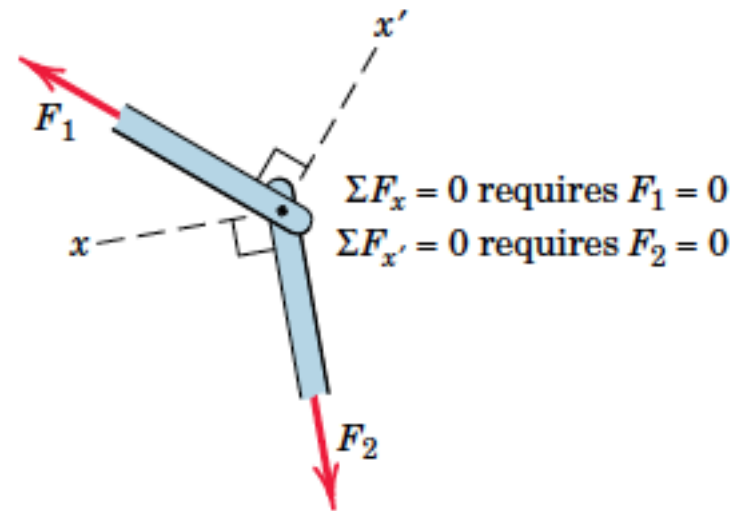
Special Cases

- When **two collinear** members are **under compression**,
- it is necessary to **add a third member** to maintain alignment of the two members and prevent buckling.
- F_3 in the third member is zero
- $F_1 = F_2$.



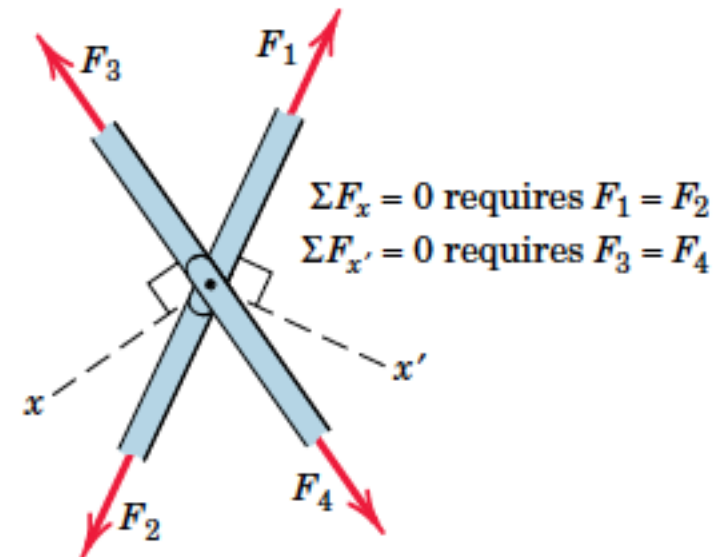
Special Cases

- When **two noncollinear members are joined** and no externally applied load at this joint,
- the **forces in both members must be zero**,



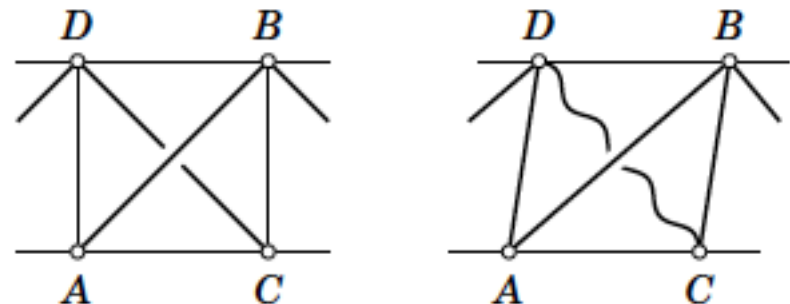
Special Case

- two pairs of collinear members are joined
- the forces in each pair must be equal and opposite



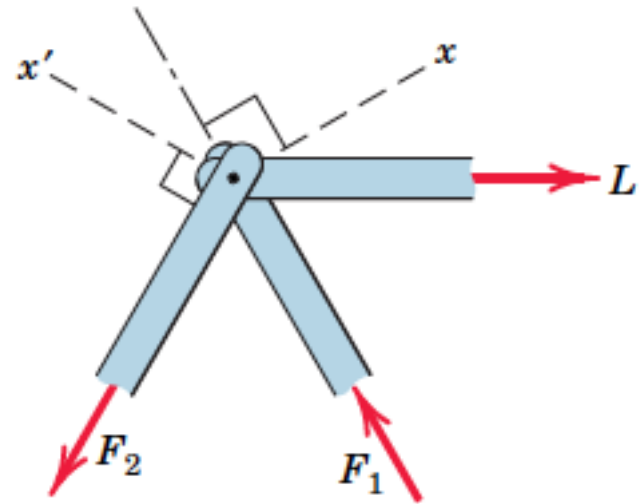
Special cases

- Truss panels are frequently cross-braced
- Such a panel is statically indeterminate if each brace can support either tension or compression.
- For flexible members capable of supporting compression, then only the tension member acts and we can disregard the other member.



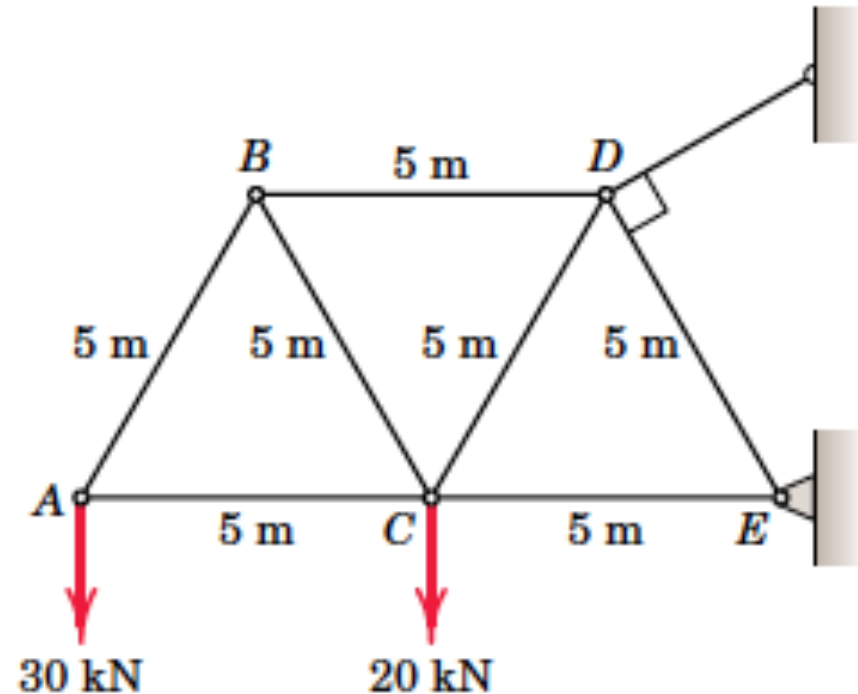
Choice of frame

- We can avoid **simultaneous solution** of the equilibrium equations for two unknown forces at a joint by a **careful choice of reference axes**

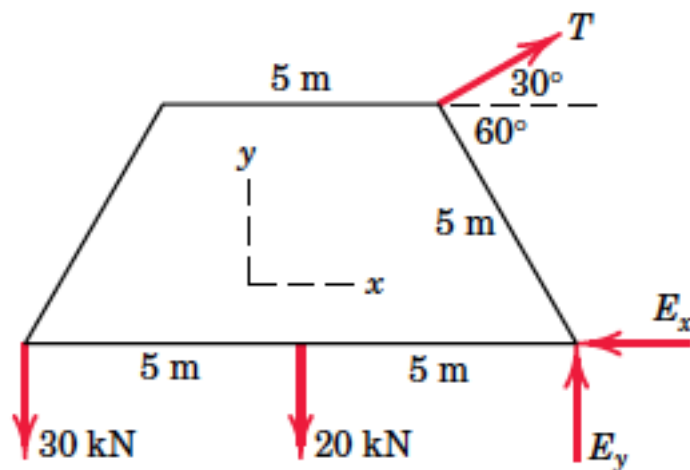
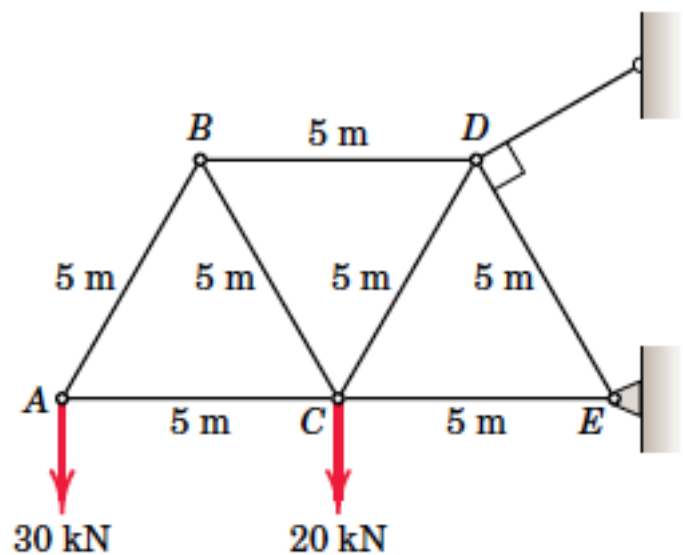


Example 1

- Compute the force in each member of the loaded cantilever truss by the method of joints.



Example 1: External reactions



$$[\Sigma M_E = 0]$$

$$5T - 20(5) - 30(10) = 0$$

$$T = 80 \text{ kN}$$

$$[\Sigma F_x = 0]$$

$$80 \cos 30^\circ - E_x = 0$$

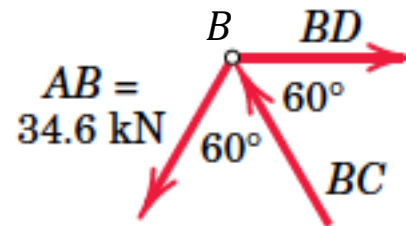
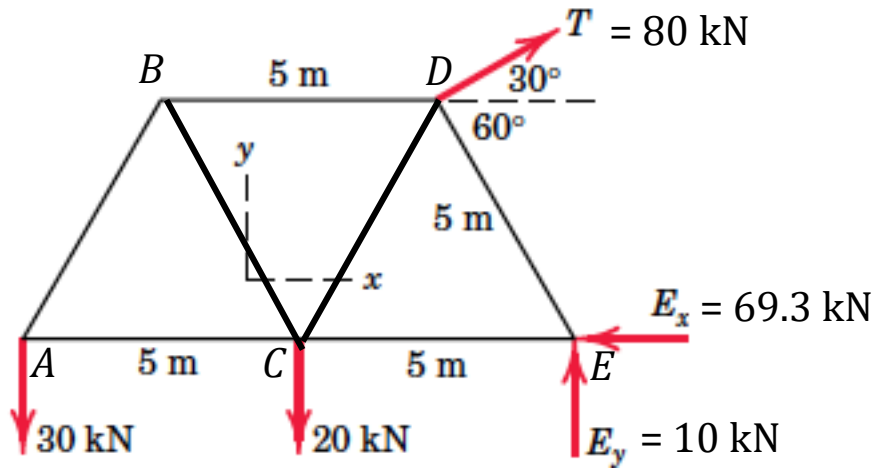
$$E_x = 69.3 \text{ kN}$$

$$[\Sigma F_y = 0]$$

$$80 \sin 30^\circ + E_y - 20 - 30 = 0$$

$$E_y = 10 \text{ kN}$$

Example 1: Forces in Members



$$[\Sigma F_y = 0]$$

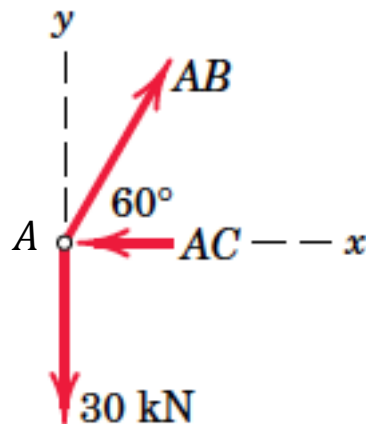
$$0.866AB - 30 = 0$$

$$AB = 34.6 \text{ kN T}$$

$$[\Sigma F_x = 0]$$

$$AC - 0.5(34.6) = 0$$

$$AC = 17.32 \text{ kN C}$$



$$[\Sigma F_y = 0]$$

$$0.866BC - 0.866(34.6) = 0$$

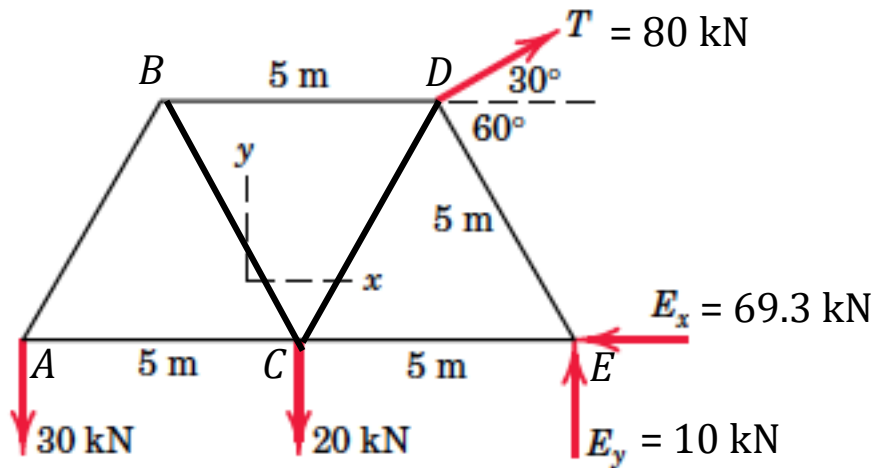
$$BC = 34.6 \text{ kN C}$$

$$[\Sigma F_x = 0]$$

$$BD - 2(0.5)(34.6) = 0$$

$$BD = 34.6 \text{ kN T}$$

Example 1: Forces in Members



$$[\Sigma F_y = 0]$$

$$0.866DE = 10$$

$$DE = 11.55 \text{ kN C}$$

$$[\Sigma F_y = 0]$$

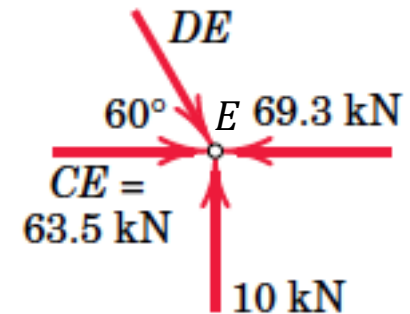
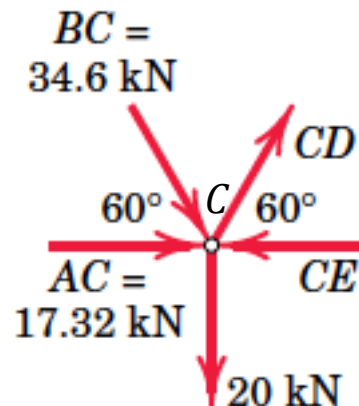
$$0.866CD - 0.866(34.6) - 20 = 0$$

$$CD = 57.7 \text{ kN T}$$

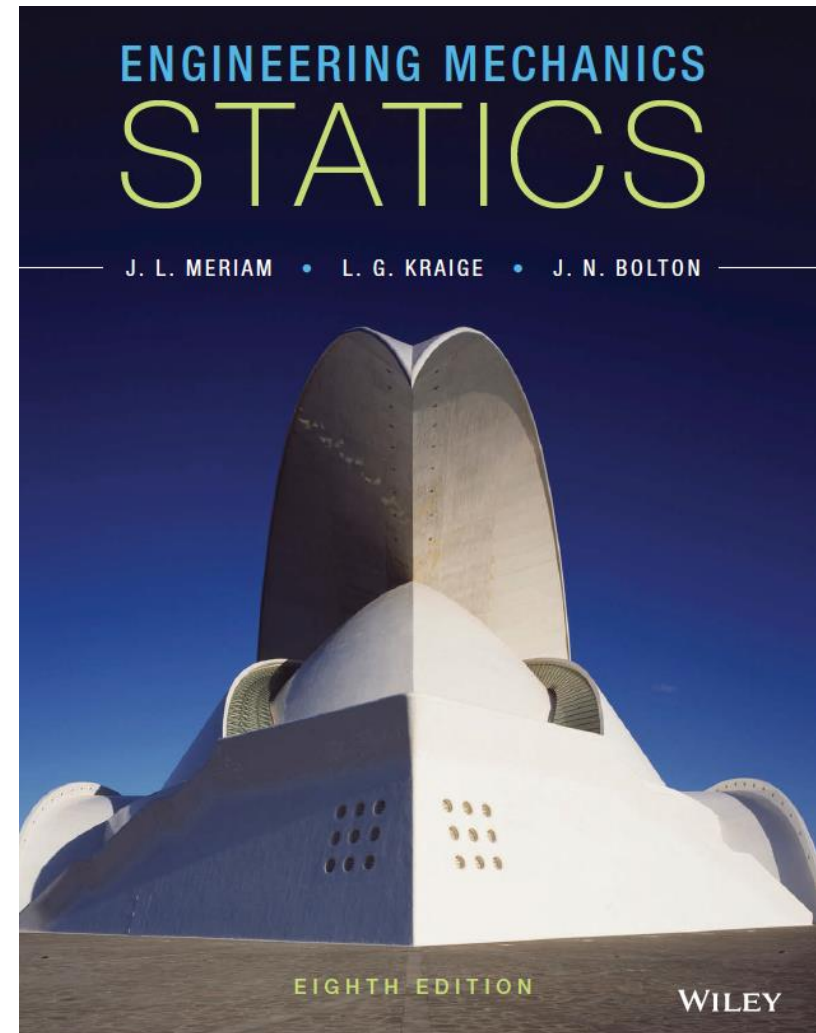
$$[\Sigma F_x = 0]$$

$$CE - 17.32 - 0.5(34.6) - 0.5(57.7) = 0$$

$$CE = 63.5 \text{ kN C}$$



References :



THANK YOU

Engineering Mechanics

Lecture 7: Truss and Frame

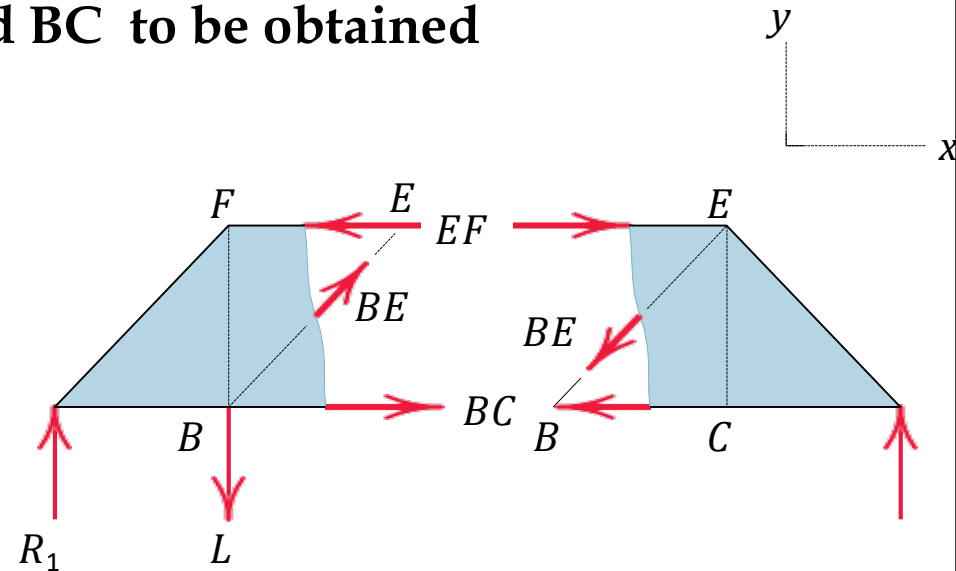
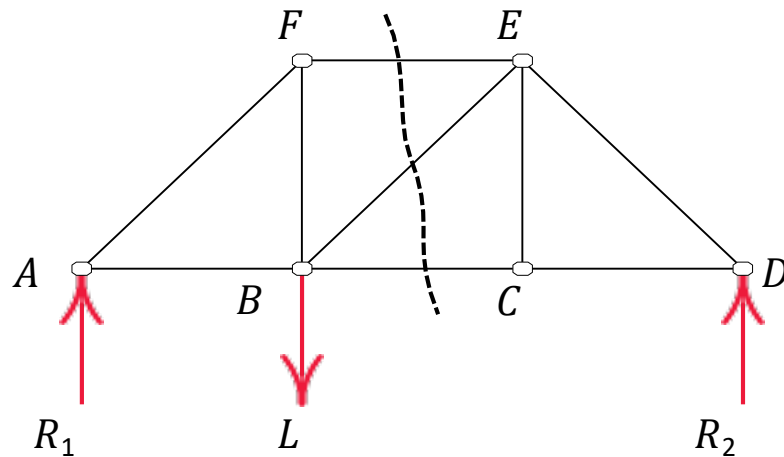
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Method of Section

- Use **Moment balance of forces** on section of Truss
- Entire section of the truss is considered a **single body in equilibrium**
- **Advantage:** Forces in **any desired member** to be found directly
- **Caution:** not **more than three members** whose forces are unknown should be cut

Method of section: Illustration

BE, EF and BC to be obtained



- Take Moment about
 - B to Find EF
 - E to Find BC
 - F to find BE

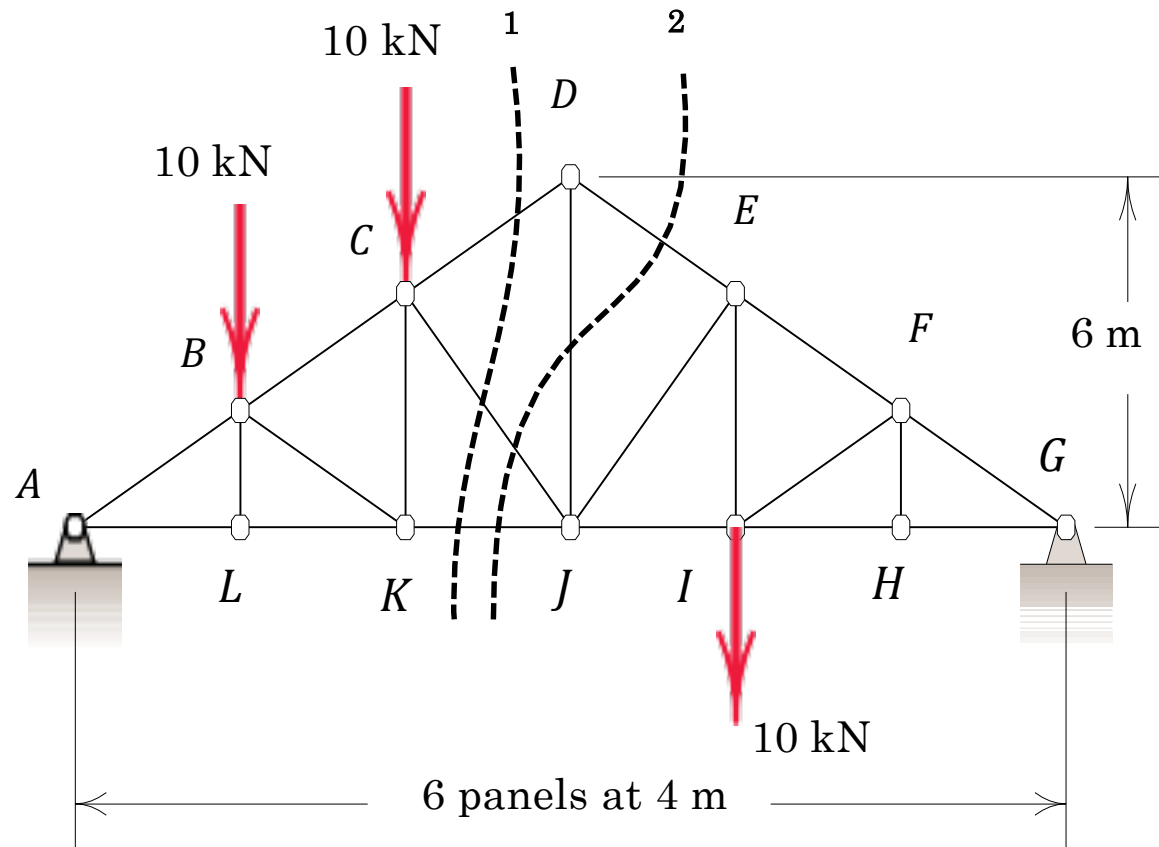
Additional Considerations

- **Moment equations** are used to great advantage
- Choose a **moment center**, either on or off the section, through which as many unknown forces as possible pass.
- A **positive answer** will verify the assumed sense, and a **negative result** will indicate that the force is in the sense opposite to that assume

Additional Considerations

- Some cases the methods of sections and joints can be **combined** for an efficient solution.
- Section through the member can not be created without **passing through at least four unknown members**.
 - Determine the **forces in nearby members** by the method of sections and then
 - Progress to the **unknown member** by the method of **joints/section**.

Example 1: Calculate the force in member DJ of the Howe roof truss illustrated. Neglect any horizontal components of force at the supports.

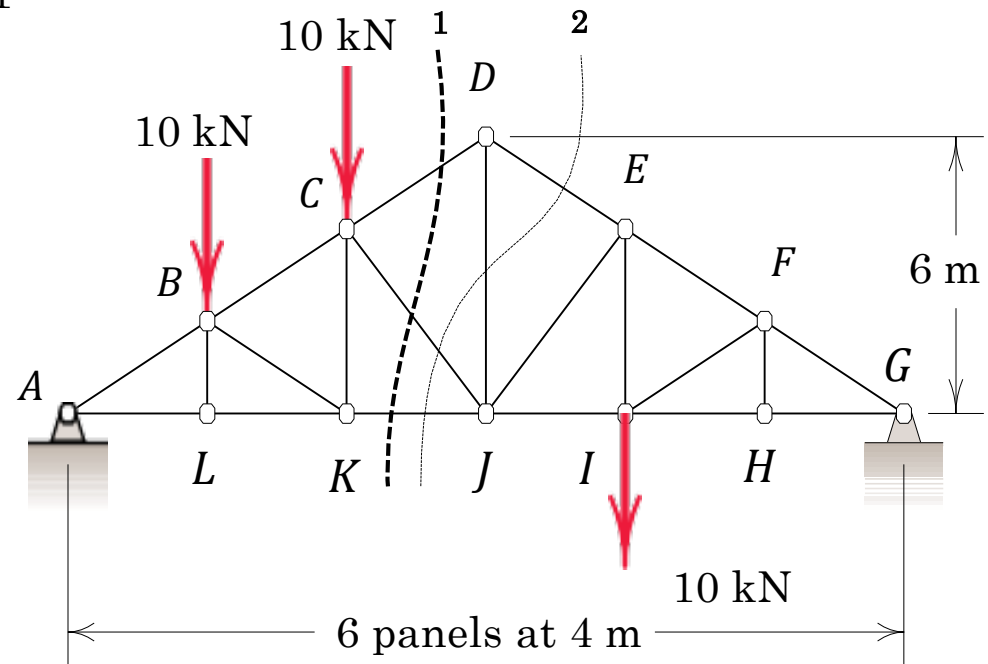
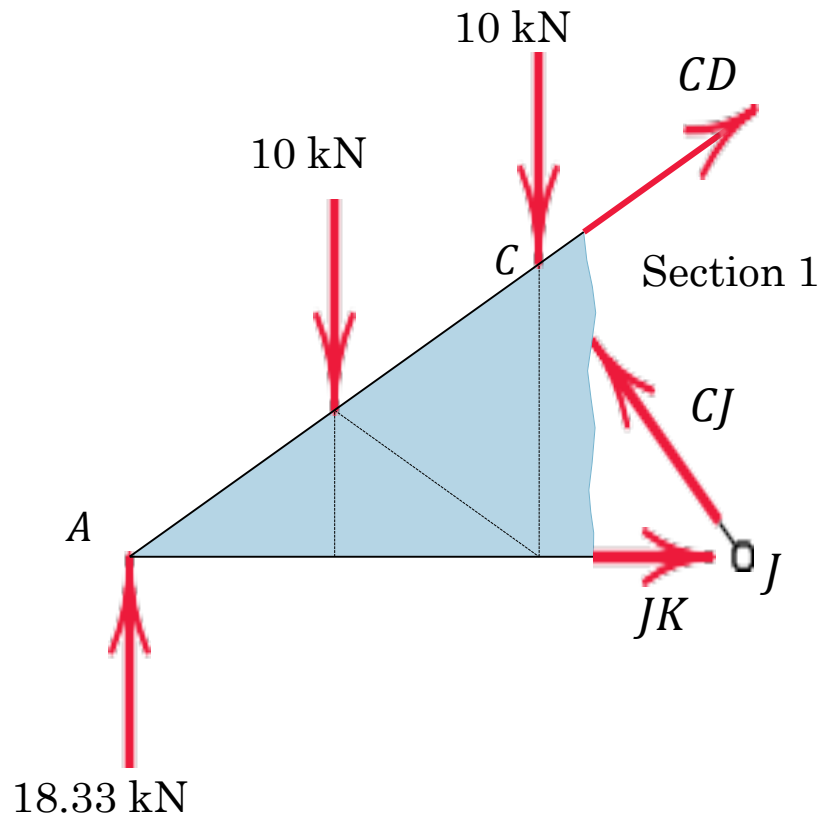


Example 1: Find CJ

$$[\Sigma M_A = 0]$$

$$0.707CJ(12) - 10(4) - 10(8) = 0$$

$$CJ = 14.14 \text{ kN C}$$



FRAMES AND MACHINES

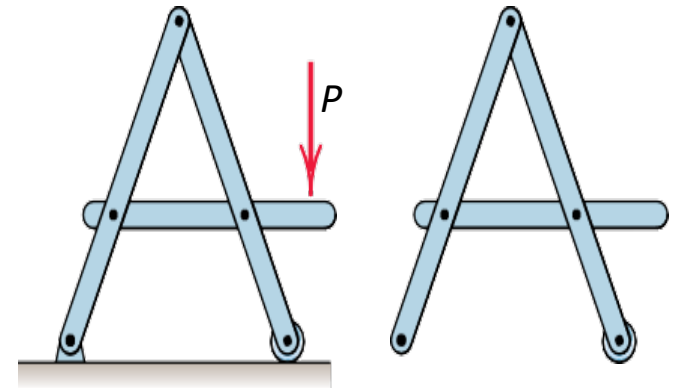
- A structure is called a **frame or machine** if at least one of its individual members is a **multiforce member, three or more forces**
- **Frames** are structures which are designed to **support applied loads** and are usually fixed in position.
- **Machines** are structures (static case) which contain **moving parts** and are designed to **transmit input forces or couples** to output forces or couples.

Interconnected Rigid Bodies

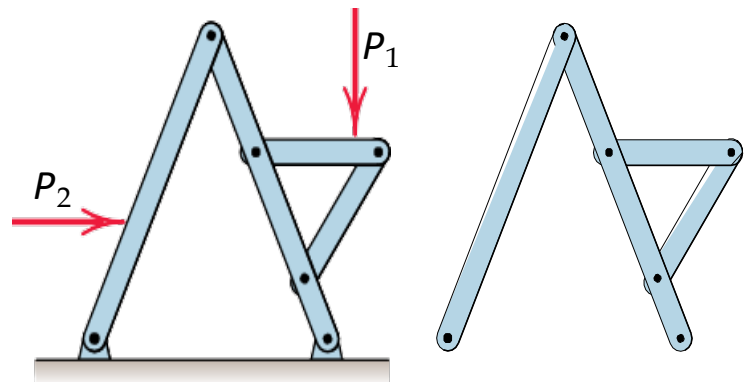
- Relies on **equilibrium of interconnected rigid bodies**
 - Members are **isolated** with a free-body diagram and
 - Apply the **equations of equilibrium** on isolated body
 - Principle of **action and reaction** must be carefully observed
- We may occasionally need to solve **two or more equations simultaneously** in order to separate the unknowns.

Collapsible vs Non collapsible frames

- **Non collapsible:** frame or machine constitutes a rigid unit when removed from its supports,
 - Analysis is begun by establishing all the forces external to the structure treated as a single rigid body
- **Collapsible:** frame or machine constitutes a non rigid unit when removed from its supports,
 - the calculation of the external support reactions cannot be completed until the structure is dismembered



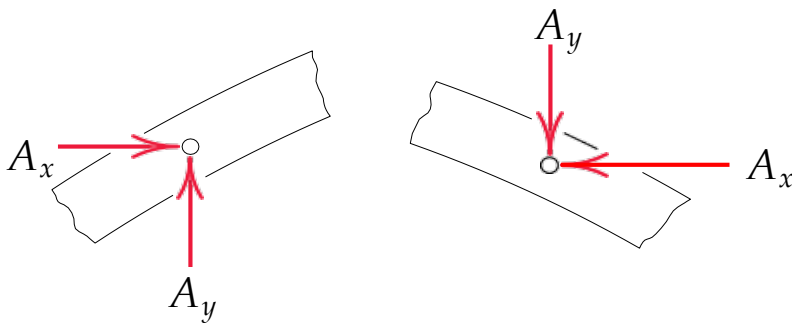
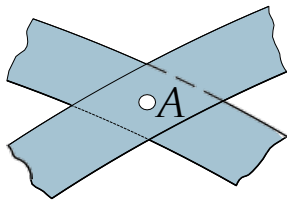
Rigid non collapsible



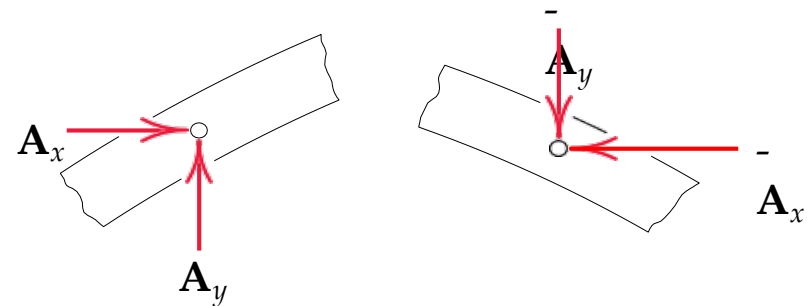
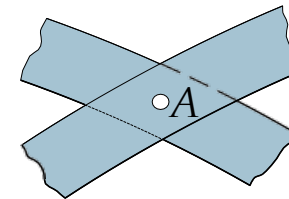
Nonrigid collapsible

Force Representation & Free-Body Diagrams

Scalar Notation

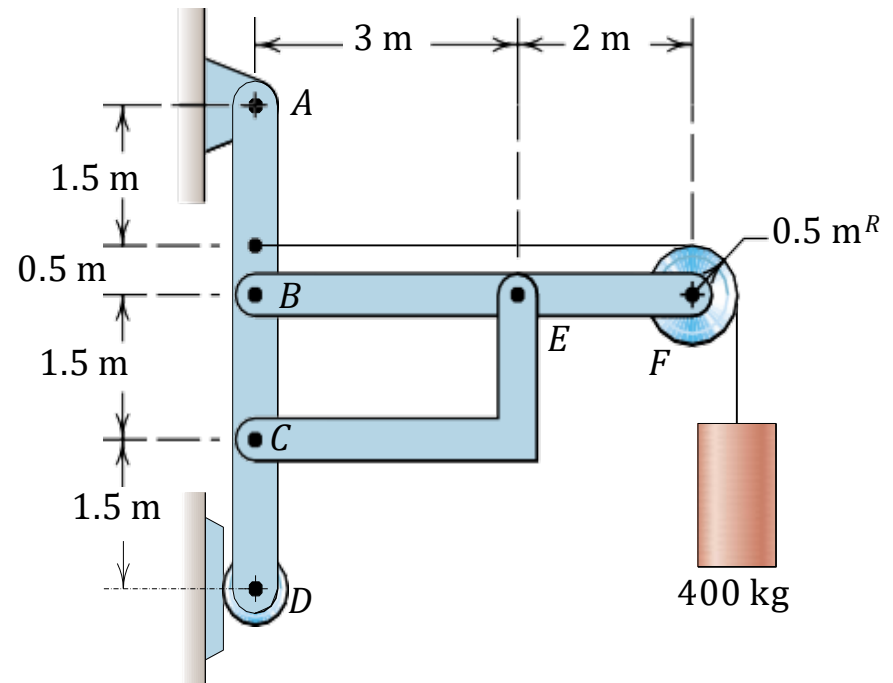


Vector Notation

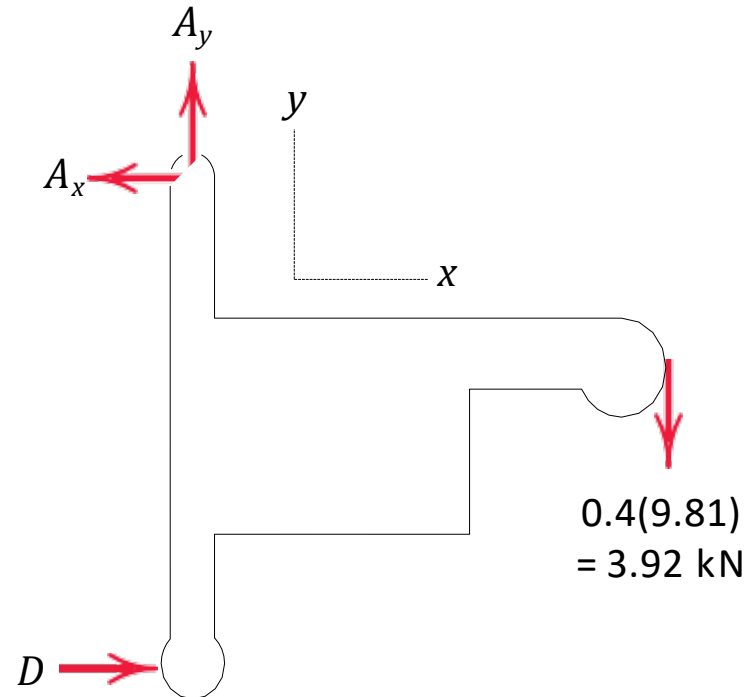
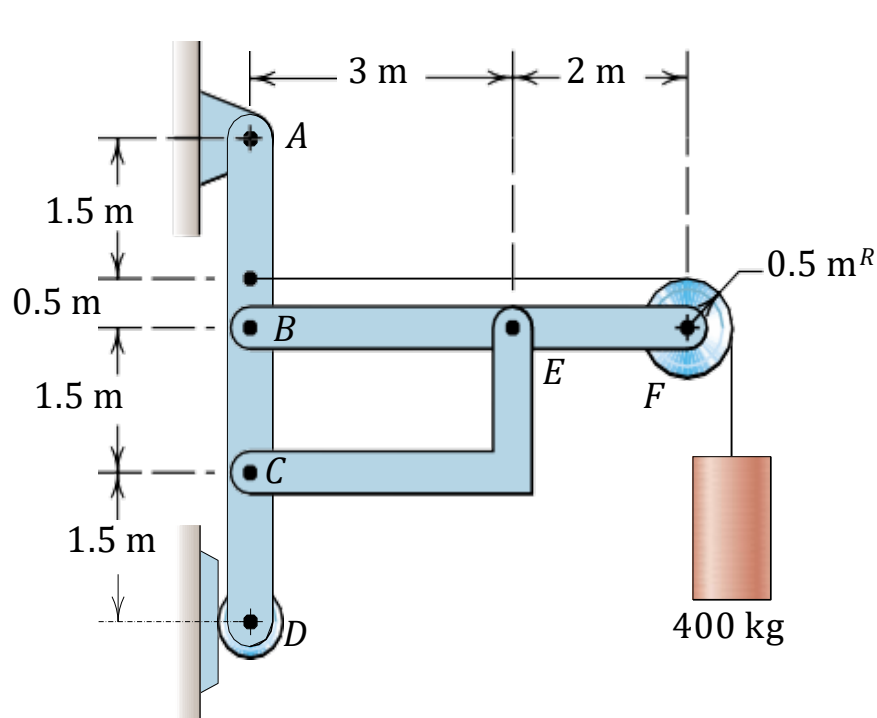


Use proper sign in vector notation

- Example 2: The frame supports the 400-kg load in the manner shown. Neglect the weights of the members compared with the forces induced by the load and compute the horizontal and vertical components of all forces acting on each of the members.



Example 2: Find Support Reactions



$$[\Sigma M_A = 0]$$

$$5.5(0.4)(9.81) - 5D = 0$$

$$D = 4.32 \text{ kN}$$

$$[\Sigma F_x = 0]$$

$$A_x - 4.32 = 0$$

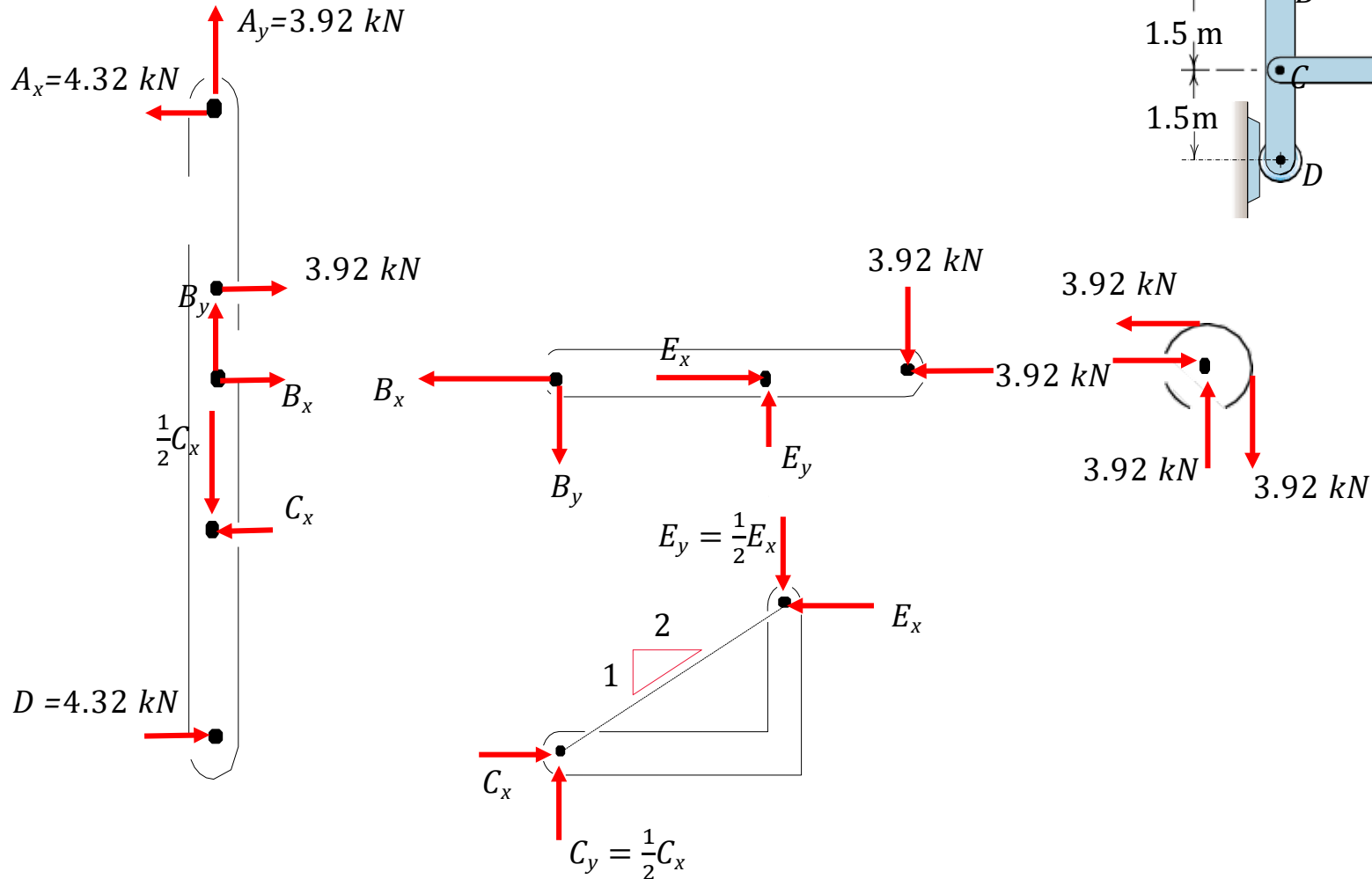
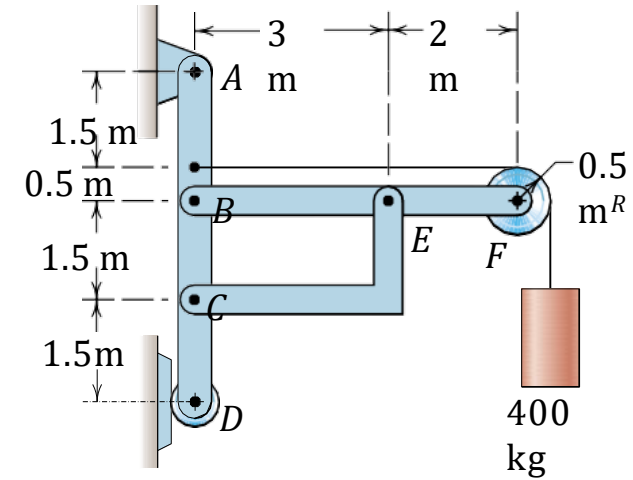
$$A_x = 4.32 \text{ kN}$$

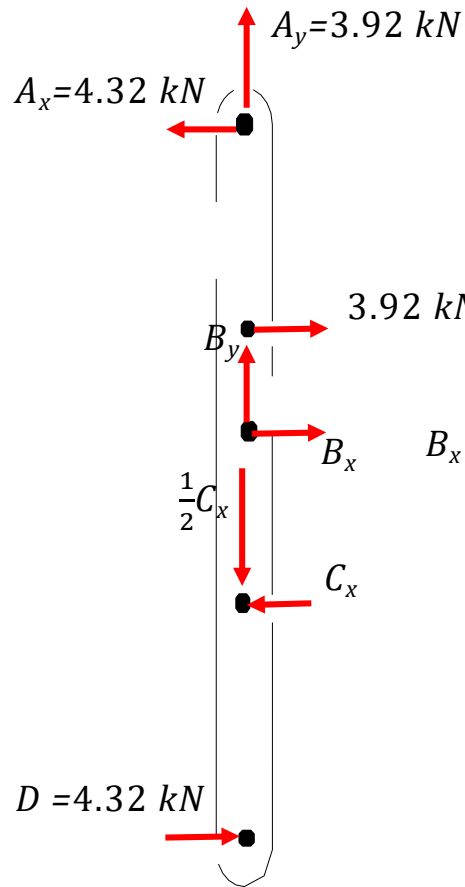
$$[\Sigma F_y = 0]$$

$$A_y - 3.92 = 0$$

$$A_y = 3.92 \text{ kN}$$

Find other reactions at joints





$$[\Sigma M_B = 0]$$

$$3.92(5) - \frac{1}{2}E_x(3) = 0$$

$$E_x = 13.08 \text{ kN}$$

$$[\Sigma F_y = 0]$$

$$B_y + 3.92 - 13.08/2 = 0$$

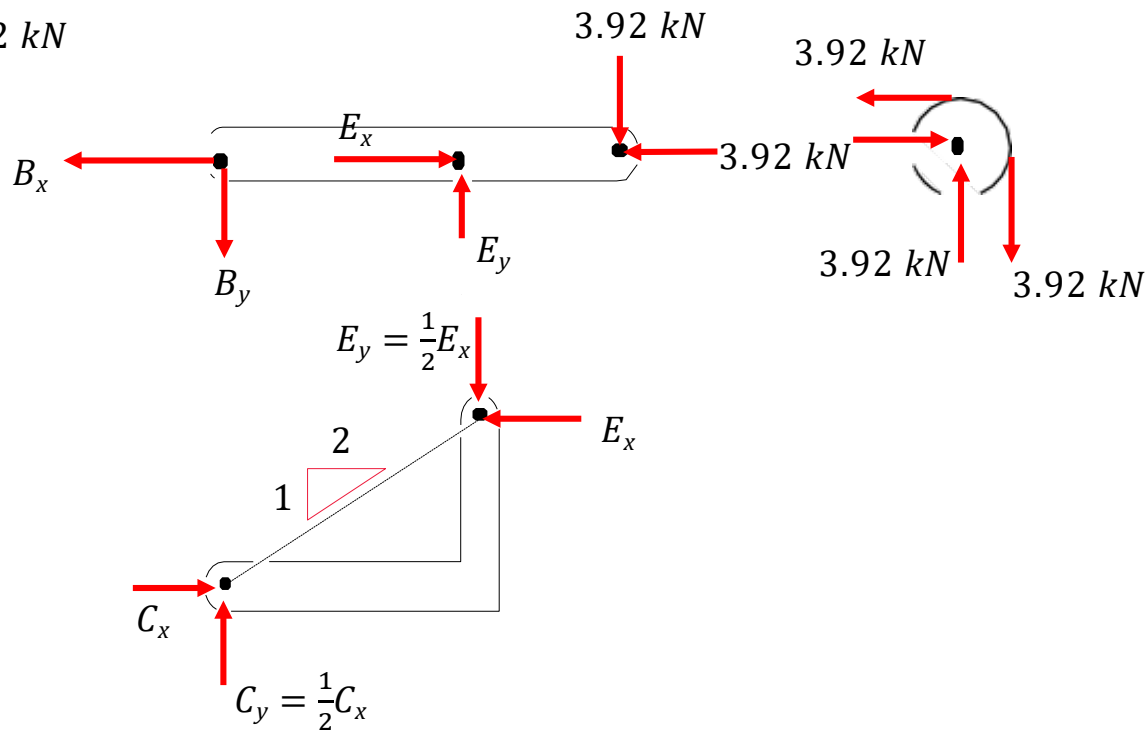
$$B_y = 2.62 \text{ kN}$$

$$[\Sigma F_x = 0]$$

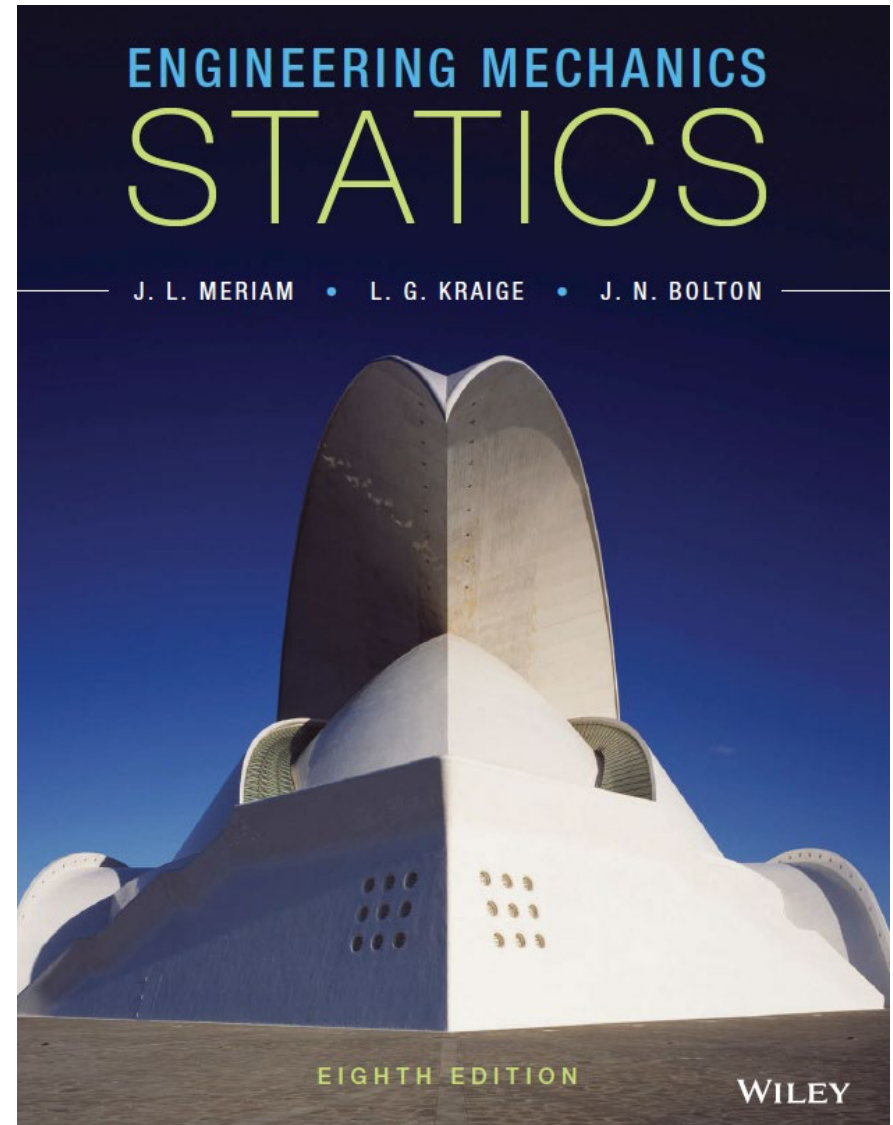
$$B_x + 3.92 - 13.08 = 0$$

$$B_x = 9.15 \text{ kN}$$

$$C_x = E_x$$



References :



THANK YOU



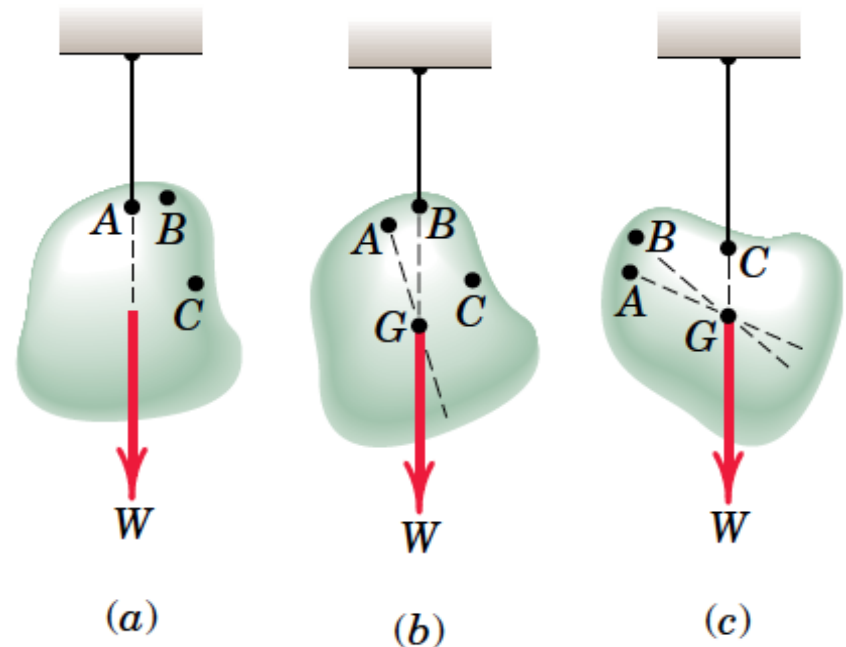
Engineering Mechanics

Lecture 8: Center of Gravity

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Center of Gravity by Experiments

- **Suspending the body** from three points A, B and C, and mark the line of action of the resultant force.
- For all practical purposes these **lines of action will be concurrent** at a single point G,
- which is called the **center of gravity** of the body.

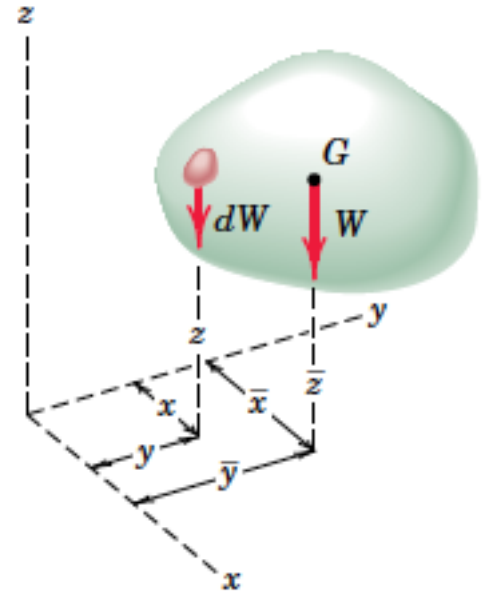


Determining the Center of Gravity

Using Principle of moments

- The **moment of the resultant gravitational force W** about any axis equals the sum of the **moments** about the same axis **of the gravitational forces dW** acting on all particles treated as infinitesimal elements of the body
- Moment about y axis

$$\bar{x}W = \int x dW$$



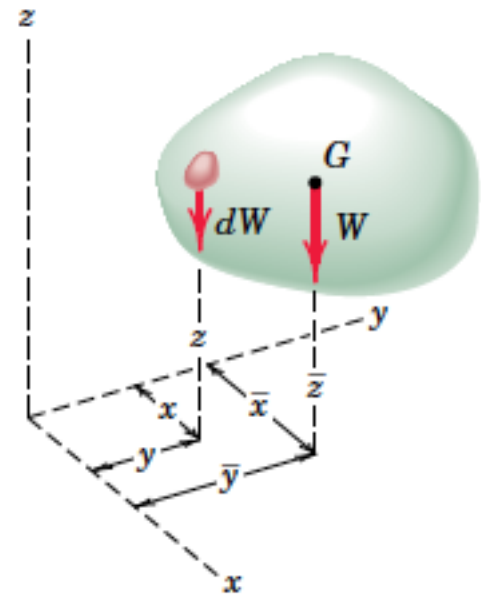
Center of Gravity

Expression of Center of Gravity

$$\bar{x} = \frac{\int x dW}{W}$$

$$\bar{y} = \frac{\int y dW}{W}$$

$$\bar{z} = \frac{\int z dW}{W}$$



Center of Mass

Center of Gravity

$$\bar{x} = \frac{\int x dW}{W}$$

$$\bar{y} = \frac{\int y dW}{W}$$

$$\bar{z} = \frac{\int z dW}{W}$$

Uniform and parallel gravity field

$$W = mg$$

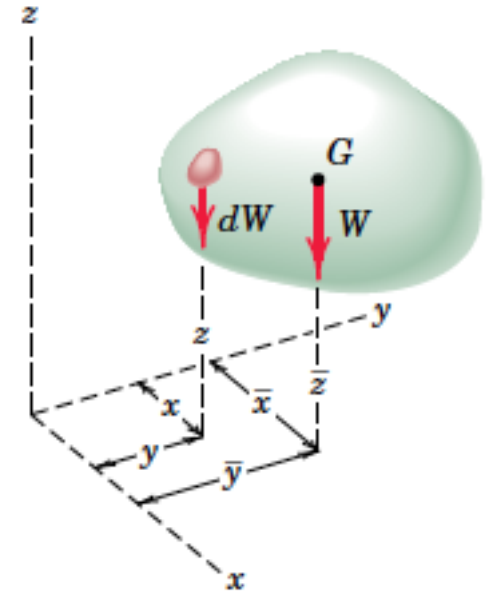
$$dW = dm g$$

$$\bar{x} = \frac{\int x dm}{m}$$

$$\bar{y} = \frac{\int y dm}{m}$$

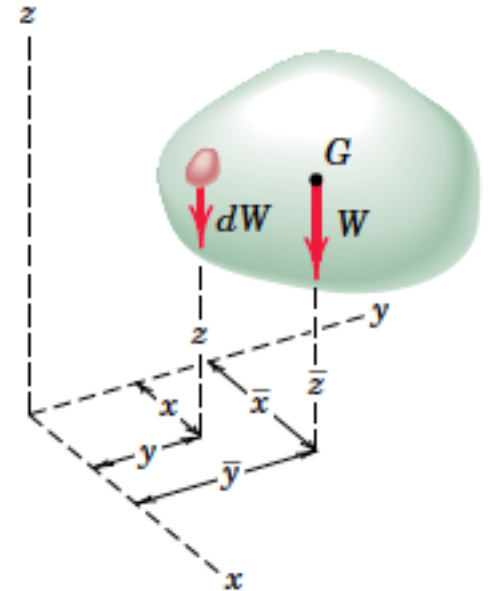
$$\bar{z} = \frac{\int z dm}{m}$$

= Center of mass



Center of Mass

- A **unique point in the body** which is a **function** solely of the **distribution of mass**.
- It **coincides with the center of gravity** as long as the gravity field is treated as uniform and parallel
- We will usually refer henceforth to the **center of mass** rather than to the center of gravity.

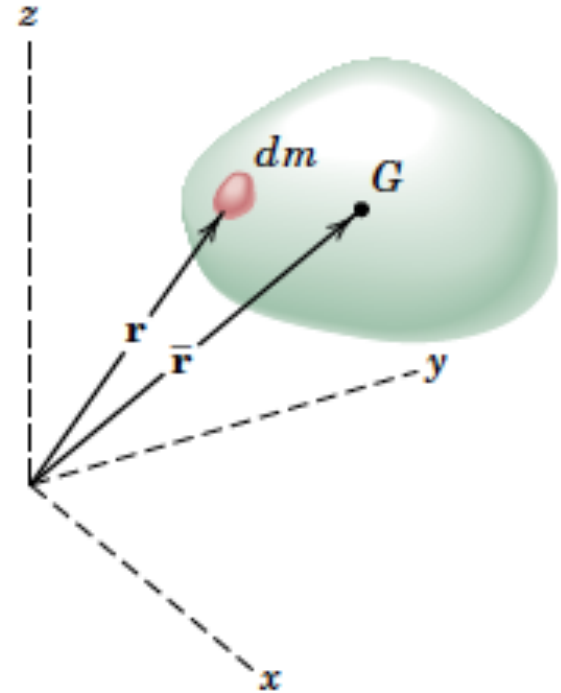


Center of Mass in vector form

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

$$\bar{\mathbf{r}} = \bar{x}\mathbf{i} + \bar{y}\mathbf{j} + \bar{z}\mathbf{k}$$

$$\bar{\mathbf{r}}m = \int \mathbf{r}dm$$



$$\bar{\mathbf{r}} = \frac{\int \mathbf{r}dm}{m}$$



$$\bar{x} = \frac{\int xdm}{m}$$

$$\bar{y} = \frac{\int ydm}{m}$$

$$\bar{z} = \frac{\int zdm}{m}$$

Center of Mass

COM

$$\bar{\mathbf{r}} = \frac{\int \mathbf{r} dm}{m}$$

$$\bar{x} = \frac{\int x dm}{m}$$

$$\bar{y} = \frac{\int y dm}{m}$$

$$\bar{z} = \frac{\int z dm}{m}$$

ρ the density of a body in mass per unit volume.

$$dm = \rho dV$$

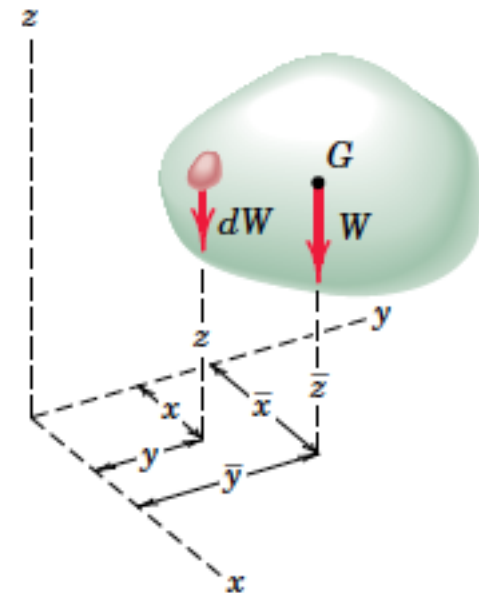
$$m = \int dm = \int \rho dV$$

For non uniform density, COM is given by

$$\bar{x} = \frac{\int x \rho dV}{\int \rho dV}$$

$$\bar{y} = \frac{\int y \rho dV}{\int \rho dV}$$

$$\bar{z} = \frac{\int z \rho dV}{\int \rho dV}$$



Centroid

COM

$$\bar{x} = \frac{\int x \rho dV}{\int \rho dV}$$

$$\bar{y} = \frac{\int y \rho dV}{\int \rho dV}$$

$$\bar{z} = \frac{\int z \rho dV}{\int \rho dV}$$

If ρ is uniform throughout,

$$\bar{x} = \frac{\int x dV}{\int dV}$$

$$\bar{y} = \frac{\int y dV}{\int dV}$$

$$\bar{z} = \frac{\int z dV}{\int dV}$$

- The expressions define Centroid , a purely geometrical property of the body, since any reference to its mass properties has disappeared.
- Uniform density COM = Centroid, whereas if the density varies, these two points will, in general, not coincide.

Centroid of Line

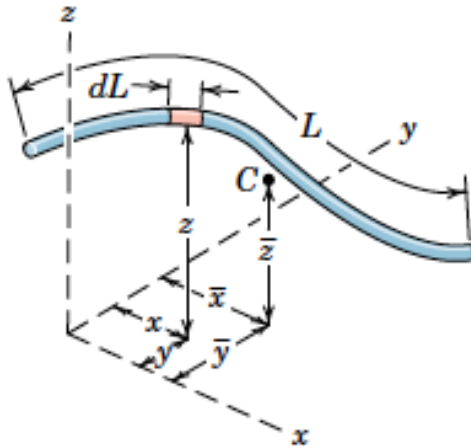
COM

$$\bar{x} = \frac{\int x \, dm}{\int dm}$$

$$\bar{y} = \frac{\int y \, dm}{\int dm}$$

$$\bar{z} = \frac{\int z \, dm}{\int dm}$$

For a slender rod or wire of length L and uniform cross section area and density ρ



$$dm = \rho A \, dL$$

Substituting in the expressions above we get

$$\bar{x} = \frac{\int x \, dL}{L}$$

$$\bar{y} = \frac{\int y \, dL}{L}$$

$$\bar{z} = \frac{\int z \, dL}{L}$$

Centroid of Area

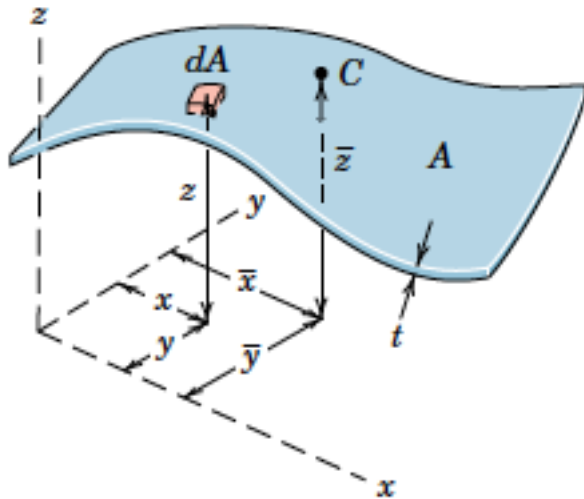
COM

$$\bar{x} = \frac{\int x \, dm}{\int dm}$$

$$\bar{y} = \frac{\int y \, dm}{\int dm}$$

$$\bar{z} = \frac{\int z \, dm}{\int dm}$$

When a body of density ρ has a small but constant thickness t , we can model it as a surface area A ,



$$dm = \rho t \, dA$$

Substituting in the expressions above we get

$$\bar{x} = \frac{\int x \, dA}{A}$$

$$\bar{y} = \frac{\int y \, dA}{A}$$

$$\bar{z} = \frac{\int z \, dA}{A}$$

Centroid of Volumes

COM

$$\bar{x} = \frac{\int x \, dm}{\int dm}$$

$$\bar{y} = \frac{\int y \, dm}{\int dm}$$

$$\bar{z} = \frac{\int z \, dm}{\int dm}$$

For a general body of volume V and with uniform density ρ

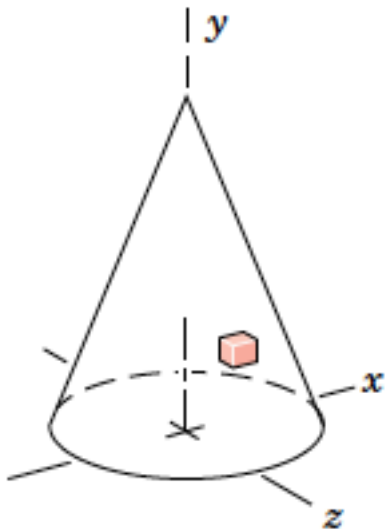
$$dm = \rho dV$$

Substituting in the expressions above we get

$$\bar{x} = \frac{\int x \, dV}{V}$$

$$\bar{y} = \frac{\int y \, dV}{V}$$

$$\bar{z} = \frac{\int z \, dV}{V}$$

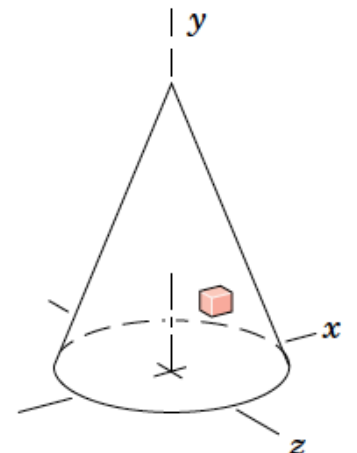
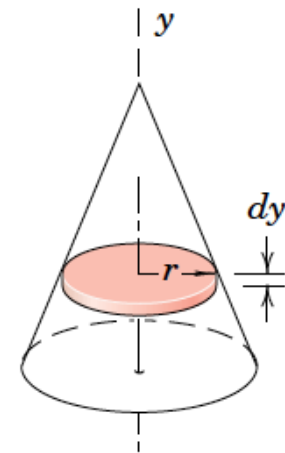
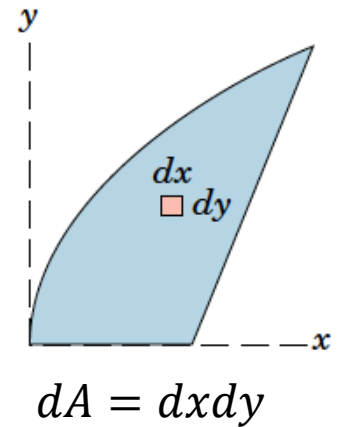
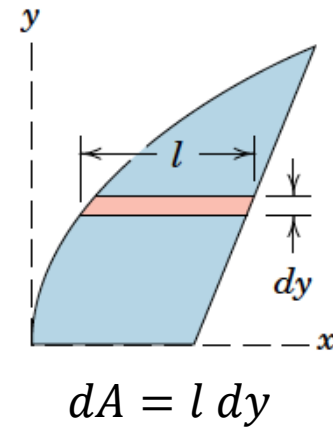


CHOICE OF ELEMENT FOR INTEGRATION

1. Order of Element.
2. Continuity
3. Discarding Higher-Order Terms.
4. Choice of Coordinates
5. Centroidal Coordinate of Element.

Order of Element

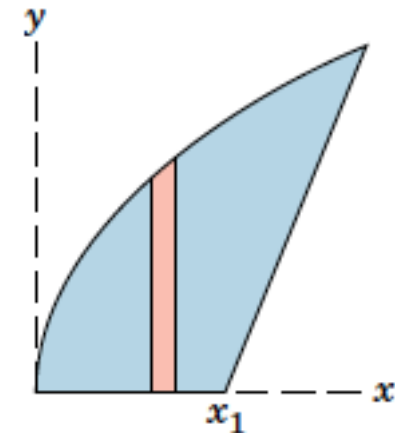
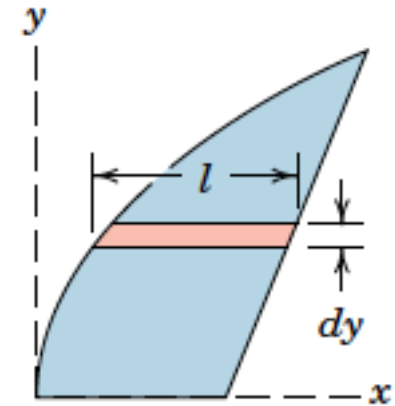
- Whenever possible, a **first-order differential element** should be selected in preference to a higher-order element
- **only one integration** will be **required** to cover the entire figure.



$$dV = \pi r^2 dy \quad dV = dx dy dz$$

Continuity

- Whenever possible, choose an element which can be **integrated in one continuous operation** to cover the figure.
- **Horizontal strip** in Fig. would be **preferable** to the vertical strip in
- Which require two separate integrals because of the **discontinuity in the expression** for the height of the strip at $x = x_1$.

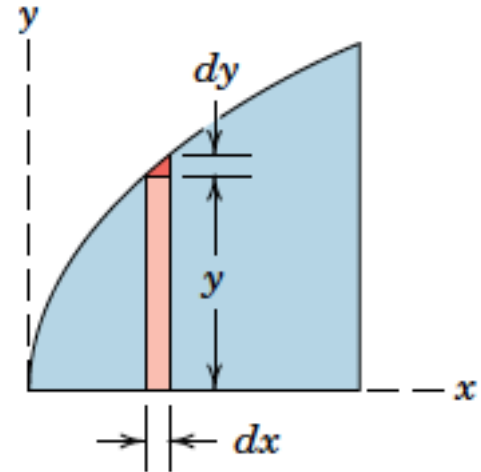


Discarding Higher-Order Terms

- **Higher-order terms** may always be dropped compared with lower-order terms

- Area of strip

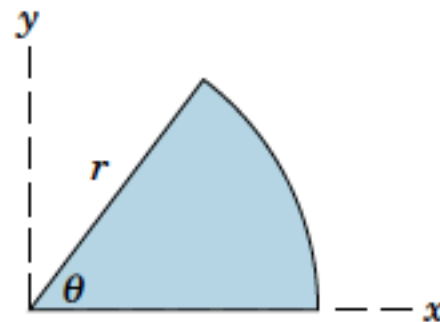
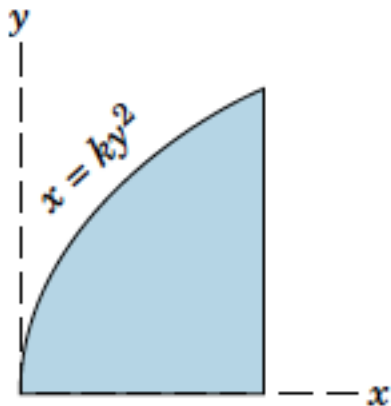
$$A = ydx + \frac{1}{2}dxdy$$



- Higher order term $\frac{1}{2}dxdy$ can be discarded

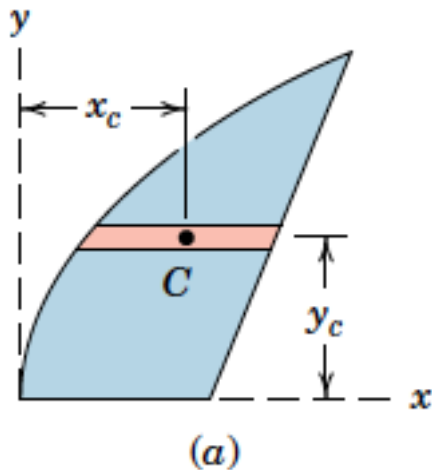
Choice of Coordinates

- As a general rule, we choose the coordinate system which **best matches the boundaries** of the figure.
- Fig 1 is easily described in **rectangular coordinates**, whereas the boundaries of the circular sector of Fig. 2 are best suited to **polar coordinates**.

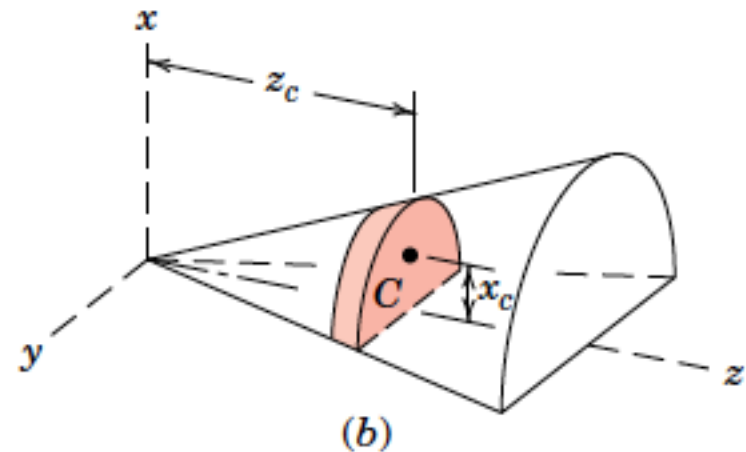


Centroidal Coordinate of Element

- When a first- or second-order differential element is chosen, it is **essential to use the coordinate of the centroid** of the element for the moment arm in expressing the moment of the differential element.

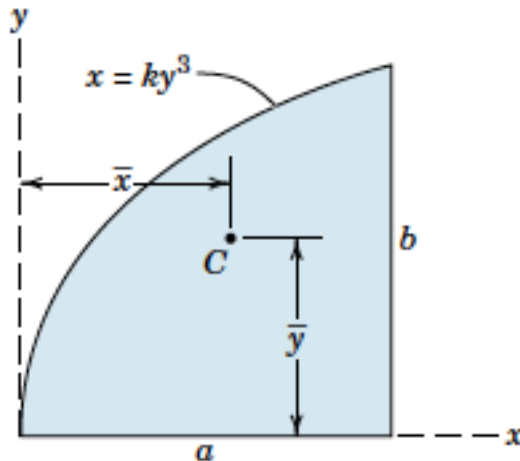


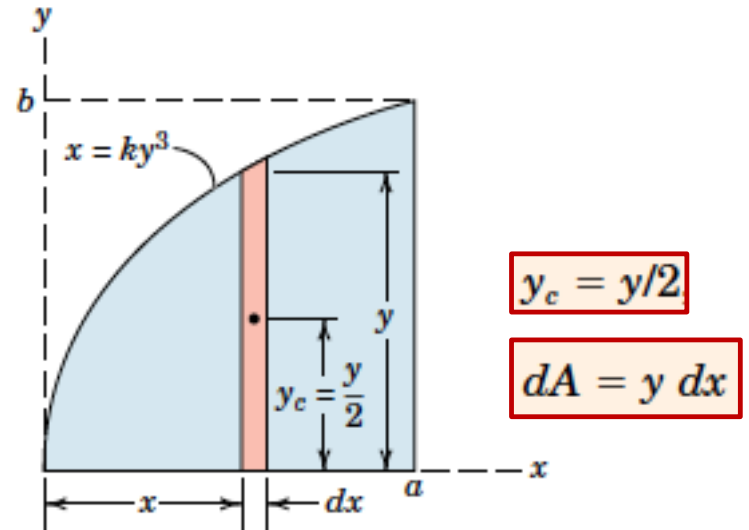
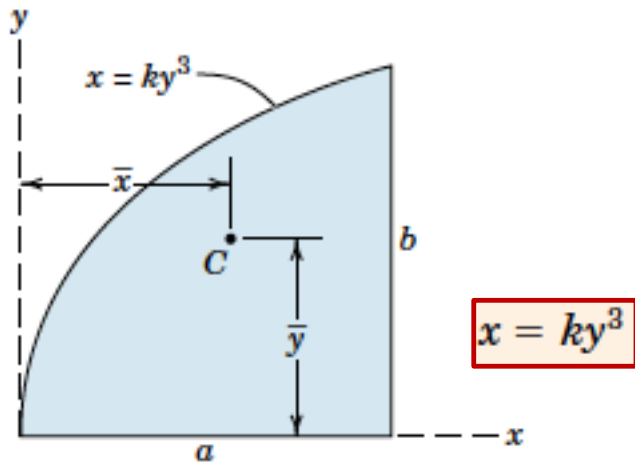
$$\bar{x} = \frac{\int x_c dA}{A} \quad \bar{y} = \frac{\int y_c dA}{A} \quad \bar{z} = \frac{\int z_c dA}{A}$$



$$\bar{x} = \frac{\int x_c dV}{V} \quad \bar{y} = \frac{\int y_c dV}{V} \quad \bar{z} = \frac{\int z_c dV}{V}$$

- **Example 1:** Locate the centroid of the area under the curve $x = ky^3$ from $x = 0$ to $x = a$.





$$[A\bar{x} = \int x_c dA]$$

$$\bar{x} \int_0^a y dx = \int_0^a xy dx$$

$$y = (x/k)^{1/3} \text{ and } k = a/b^3$$

$$\frac{3ab}{4} \bar{x} = \frac{3a^2b}{7}$$

$$\bar{x} = \frac{4}{7}a$$

$$[A\bar{y} = \int y_c dA]$$

$$\frac{3ab}{4} \bar{y} = \int_0^a \left(\frac{y}{2}\right) y dx$$

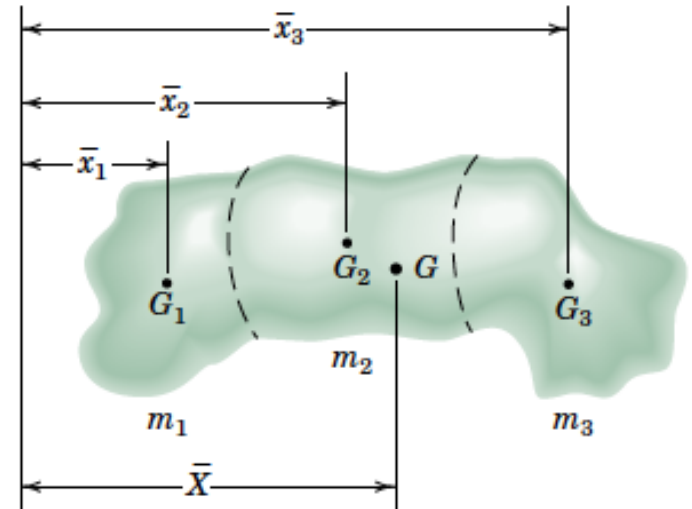
$$y = b(x/a)^{1/3}$$

$$\frac{3ab}{4} \bar{y} = \frac{3ab^2}{10}$$

$$\bar{y} = \frac{2}{5}b$$

COM of Composite Bodies And Figures

- When a body can be divided into several parts whose mass centers are easily determined,
- The moment principle gives



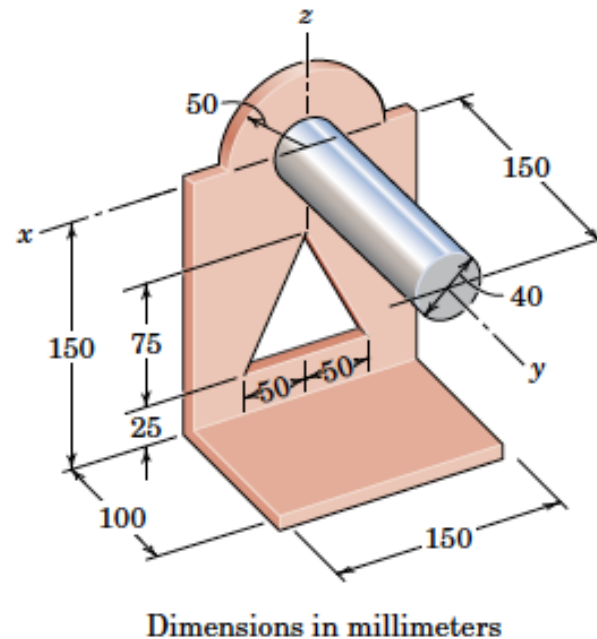
$$(m_1 + m_2 + m_3)\bar{X} = m_1\bar{x}_1 + m_2\bar{x}_2 + m_3\bar{x}_3$$

$$\bar{X} = \frac{\Sigma m\bar{x}}{\Sigma m}$$

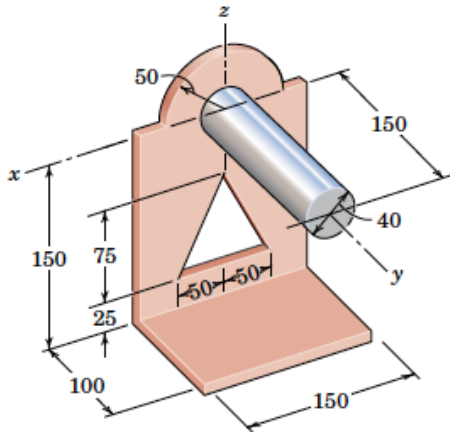
$$\bar{Y} = \frac{\Sigma m\bar{y}}{\Sigma m}$$

$$\bar{Z} = \frac{\Sigma m\bar{z}}{\Sigma m}$$

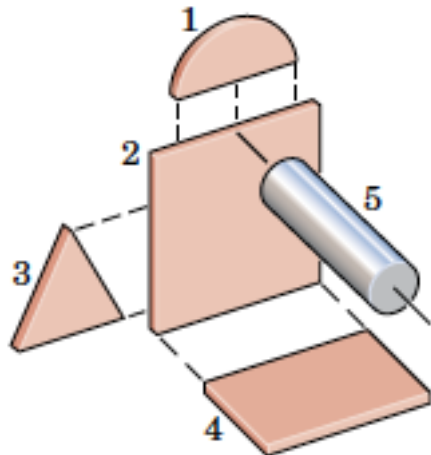
- Example 2: Locate the centre of mass of the bracket-and-shaft combination. The vertical face is made from sheet metal which has a mass of 25 kg/m^2 . The material of the horizontal base has a mass of 40 kg/m^2 , and the steel shaft has a density of 7.83 Mg/m^3 .



Example 2



Dimensions in millimeters



PART	m kg	$\bar{y}$ mm	$\bar{z}$ mm	$m\bar{y}$ kg·m	$m\bar{z}$ kg·mm
1	0.098	0	21.2	0	2.08
2	0.562	0	-75.0	0	-42.19
3	-0.094	0	-100.0	0	9.38
4	0.600	50.0	-150.0	30.0	-90.00
5	1.476	75.0	0	110.7	0
TOTALS	2.642			140.7	-120.73

$$\left[\bar{Y} = \frac{\Sigma m\bar{y}}{\Sigma m} \right]$$

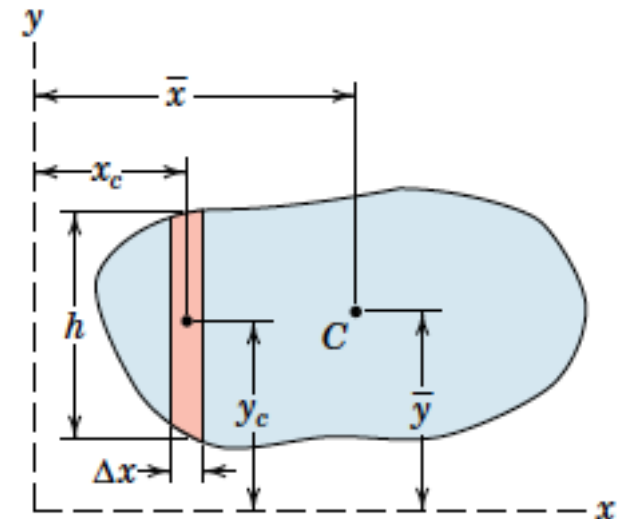
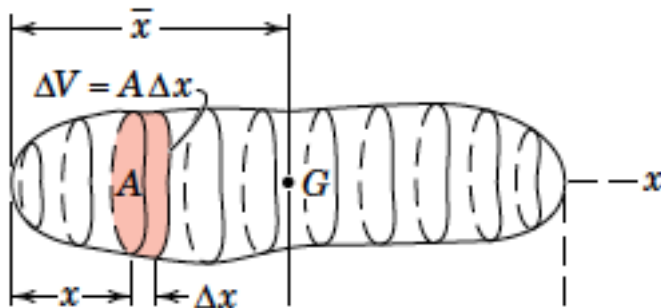
$$\bar{Y} = \frac{140.7}{2.642} = 53.3 \text{ mm}$$

$$\left[\bar{Z} = \frac{\Sigma m\bar{z}}{\Sigma m} \right]$$

$$\bar{Z} = \frac{-120.73}{2.642} = -45.7 \text{ mm}$$

Centroid an Approximate Method

- Boundaries of an area or volume might not be expressible in terms of simple geometrical shapes or as shapes which can be represented mathematically.
- For such cases we must resort to a method of approximation.

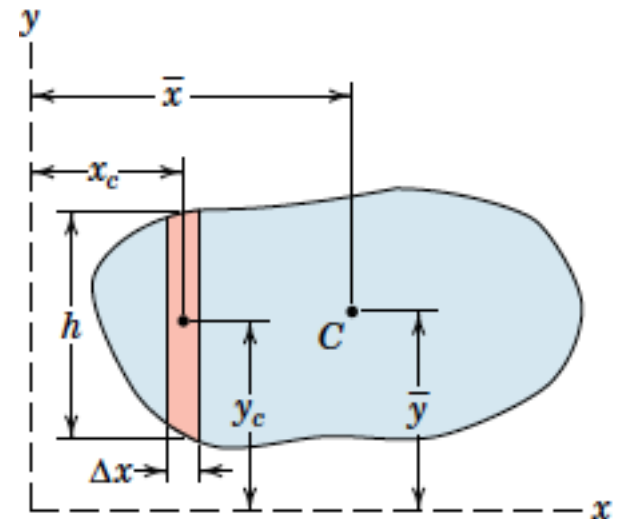


Centroid of Irregular Area

- The area is divided into strips of width Δx and variable height h .
- Area of strip $A = h \Delta x$
- Hence, Centroid
- Better accuracy with decreasing widths of the strips.

$$\bar{x} = \frac{\Sigma A x_c}{\Sigma A}$$

$$\bar{y} = \frac{\Sigma A y_c}{\Sigma A}$$



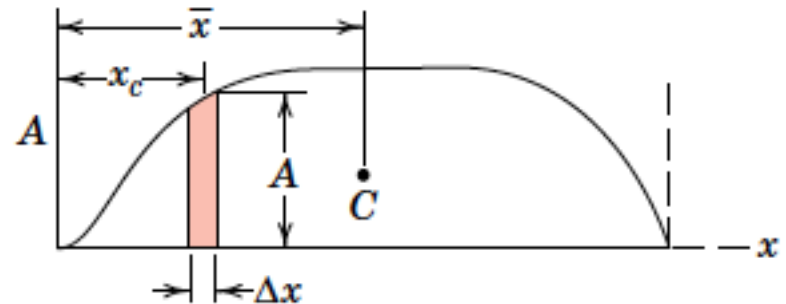
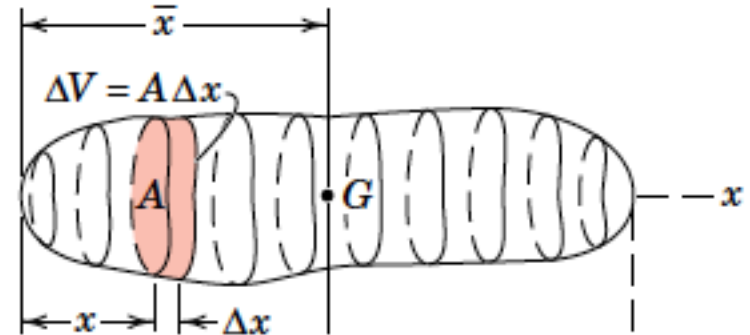
Centroid of Irregular Volume

- To locate the centroid of an irregular volume, we may reduce the problem to one of locating the centroid of an area
- Volume of the element

$$\Delta V = A \Delta x$$
- $\bar{X}$ = coordinate of centroid

$$\bar{x} = \frac{\Sigma(A \Delta x) x_c}{\Sigma A \Delta x}$$

$$\bar{x} = \frac{\Sigma V x_c}{\Sigma V}$$



THEOREMS OF PAPPUS

- A very simple method exists for **calculating the surface area** generated by revolving a plane curve about a nonintersecting axis in the plane of the curve.

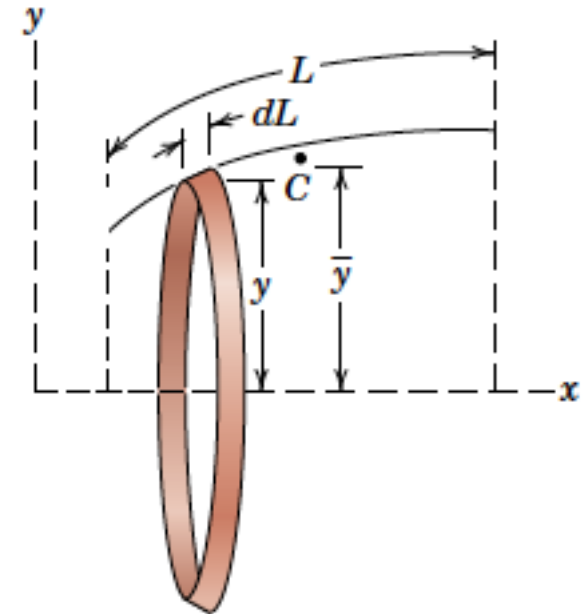
$$dA = 2\pi y dL$$

$$A = 2\pi \int y dL$$

Substituting

$$\bar{y}L = \int y dL$$

$$A = 2\pi \bar{y}L$$



THEOREMS OF PAPPUS: Volume

- In the case of a volume generated by revolving an area about a nonintersecting line in its plane,

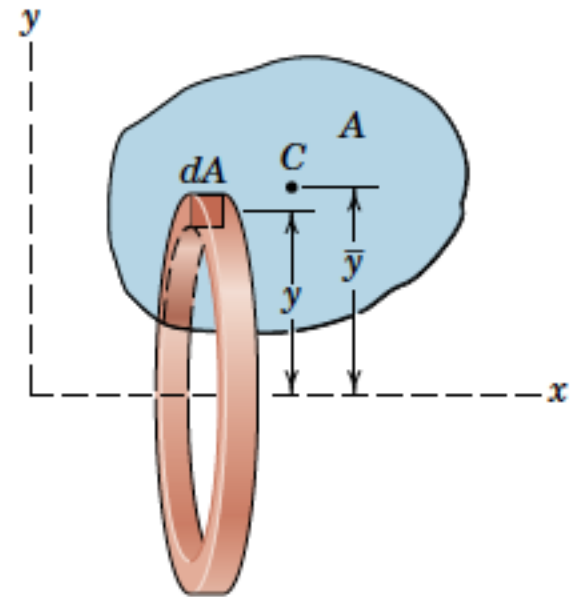
$$dV = 2\pi y dA$$

$$V = 2\pi \int y dA$$

Substituting

$$\bar{y}A = \int y dA$$

$$V = 2\pi \bar{y} A$$



THEOREMS OF PAPPUS

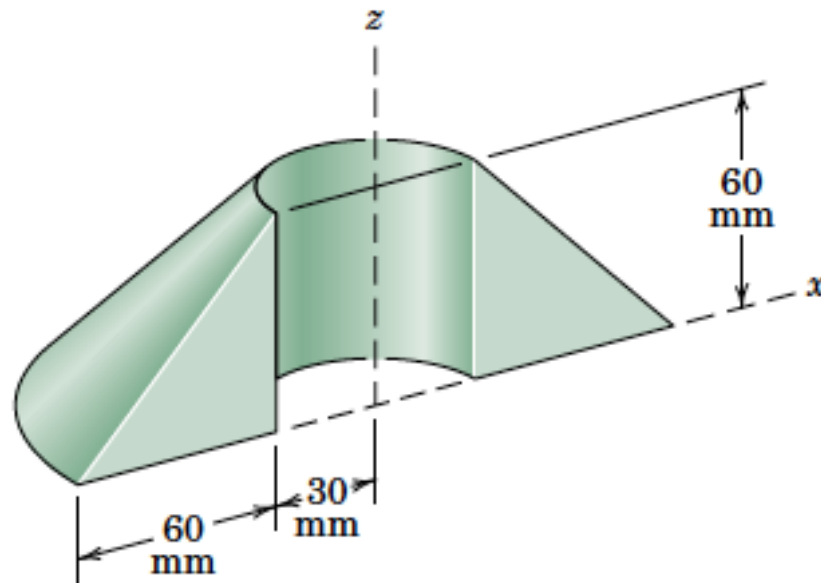
- If a line or an area is revolved through an angle less than 2π , we can

$$A = \theta \bar{y} L$$

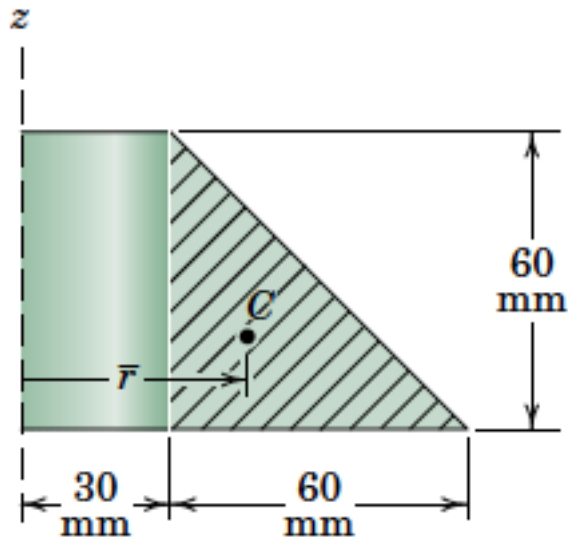
$$V = \theta \bar{y} A$$

where θ is in radian

- **Example 3:** Calculate the volume V of the solid generated by revolving the 60-mm right triangular area through 180° about the z -axis. If this body were constructed of steel, what would be its mass m ?



Example 3



$$V = \theta \bar{r} A$$

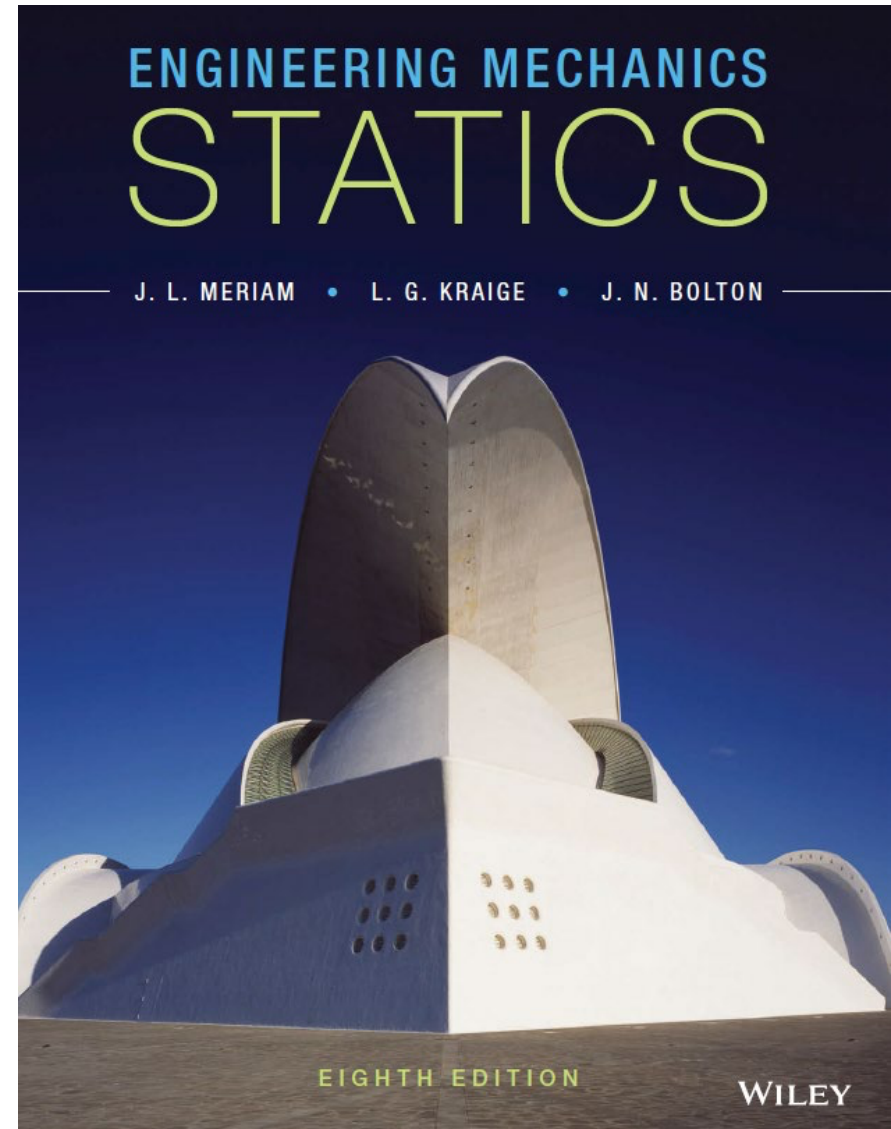
$$= \pi \left[30 + \frac{1}{2}(60) \right] \left[\frac{1}{2}(60)(60) \right] = 2.83(10^5) \text{ mm}^3$$

$$m = \rho V$$

$$= \left[7830 \frac{\text{kg}}{\text{m}^3} \right] [2.83(10^5) \text{ mm}^3] \left[\frac{1 \text{ m}}{1000 \text{ mm}} \right]^3$$

$$= 2.21 \text{ kg}$$

References :



THANK YOU



Engineering Mechanics

Lecture 9: Distributed Force and Friction

By

Dr. Bhupendra Singh

Assistant Professor

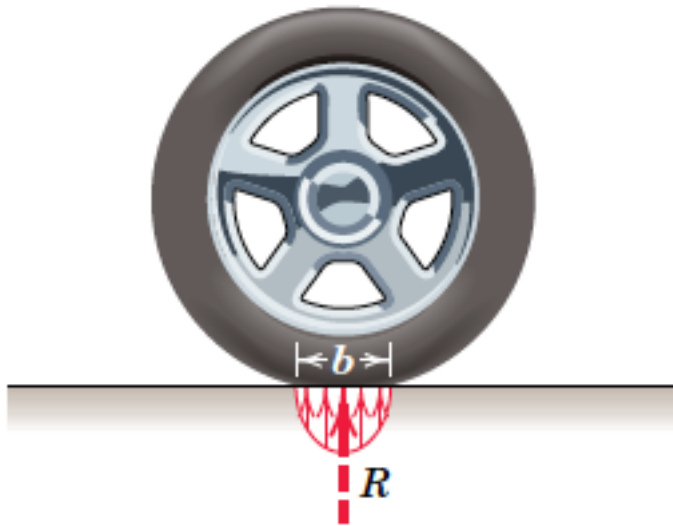
Civil and Infrastructure Engineering

Indian Institute of Technology Jodhpur

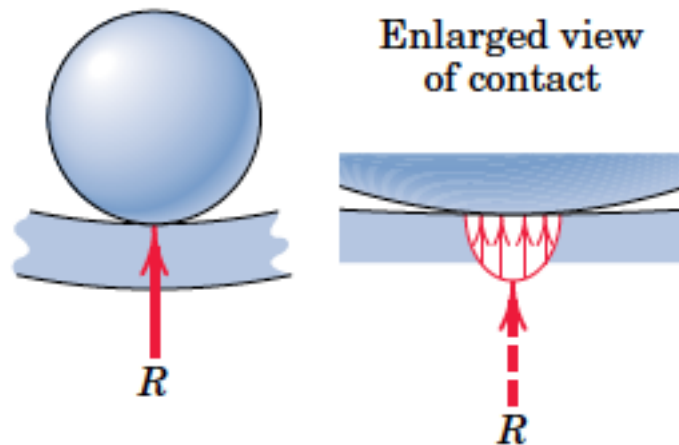
Distributed forces

- Earlier we treated all forces as *concentrated* along their lines of action and at their points of application.
- This treatment provided a *reasonable model* for those forces.
- Actually, “*concentrated*” forces do not exist in the exact sense
- Every external force applied mechanically to a body is *distributed* over *a finite contact area*, however small.

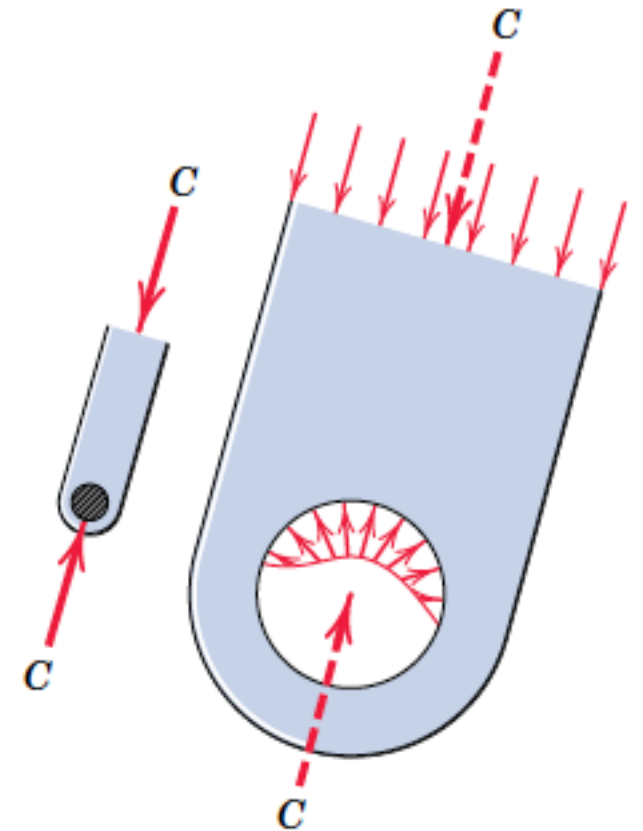
Distributed forces: Examples



An automobile tire



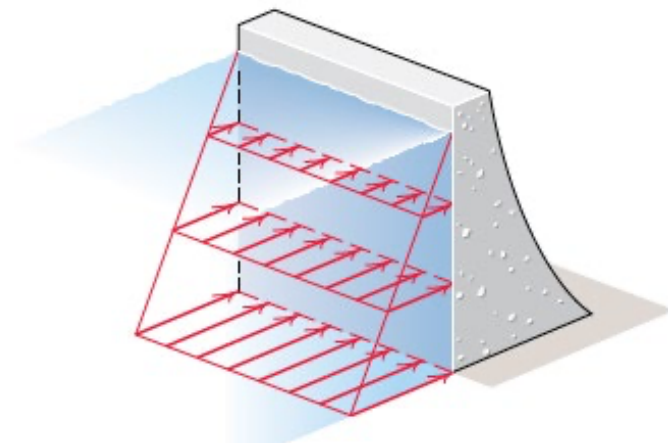
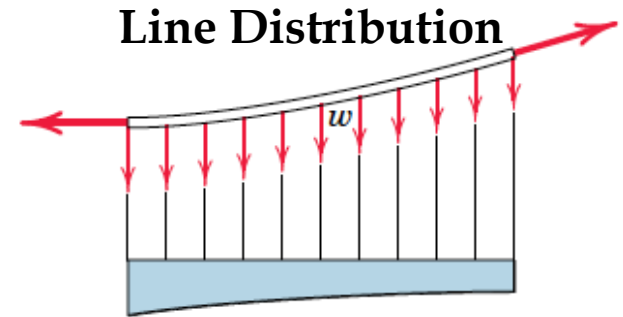
Hardened steel ball and its race in ball bearing



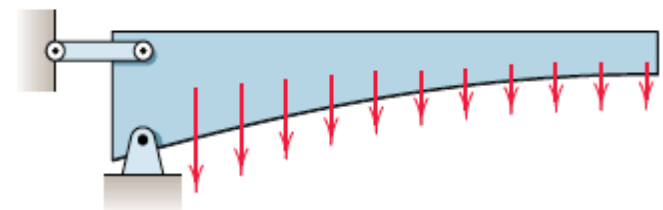
Pin at two-force member of a truss,

Type of Distribution

- **Line Distribution:** force is distributed *along a line*
 - Vertical load supported by a suspended cable
 - intensity w of the loading is expressed as *force per unit length*
- **Area Distribution:** force is distributed *over an area*
 - Hydraulic pressure of water against wall
 - Intensity is expressed as *force per unit area*
- **Volume Distribution:** force is distributed *over the volume* of a body (body force)
 - Gravitational attraction
 - Intensity is expressed as *force per unit volume*



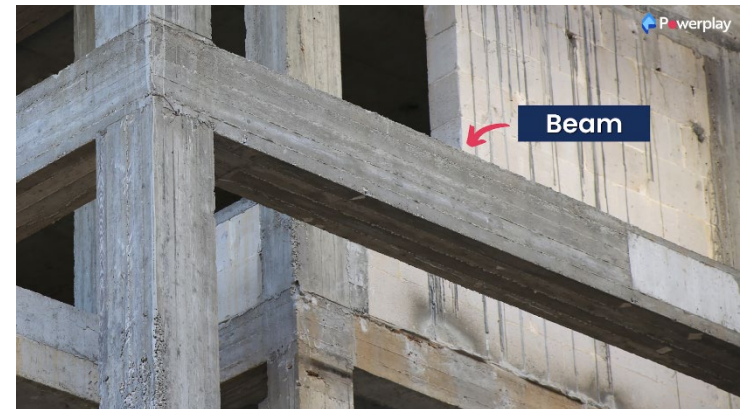
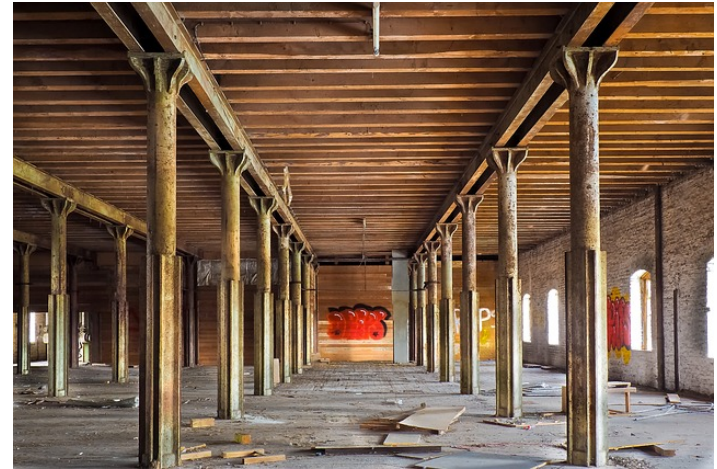
Area Distribution



Volume Distribution

BEAMS – EXTERNAL EFFECTS

- *Beams* are structural members which offer *resistance to bending* due to applied loads.
- Most beams are *long prismatic bars*, and the *loads* are usually applied *normal to the axes* of the bars.
- Beams are *often subjected to distributed load*.



Beam: A Prismatic Bar

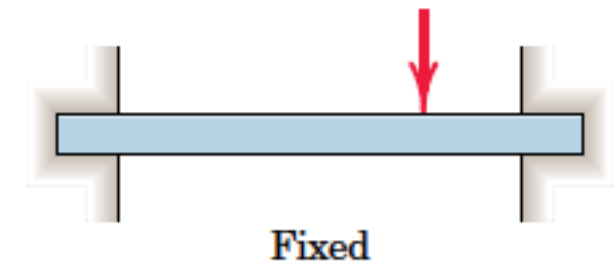
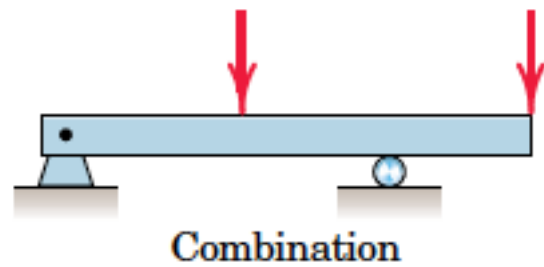
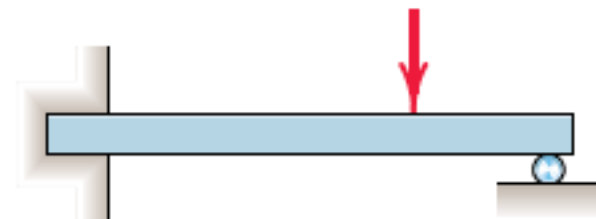
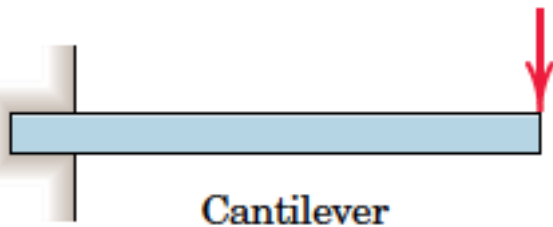
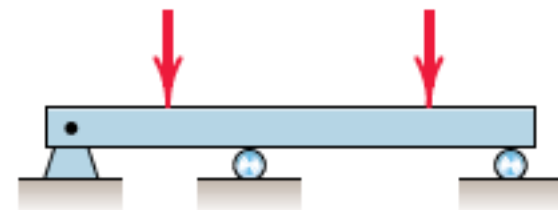
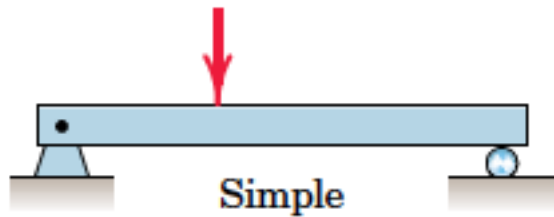


Prismatic bar

- It means the bar has a uniform cross-section.
- It is a structural member having a straight longitudinal axis and a constant cross-section throughout its length.
- The elongation δ of a prismatic bar subjected to a tensile load F is shown in the figure.
- The elongation is $\delta = \frac{FL}{AE}$
- The stress at cross-section is given as, $\sigma = \frac{F}{A}$



Types of beams



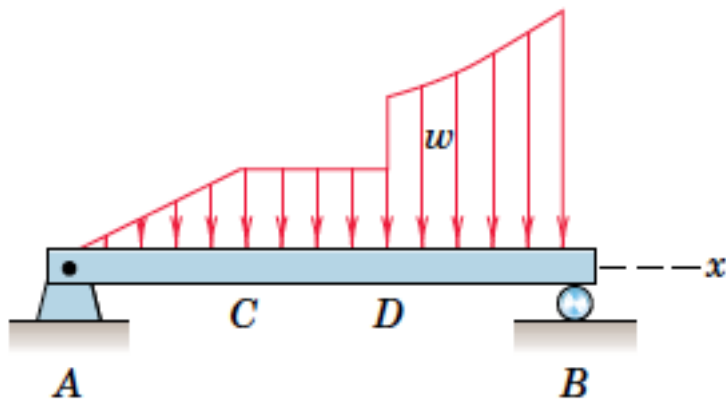
Statically determinate beams

Statically indeterminate beams

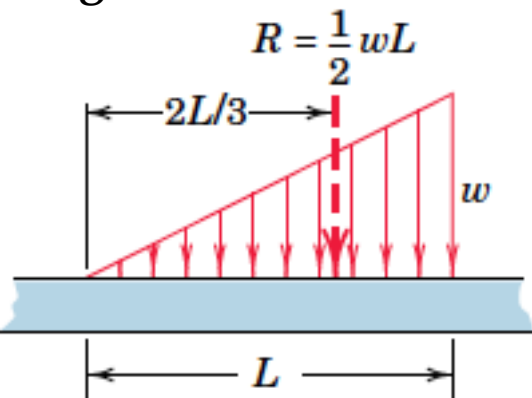
Beams under Distributed Forces

Solution:

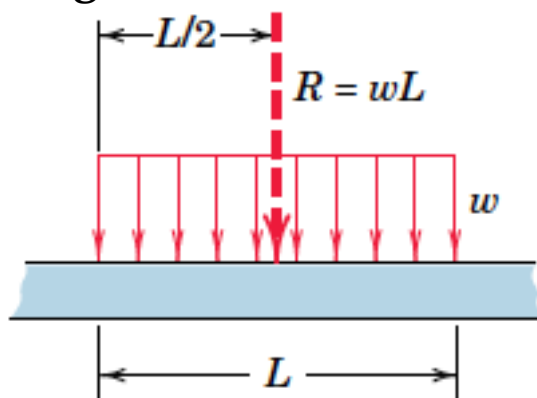
1. Given load intensity is divided into three regions
2. Finally the problem can be solved with three concentrated load **or** with overall load and centroid is obtained using method of Composite Bodies.



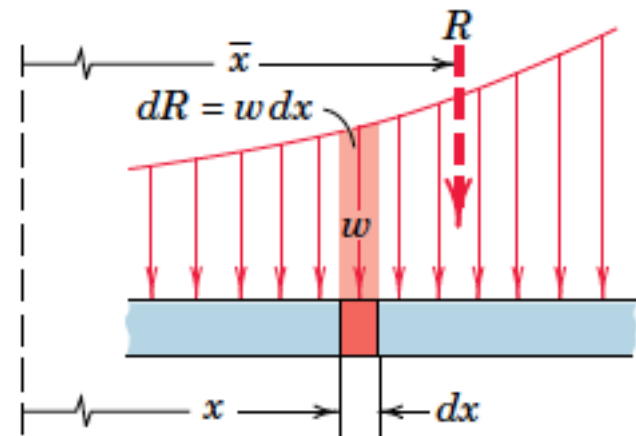
Region 1



Region 2



Region 3

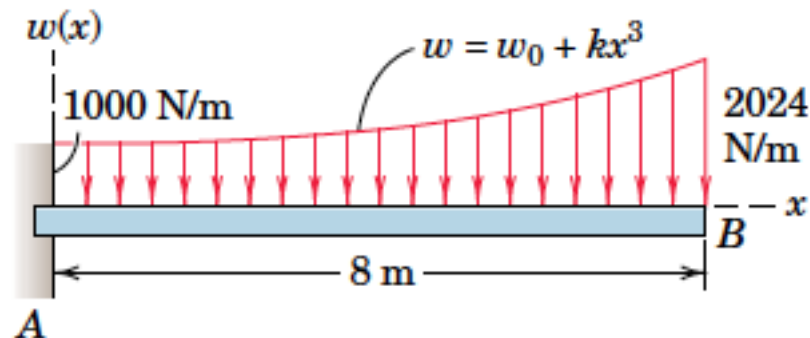


$$dR = w \, dx$$

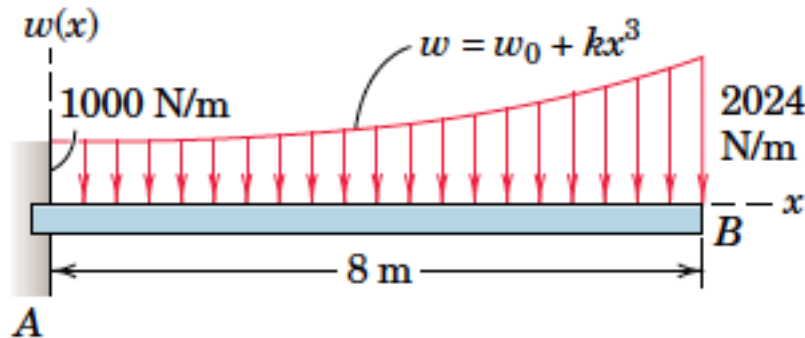
$$R = \int w \, dx$$

$$\bar{x} = \frac{\int xw \, dx}{R}$$

- Example 1: Determine the reaction at the support A of the loaded cantilever beam.



Example 1



Using boundary condition

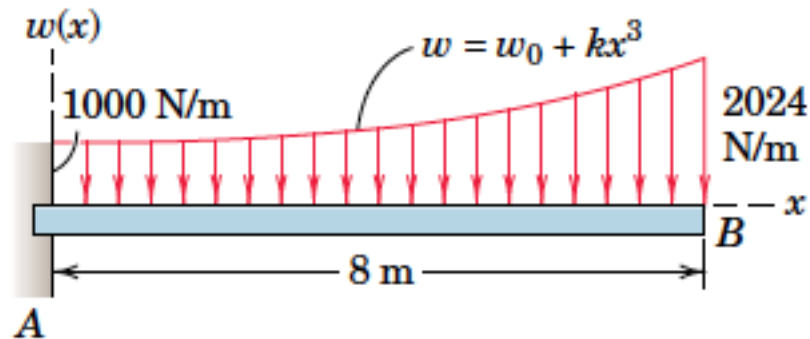
$$w_0 = 1000 \text{ N/m}$$

$$k = 2 \text{ N/m}^4$$

$$R = \int w \, dx = \int_0^8 (1000 + 2x^3) \, dx = \left(1000x + \frac{x^4}{2} \right) \Big|_0^8 = 10 \, 050 \text{ N}$$

$$\bar{x} = \frac{\int xw \, dx}{R} = \frac{1}{10 \, 050} \int_0^8 x(1000 + 2x^3) \, dx = \frac{1}{10 \, 050} (500x^2 + \frac{2}{5}x^5) \Big|_0^8 = 4.49 \text{ m}$$

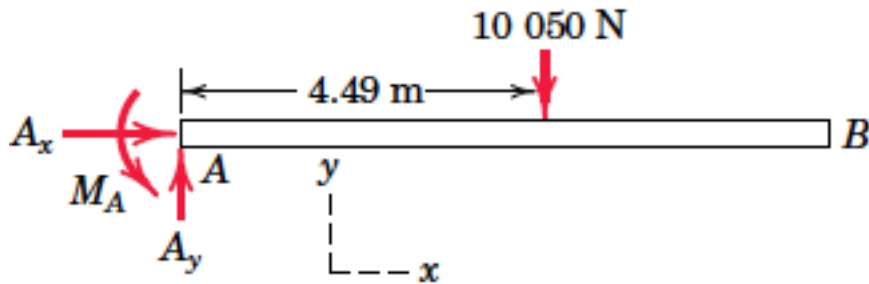
Example 1



$$[\Sigma M_A = 0]$$

$$M_A - (10\,050)(4.49) = 0$$

$$M_A = 45\,100 \text{ N}\cdot\text{m}$$



$$[\Sigma F_y = 0]$$

$$A_y = 10\,050 \text{ N}$$

$$A_x = 0$$

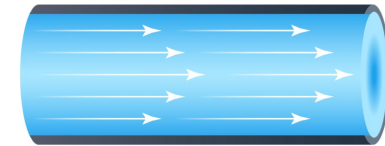
Friction

- So far we have assumed that the *forces of action and reaction* between contacting surfaces act *normal to the surfaces*.
- Many problems in which we must consider the ability of contacting surfaces to support *tangential* as well as *normal* forces.
- *Tangential forces* generated between contacting surfaces are called *friction forces* and occur to some degree in the interaction between all real surfaces.

Examples

Friction as disadvantage

- Bearings of all types, power screws, gears,
- the flow of fluids in pipes, and the propulsion of aircraft and missiles through the atmosphere



Friction as advantage

- brakes, clutches, belt, drives, and wedges



Types of Friction

- Dry Friction
- Fluid Friction
- Internal Friction

Types of Friction

- *Dry Friction*

- Dry friction occurs when the *unlubricated surfaces of two solids are in contact* under a condition of sliding or a tendency to slide
- A friction force *tangent to the surfaces* of contact
- The direction of this friction force always *opposes the motion* or impending motion.
- This type of friction is also called *Coulomb friction*.

Types of Friction

- *Fluid Friction*

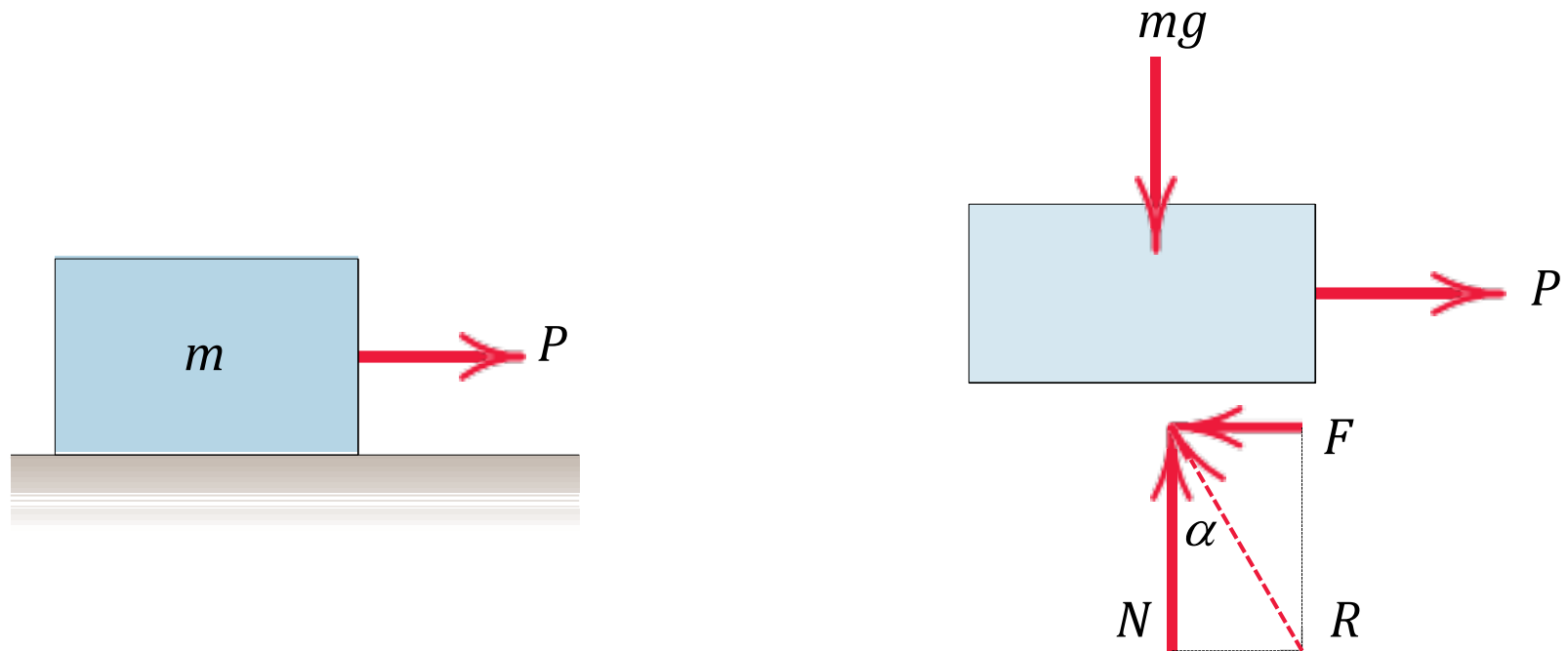
- It occurs when *adjacent layers in a fluid* (liquid or gas) are *moving at different velocities*.
- This motion causes *frictional forces* between fluid elements, and these forces depend on the relative velocity between layers.
- When there is *no relative velocity*, there is *no fluid friction*.
- Fluid friction also depends on *the viscosity of the fluid*, which is a measure of its resistance to shearing action between fluid

Types of Friction

- *Internal Friction*

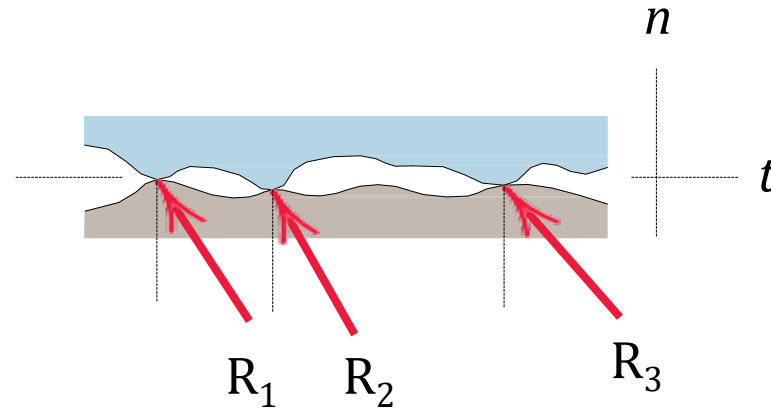
- Internal friction occurs in *all solid materials* which are subjected to *cyclical loading*.
- For *highly elastic materials* the recovery from deformation occurs with very little loss of energy due to *low internal friction*.

Mechanism of Dry Friction



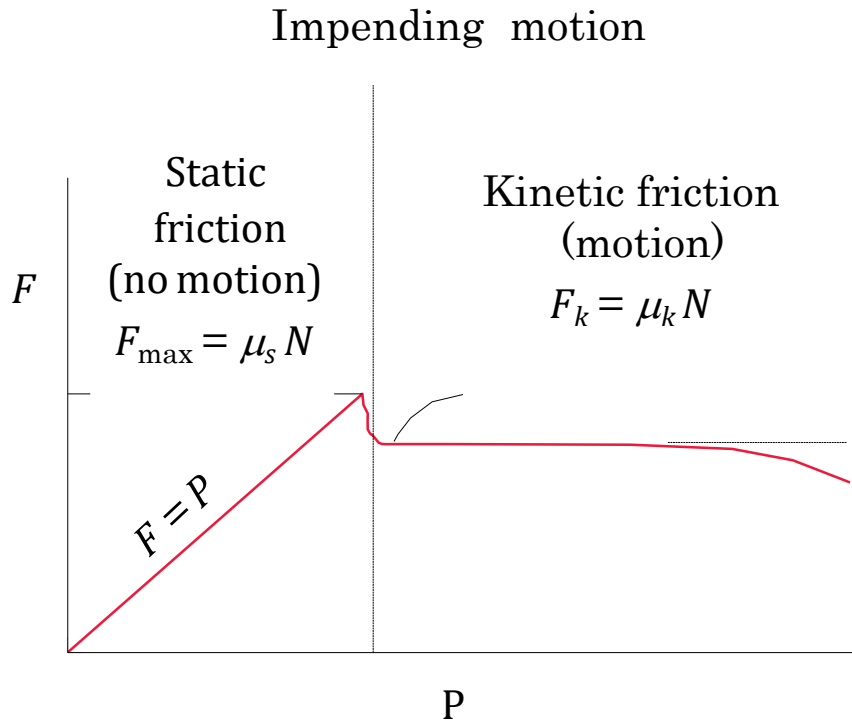
- P continuously increases from zero to a value sufficient to move the block and give it an appreciable velocity.

Irregularities of the mating surfaces



- The *total normal force* N is the sum of the *n -components* of the R 's, and the *total frictional force* F is the sum of the *t -components* of the R 's.
- Friction not only depend on *geometric profile* of the irregularities but also on the *extent of local deformation* at each contact point.

Static Friction



Motion is impending

$$F_{\max} = \mu_s N$$

Condition of static equilibrium when motion is not impending

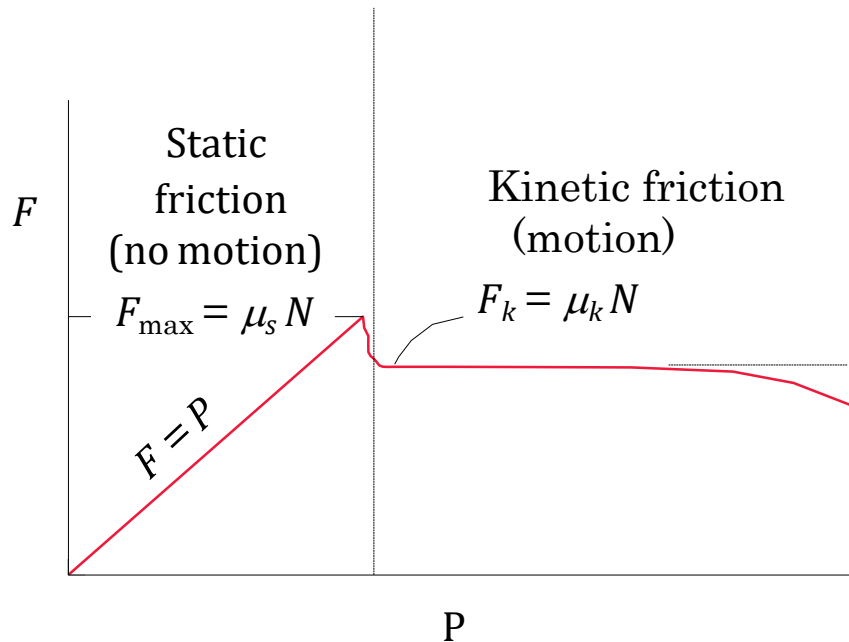
$$F < \mu_s N$$

μ_s is the coefficient of static friction

- The region in to the point of slippage or impending motion is called *the range of static friction*,
- In this range the value of the friction force is determined by the *equations of equilibrium*

Kinetic Friction

Impending motion



In Motion

$$F_k = \mu_k N$$

μ_k is the coefficient of kinetic friction

- After slippage occurs, a *condition of kinetic friction* accompanies the ensuing motion.
- Kinetic friction force is *usually somewhat less than* the maximum static friction force

Friction Angles

The direction of the resultant R measured from the direction of N is specified by

$$\tan \alpha = F / N$$

When the friction force reaches its limiting static value F_{max} , the angle α reaches a maximum value ϕ_s . Thus,

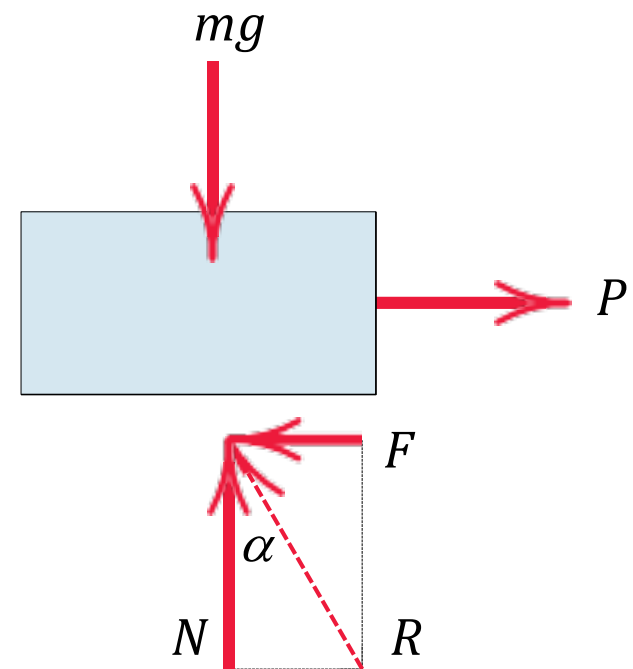
$$\tan \phi_s = \mu_s$$

When slippage is occurring, the angle α has a value ϕ_k

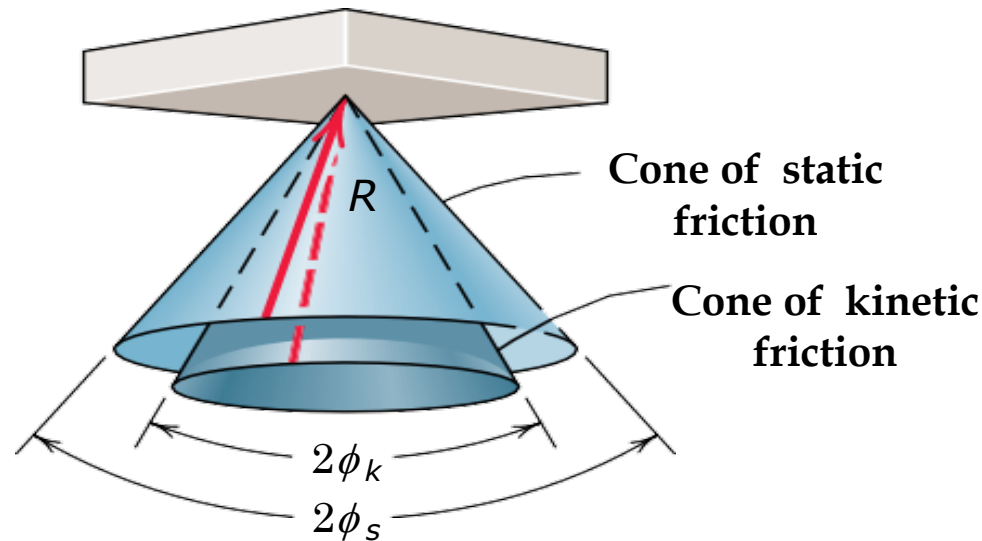
$$\tan \phi_k = \mu_k$$

In practice we often see the expression

$$\tan \phi = \mu$$



Friction cone

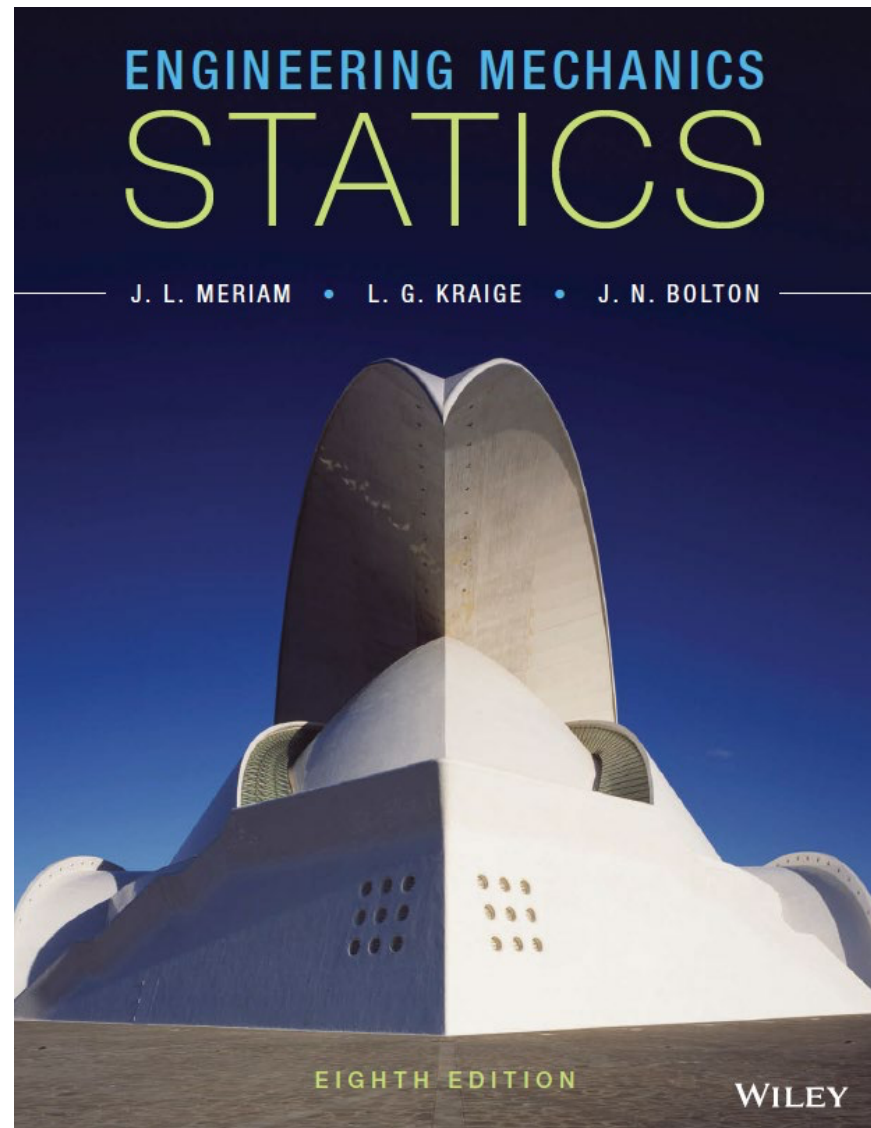


- **Motion is impending:** R must be *one element* of a right-circular cone of vertex angle $2\phi_s$,
- **Motion is not impending:** R is *within* the cone. This cone of vertex angle $2\phi_s$ is called *the cone of static friction*
- **Motion occurs:** the angle of kinetic friction applies, and the reaction must lie on the surface of a slightly different cone of vertex angle $2\phi_k$. This cone is the *cone of kinetic friction*.

Type of problems

- Case 1: Body which is in equilibrium is on the verge of slipping
 - Friction force: $F_{max} = \mu_s N$
- Case 2: relative motion is known to exist
 - Friction force: $F_k = \mu_k N$
- Case 3: neither the condition of impending motion nor the condition of motion is known
 - Under Static equilibrium: Friction force: $F < (F_{max} = \mu_s N)$
 - On the verge of slipping : Friction force: $F = (F_{max} = \mu_s N)$
 - In motion: $F > (F_{max} = \mu_s N)$, Friction force: $F_k = \mu_k N$

References :



THANK YOU



Engineering Mechanics

Lecture 10: Wedges

By

Dr. Bhupendra Singh

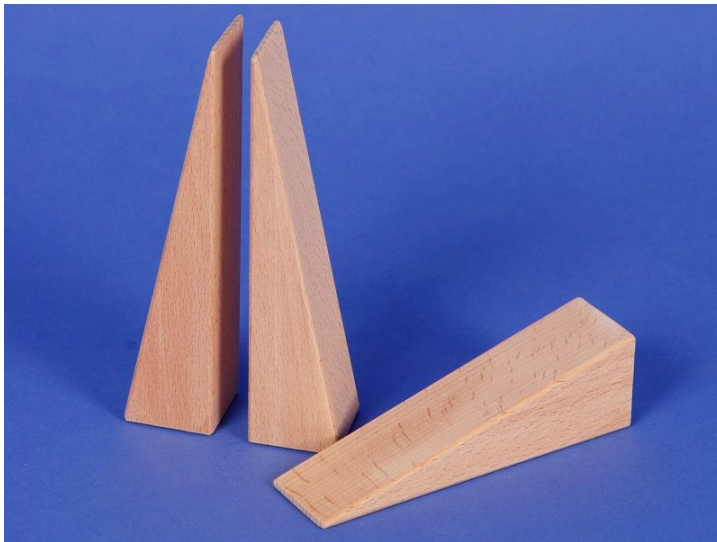
Assistant Professor

Civil and Infrastructure Engineering

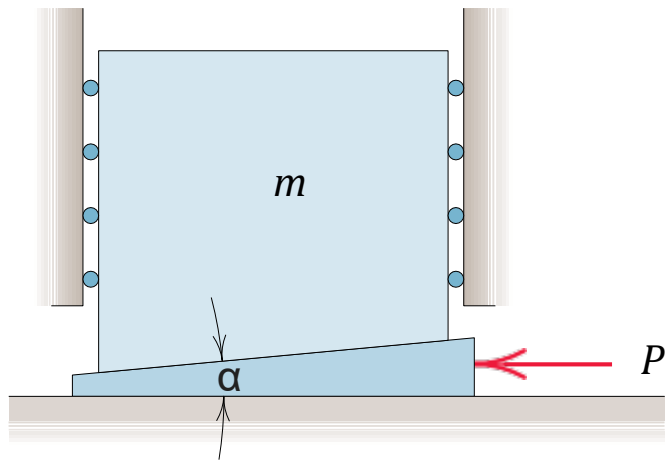
Indian Institute of Technology Jodhpur

Wedge

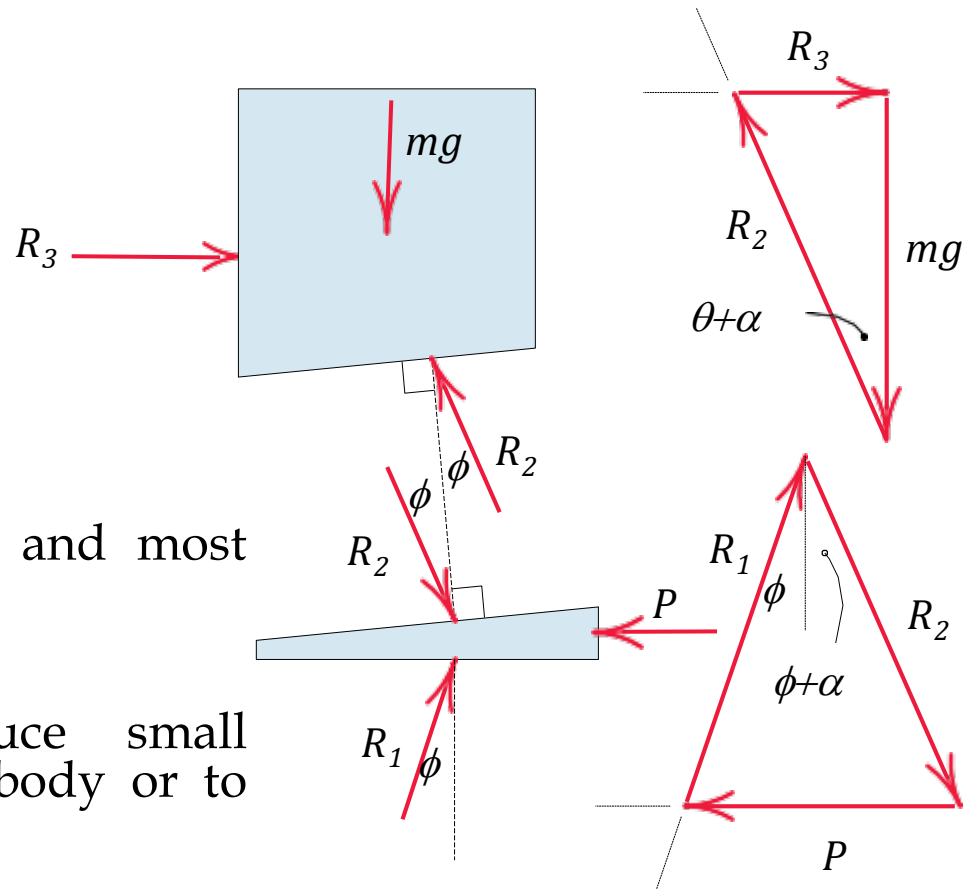
- A wedge is a triangular-shaped tool or object with a thin edge that tapers to a thicker end.
- It is used to split, lift, or secure objects by applying force, such as in cutting wood with an axe or lifting heavy objects.



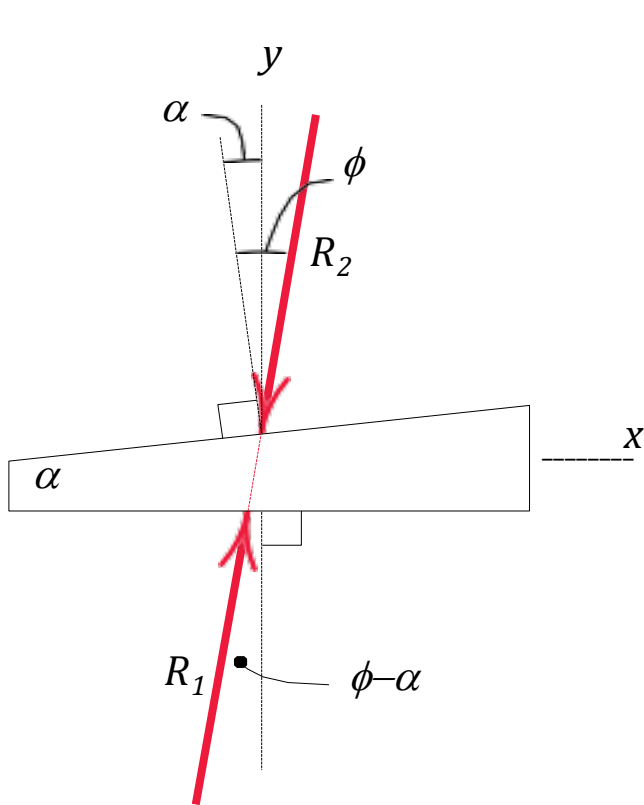
Wedge: Raising Weight



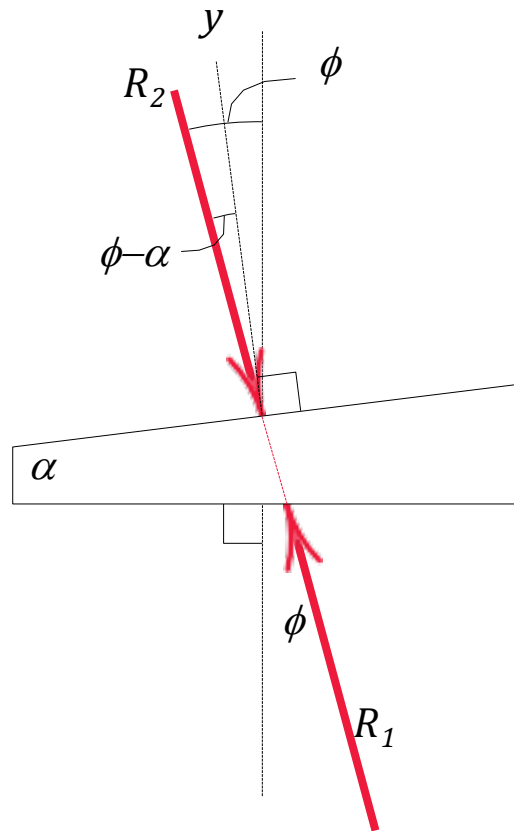
- A wedge is one of the simplest and most useful machines.
- A wedge is used to produce small adjustments in the position of a body or to apply large forces.
- Wedges largely depend on friction to function.



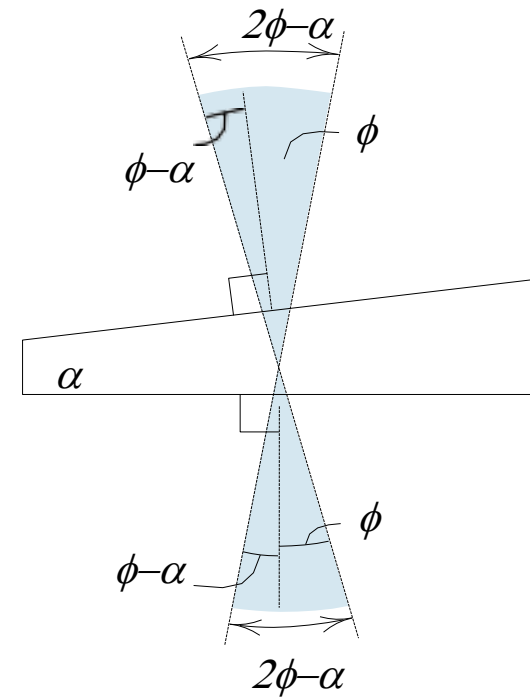
Force $P = 0$ ($R_1 = R_2$ for equilibrium)



Slipping impending
at upper surface

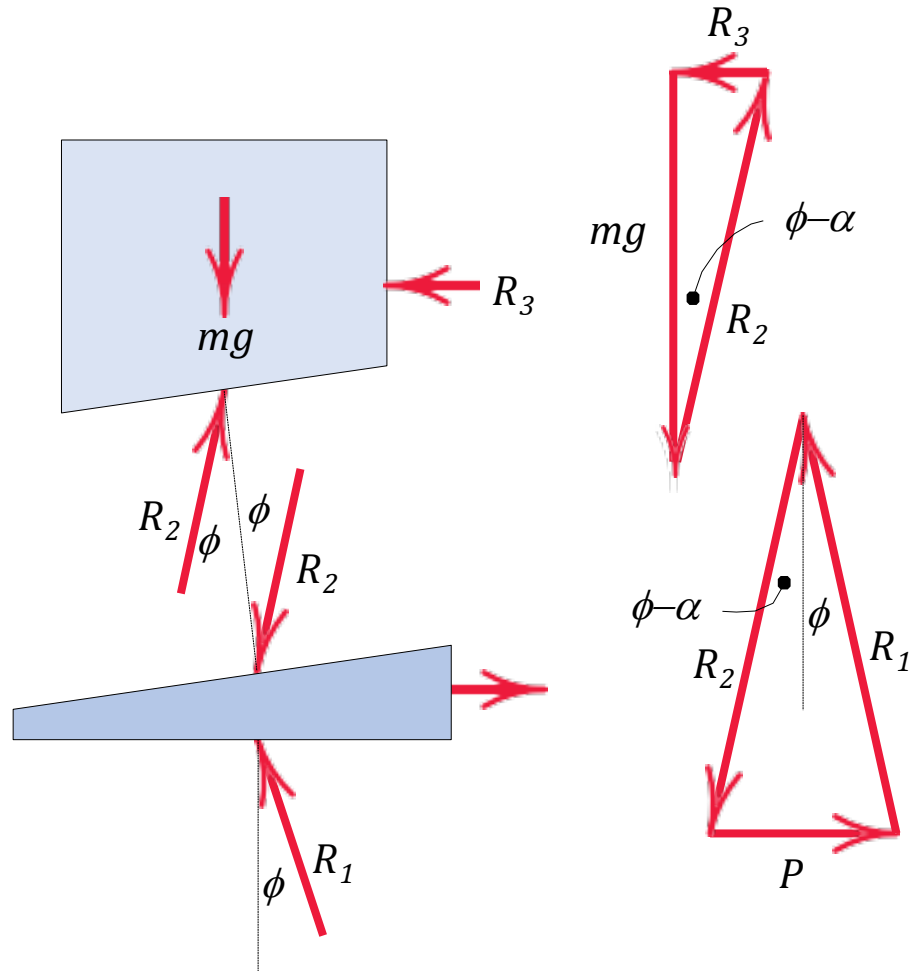
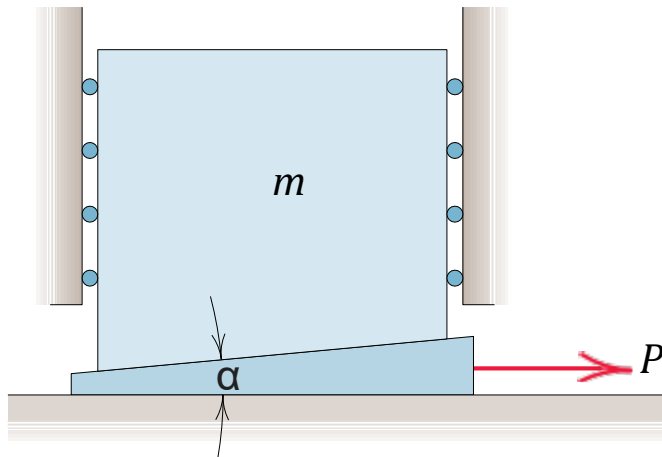


Slipping impending
at lower surface

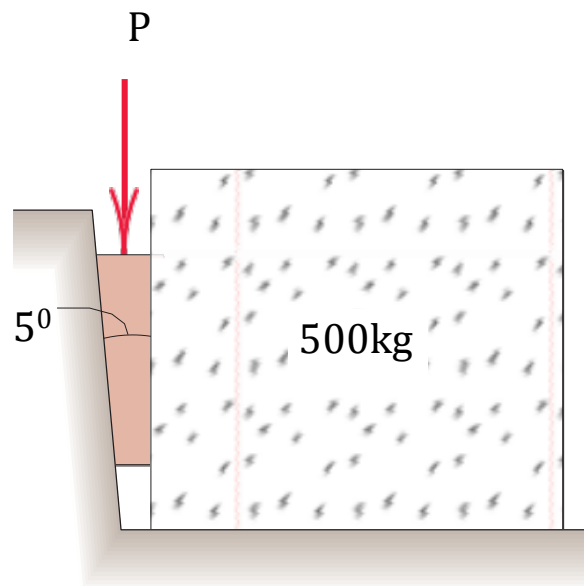


Range of $R_1 = R_2$
for no slip

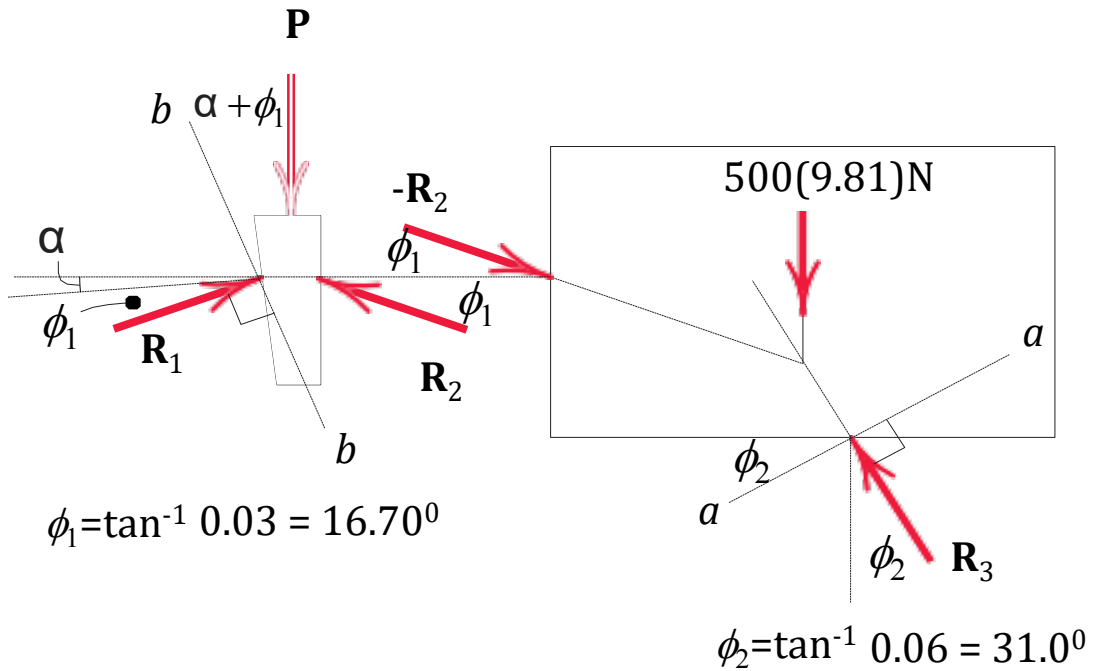
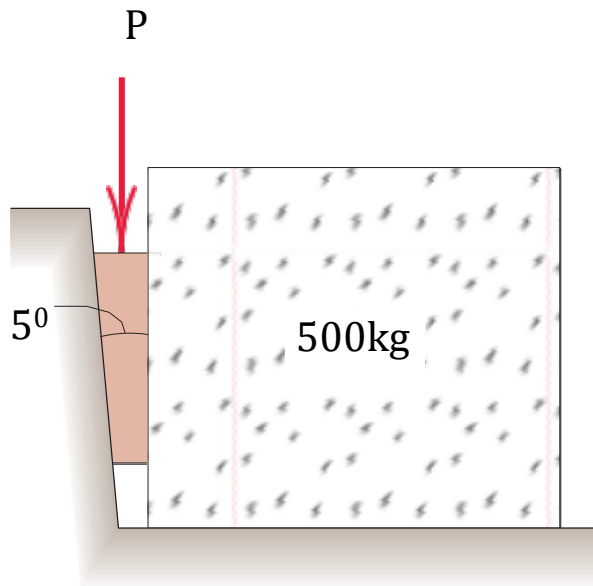
Wedge: lowering weight



- **Example 2:** The horizontal position of the 500-kg rectangular block of concrete is adjusted by the 5° wedge under the action of the force P . If the coefficient of static friction for both wedge surfaces is 0.30 and if the coefficient of static friction between the block and the horizontal surface is 0.60, determine the least force P required to move the block.



Algebraic solution



$$[\Sigma F_a = 0]$$

$$500(9.81) \sin 31.0^\circ - R_2 \cos 47.7^\circ = 0$$

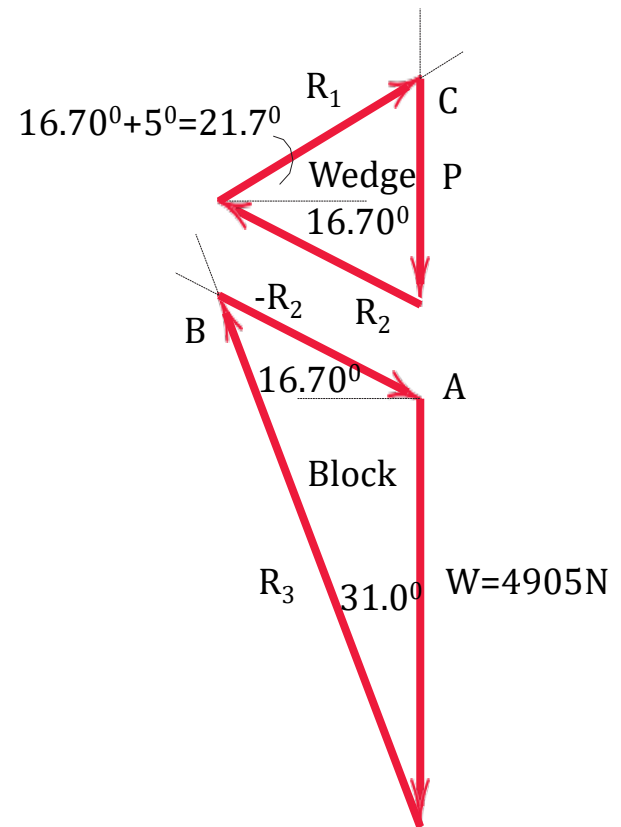
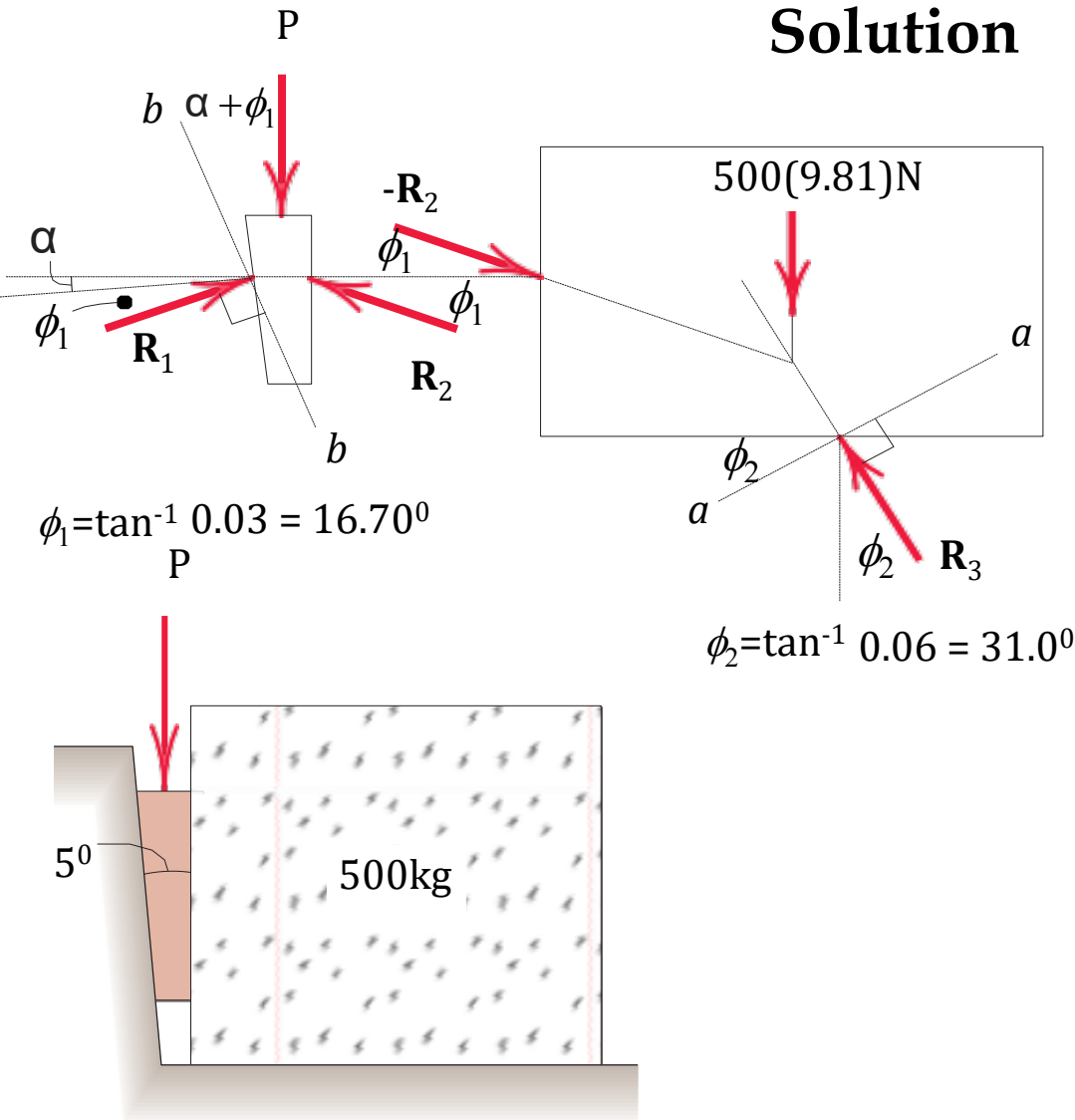
$$R_2 = 3750 \text{ N}$$

$$[\Sigma F_b = 0]$$

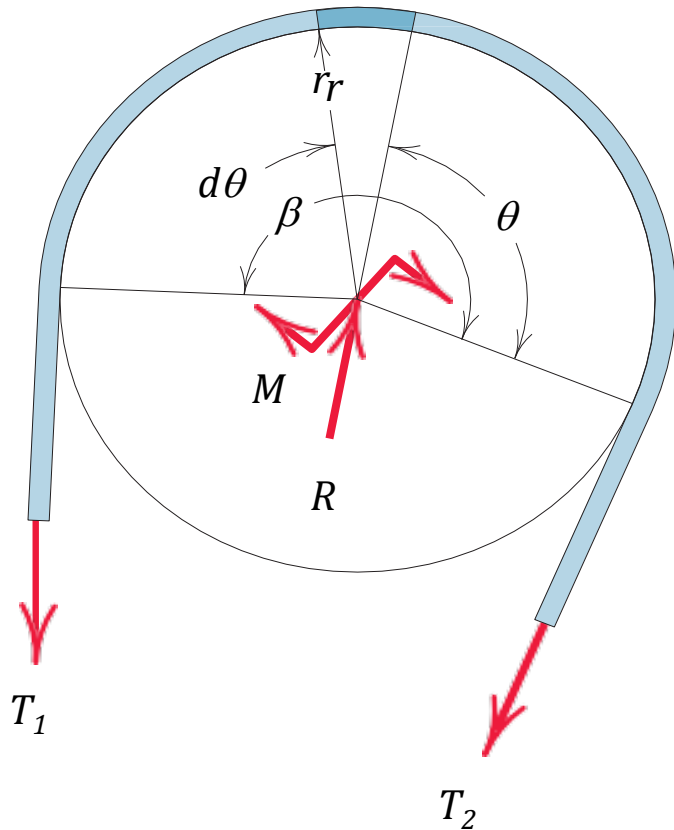
$$3750 \cos 51.6^\circ - P \cos 21.7^\circ = 0$$

$$P = 2500 \text{ N}$$

Graphical Solution

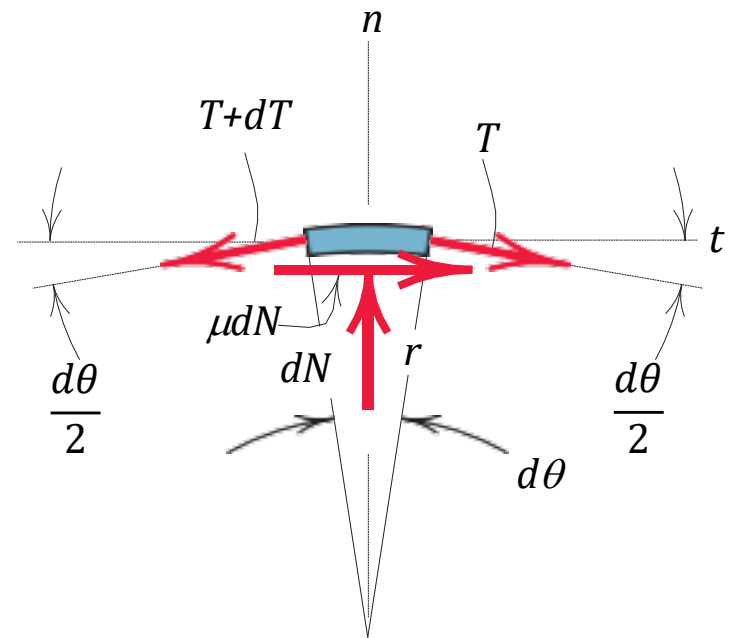
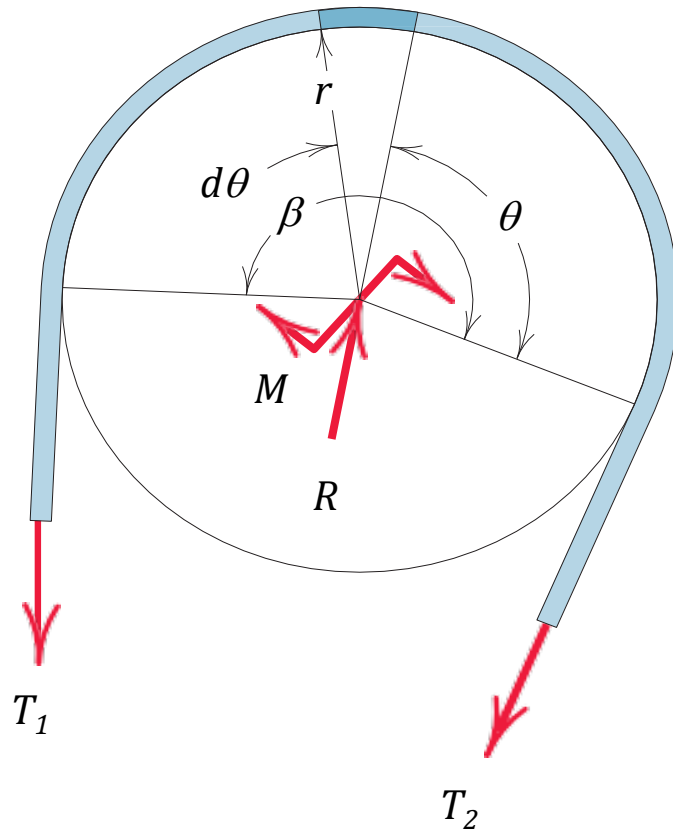


FLEXIBLE BELTS



- The impending slippage of flexible cables, belts, and ropes over sheaves and drums is important in the design of belt drives of all types, band brakes, and hoisting rigs.
- Drum is subjected to the two belt tensions T_1 and T_2 , the torque M necessary to prevent rotation, and a bearing reaction R .
- With M in the direction shown, T_2 is greater than T_1 .

FLEXIBLE BELTS



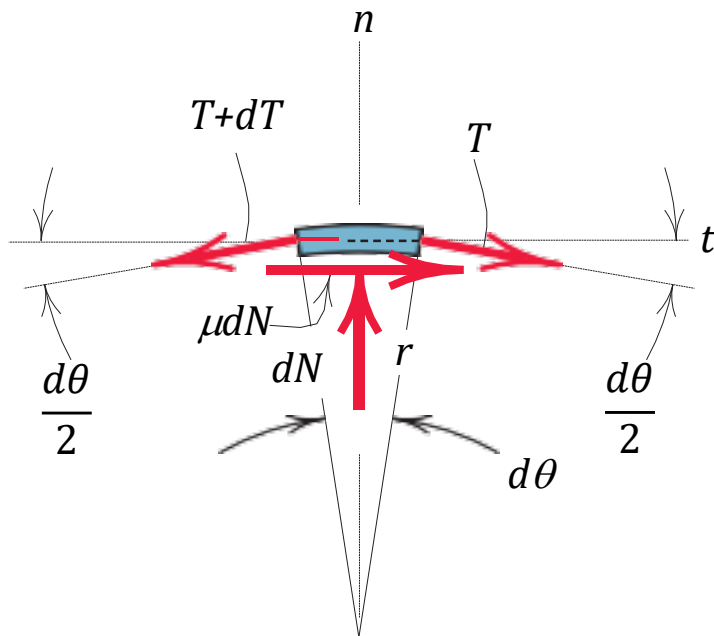
FLEXIBLE BELTS

t: $T \cos \frac{d\theta}{2} + \mu dN = (T + dT) \cos \frac{d\theta}{2}$

$$\mu dN = dT \quad \leftarrow \quad \cos \frac{d\theta}{2} = 1$$

n: $dN = (T + dT) \sin \frac{d\theta}{2} + T \sin \frac{d\theta}{2}$

$$dN = T d\theta \quad \leftarrow \quad \sin \frac{d\theta}{2} = \frac{d\theta}{2}$$



Combining both

$$\frac{dT}{T} = \mu d\theta$$

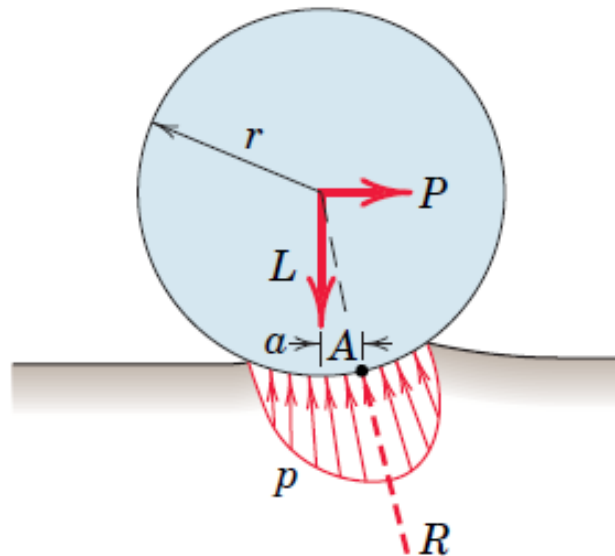
$$\int_{T_1}^{T_2} \frac{dT}{T} = \int_0^\beta \mu d\theta$$

$$\ln \frac{T_2}{T_1} = \mu \beta$$

$$T_2 = T_1 e^{\mu \beta}$$

ROLLING RESISTANCE

- Deformation at the point of contact between a rolling wheel and its supporting surface introduces a resistance to rolling,
- This resistance is not due to tangential friction forces
- Therefore is an entirely different phenomenon from that of dry friction.

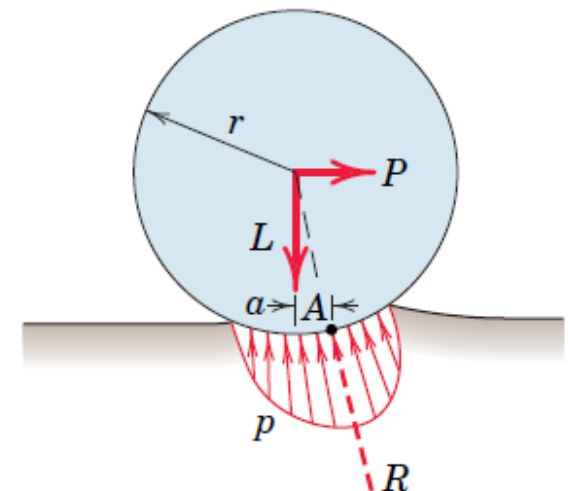


ROLLING RESISTANCE

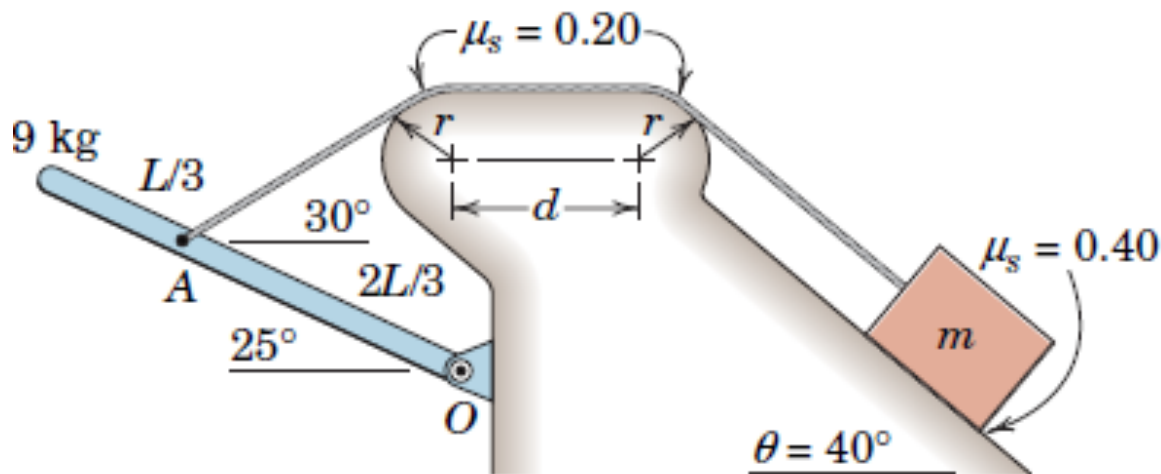
- The **resultant** R of this distribution acts at some point A and must **pass through the wheel center** for the wheel to be in equilibrium.
- We find the **force** P necessary to maintain rolling at constant speed by equating the **moments of all forces about A to zero**.
- This gives us

$$P = \frac{a}{r} L = \mu_r L$$

$$\mu_r = \frac{a}{r}$$



- Example 1: Determine the range of mass m over which the system is in static equilibrium. The coefficient of static friction between the cord and the upper curved surface is 0.20, while that between the block and the incline is 0.40. Neglect friction at the pivot O .



Motion of m impends up the incline.

$$[\Sigma M_O = 0]$$

$$-T_A \left(\frac{2L}{3} \cos 35^\circ \right) + 9(9.81) \left(\frac{L}{2} \cos 25^\circ \right) = 0$$

$$T_A = 73.3 \text{ N} = T_2$$

$$[\Sigma F_y = 0]$$

$$N - mg \cos 40^\circ = 0$$

$$N = 0.766mg$$

$$[T_2 = T_1 e^{\mu_s \beta}]$$

$$73.3 = T_1 e^{0.20[30^\circ + 40^\circ]\pi/180^\circ}$$

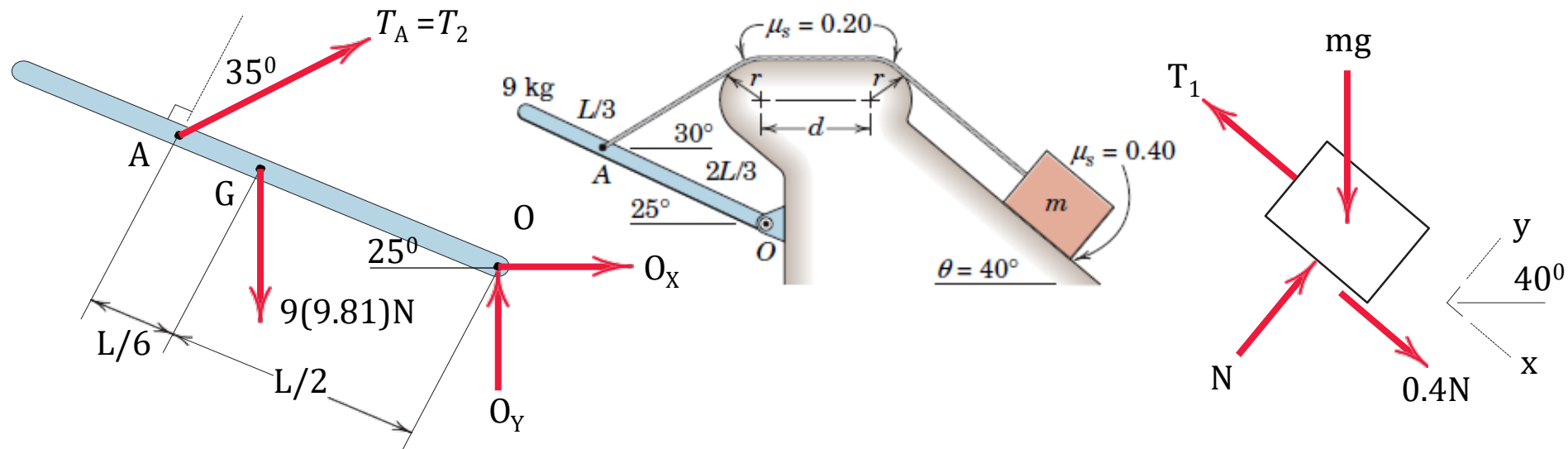
$$[\Sigma F_x = 0]$$

$$-57.4 + mg \sin 40^\circ + 0.40(0.766mg) = 0$$

$$T_1 = 57.4 \text{ N}$$

$$mg = 60.5 \text{ N}$$

$$m = 6.16 \text{ kg}$$



Motion of m impends down the incline

$$[\Sigma M_O = 0]$$

$$-T_A \left(\frac{2L}{3} \cos 35^\circ \right) + 9(9.81) \left(\frac{L}{2} \cos 25^\circ \right) = 0$$

$$T_A = 73.3 \text{ N} = T_1$$

$$[T_2 = T_1 e^{\mu_s \beta}]$$

$$T_2 = 73.3 e^{0.20[30^\circ + 40^\circ]\pi/180^\circ}$$

$$T_2 = 93.5 \text{ N}$$

$$[\Sigma F_y = 0]$$

$$N - mg \cos 40^\circ = 0$$

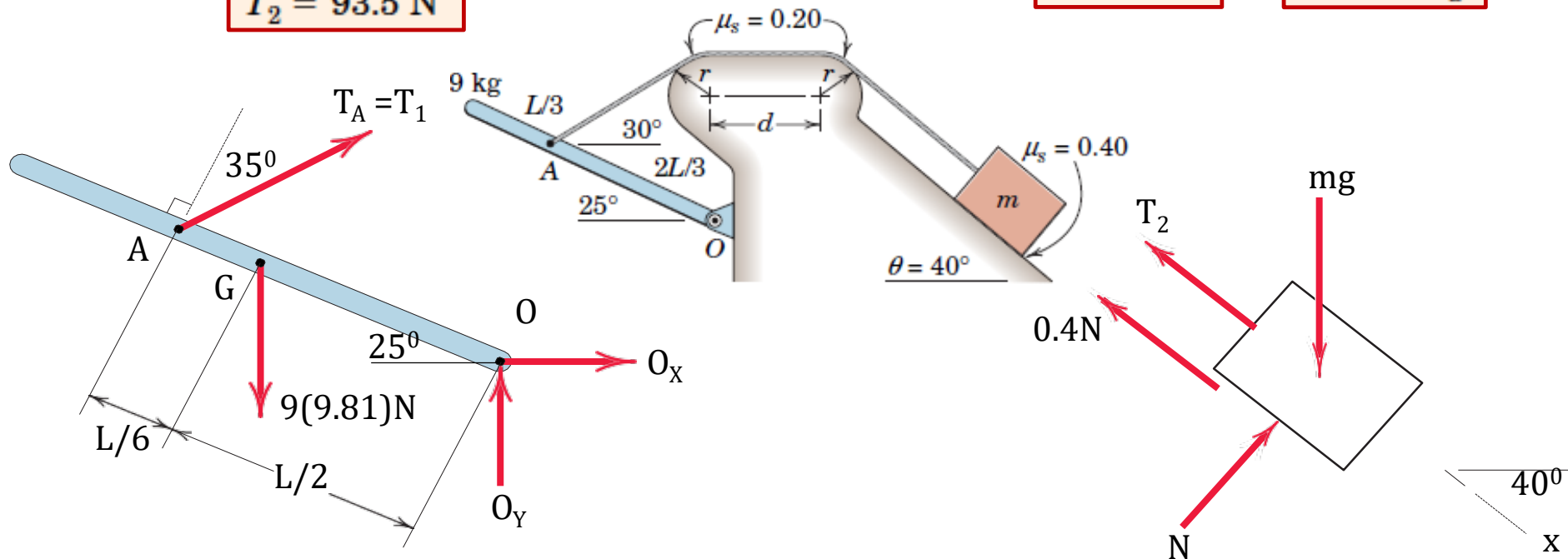
$$N = 0.766mg$$

$$[\Sigma F_x = 0]$$

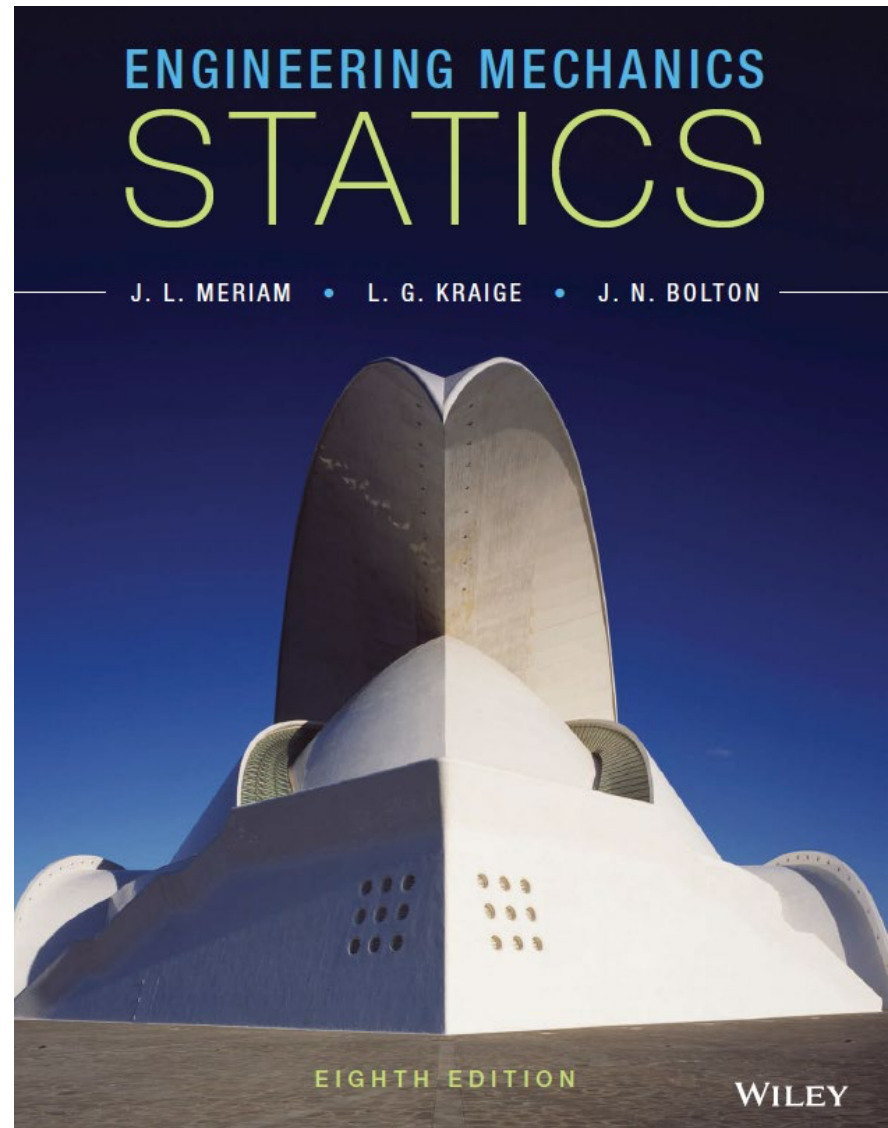
$$-93.5 - 0.4(0.766mg) + mg \sin 40^\circ = 0$$

$$mg = 278 \text{ N}$$

$$m = 28.3 \text{ kg}$$



References :



THANK YOU